

# **Photony Foundations**

Fred Lane Martin



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## ABSTRACT

Foundations of a novel theoretical framework are proposed that challenge conventional understandings of physics. The author proposes that the universe is fundamentally composed of "Photony energy filaments" that exist and interact within a structured yet flexible network called "quasi-static Planck Space." The dynamic behavior of these filaments, particularly their spin and interlinking, is posited to give rise to all known particles and forces, including gravity and electromagnetism. The theory also introduces the concept of a "compression factor,"  $4\pi$ , which is reasoned to be fundamental to numerous physical equations and phenomena. The book serves as a foundation for a series of "Photony Compendium" books that delve deeper into the mathematical and theoretical underpinnings of this new perspective on the Universe.

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# Introduction

My search for scientific “truth” began in 2014 based on a general need to learn more about science (physics) in the hope of understanding how the Creator designed the laws of nature to breathe life into the primordial Universe and govern its evolution. The theme of that effort soon had a title, Photony™ (Photony), partly because Photonics was already trademarked and partly because the sound of the term “Photony” grew on me.

Another motivator (as a student of EM field theory) was the ongoing irritation of not knowing what “charge” is or what a “line” is in the context of electric and magnetic (lines of) flux/fields. One of the driving forces for this effort was that no one I have read, studied under, or lectured could explain what “charge” is. As a teacher at the university level in electrical engineering, it was uncomfortable to teach about charge, charge flow (current), charge accumulation (voltage or potential), electric fields, magnetic fields, charge “carriers” (electrons), and forces (from Photony chains) without knowing what charge is. Based on the ill-defined definition of charge, one can only conclude that current, voltage, electric fields, magnetic fields, “carried” charge, and force are ill-defined even though most EM equations work well (accurate to several decimal places). How could this be? The answer: Many EM equations are merely testable approximations of reality.

## The Photony Discovery Story

The search for the truth required endpoint analysis, so it took me into the smallest of venues, Planck Space, and the largest of venues, the cosmos. The size and resolution of Planck Space seemed obvious to be at a scale far smaller than classical photon size. Ultimate nothingness, meaning the Planck Space structure's absence, seemed incomprehensible, so I left that as “TBD” and multiverse theorists. My best guess remains that there was always, even before time, a space structure that Photony calls “the Void.”

To paraphrase Dirac's words: *“If God exists, then he's a mathematician—not a plumber.”* [1] Au contraire, as we increase our understanding of the quasi-static Planck Space network of cells and flexible interconnecting passages for energy flow, we may begin to think that the Creator is more of a plumber.

Notwithstanding the Big Bang Theory, thoughts of our origin led to the questions: What first came “into the void” and “how did it enter the Void?” The goal herein was to get to a primordial era in space and time when primordial elementary quanta and particles appeared to precede an evolution of larger interlinked quanta and particles. So, I eventually imagined that energy transcended into the Void and hypothesized that energy was in the form of (Photony) energy filaments. So (skipping most of the details here), the filaments (keyring shaped, geodesic spinning and indexing at a common collision cell) begat spherical elemental photons as stable, static fundamental particles. These fundamental particles (static elemental photons) could become dynamic if a gap opens at the overlapping ends formed under pressure or an intra-cell collision, resulting in a high-speed forward helical motion. From the Photony point of view, these static particles and dynamic quanta became the basis of what has been universally and loosely called “charge.” The static and dynamic embodiments are the basis of dark matter and energy, respectively.

The expanding Universe began at “Creation” with energy coming “Into the Void.” This subject matter/theory called the “Photony Dawn” is further studied in the Photony Compendium. We will find that the reason why dark matter and dark energy have only recently been acknowledged, if not identified, resulting from groupthink acceptance of the Big Bang Theory. The Big Bang Theory has insisted that a quark-gluon plasma was sourced by a single singularity point (non-existent), thus the beginning of the Universe. By missing the most fundamental of quanta, Photony elemental charge photons at the beginning of the Universe, the stage was set for missing them until recent times.

The search for the physical nature and meaning of charge took me to the smallest of places, Planck Space, where I eventually modeled a quasi-static (flexible) network of cells that each have the smallest possible dimensions. Planck Length  $L_p = 1.61618148 \times 10^{-35}$  meters is cubed for cellular volume ( $4.2215232 \times 10^{-105} \text{ m}^3$ ). I did a virtual walk-through of orthogonal Planck Space after realizing that the Universe that we live within can be modeled with this structure of interconnected cells. The remaining material is too complicated to describe in an introduction. The “syrup” of the Higgs field is comparable to the flow of energy within quasi-static Planck Space with an effective viscosity described by the Photony Motion/Spin Rule, which essentially prohibits lateral motion of Photony energy filaments while favoring worm-like head-first motion in the essentially orthogonal Planck space network.

Upon determining that energy filaments maneuver within the Planck Space network, it became possible to extrapolate back to the primordial Creation (of the Universe). Planck Space is “pitch black” since classical photons are magnitudes larger than the Planck Length between cells. I adopted the phrase “Into the Void” to describe a creation theory that began with space (quasi-static Planck Space) and the birth of the Universe at the smallest of entrances (Planck Cells) with the smallest of quanta (energy filaments). Soon it was apparent that a newly conceived process was needed for which these energy filaments could become small 3-D particles and quanta. So, after theorizing several filament spins and motions, I settled on a circular geodesic motion. Still, I was unhappy with a simple looping motion since it did not lead to a real or virtual 3-D particle or quanta. So, after looking at various arc lengths, including some as less than 360-degree arcs and some arc lengths greater than 360 degrees with overlap, I settled on a keyring model with some head-to-tail overlap. Then, to obtain 3-D, I introduced longitudinal indexing at an arbitrary north pole.

This process produces geodesic motion and a second spin resulting from indexing and a spherical energy field or shell due to high-speed indexing. Note that the energy filament is spinning on its geodesic trace at much greater than the speed of light. The result of these two spin levels was a particle: a stable, static particle that I coined “elemental photon.” So, the Photony static elemental photon was born. Later I calculated its indexed spin rate.

After the formation of static elemental photons during creation, other phenomena had to happen; these elemental photons had to be capable of becoming dynamic to move effectively through space; otherwise, the Universe could not expand. This circumstance is where the keyring model came into

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vogue. Under certain conditions, including collisions and high pressure, the energy filament could open slightly at the overlap producing a gap. Once a gap is formed, the elemental photon immediately achieves traction and transforms its static motion into a helical motion through Planck Space. However, the orthogonal network enforces specific kinematics upon the dynamic elemental photon. This consideration gave rise to the Photony Motion/spin rule covered in this book. At this stage of the creation process, there was a static elemental particle and a dynamic elemental particle (energy quanta) in the primordial particle zoo. Using equations involving Planck's constant, following the lead of Miles Mathis, I calculated the transitional IR photon's spin rate to be 37.6 THz in the spectrum's infrared band several years before writing this book.

From these Photony static and dynamic elemental particles, I imagined other collisions and high-pressure scenarios whereby each of the two colliding particles' filaments could interlink as at least one filament gap opened enough to form the link. From that combination of interlinked-filament spin motions emerged a larger non-spherical particle (quanta). Yes, it was the classical photon of Einstein's thought experiments. This classical photon's combined dual filament tracings are that of a peanut shell, which can be readily polarized. Also, other combinations of sub-spins at this level resulted in neutrinos. So, the primordial particle zoo grew.

Next (after classical photons) came electrons and positrons as gamma rays (classical photons of high frequency and energy) collided, causing their forward linear motion to become rotational by adding a stacked C-spin, called particle (pair) production by mainstream physics. This phenomenon is reciprocal because a positron-electron collision can result in their annihilation, and two gamma rays result. So, the primordial zoo grows. The primordial elemental kinematic processes continued until the Universe became what we see today as old light and energy and continues to build and expand beyond our horizon. Photony explains the emergence of all particles in this book and in more detail in the Photony Compendium.

The Electricity and Magnetism, Fields, and Force {EMFF}, Photony Energy Science, and Photony Unification were sequenced (delayed) to first develop the foundations of the two other Unification Classes fully (Space, Gravity, Energy, Mass, Time {SGEMT} and Quanta, Particles, and Chains {QPC}). One of the original goals was to answer what is "charge" in the EM context. It was serendipity that the same static elemental particle and its dynamic quanta counterpart became the basis of dark matter/energy, EM, and gravitational phenomena, all integral to Unification.

For an individual such as me to claim discovery of Unification and want to write a Ph.D. dissertation on the same does not pass the "giggle test," according to one of my advisors at the University of Tennessee. That comment has only motivated me, even more, to proceed with my efforts to prove Unification based on Photony Theories.

## From Elemental Photons to Chains

(Reflecting on Photony Emergence, Interlinking, and the importance of chains)

I remember the excitement I felt when I realized that the flexible tubes of force pioneered by Faraday and Maxwell were similar to the electric and magnetic chains of Photony. The 350-year-old theory of flux lines as flexible tubes appeared to be a direct fit for Photony chains. Another excitement came when I discovered how free electrons created magnetic and electric chains in the toroidal cavity or locus of electrons. The chains emerged through the opposite face and wrapped around the electrons' bodies as magnetic chain loops or emerged as single-ended electric chains based on the free electrons' location and toroidal face-aligned orientation at the surface.

Free, windsock-aligned electrons produce a static magnetic (magnetostatic) field of chains within a conductor. Each internal free electron becomes axially aligned by the direct-current flow (wind) of dynamic elemental photons internal to a conductor. As the dynamic elemental photons flow through the toroidal cavity of an electron near the electron's point-source locus, the electron transduces (interlinks) the elemental photons into chain loops that wrap around the toroidal body of the electrons. These chains build in numbers and lengths based on the magnitude of the current producing the magnetic field. The magnetic field within the conductor increases linearly from the center to the surface of the conductor. The magnetic field (the number of magnetic chains) reduces from the conductor surface outward according to the inverse square law applied to the radial distance away from the conductor. The charge flow of Photony elemental dynamic particles produces toroidal face-alignment of free electrons in a conductor and pushes them along **slowly** due to interactions, including collisions.

Free, windsock-aligned surface electrons produce a static electric (electrostatic) field of chains at or near a conductor's surface. Each free surface electron becomes axially aligned by the charge potential of dynamic elemental photons as they collide with a surface and are converted to static elemental photons. As the dynamic elemental photons flow through the toroidal cavity of an electron near the electron's point-source locus, the electron transduces (interlinks) the elemental photons into a single-ended chain that exits through the toroidal body of the electrons as tentacles. These chains build in numbers and lengths based on the magnitude of the voltage producing the electric field. Note that there is no electric field within the conductor. From the conductor surface outward, the electric field (the number of electric chains) reduces according to the inverse square law applied to the radial distance away from the conductor. The DC (electric) charge field of Photony elemental dynamic particles maintains toroidal face-alignment of free electrons at or near the conductor surface.

Dynamic (AC) Electromagnetic Fields are produced by periodic reversals of charge flow. On one end of the spectrum is the "static" charge of DC transitioning to low-frequency signals and extending upward in frequency through radio waves and microwaves. The EM spectrum is informative regarding the frequency and wavelength of electromagnetic fields. The reader will discover that the transition between radio/microwave frequencies and far infrared photons is 37.6 THz, and this transition relates to the "light of life" tenability range in Wein's law and, more specifically, the boiling point of water. All life on earth depends on this "light of life" temperature range corresponding directly to Wein's 8-14 microns infrared light. Human body temperature is 98.6 °F which translates to 9.3 microns and 18.2 THz using Wein's Law.

# Photony Foundations

## From EM Chains to Classical Photon Chainlets

Regarding Alternating current and voltage, what a free electron in a conductor can do during AC is extrinsic rotation about the polar axis (axial end-over-end or **x-spin**). Alternating current (charge) flow results in interlinked chains of elemental photons **sinusoidal chain segments that vary in length proportionally to the electron's x-spin**. The internal electrons synchronously produce sinusoidal magnetic chain loops. In contrast, the surface electrons produce sinusoidal single-ended electric chains as tentacles exiting outward (seeking termination), essentially normal to the conductor surface.

A sinusoidally modulated field of orthogonal interconnected chain loops through space, pinched into EM loops by the electrons as they complete each 360-degree axial (end-over-end) rotation. As the AC frequency increases, the electron-transduced magnetic and electric chains decrease in length through the reduction process. Photony explains what happens with electric and magnetic chains as the frequency reaches a behavior limit in the spectrum's far-infrared portion. As the frequency transitions at this far-infrared limit, the EM wave's power and energy transition from amplitude-based to frequency-based in the optical portion of the spectrum.

**The electromagnetic field of longer chains transitions to photonic particle spin once the axial rotation spin rate exceeds the limit of the electron to interlink at least three elemental photons** (charged particles). From this electron spin rate limit (far-infrared/microwave interface and beyond), the chainlets become purely photonic, two elemental photons (energy filaments) interlinked and spinning extrinsically at a rate that determines the energy of the photons emitted. The interlocked energy filaments continue to spin in a complex spin scenario that maintains a gap at each filament overlap. However small, these gaps produce traction in Planck Space, causing constant forward motion at the speed of space ( $c$ ). This lack of state is why classical **photons have no rest mass- they simply cannot exist at rest**.

### Power/Energy Flow

Discovering how slow electrons “flow” in a conductor made it impossible to accept the mainstream definitions of current, voltage, power, and energy. The fact that electrons cannot flow at the speed of light in a conductor begs the question: What is it that IS traveling at the speed of light? The most obvious answer is LIGHT (photons). However, I could find no meaningful proof or assertions that current flow comprises classical photons of any specific frequency/energy. QED did not provide acceptable answers to this question, so I went into Planck Space (figuratively) to find the answers.

In Planck Space, I found the facility and structure for a filament of energy to flow. Finding that this filament of energy could flow within the orthogonal, quasi-static Planck Space network was a foothold leading to many discoveries. Answers to questions like what shape are energy filaments (?) and how do they maneuver (?) became fundamental to Photony theory goals, along with quasi-static Planck space structure, leading to the Manhattan Metric governed pathways (for energy flow). Planck Space compression (mass), the Photony Motion/spin rule, and the speed of light in the space model came forth.

These discoveries and more are explained clearly and precisely herein with a minimum number of equations needed. A surprise indeed was that  $PI = 4$  in Planck Space regarding perimeters (particle statics of self-confined energy filaments in geodesic motion) and forward motion (dynamics of energy filaments in helical motion) through Planck Space.

The “Poynting Field” incorporates the product of the E-field by the B-field (vector cross-product in the proximity of an antenna and beyond). Surprising to many, electromagnetic field energy primarily flows outside of conductors in circuits and antenna elements. That was unintuitive until I realized that the electric field originates and terminates on the surface of conductors. So, if  $\text{power} = \vec{E} \times \vec{B}$  and  $\vec{E}$  is essentially zero inside a conductor, then the power must also be essentially zero inside the conductor. Right? Poynting took it much further by superimposing a simple circuit's E and B fields, making it graphically intuitive that the energy flows mostly outside the conductors and components. Also, the skin effect in high-frequency circuits keeps the magnetic fields mostly outside the conductors.

So now you see the fields in your mind in ways you probably did not visualize before.

It is the way.

### Why Photony?

There is a need to find elemental particles smaller than the classical photons and supplement and provide alternatives to the mainstream theory of particles, quanta, and field lines. Photony also offers supplements and alternatives to EM theory, including **Quantum Electrodynamics (QED)**. Other quantum theories, including **Quantum chromodynamics (QCD)** and **Quantum Field Theory (QFT)**, are targeted by Photony in hopes of synergy, Unification, and a basis for power and energy at the macro scale, including the Grid.

QED is heralded as a great success<sup>1</sup> but falls short in several areas. H.R. Reiss and Jean Louis Van Belle pointed out much of the weakness of QED. A misconception of QED is the assumption that fields are primary, and potentials are secondary (not making potentials more fundamental than fields). Photony provides a clear understanding of the importance of potential as the causal factor underlying electric, magnetic, and electromagnetic fields, as electric and magnetic chains, and classical photon chainlets are transduced from elemental photons passing through toroidal electrons. According to Jean Louis Van Belle regarding the QED, it lacks a good description of what electrons and photons are physically (and without kinematics or spin mechanics). Jean Louis Van Belle articulates complex ideas in comprehensible and sensible words to intelligent folks who relate to quantum mechanics.

These and many other references concur that QED is not complete, leaving considerable room for supplementation and advancement. The need for Photony does not end with the deficiencies of QED.

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<sup>1</sup> The development of QED theory was the basis of the 1965 Nobel Prize in physics, awarded to Richard Feynman, Julian Schwinger, and Sin-ichiro Tomonaga.

# Photony Foundations

Quantum chromodynamics (QCD) is an animal of many colors. Scientists with a tongue-in-cheek conceived the colors and flavors' metaphors for "types."

Effective field theory (EFT) simplifies and approximates underlying physical theory, such as a quantum field theory. A practical field theory includes the appropriate degrees of freedom to describe physical phenomena occurring at a chosen length scale or energy scale while ignoring substructure and degrees of freedom at more minor scales. This is one way to avoid the Heisenberg Uncertainty Principle at smaller scales. However, EFT is not sufficiently firm or scaled to allow the actual construction of fundamental field theories. Photony does the opposite, embracing the smallest of scales, the Planck scale at approximately  $10^{-35}$  m. Photony provides the kinematics of energy as filaments interacting with quasi-static Planck space resulting in compression, which we perceive as mass.

The reality of non-zero neutrino mass and oscillation is one of the most significant physics examples beyond the Standard Model and one of the few places the model fails. Another way of saying the neutrinos have mass is to say they are comprised of Photony energy filaments that compress quasi-static Planck Space. Planck Space's compression due to high-speed neutrinos results in a relatively sizeable inertial mass as the Lorentz factor ( $\gamma$ ) and the Photony Motion/Spin Rule come into play. Like classical photons, neutrinos are never at rest, so they may as well have no rest mass. Neutrinos are abundant subatomic particles that pass through almost everything, infrequently interacting with matter. We have only known about all three types of neutrinos for about twenty (20) years as of 2021. Physicists have been slow to determine the presence, detection, and identification of small particles and energy quanta. Regarding dark energy and dark matter, modern theory finds them present but poorly identified and detected from the Photony point of view. This book addresses these concerns and further quantifies the phenomena in the Photony Compendium.

Gravitational Wave Velocity = Velocity of Light in the Same Medium

A gravitational wave is the motion of a compression wave in quasi-static Planck space initiated by a remote energy density event such as the abrupt merger of two black holes. The speed of light is governed by the speed of space, quasi-static Planck space, based on one divided by the square root of permittivity times permeability proximal to the relative motion. General relativity describes how clocks on massive bodies, e.g., the Earth, run slower. The Special Relativity and Lorentz Factor  $\gamma = 1/\sqrt{1 - (v/c)^2}$  describe how moving clocks run slower and how lengths contract based on velocity. Photony also captures this phenomenon at the smallest of scales- Planck space. In both cases, the quasi-static Planck space is compressed in the proximity of the clocks. The combination of special and general relativity factors is why GPS systems must be calibrated periodically to correct the accumulated time errors related to the user's position on the Earth's surface.

## My Light/Your Darkness

It is generally known that most of the energy and mass in the Universe is not visible or, more specifically, identifiable. Photony explains how static elemental photons are stable and ubiquitous, whereas dynamic elemental photons are elementary quanta (energy filaments in helical motion) and are also ubiquitous. Neutrinos are more detectable regarding mass (Planck space compression) and ubiquitous, as evidenced by the billions to trillions of neutrinos passing through your body every second (depending on the information source). The sum of the static and dynamic elemental photons, neutrinos, and undetected classical photons, along with their antiparticle and antiquanta energy motion, is the stuff of smallness and darkness.

Reportedly<sup>2</sup>, about  $2.5 \times 10^{-16}$  kilograms of “dark matter” pass through your body every second. Approximately 240 cosmic particles<sup>3</sup> per meter<sup>2</sup> per second reach ground level. So, perhaps 30 cosmic rays pass through our bodies every second. The number of neutrinos passing through our body every second is between 50 and 100 billion. These fluxes sound huge until you extrapolate the numbers from our body size to the size of the Universe, or even our galaxy, the Milky Way.

### Equations: Accurate Tools or Fascia?

EVERY EQUATION IS, AT BEST, APPROXIMATE and can only be resolved to 10-20 decimal places, far removed from the ultimate Planck resolution comprised of 36 decimal places. The difference between empirical and heuristic is that empirical is associated with experience, while heuristic relates to general strategies or methods for solving problems. When used as a noun, it is a heuristic method.

Some equations have no reality, just math with no physics applicability, while others are useful in science. The worst use of equations is when they are asserted cryptic statements to impress someone or hide behind. "What we often call laws of physics are really just consistent mathematical theories that seem to match some parts of nature," theoretical physicist Sankar Das Sarma.

Singularity points are a favorite of mathematicians and BBT/Black Hole Theorists. Many times, singularity points are real to the extent of being asymptotes taken at mathematical limits. Being purely mathematical, therefore, they can be manipulated in equations using complex algebra and other techniques. It's quite simple; a mathematical singularity point is not real, just like a point is unreal. Lines are not real, and planes are not real. Lines and planes (zero thickness) are symbols representing something else, usually boundaries. Points are location indicators with no size (the pencil dot or ink dot has the size). Lines and planes are indications of boundaries- not real things. In the case of magnetic and electric fields, flux lines are represented as Photony chains (comparable to the flexible force tubes of Faraday and Maxwell). Singularity points can never be physically reached because they are not physical.

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<sup>2</sup> This Is How Much Dark Matter Passes Through Your Body Every Second- Neutrino Science, 2021, “throughout the course of a human lifetime, you’ve been exposed to little under 1 milligram of dark matter.”

<sup>3</sup> Cosmic rays are particles - the nuclei of elements in the periodic table. They have nothing to do with the cosmic microwave background.



# Photony Foundations

So, for example, in Black Hole theory, a singularity point is merely an extremely high energy density sector within the black hole event horizon<sup>4</sup>.

Many well-known equations were tuned empirically or adjusted initially to make them work. For example, Newton's universal law of gravitation depends on  $G$ . For the mutual attraction gravitational force,  $F = Gm_1m_2 / r^2$ , a parameter such as  $G$ , perhaps a constant value that makes the equation accurate within an acceptable range of other parameters factored into the equation.  $G$  must be a constant since the assumptions that  $m_1$ ,  $m_2$ , and  $r$  (radius) are apparent variables until they are not. That said, the resulting constant such as  $G$  could be a fundamental constant of nature and eventually derivable.

The Coulomb force equation is another good example ( $F = k_e \times q_1 \times q_2 / r^2$ ) of a tuned equation based on the Coulomb constant,  $k_e$ . The two charges (accumulated charges on objects 1 and 2) are (obviously) based on "charge," which has been ill-defined. Nothing about this equation and many others, like Newton's equation for gravitational force, are fundamental; they have the terms in the right places and are tuned to produce approximately correct answers. Often, the constants involved in prevalent equations are not fixated and not fundamental constants of nature. They may vary based on proximal conditions, or they may vary based on instability or decay.

Once tuned, an otherwise unknown parameter can become a "constant," such as some  $k_x$ , or  $c_x$ . Other constants used in equations are fundamental constants of nature, such as  $e$  (2.71828...) and  $\pi$  (3.14159...), which are fixated or established by nature to a specific natural value. The ultimate natural environment is Planck Space. However, Planck Space is quasi-static, which means it can compress and expand from a uniform but elastic orthogonal structure. A gravitational field can be static (e.g., near a star with a stable energy density, or a dynamic gravitational field such as a gravitational (compression) wave from a distant black hole rotational pair or merger. How is it, and why is it, that a gravitational wave travels at "light speed?" This is one of the primary goals of this book.

Now I digress: The Bohr Train

Niels Bohr, the egocentric madman of the 1920s, left a consequential but mixed bag of accomplishments. By the era of the despicable Copenhagen Interpretation of 1927, Bohr was notably opposed to the contemporary interpretation of beta decay and believed that energy, momentum, and angular momentum were not conserved quantities. By 1929 the phenomenon of beta decay again prompted Bohr to suggest that the energy conservation law should be discarded. In opposition, Enrico Fermi's (then) hypothetical neutrino and the neutron's 1932 discovery provided yet another insight.

Bohr Train Metaphor: It seems that the hijacking of Max Planck's and Albert Einstein's quantum theory train terminated in the Copenhagen Interpretation of 1927, with the conductor, Niels Bohr, at the controls. While at the controls, Bohr said just before the derailment, "*Prediction is very difficult,*

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<sup>4</sup> Schwarzschild solution to Einstein's field equations that corresponds to the radius defining the event horizon

*especially about the future,"* so full speed ahead down the ravine! Note: It may take a few more decades to recover from this derailment. Other quotes from Bohr:

*"Your theory is crazy, but it's **not crazy enough to be true.**"*

*"Those who are not shocked when they first come across quantum theory cannot possibly have understood it."*

*"No, no, you're not thinking; you're just being logical."*

As World War II raged, Bohr was slow to agree that it was technically feasible to extract a sufficient quantity of uranium-235 to make a bomb. By 1943 Bohr had escaped from Nazi-controlled Denmark and found refuge in Britain via Sweden, whereby Lord Cherwell sent a telegram of invitation to Bohr. Later, Bohr met Churchill on May 16, 1944, but according to Margaret Gowing, Churchill found that "we did not speak the same language." Churchill disagreed with the idea of openness towards the Russians to the point that he wrote in a letter, according to Finn Aaserud, **"It seems to me Bohr ought to be confined or at any rate made to see that he is very near the edge of mortal crimes."**

Bohr said that the "deep truths" are ones for which the opposite is also true. Bohr repeatedly attempted to conjugate the forces QM into schizoid positions with a frustrating inability to describe reality at all. As Bohr emphasized repeatedly, the Copenhagen interpretation of quantum theory cannot, as a matter of principle, answer any question of the form: "What is really happening when - - - ?" However, those questions are exactly the ones that a physicist should be trying to answer to understand the real world. With the faux reality of Bohr, there could be no point of physics or any other science.

An outline of the Bohr atom was introduced during the first Solvay Conference in 1911. [2] It was modeled as a system similar to the structure of the Solar System, consisting of a small, dense nucleus surrounded by orbiting electrons but with electrostatic attraction forces instead of gravity. It was modeled after the solar system Joseph Larmor model (1897). Other preceding atomic models included the cubical model (1902), the Hantaro Nagaoka Saturnian model (1904), the plum pudding model (1904), and the quantum Arthur Haas model (1910). The Bohr model was promoted in 1913 with the help of Rutherford, but predictably, Rutherford's contributions to the model were swallowed by Bohr's ego. Sommerfeld's expansion of the 1913 model advanced the Bohr model by showing the maximum electrons per shell.

Bohr's model of the atom was completely wrong, both conceptually and in terms of agreement with experiment. John Archibald Wheeler, a colleague of Rory Coker, was at the University of Texas in the latter part of his career, had worked closely with Bohr in the 1930s, and remained a close friend after that.

According to Coker, the standard cartoon of an atom is the Bohr atom. Faced with the fuzzy probability blob of an actual atom, no one except physicists would know what the image was supposed to be. In the 2020s, introductory texts that use calculus make some token effort to introduce quantum physics, or at least the results of the application of quantum physics... a century late!...

# Photony Foundations

Regarding Richard Feynman's efforts to derive quantum mechanics, Feynman encountered serious flaws in the Copenhagen interpretation of quantum mechanics, which Feynman did not dare to challenge. Why? Too little critical thinking, common sense, awareness, and courage to speak out. Also, remember the thing called “groupthink.” When a few outspoken members are wrong, the group can evolve to be uniformly wrong. Misdirected groupthink can result in prejudice or a collective belief that prior conclusions are beyond contestation.

The fact that quantum mechanics could accommodate the likes of Bohr, and the free spirit of Feynman, reinforces Bohr’s assertions of the craziness of quantum mechanics in its infancy. Of course, the crazy parts have been rendered sane with further study since 1927 or placed into a historical (but untrue) dustbin of records. Unfortunately, many of the crazy parts are still being taught in universities without the historical context mentioned. There is nothing wrong with teaching historically false or disproven theories as long as the historical context is prefaced.

## A Look Inside This Book:

The popularized equations, e.g., the Einstein equation/relation,  $E = mc^2$  (although misrepresented in origin and scope in most literature), are essential and fundamental to the explanation of the Photony Theory. This equation and its variants are presented in detail with new insights associated with Photony regarding “equivalent mass” and energy-mass and mass-energy conversion dynamics. To fully comprehend the Einstein equation, Photony examines causality and how energy, mass, and the velocity of light relate in the smallest of venues, quasi-static Planck Space.

The Photony Unification Classes comprise the essential material of the Compendium as follows:

- |                                                    |                       |
|----------------------------------------------------|-----------------------|
| • Space, Gravity, Energy, Mass, Time {SGEMT}       | Compendium Volume I   |
| • Quanta, Particles, and Chains {QPC}              | Compendium Volume II  |
| • Electricity, Magnetism, Fields, and Force {EMFF} | Compendium Volume III |
| • Photony Energy Science                           | Compendium Volume IV  |
| • Photony Unification                              | Compendium Volume V   |

Chapter Material Assignments:

Space, Gravity, Energy, Mass, Time {SGEMT}- Chapters 1-2

Quanta, Particles, and Chains {QPC}- Chapters 3-6

Electricity, Magnetism, Fields, and Force {EMFF}- Chapters 7-9

Photony Energy Science- Chapter 10, Parts I-III

Unification- Unification is dedicated to rigorously advancing the Photony Unification and “Big ToE”. From Venn-diagram and set theory points of view, Photony intersects the Unification Classes (above) and associated subject matters. The Unification Classes unite the theory into a “big” theory of everything while supporting most classical and quantum theories. Rather than refute the standard model outright, Photony provides the models, mechanics, and kinematics needed to bolster generally accepted theories

without hiding behind complex math or needlessly mysterious concepts. **Nature has already united everything.** We must try to comprehend and articulate nature's unification work, including spacetime, quanta, particles, waves, forces, and fields.

The Big Theory of Everything (Big ToE)- Photony approaches the theory of everything (ToE) quite differently from the mainstream. Feynman's trolling in this regard, "write all the laws and equations of physics in a two-dimensional matrix and add them all to get the Theory of Everything." Sounds ridiculously odd, impractical, and to the point of being hilarious? It should; He (Feynman) was joking when he wrote this. If the Photony Compendium content sufficiently covers the intersection of the associated Unification Classes, then Photony Unification has been accomplished.

My Writing Guidelines for this introductory and foundational book:

- Selected equations/few derivations- a priority of this introductory book of fundamentals (foundations and previews) is **to avoid complicated equations, derivations, and advanced operators as much as possible** and defer them to the **Photony Compendium**.
- Reference heavily- another priority of this book is **to present, descriptively and pictorially, heavily referenced material** that is synergistic with Photony Theory.
- Make it surprisingly informative if not enlightening- The subject matter is broad, with in-depth analysis of Photony essential material deferred to the **Photony Compendium**.
- Support as much standard theory as possible- this book is not intended to curse the darkness of accepted theories but to light a candle for alternative, if not supportive, thinking.
- Use a wide range of views from exceptional people- connect as many diverse theories as possible through Photony as a unique collective. Glean from the diversity of thought over the ages.
- Expose and disrupt incorrect dogma and groupthink.

Groupthink and Lack of Critical Thinking:

According to Martin Lopez-Corredoira;

*"Specific observational cosmology results cast critical doubt on the foundations of standard cosmology but leave most cosmologists untroubled. Alternate cosmological models that differ from the Big Bang have been published and defended by heterodox scientists; however, most cosmologists do not heed these. This may be because standard theory is correct, and all other ideas and criticisms are incorrect. Nevertheless, it is also primarily due to sociological phenomena such as the 'snowball effect' or 'groupthink.' We might wonder whether cosmology, the study of the Universe as a whole, is a science like other physics branches just a dominant ideology."*

# Photony Foundations

Occam's razor is an epistemological principle (with distinction between justified belief and opinion) about how things are known, whereas Feynman addressed the fundamental nature of reality. Occam's razor is a heuristic<sup>5</sup>, not a metaphysical principle about the simplicity and beauty of theories. Occam's razor provides an intuitive guide to compare different hypotheses with a preference for simpler ones.

Example: "When you hear hoofbeats, think horses, not zebras."

Caution: Ignorance often disguises itself as Occam's razor when beliefs are unjustified, or opinions are uncorroborated hearsay. Circumstantial evidence can result in false conclusions.

## Addressing Groupthink and Occam's Razor in Photony Context:

Photony provides critical thinking and a combination of inductive, deductive, and abductive reasoning for many subjects and scientific principles. Photony looks at the Big Bang Theory with a critical eye from multiple perspectives, such as: Would the Deity start our Universe with particles (plasmatic quarks and gluons) that are not the smallest, most fundamental particles? After sifting through inductive and deductive reasoning, applying abductive reasoning regarding charge, force, singularity points, and flux/lines frequently results in a more straightforward and beautiful fundamental theory, Photony Theory.

Walter Babin; *"In my early years of study, I was impressed with the publications of leading physicists who claimed that the principal motivation behind their efforts was the search for truth. After many fruitless attempts to have my own 'eminently' truthful paper published a quarter-century ago, I was forced to consider that their purpose might have less to do with exalted goals and more to do with elevating the perception of their own works."*

Photony can explain most scientific principles and phenomena in simple terms and visual effects with only a few fundamental or elemental particles, quanta, and chains. Using only these three entities, Photony can build (with a concept called emergence) all other less fundamental particles (like electrons), other quanta (like classical photons and neutrinos), and electric, magnetic, and electromagnetic chains (like radio waves and microwaves). Photony explains how electrons transduce elemental photons into chains. Photony also explains how the frequency of electron rotation (x-spin) transitions from transducing longer chains at lower radio wave frequencies to shorter chains in the microwave spectrum and ultimately to chainlets of two interlinked elemental photons starting in the far-infrared region of the spectrum.

Photony is a theory supported by many facts, both referenced and self-supported. The most significant discovery aspects of Photony are theoretical discoveries in search of ongoing experimental and computational verdicts. Photony theory herein has openly presented material further developed in

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<sup>5</sup> involving or serving as an aid to learning, discovery, or problem-solving by experimental and especially trial-and-error methods

the Photony Compendium. Each of the Photony Compendium books proves or disproves previous claims and strives to make a strong case for Unification and the Big ToE (big theory of everything) introduced in this book. Photony has attempted to support as much mainstream theory as possible, not to contradict or disprove as much as to fill in the blanks, provide models, define the undefined, and mitigate areas where the accepted theory admits to being weak.

I eventually rejected string theory (in Planck Space) for several reasons: I found that most physicists agreed it was at a dead end with little hope of unifying anything. Rolled-up extra dimensions as a physical reality were as much of a stretch as the strings between them (pun intended). Suppose you can believe in the reality of string theory, rolled-up extra dimensions, or that the entire mass and energy of the Universe in an initial quark/gluon plasma was ejected from a singularity point in less than a second. If so, you can most certainly believe in portions of Photony theory.

### Introductory/Preparatory Thoughts

In *The Electrician* in 1886, Oliver Heaviside wrote that readers should not hastily conclude that an author of, say, Photony has revolutionized some aspect of science or is **talking nonsense**.

*“The very first step to the understanding of a writer is to find out what he (/she) means. Before that is done, there cannot possibly be a clear comprehension of his utterances. One may, by taking his language in its ordinary significance, hastily conclude that he (/she) has either revolutionized the science of induction or that he (/she) is talking nonsense. But to do this would not be fair. We must not judge by what a man (or woman) says if we have good reason to know that what he (/she) means is quite different. **To be quite fair, we must conscientiously endeavor to translate his (/her) language and ideas into those we are ourselves accustomed to use. Then, and then only, shall we see what is to be seen.**”*

It would be good if Photony delivered a few new insights that inspire awareness or an aha moment for the reader. If only a “success of one,” it may be worth it, so I ask the reader to read the whole of the book with an open, imaginative mind before making too many conclusions. The latent result may be learning many new things or seeing nature in a new way. Warning: You may get claustrophobic in Planck Space or freak out when you see that PI equals four (4) for Planck Space static perimeters, geodesics, and dynamics of particles and quanta involving spins and cycloid motion. The Photony compression factor,  $4\pi$ , is a primary Photony prominence since it comprehends the interaction between energy and quasi-static Planck space, resulting in compression as the basis of curvature and mass with underpinnings of General Relativity.

You can learn about thought experimentation and critical thinking by reading this book.

# Photony Foundations

This book of  
Photony  
Foundations  
and  
Compendium  
Previews is  
organized as  
follows:



- I. Introduction and Front Material
- II. Selected Foundational Highlights/Previews
- III. Photony Compendium Abstracts
- IV. Bibliography

# Photony Foundations

“A theory can be proved by experiment,  
but no path leads from experiment to the  
birth of a theory. “

Albert Einstein

The “Old One” and the Void are where we start our journey  
into a “grand thought experiment,”  
one that requires an open mind  
and a fearless curiosity  
leading to the birth  
of a new theory.

**PHOTONY™**



## What is Photony?

"Spacetime tells energy how to move; energy tells spacetime how to curve, and curvature is what we perceive as mass." - Photony

Photony is the study, theory, and sentience of various quantum, particle, and chained entities, including constituent Photony Filaments (energy filaments) operating in quasi-static Planck Space as the basis for natural properties and phenomena. The Photony Model showcases energy filaments as primary effectors acting as substantially circular spring-like strands of energy maneuvering in a structured yet flexible, orthogonal cellular network called Planck Space.

Filament spin behavior in Planck Space is the nexus of all particles' structures, including Photony spin motions, spin levels, orientations, and interlinking (chained) filament principles. The integrated filament spin tracings form an outer virtual shell, thus defining the particle's extent while compressing the local Planck Space quasi-static network in and around it.

Photony approaches science from the Planck scale with residential elemental particles and transitory quanta following the "rules of the road," so to speak. The rules of the road here are, more specifically, the Photony Motion/Spin Rule (see Chapter 2). In the Planck cell world of Planck space, the motion of individual energy filaments and their interlinkings dictate motion. If the number of interlinked energy filaments is small, such

as classical photons, the motion is determined by the spins and results in a speed dictated by the medium, more specifically, the medium's permittivity.

Photony is a new way of viewing the Universe. From the Planck Scale to the Cosmic Microwave Background, Photony seeks to provide a mechanical basis for the standard model without needless controversy while applying a maximum amount of critical thinking and thought experimentation.

# Photony Foundations

*“nothing defines the search for truth more than the resistance to it”*

- Walter Babin

Photony™ (“Photony”) is a relatively new theory that boldly looks into the smallest and the largest of physical milieu and the intertwined relationships between Space, Gravity, Energy, Mass, Time, Quanta, Particles, Chains, Electricity, Magnetism, Fields, and Force. Photony provides a theoretical basis for universal Unification originating at the beginning of time when energy first arrived “Into the Void” and began to compress Planck Space into what we perceive as mass, as larger particles emerged based on energy filament kinematics.

Photony is founded upon the natural order in the Universe. The natural laws have not changed with time and location in the Universe. There was a Void (space), a starting point for time as energy came “Into the Void.” Time has passed as the Universe has evolved, wherein the energy is divergent, and the mass is convergent. This phenomenon occurs because the mass is just the compressive impact of an energy density (a collection of interlinked Photony elemental filaments as sub-particles) on quasi-static Planck Space with a slow collective net forward velocity. The energy density is compressive to quasi-static Planck Space, producing curvature, which we perceive as mass. So, mass is a metric and a shape-shifted spatial gravitational aberration.

The following chapters collect selected highlights of Photony theory. Subsequent chapters source these highlights and other materials as foundations and previews of Photony fundamentals for a series of books called the Photony Compendium.

The foundations of Photony support the theories expressed and the great accomplishments of past and present scientists, encompassing quantum theory and classical theory. Photony foundations and fundamentals rely on many of the early scientists’ accomplishments and their substantial and enduring insights into our time's scientific questions.

Photony, whenever possible, provides supporting theory and analysis for classical and contemporary theories, including circuit/electromagnetic field theory, relativity, QED, QCD, and QFT. This supplementation of these theories with mechanics, kinematics, and physical models provides alternatives to questionable, unproven, or tentative theoretical dogma. The bulk of the supporting details are presented in the Photony Compendium books.

“What you can’t imagine, you can’t discover.”

Albert Einstein

# Chapter 1 Planck Space and Spacetime

Quasi-static Planck Space is a compressible but quasi-orthogonal structure wherein Photony energy filaments maneuver as geodesics in compromise with the orthogonal governance of space resulting in a significance of the product of classical  $\pi$  ( $= 3.1415\dots$ ) and the governing Manhattan Metric  $\pi$  ( $=4$ ) seen ubiquitously in natural equations of physics as  $4\pi$ , the **Photony Compression Factor**. Spacetime in physics is a mathematical model and metric which merges the three spatial dimensions of space and the one dimension of time into a single four-dimensional manifold.

## A. Preeminent Spacetime Research

René Descartes- in 1644, René Descartes proposed that no empty space can exist; a continuum of matter would cause every motion to be curvilinear [3]

Hermann Minkowski proclaimed in a 1908 colloquium that space and time could be seen as components of a single four-dimensional spacetime fabric with the dramatic words;

*"Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality".*

Michael Faraday- in 1798 correspondence with William Whewell, is thought to have originated the term dielectric indirectly by calling material bodies wherein electricity goes "diaelectric" [4] to correlate with "diamagnetic," neither of which have survived as useful terms in physics. They were soon replaced by dielectric and diamagnetic as follows. William Whewell - a polymath at Cambridge (UK), suggested dielectric (considering dia-electric a bit clumsy).

Regarding the terms permittivity and permeability, the credit goes to Oliver Heaviside, in 1885;  
*"In theoretical investigations disconnected from special applications, the unit-volume properties conductivity, inductivity, and permittivity, are generally much more useful than their reciprocals, resistivity, reluctance, and elasticity."*

Heaviside also coined the following terms of art in electromagnetic theory: [5]

- conductance- real part of admittance (1885);
- elastance (reciprocal of permittance, reciprocal of capacitance) (1886);
- inductance and impedance (1886);
- permittance (now called capacitance) and permittivity (1887);
- admittance (reciprocal of impedance) (1887); and
- reluctance (1888)

Heaviside not only cleaned up and reduced Maxwell's equations, but he also independently discovered what we call the Poynting Vector. [6] The Poynting Vector and **the Poynting field are essential for understanding the flow of energy in and mostly around a circuit.**

# Chapter 1

## B. Spacetime Quasi-static Planck Space

The metric used to quantify the compression/curvature of Planck Space is “mass.” From a progressive context, it’s “space” first, energy (filaments) second, and compression/curvature of space third as the metric underlying the equivalence of mass and energy. John Wheeler, the renowned black hole physicist, later summed up general relativity [7] as;

*“Spacetime tells matter how to move; matter tells spacetime how to curve.” - Wheeler*

Photony replaces the term “matter” with **energy** based on the Einstein equivalence principle plus another important distinction:

***“Spacetime tells energy how to move; energy tells spacetime how to curve, and curvature is what we perceive as mass.” - Photony***

Spacetime can be viewed as space at a given time with an array of mass and energy entities that change from moment to moment. In other words, the distribution of energy and mass changes based on time. However, space itself changes structurally as the distribution of matter and energy changes. More specifically, the permittivity changes as energy densities change in position, speed, and density. When the permittivity changes, so does the speed of light since the speed of light depends on the permittivity and permeability of a given medium. So, spacetime is far more complicated than a simple addition of a time dimension in the spacetime equations.

Spacetime of free space would seem simple until you realize there is a paradox. There are no energy densities within or nearby free space, no mass (no quasi-static Planck space compression), and a relative permittivity of 1.000..., so “free space” is not found in our Universe. So why do we need it?

The concept of free space is similar to that of a singularity point in that neither exists, but when viewed as asymptotes, they provide some real meaning. An unattainable goal is nevertheless a goal where closeness is an important factor.

Mathematically we can assign a value of zero (0) to the width (diameter) of a singularity point as we can assign a value of 1.000... to a relative permittivity or permeability. It can be used as a reference value just as a zero-point singularity can be used as an attractor (sink) in case of black hole theory or source in the case of the popularized big bang theory.

The big bang theory has evolved ever since 1948. According to Wikipedia, the modern, most widely accepted cosmological model is called the “lambda cold dark matter” model, in which the universe contains three major components: first, a cosmological constant denoted by Lambda (Greek  $\Lambda$ ) associated with dark energy; second, the postulated cold dark matter (abbreviated CDM); and third, ordinary matter. This has been called the “standard model of big bang cosmology” which comprehends dark matter, dark energy, and the cosmic microwave background.

# Photony Foundations

## C. Photony Quasi-static Planck Space

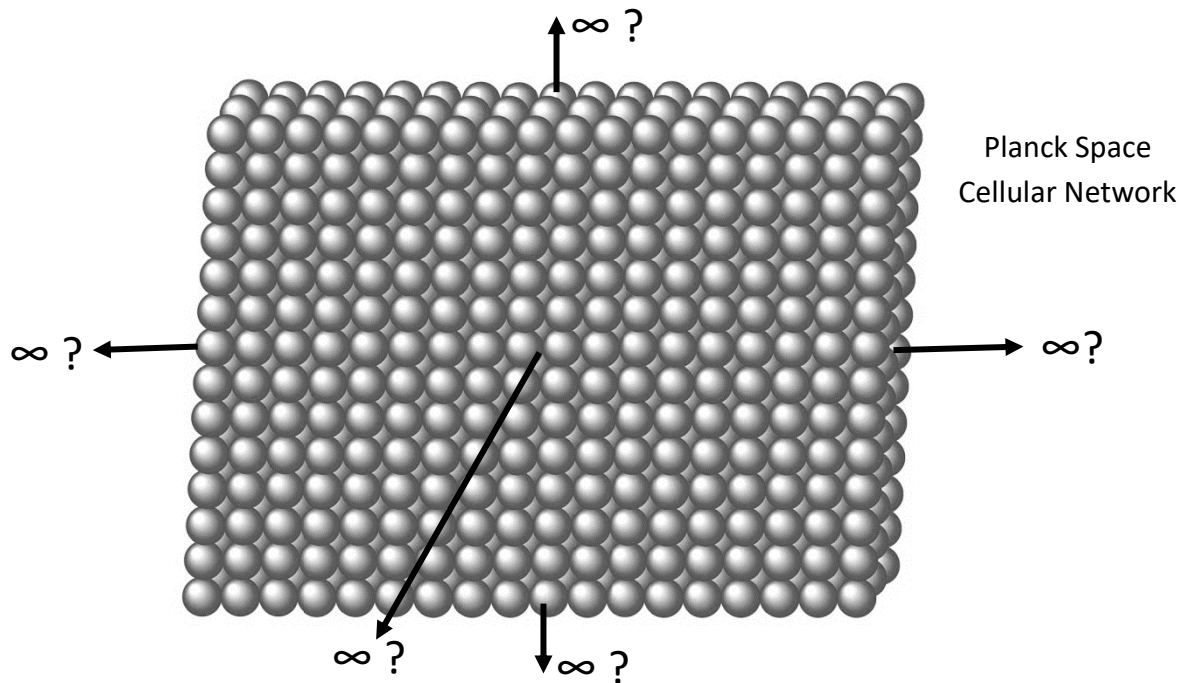


Figure 1: Quasi-static Planck Space

The depiction of Planck Space above may be illustrated as an infinite array of 3-d interconnected spherical cells. The illustration depicts the fabric of space as a possible representation of the void before creation and as a containment system to manage the separation of our reality from that of higher dimensions (the outer cells do not show the outer portals). In a Photony modeled three-dimensional (3-D) universe, the number of interconnecting channels for each Planck cell is six. **Life (our natural universe) exists inside the interconnected flexible cellular network.**

ESA: ***"Spacetime is an elastic medium that remains stubbornly stiff."*** [8]

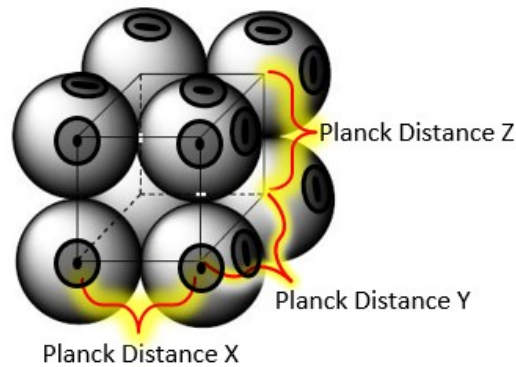
The popularized "fabric of space" is comprehended herein as "Planck space." The Planck Size is the smallest resolution distance or unitary size that can be resolved when quantifying the smallest intervals of distance in space (including the interior of particles). "Planck Space" was termed and modeled by **PHOTONY™** to comprehend an underlying structure of space.

Planck Space compression (mass) is realized through the interlinking of energetic Photony filaments within all larger particles and matter (masses), incrementally compressing space within and extending around the masses. Photony theory explains how the elemental photons (field particles) can interlink within electrons and form elastic chains as the basis of magnetic and electric flux lines, comparable with the tension of spinning tubes, as Maxwell suggested.

# Chapter 1

Flexible Planck network cells below illustrate the Photony form of interconnectivity via cell junctions of Planck space. Each spherical Planck Cell model connects six neighbors (four sides plus top and bottom) for an outer cubical structure with orthogonal inter-cellular portals. The interconnected centroids of the eight adjacent Planck Cells form a Planck Cube. The Planck Cells are flexible, somewhat like a structured membrane. The relaxed, minimum energy, non-stressed intercellular distance, or Planck length  $L_P$  [9], is defined by combining other constants of nature such that their units combine to give a length.

Planck Space Cubic  $L_P$  Section



**Figure 2: Planck Space X, Y, Z Cubic  $L_P$  Section**

As energy filaments pass through the cellular pathways, stress-energy is acquired in the quasi-static (flexible) Planck Space network in the form of radial compression. After entering a cell, energy filaments can flow orthogonally between the Planck network cells with 5 degrees of freedom. The Photony Motion/Spin Rule (see Chapter 2) governs the flow of energy filaments through quasi-static Planck space. In modern physics, the Planck length ( $L_P$ ) is the resolution wherein classical ideas about gravity and space-time become invalid, and quantum effects prevail. Alternatively, Photony provides validity at the Planck scale based on a mechanical model with kinematics (the description of motion) supporting the Photony model. Photony variants based on the  **$4\pi$  compression factor**<sup>6</sup> as follows:

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^2 \approx 1/(16\pi \cdot c)$$

Equation 1: Gravitational Constant with hidden  **$4\pi$**

$$G = 1/(4\pi \cdot 4c) = 6.636 \times 10^{-11}$$

Equation 2: Gravitational Constant with  **$4\pi$**

$$l_p = \sqrt{hG/(2\pi c^3)}$$

Equation 3: Planck Length with hidden  **$4\pi$**

$$l_p = \frac{1}{4\pi c^2} \sqrt{h/2} = 1.6212 \times 10^{-35} \text{ m}$$

Equation 4: Planck Length Photony Variant with  **$4\pi$**

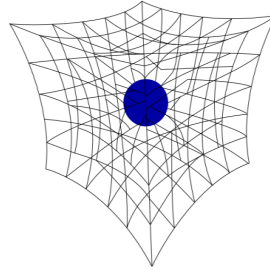
<sup>6</sup> The  **$4\pi$**  compression factor is explained in detail in later sections of this book.



# Photony Foundations

## D. Compression of Planck Space

**Compressive Energy Density:**  
*Gravitational Curvature results  
from the interplay between  
space, motion, and energy  
(Photony filaments)*



**Figure 3: Compression of Planck Space**

Photony comprehends that **mass is a measure of the extended compression or curvature** in the otherwise regular (evenly spaced, orthogonal, and interconnected) and quasi-static Planck Space background. The Planck Space compression/curvature results from, and is proportional to, **filament energy density; this is the basis of Einstein's General Relativity**. According to Walter Isaacson in Einstein, His Life and Universe, regarding Einstein's curvature [10]:

Albert Einstein,

*"Gravity is a warping of space and time, and he came up with the equations that describe the dynamics of this curvature result from the interplay between matter, motion, and energy."*

Photony identifies what we perceive as **mass as a metric of the compression or "curvature"** corresponding to a central filament (energy) density and motion in Planck Space. Furthermore, **energy is the superposition of interlinked or localized energetic filaments, causing curvature**. This mass description is similar to **Einstein's explanation of gravity above**; *"interplay between matter, motion, and energy."* These two concepts' unifier relies on mobile energy filaments to naturally travel "downhill" toward higher density Planck Space cells based on gravitational gradients (compression/curvature) resulting from proximal **compressive energy density**.

The energy density of Photony filaments distorts space (and time), attracting other localized (or passing through) filaments. **Filaments are forever swimming in Planck Space and naturally swim downhill relative to the spatial compression gradient**. The effect of this compression phenomenon is what we call gravity and underlies Newton's Law of Gravitation. According to Yuan K. Ha, regarding **An Underlying Theory for Gravity** [11];

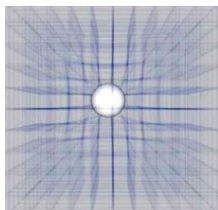
*"We recognize that the quantum gravity domain is naturally of Planck size  $10^{-33}$  cm. Furthermore, that quantum gravity domain must be physical and contain real degrees of freedom, and that those degrees of freedom should also carry energy, momentum, spin and other attributes like any other excellent degrees of freedom in physics."*

Quasi-static Planck space can **"carry energy, momentum, spin ..."** by providing traction in orthogonal pathways for self-propelled quanta such as photons comprised of interlinked energy filaments.

# Chapter 1

## E. Energy Compression/Mass Perception

Einstein also apparently anticipated that matter is derived from the “physical states of space”. In 1930 he said, "Now it appears that space will have to be regarded as the primary thing and that matter is derived from it, as a secondary result." [12], [13]



**Figure 4: Energy Compression/Mass**

Energy compresses quasi-static Planck Space in support of general relativity, which was proven by astronomers for Einstein on May 29, 1919, by observing the deflection of starlight passing near the sun during an eclipse [14]. The sun's energy (density) produces compression of space, which is maximal at the sun's center but is optically detectable in space near its surface during an eclipse.

Photony filaments of energy are compressive to Planck Space, causing the compression/curvature of Planck Space, which Photony now identifies as a metric of mass. The compression/curvature is assigned to a “**gravitational field**” (gravity field gradient) since it attracts neighboring objects based on energy filament density/motion dynamics. Filaments always spin and **naturally lean/move unilaterally toward higher density Planck Space cells based on the Photony Motion/Spin Rule**. This gravitational attraction phenomenon through energy filaments, interlinked as energy density flowing individually and superpositionally as a collective toward relatively higher compression gradients in quasi-static Planck space; this is what gravity is!

What many folks perceive as mass, based on touch, is interlinked energy filaments (energy density). Thermal energy is transferred via conduction (touch), convection, and radiation. With the sense of touch, we sense both temperature and force/area (pressure). Furthermore, **mass (m)** is a perception and a metric of Planck Space compression/curvature caused by energy density (resulting in gravitational mass =  $E/c^2$ ) and motion (resulting in inertial mass). Inertial mass is a velocity-based exacerbation of the gravitational mass, causing increased Planck Space compression at the leading edge of the body, object, or particle in motion, somewhat like the shock wave of a sonic boom.

Gravitational waves and electromagnetic waves are fundamentally dissimilar. Electromagnetic waves travel in space, whereas gravitational waves, created by mass acceleration, are waves of the spacetime ‘fabric’ itself. [15] Spacetime is an “elastic medium,” elastic but (stubbornly) stiff! Hence, the Photony proposition of quasi-static Planck space.

# Photony Foundations

## F. The Speed of Space



$$C_0 = \sqrt{1/\mu_0 \epsilon_0} \text{ Equation 5}$$

THE SPEED OF SPACE  $\equiv$  THE  
SPEED OF LIGHT IN SPACE

Where:  $\epsilon_0$  = permittivity and  $\mu_0$   
= permeability of free space:  $C_0$  is  
a scalar that **does not depend  
on frequency (photon energy)**,  
which means that there are no  
dispersive effects

**Figure 5: Speed Limit of Free Space (The Void)**

### 1. A History of Light Speed

Before the 17<sup>th</sup> century, “scientists” thought the speed of light was infinite and could travel from point A to point B instantaneously. Galileo tried to measure the speed of light with an experiment in which two people with covered lanterns stood a known distance apart. A first-person uncovered a lantern, and when a second person saw the light and uncovered his/her lantern. Galileo failed to record the time between lantern signals because the distance between the first and second person was too short, and the lantern light traveled too fast to be measured and was too dim to be further separated this way. Danish astronomer Ole Roemer was the first to prove that light travels at a finite speed in the mid-1600s.

The phrase “speed of light” has been used perhaps more than any other phrase in physics. Now it is Photony’s responsibility to explain why this phrase is misleading and misused. Light (self-propelled photons) travel at a speed or velocity in space called “the speed of light.” The typical symbol is  $c$ , but **Photony promotes the use of  $c_0$  when it pertains to the speed in free space**, wherever that might be located or visualized. Note, it is reasonably safe to say, few locations in space are devoid of dark energy or dark matter.

However, different mediums have a different speed of light based on permittivity for which photons travel within that medium, including nonuniform space itself. Furthermore, **photons are self-propelled and do not exclusively control the speed at which they travel from a distant observer’s point of view!** How is it that the medium, particularly the permittivity of the medium, controls the speed of photons?

# Chapter 1

## 2. Real Space as a Speed Governor

In real space:  $c \neq c_0$  since  $\mu \neq \mu_0$  and  $\epsilon \neq \epsilon_0$

$c = v(1/\mu \cdot \epsilon)$  Equation 6: Speed of light in Real Space

The commonality of EM waves and Gravity Waves: For spacetime entities comprising energy quanta/EM waves, particles, and ponderable matter (objects), the Photony Motion/Spin Rule for filaments maneuvering in Planck Space acts as a forward velocity throttle (perhaps comparable to “Higgs syrup”). Space has a natural speed that is a function of its permittivity and permeability, which varies based on compression. The compression varies based on proximal energy density affecting both permittivity and permeability with a resolution of a Planck length in quasi-static Planck space. **A medium's impedance and energy density characteristics dictate a specific photon and maximal EM wave propagation speed.** [16]

**The Photony Motion/Spin Rule governs motion and speed limitations of all quanta and particles**, starting with compact, high-speed particles (elemental and classical photons) through lower speed larger multi-filament objects, or as Newton put it, “ponderable matter.” The effect of the originator’s velocity at launch and the receiver’s velocity at impact also does not affect the compact particles’ actual speed (Special Relativity) during transit. So, what affects the speed of light (photon speed), and what governs its speed limit? Speed of light in space<sup>7</sup> = speed of space: **Photons are self-propelled**- the speed of light (photon motion) is mechanically established and maintained by the traction of Photony energy filaments maneuvering in space as photons filaments spin. Photony elemental photons, classical photons, and neutrinos **self-propel at approximately  $c_0$**  (or  $\approx 3 \times 10^8$  m/s). We should anticipate that whatever the self-propelling photonic phenomenon, it also applies to radio waves and microwaves (EM chains with  $> 2$  interlinked elemental photons).

The speed of photon propagation ( $v_p$ ) is determined solely by the dielectric permittivity of the space wherein the photon is propagating. In the case of quasi-static Planck space, propagation velocity is the speed of light in space (the speed of space).

$$v_p = c * vf = c/\sqrt{\epsilon_r} \quad \text{Equation 7: Photon Propagation dependence on permittivity}$$

where  $\epsilon_r$  = the relative permittivity and  $c$  = velocity of light (in real space),

$$vf = 1/\sqrt{\epsilon_r} \quad \text{Equation 8: Photon Propagation Velocity}$$

**The velocity factor varies based solely on real (quasi-static Planck) space variability in permittivity in our Universe.** The velocity factor changes as the gravitational field changes due to the compression of quasi-static Planck space resulting from energy densities (typically conflated with “mass”). This is a simplified but insightful restatement of general relativity.

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<sup>7</sup> Photony associates the term “space” with spacetime and quasi-static spacetime as equivalent with the amount of particulate and energy present and associated spacetime distortion/compression/curvature depending upon the context.

# Photony Foundations

## 3. The Nature of Light Speed

The term **“speed of light”** has been used loosely for centuries. Today, when scientists and engineers speak of an electromagnetic wave's speed, they refer to a speed or velocity factor, such as 0.95 of the speed of light (0.95c). So, what exactly is c? The “speed of light,” c, is typically used **with questionable regard for context**. In most references, the assumed context appears to be “in free space.” A dictionary definition of free space; is a zone with no gravitational and electromagnetic fields used as an absolute standard.

The speed of light standard should have always been assigned to  $c_0$  instead of “c,” in a free space medium where the relative permittivity and permeability are ideal with the value of one (1). This assignment more clearly articulates a fixed maximal value of the velocity of light (photons). Free space is ideal, so the typical reference to light speed is also ideal (not exact). A close association to light speed is characteristic impedance  $Z_0$ , whereby the impedance of free space is assumed and associated with the naught subscript. Similarly, an association regarding relative permittivity and relative permeability exists for the relative speed of light,  $c_r$ , whereby  $c = c_r \cdot c_0$ . The speed of light is directly affected by the relative permittivity,  $\epsilon_r$ , and relative permeability,  $\mu_r$ , calculated by Mathcad as follows:

$$c_0 = (2.9979 \cdot 10^8) \frac{m}{s} \quad \frac{1}{\sqrt{\epsilon_0 \cdot \mu_0}} = (2.9979 \cdot 10^8) \frac{m}{s}$$

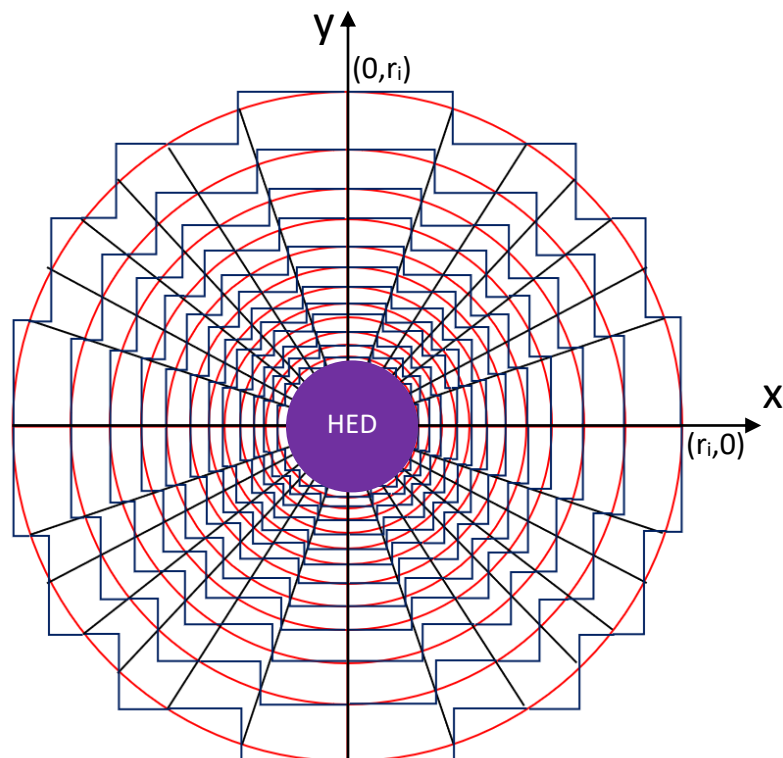
Light (classical photons of all frequencies/energies) and electromagnetic (EM) waves are obliged to travel at the speed of space to minimize their compression of Planck Space and remain in the least energy mode of travel, thus appearing to have zero mass (i.e., having no apparent compression → no mass). Since photons travel at the speed of space, the apparent mass is zero since quasi-static Planck Space compression is minimal, transient in nature, and unmeasurable.

Photony teaches that classical photons are interlinked pairs (chainlets) of elemental photons with synchronized a1 and a2 spins wherein the energy filaments interact in a manner that results in Photony B-spin and forward motion. The photon motion mechanics is based on energy filament traction in quasi-static Planck space.

The energy of classical photons is based on their spin frequency. Photon energy increases as the frequency increases while the forward speed does not increase. The Photony Motion/Spin Rule supports this, which describes quasi-static Planck space as a compressible medium acting as a speed governor. So self-propelled photons travel in the least energy mode by complying with space's natural but variable speed based on permittivity and permeability variations. The least energy travel/propagation results in minimal compression of quasi-static Planck space as the photons use energy filament traction for propulsion.

# Chapter 1

## G. Manhattan Metric vs. Circular Pathways in Planck Space



**Figure 6: Planar View of Manhattan vs. Circular Pathways in Planck Space**

The figure above is a cross-sectional view of Planck space around a central massive<sup>8</sup> (high energy density) particle, object, or body. Here, we refer to it as “HED.” The space around the HED is relatively low in energy density and relative permittivity.

This figure above illustrates orthogonal Manhattan (dark blue) pathways in Planck space without compression. The segmented Manhattan circles each have a circumference of  $2\pi r_i = 8r_i$  where  $\pi = 4$  (proof in a later section).

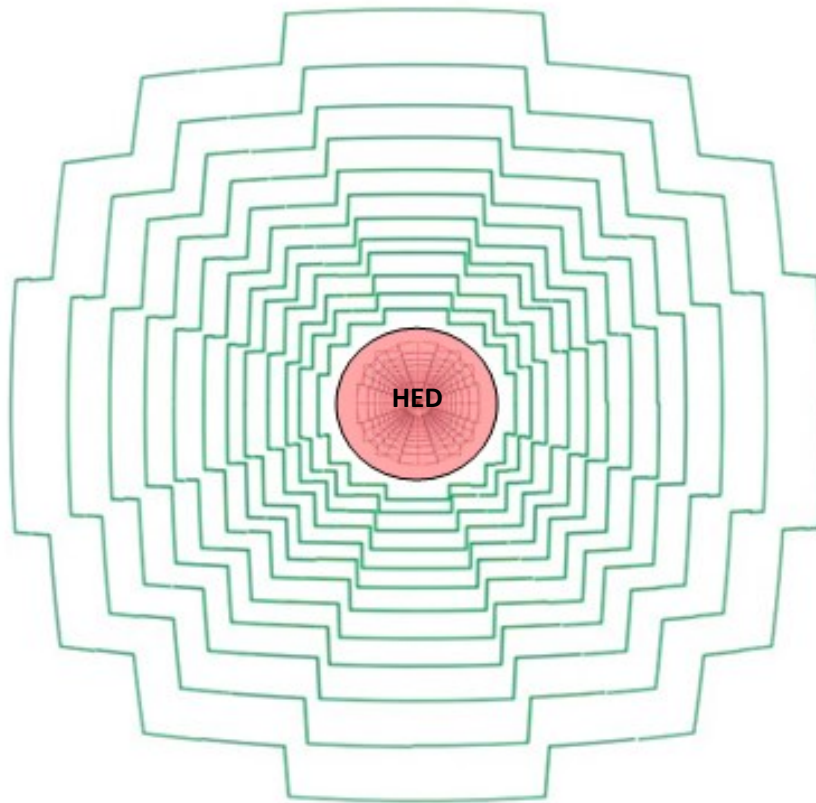
The planar geodesic flows of energy (red) are shown as continuous circular pathways with smooth perimeters and circumferences of  $2\pi \cdot r_i = 6.283 r_i \dots$  where  $\pi = 3.1415\dots$  The energy (comprised of Photony energy filaments) flows in a least-energy mode in 3D space as seeks geodesic pathways (explanation in a later section).

In free space, the relative permittivity would be 1.0, as depicted above. However, this is inaccurate in real space around or near the HED. The interaction of the energy and the quasi-static Planck space compresses Planck space according to a compression factor,  $\pi \cdot \pi$ , or  $4\pi$ .

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<sup>8</sup> Called “ponderable” mass by Newton

# Photony Foundations



**Figure 7: Energy Pathways in Quasi-static Planck Space**

This figure above illustrates quasi-orthogonal Manhattan pathways in Planck space with HED compression shown as distorted (green) pathways. These pathways illustrate how the flexible quasi-static Planck space becomes distorted by energy attempting to flow as continuous geodesics. The net result shown above in green illustrates the Photony compression factor  $4\pi$ .

The compression factor  $4\pi$  is a factor in all fundamental equations associated with spacetime. Sometimes not obvious but based on permeability, the compression factor (sometimes called the induction constant) is the basis of the origin and effects of energy in Planck space and many scientific equations.

European Space Agency (ESA)- **Dents in Spacetime** [15];

*“Spacetime is an ‘elastic medium’... Massive objects produce ‘dents’ in the fabric of spacetime. Other objects move in this curved spacetime taking the shortest path, like billiard balls on a springy surface. So spacetime is an ‘elastic medium’. Elastic but stiff!”*

It is noteworthy that the term quasi-static used throughout Photony theory in describing Planck space is equivalent to ESA’s “elastic but stiff.” Also noteworthy, the Photony compression factor  $4\pi$  represents multiple significant digits (12.5664) of the permeability of free space,  $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$ .

# Chapter 1

## H. Pondering Mass (and Energy)

In a physics context, the difference between “mass” and “ponderable mass” is more edifying than precise articulation. Newton spoke of mass as “*The quantity of matter is the measure of the same, arising from its density and bulk conjunctly.*” [17] Ponderable is having a detectable amount of matter; having a measurable mass. So, it follows that Photony distinguishes the concept of mass from the concept of a particle or object with mass in light of energy.

The transition between energy and mass or ponderable mass inevitably leads to the popularized mass/energy ( $E = mc^2$ ) transition. So, in the absence of a concise accepted definition for ponderable mass, Photony uses Einstein’s preferable  $\Delta m = L/c^2$  which others morphed into the  $m = E/c^2$  variant, where  $L$  = light energy, or more specifically, photon energy. Einstein [18], after deriving that a change in kinetic energy of a ponderable mass  $\Delta K$  equals the photon quantum energy  $L \times v^2/(2 \cdot c^2)$ , concluded that **“If a body emits energy  $L$  in the form of radiations (photons), its mass decreases by  $L/c^2$ .”**

Starting with ponderable mass, Photony defines a ponderable mass as having an equivalent energy density consisting of interlinked energy filaments of multiple numbers of filaments first interlinked into photons (two elemental charge photons each) and further linked into other interlinked entities. Atoms are built with electric and magnetic chains of bonded nucleons and electrons. An energy density is a mass of particles bonded in classical ways including covalent and ionic bonding, but Photony chains are the means of electron passage.

So, the progressive transition to electrons and positrons then to pions, quarks, protons, and neutrons, (partial list) then with combinations to atoms, ions, isotopes, and molecules bound by Photony electric and magnetic chains. So, the full range is made from the most fundamental quanta, Photony elemental charge photons, each with a single energy filament.

Photony has identified electrons riding on chain loops in atomic shells wherein the electron breaks loose under certain collision conditions transition to a lower energy shell (chain) while emitting a photon which was temporarily trapped in the vortex of the toroidal electron. As atoms experience changes in chains as electrons and holes jump around, if certain photons are absorbed (linked into chains the atomic mass increases whereas if certain photons are emitted and lost outward from or through the atom’s valence shell, the atom will lose mass.

Wein’s law teaches how to find the spectral peak (frequency/energy) of Photon emission at any given temperature of a ponderable mass. So, it is temperature, atomic and molecular vibration, that ultimately governs the emission (loss) of photons from a mass as electrons with x-spin at frequencies greater than 37.6 THz emit photons.



# Chapter 2      Curvature and Pi

## A. Gravitation Preeminent Research

Gravitation theory has a storied and interesting past, with no shortage of great scientists offering their theories and always coming up short. Many historical gravitational theories are creative and interesting, but none are correct. The connection between the fabric of space and gravity has never been fully comprehended and articulated.

Galileo Galilei- Galileo has been called the "father of observational astronomy," [19] the "father of modern physics," [20] [21] the "father of the scientific method," [22] and the "father of modern science." Regarding gravity, Galileo discovered that all objects in free fall tend to accelerate equally concerning the duration of their fall. [23] In 1604, Galileo correctly hypothesized that a falling object's distance is proportional to the square of the elapsed time.

Robert Hooke- Robert Hooke published the System of the World and, in 1666, presented a paper "*concerning the inflection of a direct motion into a curve by a supervening attractive principle*" to the Royal Society. In 1671, Robert Hooke preempted Newton by speculating that gravitation results from bodies emitting waves in the aether. [24] In 1674 Hooke proclaimed that; [25]

*"all Celestial Bodies whatsoever, have an attraction or gravitating power towards their own Centers" and "also attract all the other Celestial Bodies that are within the sphere of their activity."*

Isaac Newton- in 1675, argued that aether streams attract all bodies to one another. In 1687, Newton published *Philosophiæ Naturalis Principia Mathematica*, which hypothesizes the inverse-square law of universal gravitation. In his own words: [26]

*"I deduced that the forces which keep the Planets in their orbs must be reciprocally as the squares of their distances from the centres about which they revolve..."*

Nicolas Fatio de Duillier, Georges-Louis Le Sage- Nicolas Fatio de Duillier (1690), and Georges-Louis Le Sage (1748) proposed a corpuscular model using a screening or shadowing mechanism between bodies to evoke an inverse square law gravitational relationship.

The Genius (and Humor) of Einstein- We should appreciate Albert Einstein for more than just his life-changing theories; his humor and way of thinking, including his heralded "**Thought Experiments**," were key to his accomplishments. Einstein did not begin with tensors, much less the Einstein tensor (also known as the trace-reversed Ricci tensor) to convey the curvature of a pseudo-Riemannian manifold or, in General Relativity, the Einstein field equations for gravitation describing Minkowski spacetime. No, **Einstein used imagination and thought experiments as the driving force of his theory of general relativity**. The corrected equations and accurate answers came during and after the astronomers' starlight deflection (during a solar eclipse) data proving the theory.

# Chapter 2

## B. Gravity vs. Gravitational Waves

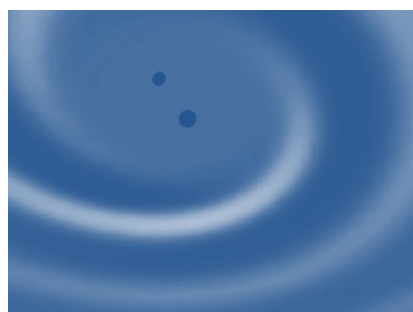
Planck Space's curvature empowers gravity from compression caused by proximal collectives of interlinked Photony energy filaments, each of which contributes a small compression and contributes to the overall energy filament compression and energy density which we perceive as mass. The compression of flexible Planck Space can result in a gravitational center. Gravitational waves and EM waves are somewhat quite different phenomena. Gravitational compression waves carry energy, and a system emitting gravitational waves loses energy.

- **Gravitational fields (gradients) result from the compression of quasi-static Planck Space** based on the presence of Photony filaments of energy. They are dependent upon filament energy density in nearby objects.
- The **propagation of gravitational waves is varying gradients of quasi-static Planck Space itself** (which is known to transmit waves from distant locations in space at light speed<sup>9</sup>)

Once the compression-induced curvature of space around a gravitational center is present, a gravitational mass ( $m_1$ ) field is established. Their gravitational fields become mutually coupled as nearby energy filaments within a separate mass ( $m_2$ ) maneuver in Planck space. **Each filament spin in each body is influenced as if going downhill towards the other body. This is the essence of Photony gravitation.**

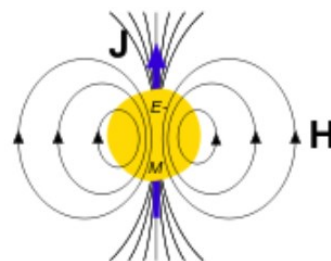
Formations of interlinked and proximal energy filaments of a “mass” with various stacked spin levels and relatively low collective en masse velocities are the basis of gravity (attraction) with “pull.” In contrast, divergent dynamic energy quanta (e.g., classical photons) are repulsive with “push.”

Depiction of two black holes (below) spiraling into a union as gravity waves are emitted as spiral space compression waves (LIGO Lab Caltech).



**Figure 8: Rotating Energy Density**

Below, the rotating mass-energy distribution with angular momentum,  $J$ , generate GEM fields. [83]



**Figure 9: Rotating Mass-Energy**

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<sup>9</sup> The speed of light is more accurately referenced as the speed of light in free space,  $c_0$ , a theoretical constant. However, space is largely not free space and therefor the speed of light in that context is a variable based on general relativity. So, the speed of light in space which proven now by LIGO to be the speed of space itself which is likewise variable.

# Photony Foundations

## C. Curvature of Spacetime

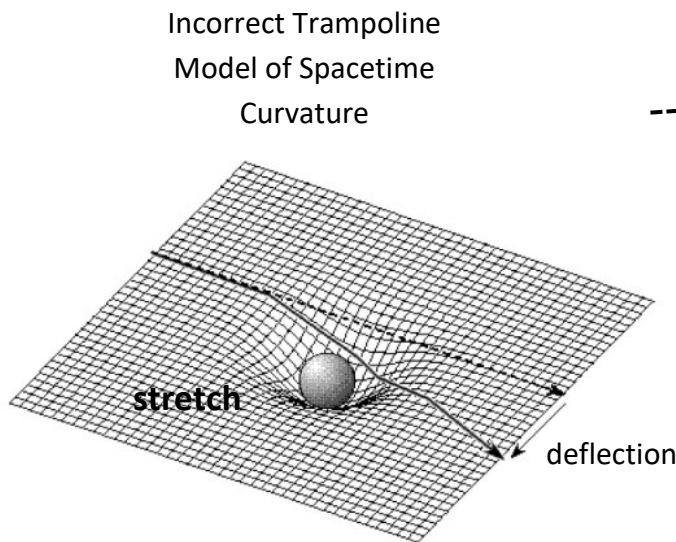


Figure 10: Trampoline Curvature Model

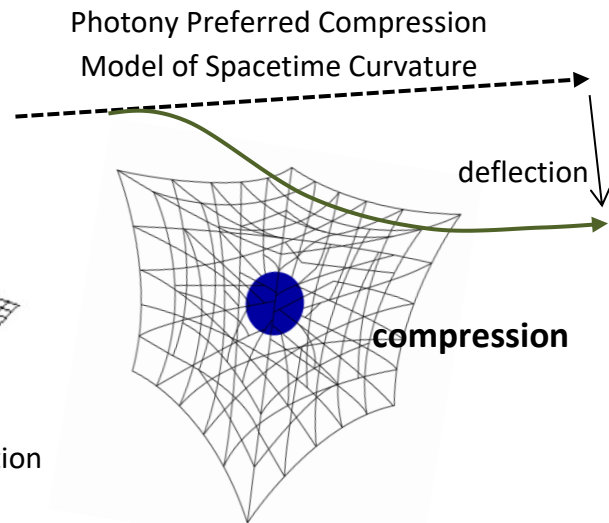


Figure 11: Compression Curvature Model

The diagram on the above left misrepresents gravitation as the stretching of spacetime (fabric) near an energy source is evidenced by larger (stretched) proximal cells. The above right diagram is preferred by Photony gravitation theory as the compression of spacetime near an energy source evidenced by reduced proximal cells caused by Photony energy filament compression! The compression of spacetime is due to the energy of the “massive object.” The **mass** of the “massive object” is a measure of spacetime compression related quantitatively by the  $c^2$  transform. According to Morgan D. Rosenberg [27] , in his paper; “Lorentz Contraction of Space and the Gravitational Field,”

*“By considering the gravitational field as an optical medium with a radially dependent index of refraction, **we are able to show that a physical model of space being radially compressed by mass<sup>10</sup>**, rather than curved by mass, as in general relativity, yields the same results predicted by Einstein’s theory of general relativity. **We are further able to show that this spatial compression is equivalent to the Lorentz contraction of special relativity.**”*

From Rosenberg, the gravitational field can be viewed as an optical medium (as in Gravitational Lensing). Rosenberg failed to recognize that energy density, not the mass, is the root cause of space compression. Spacetime compression follows Lorentz’s length contraction (and associated time dilation) teachings as well as general relativity curvature principles.

<sup>10</sup> Photony provides the correction of replacing mass hereby with energy as Einstein would likely concur.

## Chapter 2

### D. Photony Basis of General Relativity- Space U Energy

In general relativity, mass-energy warps spacetime (**Einstein tensor G**) [28] shown below in a 2-D slice of Planck Space.

The Einstein Tensor  $G$ , of his general relativity theory, confirms the **compression of Planck Space due to Photony filament energy** and motion as a basis for energy and mass, ("E-M") as illustrated (right) Maschen-Creative Commons CC0 1.0 Universal Public Domain Dedication

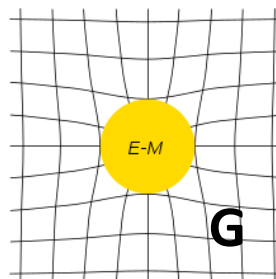


Figure 12: Einstein Tensor G

The radial compression of quasi-static Planck Space by energy is the basis of General Relativity and scalable. The motion and geodesic force applied to quasi-static compressed Planck Space, whereas the orthogonal nature of Planck Space distorts the otherwise geodesic shape of the energy filament. The figure below illustrates how the smallest particle energy density compresses quasi-static Planck space (e.g., Photony static elemental photons) ranging up to ponderable mass equivalence as high energy density (HED) particles or objects as indicated below (right).

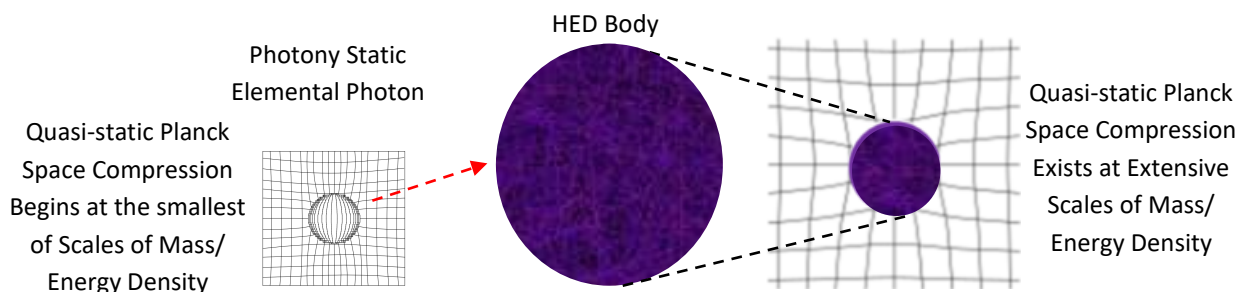


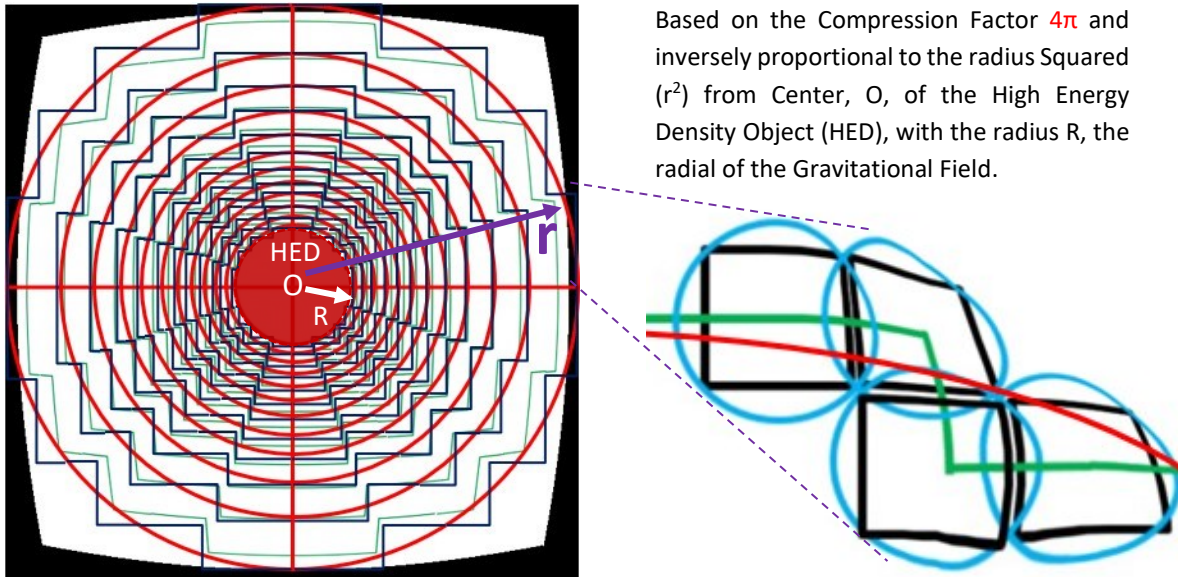
Figure 13: Planck Space compression Basis of General Relativity

The PHOTONY Motion/Spin Rule for energy filaments in Planck Space requires that filaments cannot transit laterally (broadside) through Planck Space without fracturing the filament or excessively compressing quasi-static Planck Space. Instead, the filament moves through the Planck Space network more like a worm with head-first motion. Therefore, the photon filament does not rotate or vibrate laterally like a jump-rope (disallowed and counter to string theory), but for a static elemental (charge) photon, the energy filament rotates by longitudinally shifting paths (indexing) at or near a polar index zone of Planck cells. Indexing of multiple geodesic filament pathways traces and embodies a virtual shell (a spherical elemental force field), making it an elementary particle.

# Photony Foundations

## E. Planck Space Compression Factor

$4\pi$



**Figure 14: Planck Space Compression Factor**

The composite cross-sectional view of Planck space around a high energy density (mass) with radius  $R$  with a break-out view of a few distorted quasi-static Planck space cells is shown.

The interaction of energy filaments and quasi-static Planck space is comparable to the interactive product of  $4\pi$  whereby the “4” represents the attempt at non-distorted Manhattan Metric orthogonal pathways and “ $\pi$ ” represents the energy filament’s smooth, attempted/preferred geodesic pathway shown in red in the above figure. The result of the interaction with Planck space is the distorted green pathway. The orthogonality of Planck space is also distorted, as shown in black. In the compression (interaction) factor  $4\pi$ , the 4 is PI according to the Manhattan metric, and  $\pi$  is the classical  $\pi = 3.1415...$  Since both are affected by the interaction, the product of the two ( $4\pi$ ) becomes theoretically fundamental.

The Photony compression factor,  $4\pi$ , is perhaps the most important of Photony discoveries since it comprehends the interaction between energy and quasi-static Planck space at the Planck Length scale. The interaction or product of  $4\pi$  is prolific in scientific equations and may be fundamental to nature. As an energy density (a collective mass/spacetime compression) attempts to move through quasi-static Planck space, the energy filaments within the energy density each follow the Photony Motion/Spin Rule as they maneuver in Planck space. Each energy filament is distorted along with proximal Planck space accordingly.

# Chapter 2

## F. Lorentz Length Contraction and Planck Space Compression

The compression of quasi-static Planck space due to filament energy density and motion is equivalent to, and the basis of, length contraction and time dilation. The Photony Motion/Spin Rule for energy filaments governs quasi-static Planck Space compression based on the common velocity of all the energy filaments, resulting in inertial mass.

The hypothesis is proposed that the ether is the permittivity and permeability of quasi-static space, including free space. Since the ether is defined, the stationary state is also secured, making the Planck space network simpler and more intuitive. The hypothesis of complete frame-dragging by the gravitational field is also considered, which varies the values of the permittivity and permeability of free space to satisfy the time dilation by the gravitational potential. The values of the permittivity and permeability of quasi-static Planck space become larger due to gravity (acceleration) **to decrease the speed of light,  $c$ .**

There is a clear distinction between solid and non-solid mediums regarding relativity. For some length to be contracted, it must be solid. For a relative velocity between a sender of photons and a receiver of photons, they must be in a non-solid such as space or atmosphere. **By showing how length contraction and time dilation are affected in gravitational fields, we can suggest that length contraction and quasi-static Planck space compression are the same or equivalent phenomena.**

According to Masanori Sato;

*“... the gravitational field determines the values of the permittivity of free<sup>11</sup> space and the permeability of free space needed to satisfy time dilation by the gravitational potential. That is, the values of the permittivity and permeability of free space are modified by the gravity to isotropically decrease of the speed of light (that is, the increase of a refractive index).” [29]*

Inertial effects associated with velocity are related to “inertial mass” and special relativity since it is about energy in motion at a constant relative velocity. However, all entities in the Universe have some absolute velocity relative to the origin of the Universe and, therefore, some acceleration history.

Regarding the length contraction from the Photony point of view, **the energy filament originated quasi-static Planck Space compression is what Photony equates with length contraction.** This Photony energy-density-based Planck Space compression directly results in **gravitational potential** within, on the surface, and extending from the surface as a gravitational gradient around an entity, for example, Earth. On Earth, gravity slows down clocks (time dilation) and shortens lengths (length contraction) based on Photony energy filament density hiding within the law of general relativity.

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<sup>11</sup> The first two instances of the term “free space” (yellow highlighted) are incorrectly used since “free space” is defined as being free of gravitational and electromagnetic potential; thus, the relative permittivity and permeability each equal 1. The third instance of the term “free space” (green highlighted) is correct since changes to permittivity and permeability apply.

# Photony Foundations

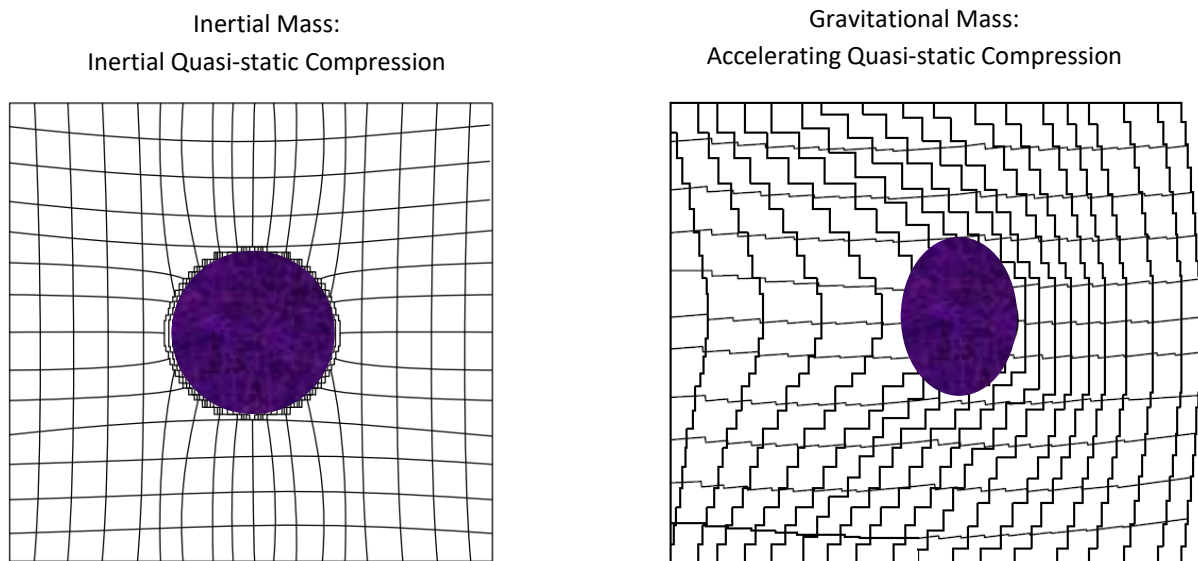
## G. Motion-Based Planck Space Compression

Everything in the Universe is in motion, either accelerating or moving at a drift velocity which we refer to as inertial velocity.

A frame of reference is constant or uniform when in rest (never the case), or motion within a common coordinate system is known as an inertial frame of reference. This means that all the particles and objects of ponderable mass share a common velocity vector component as viewed by an observer outside the reference frame.

An inertial velocity is any fixed or constant velocity (no acceleration). The occupants and equipment within a vehicle share a common frame of reference. However, objects within the vehicle can be in motion and even accelerated based on proper time motion. The occupants share the vehicle's velocity and acceleration vectors with their own velocity and acceleration vectors in superposition.

A cross-sectional (2-D) view of the inertial quasi-static compression (below left) illustrates how a zero to low-speed high-energy-density (HED) affects compression. A cross-sectional (2-D) view of the accelerating or high-speed HED quasi-static compression (below right).



**Figure 15: Inertial vs. Accelerating Quasi-static Planck Space Compression**

In the following figure, the high-speed object shown is an exemplary flashlight with an LED light source and a lens (not shown) to direct the photons away from the object in the Z-direction. Each of the photons in the light-ray is going at the speed of space, which is a function of the permeability and permittivity and is independent of the flashlight's velocity.



## Chapter 2

A perspective view of the inertial quasi-static compression (below left) with a smooth gravitational gradient illustrates how a high-speed or accelerating energy density (HED) affects compression. From the example (below), three zones are identified as quasi-static Planck space reacts at the speed of space,  $c_0$ , (the speed of light in free space), to a light-speed object compression wavefront, a compression zone, and a compression trail. A similar view (below right) illustrates a segmented quasi-static Planck space gravitational gradient.

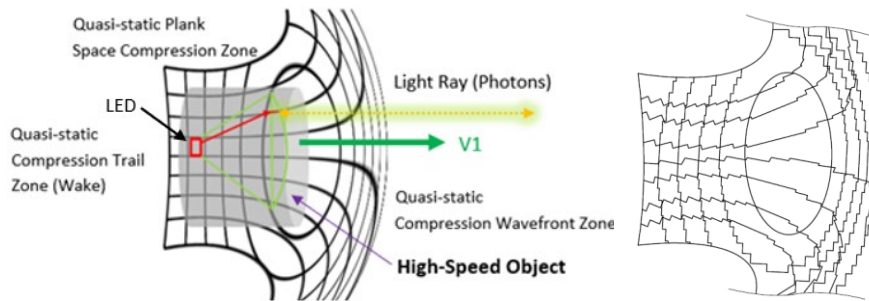


Figure 16: Inertial and Accelerated Gravity Fields

Several aspects of this scenario are important to comprehend a static gravity field gradient of a low-speed object (gravity mass), and an accelerated gravity field associated with the inertial mass contributes to the overall gravity field strength and gradient.

The wavefront zone comprises a quasi-static Planck space compression wave, not unlike a gravity wave based on variations in speed (accelerations). It is in the compression zone that the high-speed object aligns the quasi-static Planck space network. The compression zone extends proximally to the object in the X-Y plane at all velocities in the z-direction, where compression occurs based on the Photony Motion/Spin Rule. The result is called length contraction. The inertial quasi-static Planck space compression trail/zone (wake)- a funnel-shaped compression zone (wake) trails a high-speed object as its relativistic mass increases and its length decreases.

To the outside observer, the length of a high-speed solid transparent object contracts (shortens), and the time within the inertial medium dilates exactly to maintain light speed and proper time within the medium. The permittivity  $\epsilon$  and permeability  $\mu$  adjust accordingly based on  $c^2 = 1/\epsilon\mu$  to maintain the constancy of light speed. The length contraction is further explained by the Photony Motion/Spin Rule, whereby Photony energy filaments exert a compressive force on quasi-static Planck space that increases during the motion of the entity (object) comprised of the energy filaments.

In the general theory of relativity developed by Albert Einstein, the Gravitational Equivalence Principle comprises the equivalence of increasing inertial (accelerating) mass and gravitational mass. This principle states that mass determined by inertia is the same as mass determined by gravitation.



# Photony Foundations

## H. Gravitational Wave Motion

Energy quanta pass through media (space) at the same speed as a gravitational wave would pass through. In the figure below, two high energy density bodies (e.g., black holes) gradually spiral into merger. During this period, gravitational waves are emitted and travel through space at a speed determined by permittivity and permeability which can vary along the way in real space as opposed to free space.

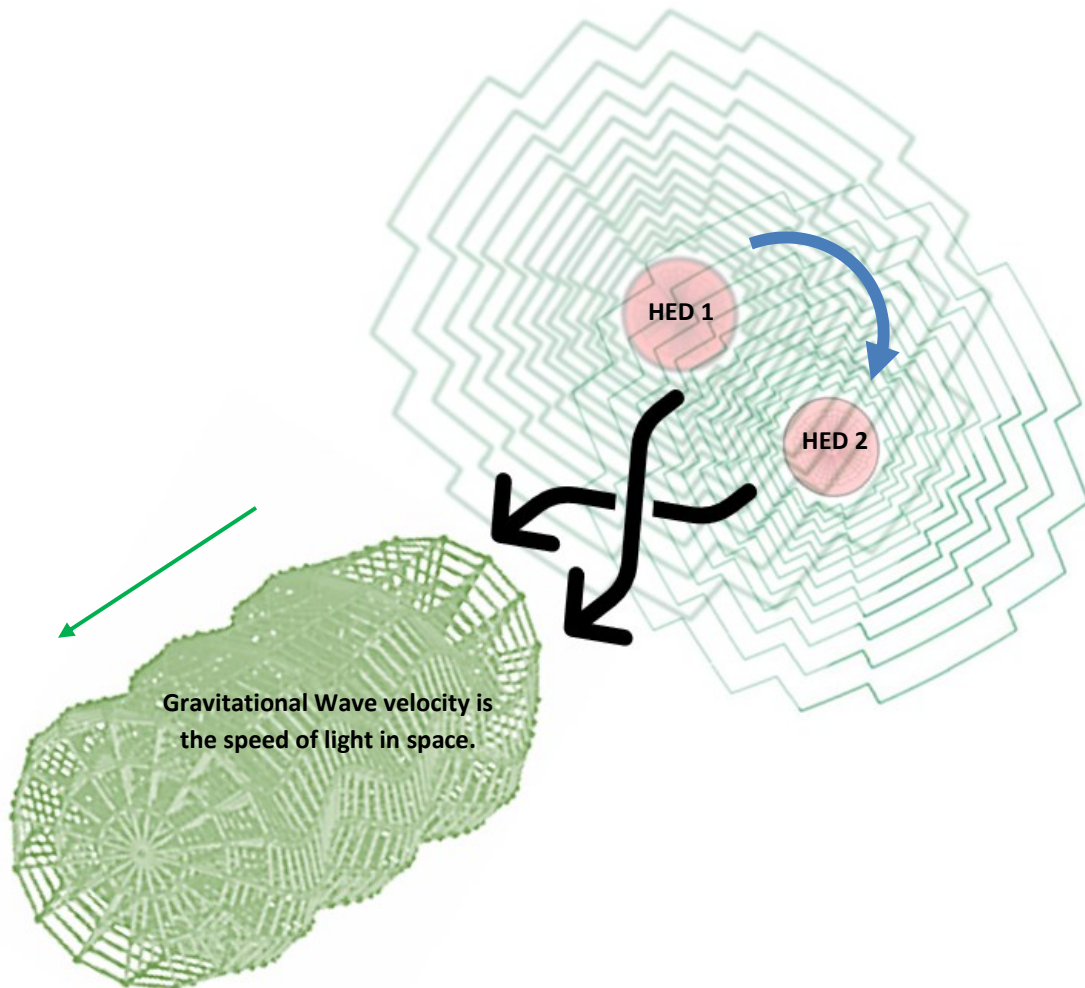


Figure 17: Gravitational Wave Motion

# Chapter 2

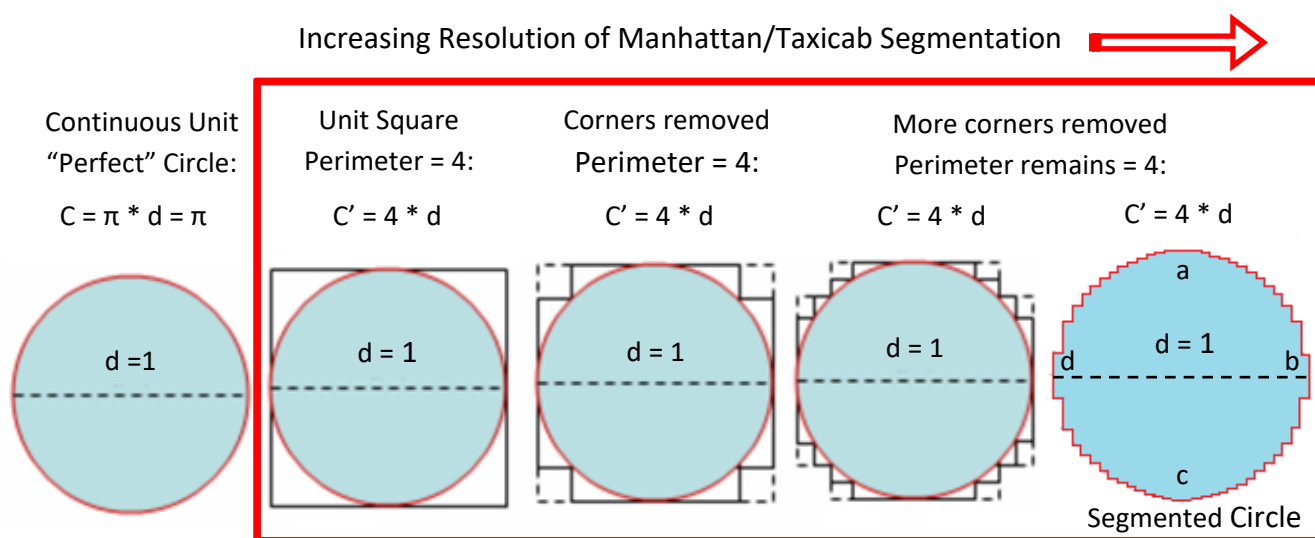
## I. Here's PI in your face

To avoid confusion, the upper-case PI and Greek symbol,  $\Pi$ , are used for the ratio of the segmented Manhattan/taxicab circumference (perimeter)  $C'$  to the diameter;  $C' = \Pi * d = \Pi * 2 * r$ .

### 1. The Manhattan Circle

See in Figure 18 below a graphic illustration of how  $\Pi = \Pi = 4$  for Manhattan/taxicab circumference (perimeter)  $C'$ . As the segmented perimeter  $C'$  approaches the smooth circular perimeter with increasing resolution of the segments, the need for  $\Pi = 4$  in the Manhattan or taxicab metric becomes apparent.

Circumference of a smooth spherical object geodesic (circle) is pi (3.14159...) times the diameter, whereas the circumference of a "taxicab circle" is 4 times the diameter.



**Figure 18: Continuous vs. Segmented Perimeters**

The highly resolved segmented circle (above right) represents a Photony geodesic energy filament in Planck Space.

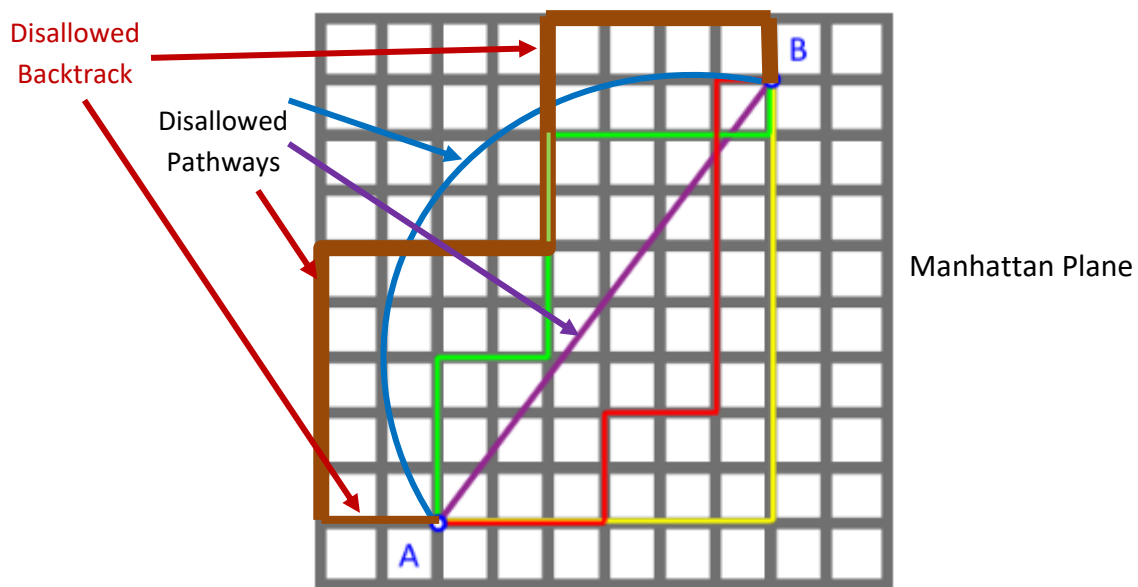
Note: there is no requirement for the removed corners to be of equal size in either dimension, as shown above. Different segmented pathways would satisfy the  $C' = 4 * d$  ( $\Pi = 4$ ) kinematic motion condition. Each allowable segmented path must follow two rules:

- No backtracking- deviations between endpoints due to collisions must not exceed the monotonic motion limits, thus avoiding backtracking
- Quadrant endpoints a-b-c-d must be intersected.

# Photony Foundations

## 2. Allowable Manhattan Pathways

The 2-d matrix below represents city blocks from a Manhattan/taxicab metric point of view. A taxi starts at location A and proceeds to drive to point B. There are several possible and allowable routes; the yellow, red, and green routes are possible and allowable. Disallowed routes include purple since it is diagonal, and the taxi must stay on the streets and blue since it is an arc that would take the taxi off the streets. There is also a disallowed route (20 blocks) that is orthogonally possible; (brown pathway) since the taxi is running up the meter by taking a longer route than necessary (14 blocks) without backtracking. This routing relates to the Photony conservation of filament energy (no backtracking)!



**Figure 19: Allowable Manhattan Pathways**

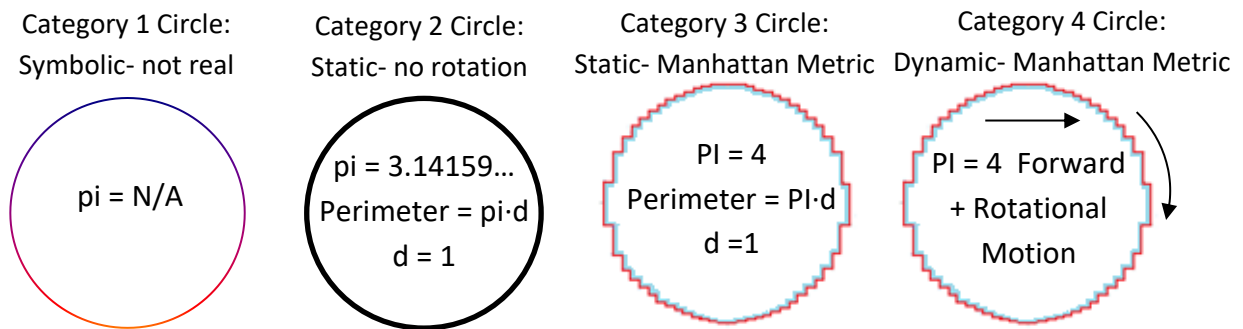
It is also shown beyond any doubt that  $\Pi = 4$  for Manhattan/taxicab circumference. [30]  
Consider the point of view of Miles Mathis as found in [The Extinction of  \$\pi\$](#) ; [31]

*“Abstract: I show that in all kinematic situations,  $\pi$  is 4. **For all those going ballistic over my title, I repeat and stress that this paper applies to kinematic situations, not to static or geometric situations.** I am analyzing the equivalent of an orbit, which is caused by motion and includes the time variable. In that situation,  $\pi$  becomes 4. I will also remind you this is not just a theory: it has been indicated by many mainstream experiments, including rocketry tests and quantum experiments ...”*

It can be uncomfortable to let go of assuming that classical  $\pi$  ( $\pi = 3.1415926535\dots$ ) applies to perimeters and rotational (spin) motions (e.g., filament rotations) in dynamic and kinematic scenarios (e.g., spin during forward motion), especially in describing quantum motion within the Planck Space environment.

# Chapter 2

## 3. Manhattan Metric Statics and Dynamics



**Figure 20: Categories of Circles**

Off we go into subject matter that may require as much rethinking as the discovery that the world is not flat, the Earth is not the center of the Universe, and black hole theory is full of holes. We start with unit (diameter) circles; herein, we ‘reinvent the wheel.’ There are four categories of interest:

a. The Symbolic Circle

The first category of the unit circle is a theoretical or “perfect circle” with no physicality and an infinitesimally small perimeter width, and a perfectly smooth and continuous boundary. The unit circle's associated drawing merely invokes a perfect circle's attributes, much like a symbol or icon.

b. The Static Circle

The second category of the unit circle is the common “static circle” that is drawn or plotted with the circular boundary line symbolizing a real circular boundary. The circumference of this circle is measured without regard to rotation. The static circle can be dismissed as being ornamental or a spherical cross-section, having a boundary representing a circle without exactly being one.

c. The Manhattan Circle

The third category of the unit circle is a “segmented circle” routed according to the Manhattan or taxicab metric. Kinematics in Planck Space requires segmented orthogonal routing with a preference for geodesic filament pathways to support the least action/energy principle. **For statics, Photony emphasizes this category of circles.**

d. The Dynamic Circle

The fourth category of unit-circle begins as a Category 3 Circle and is a unit ( $d=1$ ) **dynamic Manhattan circle** with forward motion in superposition with the rotating angular motion whereby the tangential velocity equals the forward velocity.

# Photony Foundations

## 4. Cycloidal Motion- Dynamic Circles

Now we turn our attention to cycloids. [32] According to Weisstein's work, *Cycloid*;

*"The cycloid is the locus of a point on the rim of a circle of radius a rolling along a straight line. It was studied and named by Galileo in 1599. Galileo attempted to find the area by weighing pieces of metal cut into the shape of the cycloid. Torricelli, Fermat, and Descartes all found the area. The cycloid was also studied by Roberval in 1634, Wren in 1658, Huygens in 1673, and Johann Bernoulli in 1696. Roberval and Wren found the arc length (MacTutor Archive)"*

A detailed look at the cycloid comprised of arcs with humps and adjoining cusps follows.

- The arc length [33] for each hump: Cycloid Area  $A_c$

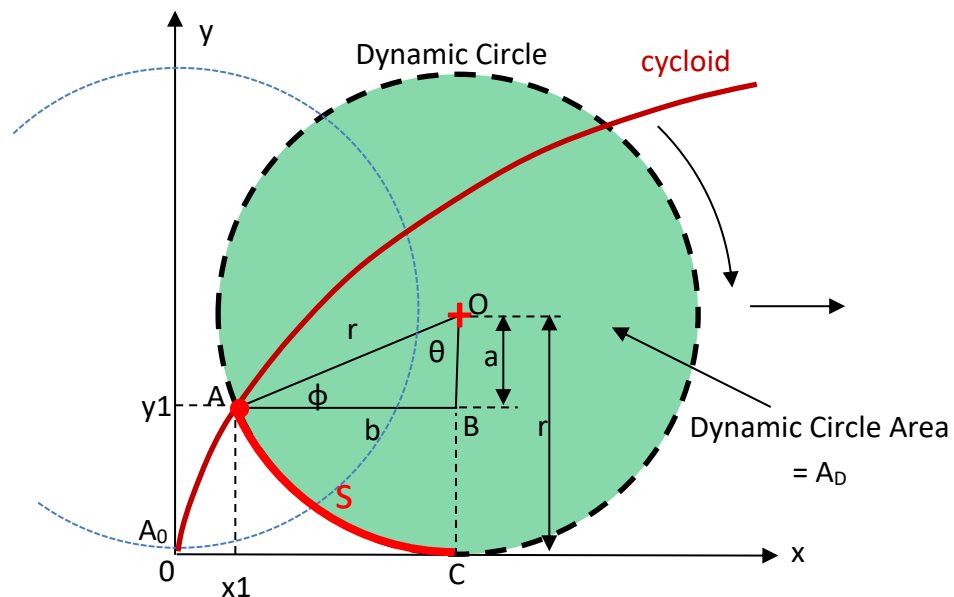


Figure 21: Cycloid Trig

From right triangle ABO:  $a = r \cos \theta$  and  $b = r \sin \theta$   
 $S = \text{arc length} = \theta \times r$   
 $x = s - b = \theta r - r \sin \theta = r(\theta - \sin \theta) \quad dx = r(1 - \cos \theta) d\theta$   
 $y = s - a = \theta r - r \cos \theta = r(\theta - \cos \theta) \quad dy = r \sin \theta d\theta$   
 $dy/dx = r \sin \theta d\theta / r(1 - \cos \theta) d\theta = \sin \theta / (1 - \cos \theta)$

$$s = \text{arc length} = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx =$$

$$= \int_0^{2\pi} \sqrt{1 + \left(\frac{\sin \theta}{1 - \cos \theta}\right)^2} r(1 - \cos \theta) d\theta$$

solving, the cycloid arc length  $s = 8r$  felt by the red dot

$L = 8r = 4d$  Equation 9: Arc length so,  $PI = 4$  for cycloid

The area under each cycloid arc  $A_c$ :

Similarly, cycloid area:

$$A_c = \int_{x_1}^{x_2} y dx =$$

$$A_c = r^2 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = 3\pi r^2$$

$A_c = 3\pi r^2 = 3A_D$  Equation 10: Area under cycloid arc  
 where  $A_D$  is the area of the static unit circle

Example:

for unit circle,  $d = 1$ ,  $r = 0.5$ ,  $A_D = \pi r^2 = 0.7854$

so. cycloid area  $= 3\pi \cdot (0.5)^2 = 2.3562$

**cycloid length  $= 8r = 4$**

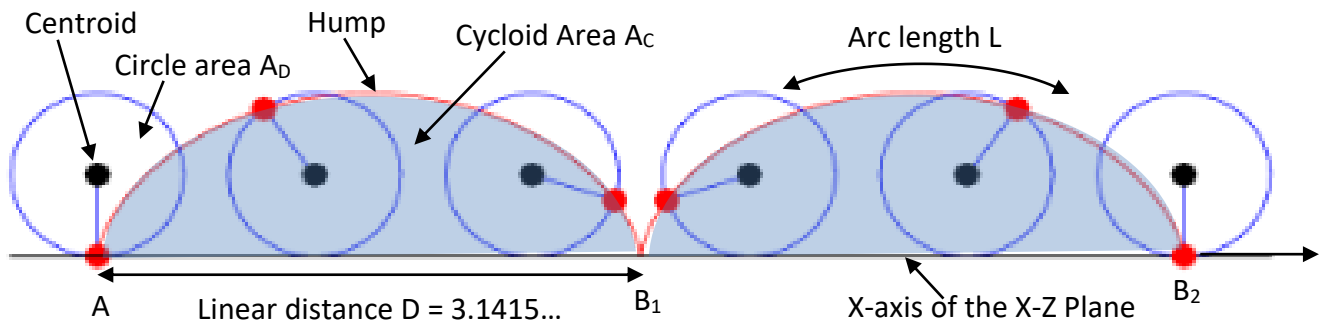
# Chapter 2

## 5. Dynamics and Motion- Cycloid of the **Category 4 Dynamic Circle**

Dynamic unit-circle in rotational motion without slippage on the XZ plane. Linear distance (below), the distance traveled of centroid:

$D = \pi \cdot 2r$  Equation 11: Linear distance traveled  $AB_1$  without rotation

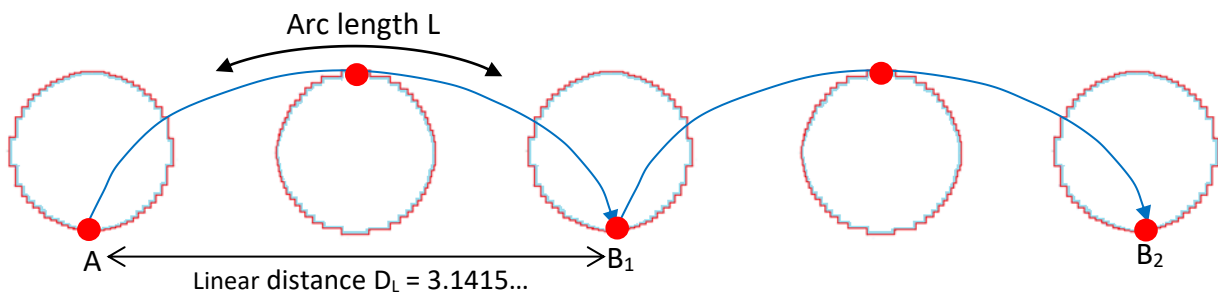
Where  $\pi = 3.1415...$  and  $d = 2r$  (unit circle) distance traveled by the centroid for one full rotation of the dynamic circle. The dynamic circle represents either a smooth unit circle or a geodesic cross-section of an essentially spherical particle or object in motion.



**Figure 22: Category 4 Dynamic (Mathematical) Circle and Arc**

The distance traveled by a point at the perimeter of a smooth unit-circle rolling between points A and  $B_1$  on the X-Z plane, is the arc length  $L = 4d = 4$ , not the A- $B_1$  linear distance, D. This **arc length is due to the Photony filament spin with traction in quasi-static Planck Space**).

**Particle with forward and rotational motion-** The segmented cross-sectional view of a spherical particle spinning through open space (a virtual X\_Z plane) is shown below. A small section (depicted by a red dot) at the perimeter of the particle begins at location A and maneuvers through quasi-static Planck space to location  $B_1$ , pulled by the particle at large as it spins and moves forward at some velocity  $v$ .



**Figure 23: Smooth Cycloidal Pathway- Segmented Particle Perimeter**

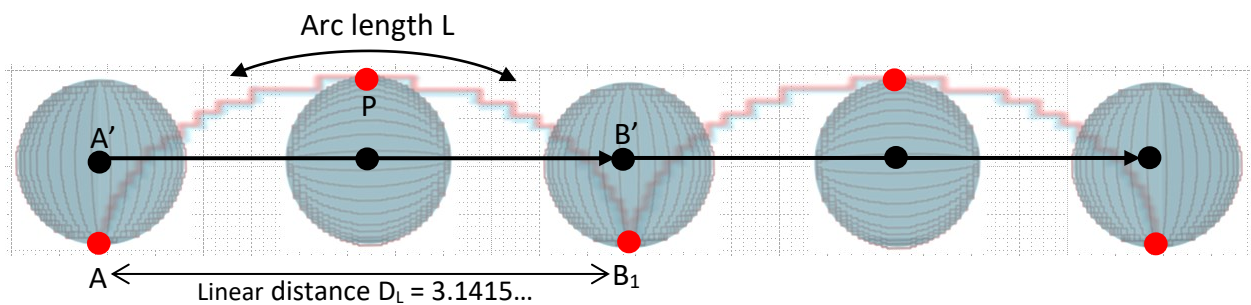
# Photony Foundations

## Particle with forward and rotational motion:

The segmented cross-sectional view of a spherical particle spinning through quasi-static Planck space (in a straight-line trajectory) is shown below. A small particle section (depicted by a red dot) at the perimeter of the particle cross-section begins at location A and maneuvers through quasi-static Planck space to location B<sub>1</sub>, pulled by the particle at large as it spins and moves forward at some velocity v.

The particle can be as small as energy quanta (Photony elemental photon, or a classical photon) traveling at the speed of space (c), or as large as a particle traveling at less than c, including sub-relativistic velocity for larger objects of ponderable mass. The differences between the quanta and the massive particles and objects include:

- Energy Quanta include Photony elemental photons, each with one Photony energy filament, and classical photons, each with two interlinked energy filaments.
- Quanta travel at the speed of space ( $c = 1/\sqrt{\mu\epsilon}$ ) and have tangential velocity c.
- Particles and objects travel at a velocity v, which can be relativistic.



**Figure 24: Photony Segmented Cycloid Pathway**

Pathways- unit circle (d = 1):

The cycloidal arc distance between A and B<sub>1</sub> (APB<sub>1</sub>) is L, whereas the AB<sub>1</sub> linear distance is  $D_L = \pi d = 3.1415...$ . The segmented cross-section is depicted as rotating in space with the tangential velocity equaling the forward velocity. The cycloidal arc distance for red-sectional reference point on the segmented cross-section is  $L_C = 4d = 4$ .

Manhattan segmented distance (independent of pathway- no backtracking) between A and B<sub>1</sub>:

$D_S = 2AA' + A'B' + 2B'B = 2r + D_L + 2r = 5.1415...$  This distance would apply for a non-rigid sensing point following the guidance of orthogonal Planck space during the motion. However, quasi-static Planck space is distorted by the Photony compression factor =  $4\pi = 12.56$  and the compression ratio =  $4\pi/16 = 0.785$ , and  $D_S$  is subject to the compression factor:

$D_S 0.785 = 4.04$ , which is close to the smooth cycloid length  $L_C = 4$ .

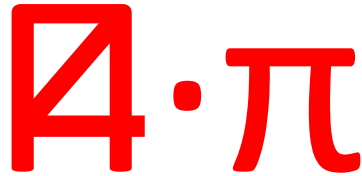
## Chapter 2

(This) Page left essentially blank on purpose to give the reader a moment to comprehend and adjust to the idea that most scientific phenomena involve the Photony Energy/Space Compression Factor,  $4\pi$ , and at the ultimate resolution of space,  $PI = 4$ .



# Photony Foundations

## J. The Battle of Pi(s)



Filaments of energy flow through the Planck space fundamental network of cells and interact with it as each experience some degree of strain and distortion. Planck space wins this distortion battle by enforcing the Manhattan Metric while the distortion is exhibited as compression (mass). The battle of strain and distortion is fundamentally reduced to the “Battle of Pi’s” or  $\text{PI} \cdot \pi = \text{PI} \cdot \pi = 4\pi$ , where 4 is the kinematic PI (Manhattan metric’s segmented perimeter and cyclonic forward motion).  $\pi$  is the classical (smooth geometric and trigonometric angular motion) pi. The product,  $4\pi$ , is found in many equations and hidden in certain constants/parameters, making it worthy of further study.

It is not difficult to visualize a compressed quasi-static (flexible) Planck Space as a tug of war between an energy filament trying to rotate smoothly on a curved circular (geodesic) pathway but constrained by the orthogonally interconnected Planck cells. In the previous illustrations, “With Planck Space Compression,” stress is identifiable in both the blue Planck cells and the green energy filaments. With some visualization, the compression of space (mass) can be seen **as a battle between the classic pi and the Manhattan PI.**

We may find pi and PI products, i.e.,  $4\pi$ , in some physics equations; the 4 therein may not originate from four quadrants as  $2\pi$  radians in the derivation but from the Manhattan metric  $\text{PI} = 4$ .

The  $4\pi$  product is defined by Photony as the “compression factor.” Photony examines  $4\pi$  in this context of interactive energy compression of quasi-static Planck space: Spacetime compression is a phenomenon of General Relativity governing gravitational wave propagation and spacetime gradients, and compression phenomenon of Special Relativity as a function of velocity (referred to as length contraction).

### 1. What about all the Equations with $4\pi$ ?

It depends; areas and volumes with  $\pi = 3.14\dots$  are correct as is. Some are possibly wrong involving kinematic/dynamic motion or static perimeter-based phenomena in Planck Space. The Manhattan Metric essentially governs and interacts with energy filaments motion in Planck Space. All things real are composed of interlinked energy filaments attempting to follow the Manhattan ( $\text{PI} = 4$ ) rule. Mass is the measure of energy based on quasi-static (flexible) Planck Space compression/curvature resulting from a density of these interlinked energy filaments.

## Chapter 2

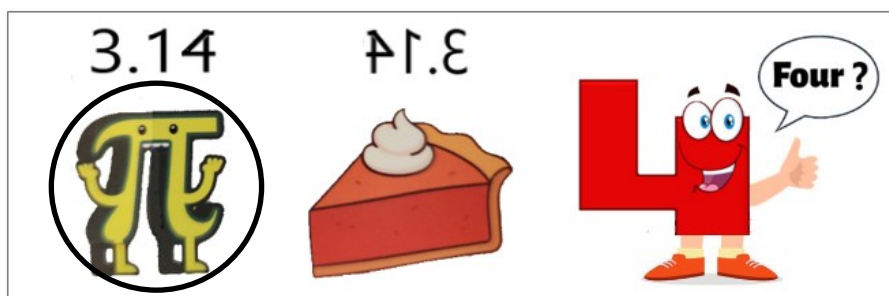


Figure 25: Pie Talk (Alton Creed)

The  $2\pi$  factor, including  $2\pi \times 2 = 4\pi$  are ubiquitous in science and mathematics. The  $2\pi$  factor seen in geometry and trigonometry is based on radians as natural angular units as opposed to degrees whereby  $360^\circ = 2\pi$  radians. A  $2 \cdot 2\pi$  factor is associated with the surface area and volume of a sphere of radius  $r = d/2$  (area =  $2\pi \cdot 2r^2$  and volume =  $2\pi \cdot \frac{2}{3}r^3$ , respectively) but is debatably outside the context of spacetime compression. Using diameter,  $d$ , area =  $\pi d^2$  and Volume =  $\pi d^3/6$ .

An example of  $2\pi$  used at the fundamental level of physics is Planck's constants,  $h$  and  $\hbar$ , whereby the energy quantification  $h$  is based on frequency,  $f$ , and  $\hbar$  is based on angular frequency,  $\omega$ . So, the  $2\pi$  converts between the two Planck constants,  $\hbar = h/2\pi$ . (no spacetime compression).

Permeability and permittivity are fundamentals of spacetime, the basis of the speed of light and the impedance of space. When these parameters are assigned to free space, the relative permittivity  $\epsilon_r$  and permeability  $\mu_r$ , result in a net zero compression as follows:

- Permeability:  $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$  Equation 12  
The compression factor is front and center here. Widely known!
- Permittivity:  $\epsilon_0 = 8.854 \times 10^{-12} \approx 4\pi / \sqrt{2} \times 10^{-12} \text{ F/m}$  Equation 13  
The compression factor may be hidden within the standard numerical value here.
- $Z_s = Z_p = 1/4\pi \epsilon_0 c = 29.98 \Omega (\approx 30)$  Equation 14  
The Planck impedance equals the strain impedance of spacetime.
- $Z_o = E/H = \mu_0 c = \frac{1}{\epsilon_0 c} = \sqrt{\mu_0/\epsilon_0} = 367.7 \Omega$  Equation 15

This intrinsic impedance encountered by EM waves ( $Z_o$ ) is  $4\pi$  times the impedance encountered by gravity waves:  $Z_o = 4\pi \cdot Z_s$

- $Z_o = 4\pi \cdot Z_s$  Equation 16  
 $Z_s = Z_o/4\pi = 376.73/4\pi = 29.98 \Omega$  Equation 17

# Photony Foundations

Electron charge:

$$m_e = \frac{\mu_0}{4\pi} (q_e^2 / r_e) \quad \text{Equation 18}$$

$$\text{so, } q_e = \sqrt{\frac{4\pi m_e r_e}{\mu_0}} = \sqrt{\frac{4\pi m_e r_e}{4\pi \times 10^{-7}}} = \sqrt{m_e r_e c / 30} \quad \text{Equation 19}$$

solving for  $r_e$ :

$$r_e = 30 q_e^2 / m_e c = 2.81 \times 10^{-15} \text{ m} \quad \text{Equation 20}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \quad q_e = 1.6 \times 10^{-19} \text{ coulomb}$$

$$m_e = \frac{4\pi \times 10^{-7}}{4\pi} \left( \frac{q_e^2}{r_e} \right) = \frac{q_e^2}{r_e} \times 10^{-7} = 9.109 \times 10^{-31} \text{ kg} \quad \text{Equation 21}$$

*Note (above) that the Photony compression factor,  $4\pi$ , is uncoupled to charge. It is deterministic to mass and permeability as indicated above. The search for answers to the question, "What is charge?" is partially answered here: it does not compress Planck space.*

We continue to search for contenders for equations comprising a  $4\pi$  compression factor, where in spacetime compression exists, here are a few:

- Heisenberg Uncertainty Principle:

$$\Delta x \cdot \Delta(mv) = \Delta x \cdot \Delta p \geq h / 4\pi \quad \text{Equation 22}$$

$$\Delta t \cdot \Delta E \geq h / 4\pi \quad \text{Equation 23}$$

Where  $\Delta x$  = position uncertainty,  $h$  = Planck's constant,  $\Delta p$  = momentum uncertainty,  $\Delta t$  = time/position uncertainty,  $\Delta E$  = energy uncertainty

*The root cause of uncertainty is the photonic energy it takes to illuminate and measure a particle alters the momentum of that particle.*

- Coulomb's Law:  $F = k q_1 q_2 / r^2 = q_1 q_2 / (4\pi \epsilon_0 r^2)$  Equation 24

Note- Coulomb's constant  $k = 1 / (4\pi \epsilon_0) \text{ N} \cdot \text{m}^2 / \text{C}^2$

Where  $q_1$  and  $q_2$  are the charge of objects 1 and 2, respectively and  $r$  = distance apart

Checks:

$$K = 8.9875517923 \times 10^9 =$$

$$K = 1 / (4\pi \epsilon_0) = 1 / (4\pi \cdot 8.854 \times 10^{-12}) = 8.987742 \times 10^9 \quad \text{Equation 25}$$

$$\text{Using } \epsilon_0 = 4\pi / \sqrt{2} \times 10^{-12}:$$

$$K \approx 1 / ((4\pi)^2 / \sqrt{2} \times 10^{-12}) = 8.895 \times 10^9 (\approx \text{check})$$

*This result is close enough to keep alive the possibility of a built-in  $4\pi$  in the permeability of free space.*

## Chapter 2

- Effective Mass density of space:  $\rho_{vac}$

$$\text{critical density [34]} \quad \rho_c = 3H^2/8\pi G \quad \text{Equation 26}$$

where the Hubble constant is  $H = 70 \times 10^3 \text{ m/s}/10^6 \text{ Parsec}$

universal gravitational constant:  $G = 6.67430 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

$$\begin{aligned} \rho_c &= 3H^2/8\pi G = 3(70/3.0857 \times 10^{19} \text{ km/MParsec})^2 / (8\pi 6.67430 \times 10^{-11}) = \\ &= 3(5.444 \times 10^{-36})/1.6763 \times 10^{-9} = 9.743 \times 10^{-27} \text{ kg/m}^3 \end{aligned}$$

$$\rho_{vac} = 0.76 \times \rho_c = 7.4046 \times 10^{-27} \text{ kg/m}^3 \quad \text{Equation 27}$$

*Note that  $\rho_c = 3H^2/8\pi G$  could be reconfigured to be  $\rho_c = 1.5H^2/4\pi G$ , so the presence of  $4\pi$  is worthy of further study regarding the Photony compression factor.*

$$\begin{aligned} \text{Equivalent energy } E_{vac} &= mc^2 = 7.4046 \times 10^{-27} \text{ kg/m}^3 c^2 = \quad \text{Equation 28} \\ E_{vac} &= 6.664 \times 10^{-10} \text{ J/m}^3 \end{aligned}$$

- Cosmological Constant:  $\Lambda = 8\pi \rho_{vac} G / c^4 = 4\pi \rho_{vac} G / 2c^4$  Equation 29

Where intrinsic energy density of the vacuum =  $\rho_{vac}$

$$\begin{aligned} \Lambda &= 8\pi \rho_{vac} G / c^4 = 8\pi (7.4046 \times 10^{-27}) 6.67430 \times 10^{-11} / (2.998 \times 10^8)^4 = \\ &1.24 \times 10^{-35} / 8.078 \times 10^{33} = 1.536 \times 10^{-69} / \text{m}^2 \end{aligned}$$

*Note that  $8\pi$  ( $2 \times 4\pi$ ) is called the “proportionality factor.” (Wikipedia- cosmological constant)*

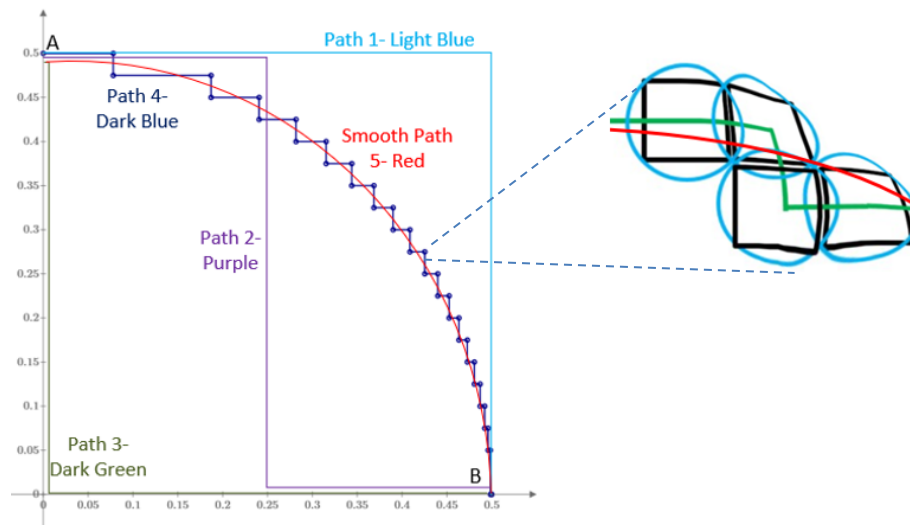
- Newton’s Gravitational Constant: Force,  $F = Gm_1m_2/r^2$  could be written as  $F = G'(4\pi) \cdot m_1m_2/r^2$  where  $4\pi$  is hidden in  $G$ :  
 $G = 6.674 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^2 \approx 1/(16\pi \cdot c) = 1/(4\pi \cdot 4c) = 6.636 \times 10^{-11}$

*Note: **G is determined by  $4\pi$  and  $4c$  ( $4 \times$  the speed of space)** since gravitational information points from  $m_1$  to  $m_2$  and from  $m_2$  to  $m_1$ , and each G-wave travels at the speed of space!*

We see that the interaction or product of  $4\pi$  is prolific and may be fundamental to nature. As an energy density (a collective mass/spacetime compression) attempts to move through quasi-static Planck space, the energy filaments follow the Photony Motion/Spin Rule. Each is distorted along with proximal Planck space accordingly. The interaction of energy filaments and quasi-static Planck space is comparable to the interactive product of  $4\pi$ , whereby the “4” represents the attempt at non-distorted Manhattan Metric orthogonal pathways and “ $\pi$ ” represents the energy filament’s smooth, attempted/preferred geodesic pathway.

# Photony Foundations

## 2. Geodesic Compression in Planck Space:



**Figure 26: Manhattan Paths 1-4 vs. Smooth Path 5**

Planck Surface Differential Element(s), Cross-sectional (side) View:

Red line- the ideal geodesic path of the Photony Filament- it would take this path if space was flaccid or infinitely resolved to satisfy its  $\pi = 3.14159...$  goal

Green line: the Manhattan Metric directed path of the filament through Planck Space- energy would take this path of space was static (rigid) and complete its  $\pi = 4$  goal.

Black boxes- the Planck length orthogonal directors

Blue circles- the Planck Space Cells ("Photony spacetime foam")

The diagram above shows four paths, including four Manhattan/taxicab paths (1-4) and one smooth, infinitely resolved circular arc (Path 5, red). The classical  $\pi = 3.14...$  clearly applies to the red path. Photony energy filaments attempt to travel as geodesics by attempting to have a smooth circular path, as shown in red wherein  $\pi = 3.14...$ , but are constrained by quasi-static Planck space to travel in a segmented Path 4 (Dark Blue) path 2 resulting in  $\pi = 4$ . The energy filament path becomes somewhat warped away from orthogonality as the adjacent quasi-static Planck space cells compress (see pullout above). The combination of the two warpages is captured/quantized by the product,  $4\pi$ . Photony hypothesizes that the filaments and any Planck cells that they pass through interactively experience stress. This makes for an interesting transformation of rectangular Planck space to curved quasi-static Planck space.

Note that without curvature, for  $N = 1$  horizontal segments and  $r = 0.5$ , the single quadrant shape reverts to a partial circumference of 1, as with all other resolutions/shapes. The path 1 perimeter length of the light blue above is the same as the segmented blue circle. Assuming a radius of 0.5 for the circle, the perimeter of full 4-quadrant shape is 4.

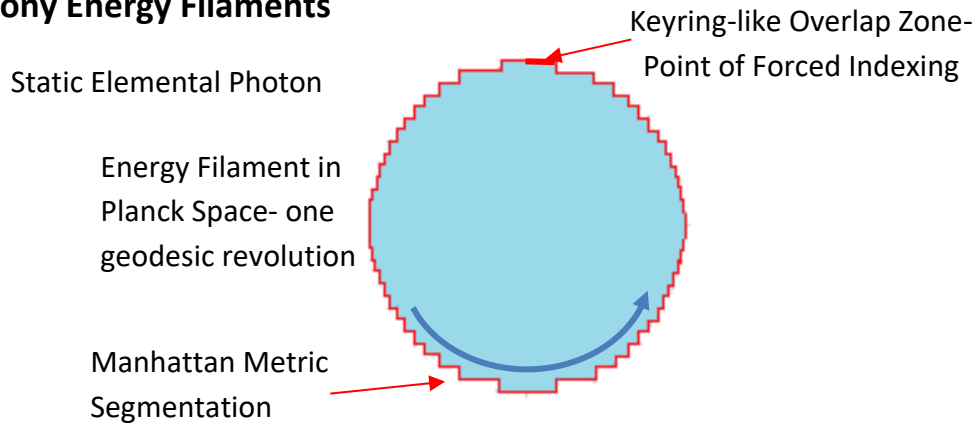
## Chapter 2



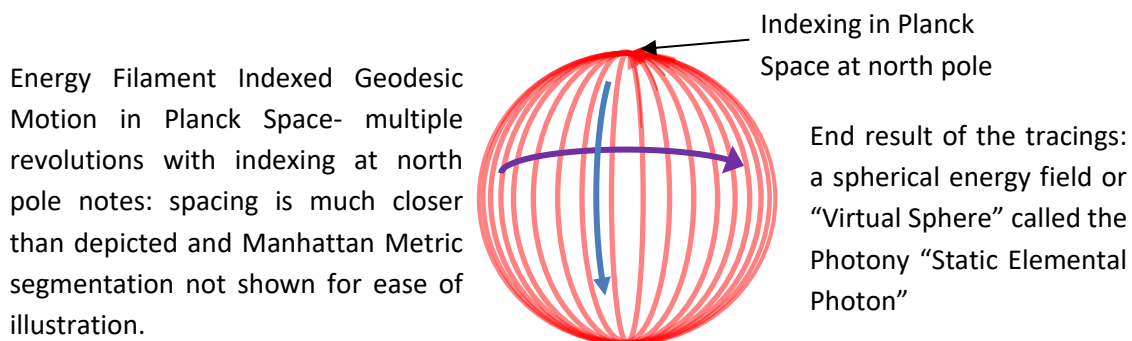
# Chapter 3

## Photony Filaments and Elemental Photons

### A. Photony Energy Filaments



**Figure 27: Geodesic Energy Filament**



**Figure 28: Energy Filament Indexed Geodesic Motion**

A Photony Filament is a strand of energy flowing through Planck Space modeled with motion in the shape of a keyring with an overlap zone for the ends. The filament motion is geodesic for minimum energy filament traversal through Planck Space. Indexing at a pole is due to the head of the filament encroaching on the tail of the filament. This collision results in a minimal deflection to the next geodesic track. The static elemental photon (dark matter) can become dynamic if a gap occurs at the overlap zone leading to a helical high-speed forward motion (dark energy).

Photony has provided mechanics and kinematics of a new particle, the Photony elemental photon. The Photony static elemental photons exist universally and are quite happy to nudge each other as a field or aether as described in Higgs theory as a syrupy medium; Photony Motion/spin rule for filaments operating in quasi-static Planck Space. The Photony dynamic elemental photons also exist universally as high-speed quanta, moving about the Universe at the speed of space.

# Chapter 3

## B. Why Energy Filament Geodesics?

Geodesics are "straight lines in curved space."

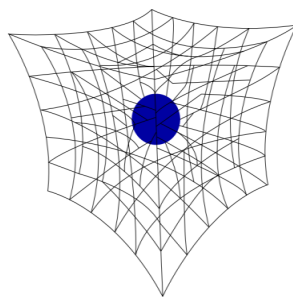


Figure 29: Straight Lines in Curved Space

Photony energy filaments with a preference for geodesic pathways to support the principle of least action/energy. Soon after Einstein published his theory of special relativity in 1905, including gravity, **gravity was not a force but a geometric property of spacetime**, curved around massive objects. Einstein famously called this "**the happiest thought of my life.**"

Geodesic motion is simply a statement of Newton's first law that inertial objects remain at rest or in uniform motion in a straight line in an inertial reference frame. According to Charles Francis, "a geodesic has been defined by parallel transport of a vector in the direction it specifies. Thus, the 4-velocity determines a geodesic in spacetime. The curvature of spacetime means that, on larger distance scales, inertial objects follow curved paths, like the orbit of the moon about the Earth."

A geodesic is a curvature that results from applying the principle of least action in the context of a pseudo-Riemannian manifold. It is widely known that geodesics are "straight lines in curved space." By applying the Hamilton–Jacobi method to the geodesic equation, this idea acquires an intuitive meaning: geodesics describe particles' motions that do not experience any forces. In flat space, it is well known that a particle moving in a straight line continues to move in a straight line if it experiences no external forces; this is Newton's first law.

Photony energy filaments operating in quasi-static Planck Space cause the proximal curvature around physical objects, which we perceive as mass. Other masses comprised of other energy filaments **communicate through mutual curvature/compression** with the original masses of energy filaments, hence, gravity. The **mass** of the "massive object" is a measure of spacetime compression related quantitatively by the  $c^2$  transform from E. According to Morgan D. Rosenberg [27] in his paper, "Lorentz Contraction of Space and the Gravitational Field."

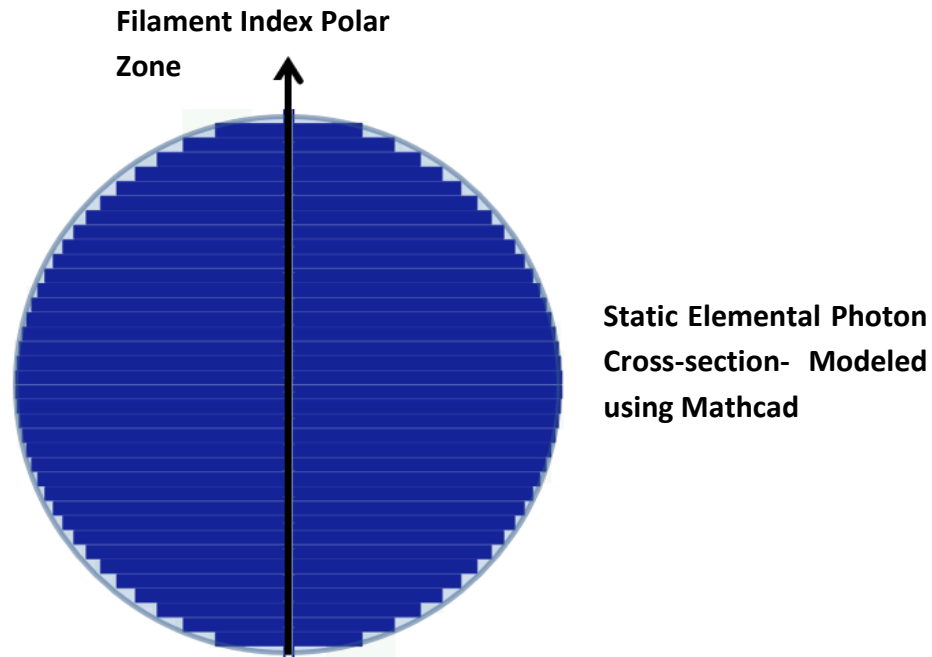
The energy filaments are routed based on the Manhattan or taxicab metric since Planck Space's kinematics require segmented orthogonal routing or pathways. The surprise hiding in the Manhattan Metric is the inevitable and surprising proposition that PI equals 4 for perimeter and other calculations, as further studied in the Photony Compendium.



# Photony Foundations

## C. Static Elemental Photon Surface Structure:

The Photony static elemental photon model features an essentially spherical shape. However, a creation event for elemental photons with a spherical shape is not as likely as a filament shape originating “Into the Void,” a process comparable to extrusion. The cross-section of the extruded filament is conceived as somewhat less than the Planck cell size to allow approximately orthogonal intercellular mobility in the Planck space network according to the Photony Motion/Spin Rule.



**Figure 30: Static Elemental Photon Cross-section**

The circumference of the segmented arc circle is calculated as  $2 \times 4 = 8$  times the radius as opposed to  $2\pi$  times the radius. The segmented stair step resolution is ultimately limited to Planck space resolution, but **regardless of the resolution, the circumference remains at 8 times the radius or 4 times the diameter**. This means that for stair-stepped circles, the role of pi as the transform from diameter to circumference transitions to 4. This is not to say pi equals 4 instead of 3.14159..., but it does say that pi is reserved for analog mathematical circles and imprecise physical analogs, such as matching a belt length to a waist size. The use of  $\text{PI} = 4$  instead of  $\text{pi} = 3.14159...$  is required for kinematics where motion and shapes are constrained or otherwise affected by perimeter resolution or motion along with rotations which ultimately get down to the resolution of Planck space at approximately  $10^{-35}$  m.

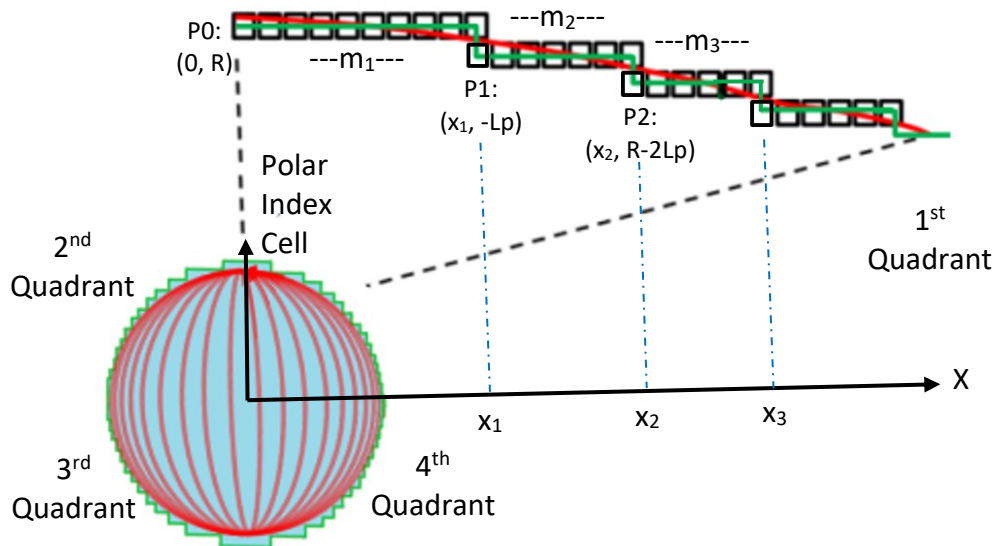
# Chapter 3

## D. Static Elemental Photon Manhattan Surface

The jagged (segmented) “Manhattan” surface in the previous illustration of an exemplary elemental photon's filament tracings is visualized from the superposition of filament tracings in orthogonal Planck Space. The filament traces longitudinal pathways with polar indexing where the overlapped head and tail of the filament cannot occupy precisely the same polar Planck Space cell.

The green filament head attempts to extend in a geodesic form in a highly resolved Manhattan pattern below. From a new polar index cell (P0), it traverses multiple cells ( $m_1$ ) with a distance of  $X_1 = m_1 \times L_p$  ( $L_p$  is the Planck Length) until the curvature of the filament downward to **one Planck Cell distance** ( $L_p$ ). At this location (P1), the natural curvature of the Photony Filament exceeds one cell Planck length in height, so the next row begins. At this location, the smooth/curved nature (in red) of the filament is forced by the orthogonal template of the Planck Space cells to make a 90-degree turn (downward), as the quasi-static nature of Planck Space bends (not shown).

Similarly, in the second row, the filament head extends from P1 multiple cells ( $m_2$ ) to a coordinate position of  $X_2 = (m_1 + m_2) \times L_p$  until the curvature of the filament deflects downward to a second Planck Cell distance ( $2L_p$ ) at P2. Similarly, in the third row, the filament head extends from P2 multiple cells ( $m_3$ ) to a coordinate position of  $X_3 = (m_1 + m_2 + m_3) \times L_p$  until the curvature of the filament deflects downward to a third Planck Cell distance ( $3L_p$ ) at P3, and so on. This 2-D analysis is extendable<sup>12</sup> to 3-D by adding the Y-Axis and adopting spherical coordinates as follows.



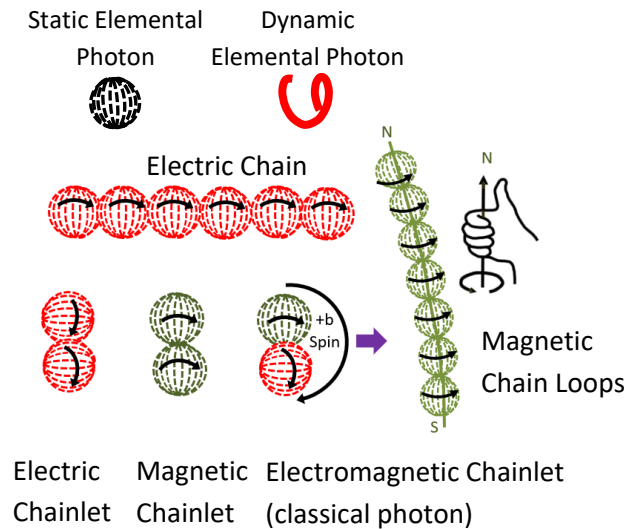
**Figure 31: Segmented Surface Cross-section**

<sup>12</sup> Fully extended to 3-D in the Photony Compendium

# Chapter 4 Photony Emergence

## A. Photony Quanta, Particle, and Chain Inventory

In certain circumstances associated with electrons, multiple energy filaments may interlink into chainlets or extended chains. A 2-link chainlet is a classical photon and launches at the speed of light as the filament overlap zone gaps widen to produce a complex helical spin motion, including end-over-end spin at a frequency that determines the photonic particle's palpable energy. The classical photons can be aligned to produce polarized light.



**Figure 32: Photony Inventory**

Statics: particles are static elemental photons, whereas chains are static only when not growing (interlinking). Dynamics: quanta are dynamic elemental photons, whereas chains are dynamic when growing in numbers or lengths. Chainlets of two interlinked elemental photons are classical photons and are always dynamic (moving at the speed of light within a given medium).

A voltage/potential exists when elemental dynamic photons flow to surfaces while aligning collections of electrons normal to a surface as they produce an electrostatic field (of single-ended chains, initially). In contrast, a current/elemental photon flow through electrons within a conductor produces a magnetostatic field (of chains in loops). Static (including quasi-static) electric and magnetic fields and electromagnetic fields consist of distributions of Photony elemental photon chains interlinked in electrons' toroidal cavity. The electric and magnetic chains are identified as lines (the "flexible tubes" of Faraday and Maxwell) of electric and magnetic flux. As the sourced sinusoidal frequency of the voltage/current flow increases from DC (0 Hz), resulting in extrinsic end-over-end axial electron spins, these electrons' chain products, chains, naturally reduce in length. Electromagnetic fields are comprised of Photony electric and magnetic chains. The length of the chains depends on the frequency of the AC potential. The AC potential causes dynamic elemental photons (dynamic charge) to flow within a conductor in alternatively reversed directions. **The electromagnetic field transitions to photons at a frequency (37.6 THz in the infrared section of the spectrum), whereby the chain length is reduced to two, which Photony lexicography calls out as chainlets.**

# Chapter 4

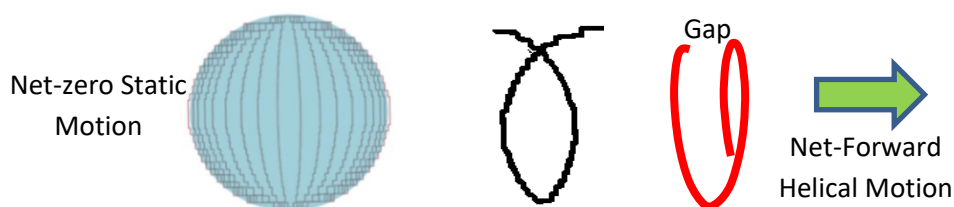
## B. Elemental Photons

Einstein considered particles a "secondary result" derived from physical space, while most contemporary physicists considered certain particles fundamental. The Photony emergence begins with elemental photons maneuvering in Planck Space upward in size and energy through classical photons and electrons to larger particles (e.g., leptons, quarks, gluons, etc.) hadrons. Many other fermions and larger particles like mesons are left out of this list, but most are unstable (e.g., mesons) with short lifetimes.

We question the Standard Model's reliance on quarks (the quark and gluon models) unrelated to electrons and positrons as potential constituents. Instead, we focus on the most impactful and prominent particles (classical photons, electrons/positrons, and protons, along with their antiparticles) for emergence. We also identify some new quanta and chains missing from the Standard Model. We include quarks since they comprise protons and neutrons. Quarks, gluons, and neutrons are also reported to be unstable outside of the nucleus.

A vast number of Photony energy filaments comprising static and dynamic elemental photons entered at creation and remain in universal spacetime. Larger quanta and particles such as classical photons and neutrinos emerged as pressures and temperatures evolved during expansion.

We begin with the following inventory of static and dynamic elemental photons in Planck space:



**Figure 33: Photony Emergence**

The static elemental (charge) photon illustrated above-left is a static particle due to its zero-gap indexing resulting in rotational motion of the energy filament with no inherent forward motion and no extrinsic (end-over-end) spin. The static (charge) photon gains momentum based on collisions and pressure, forcing the gap to open, resulting in forward traction in quasi-static Planck space. The dynamic elemental (charge) photon illustrated above-right is a quantum of energy (the most fundamental quanta). The forward motion in free space is at  $c_0$  or greater since the tangential speed of the filament is much greater than  $c$ . The dynamic charge photon can also acquire an extrinsic end-over-end spin (not shown).

The energy filament motion in both the static and dynamic embodiments maneuver in quasi-static Planck space according to the Manhattan Metric in tension with the preferred smooth geodesic motion of the filament. Planck space acts as an imperfect ubiquitous analog-to-digital converter.

# Photony Foundations

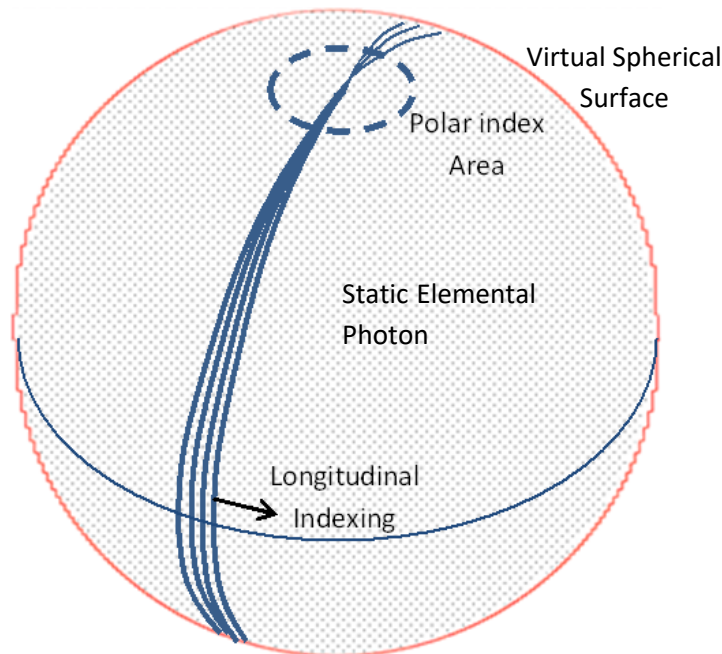
## C. Static Elemental Photon Parameters

Confirming the Photony assertion that a spherical elemental shape is likely, Neil DeGrasse Tyson, in his article, On Being Round, [35] addresses this subject matter as follows:

*“While many objects have peculiar shapes, **the list of round things is practically endless** and ranges from simple soap bubbles to the entire observable universe. **Spheres tend to take shape from the action of simple laws.**”*

The following case study regarding Photony/Manhattan Spheres showcases the Photony elemental photon, a boson. The elemental photon radius  $r_0 = 2.82467 \times 10^{-24} \text{ m}$  (follows). The elemental photon radius, although quite small, is approximately 175 billion times larger than the Planck Network Cell centers or Planck length,  $1.61618 \times 10^{-35} \text{ m}$ . Another size reference of interest is that of a classical electron radius (approximated) at  $2.82 \times 10^{-15} \text{ m}$  is approximately 1 billion times larger than the elemental photon radius.

The 3-D depiction of the Photony static elemental photon below illustrates the indexed longitudinal geodesic tracings of segmented Photony filaments forming a virtual spherical surface. The blue tracings (geodesic taxicab circles) are shown below to simplify the illustration without orthogonal segmented effects of the Manhattan or Taxicab Metric.



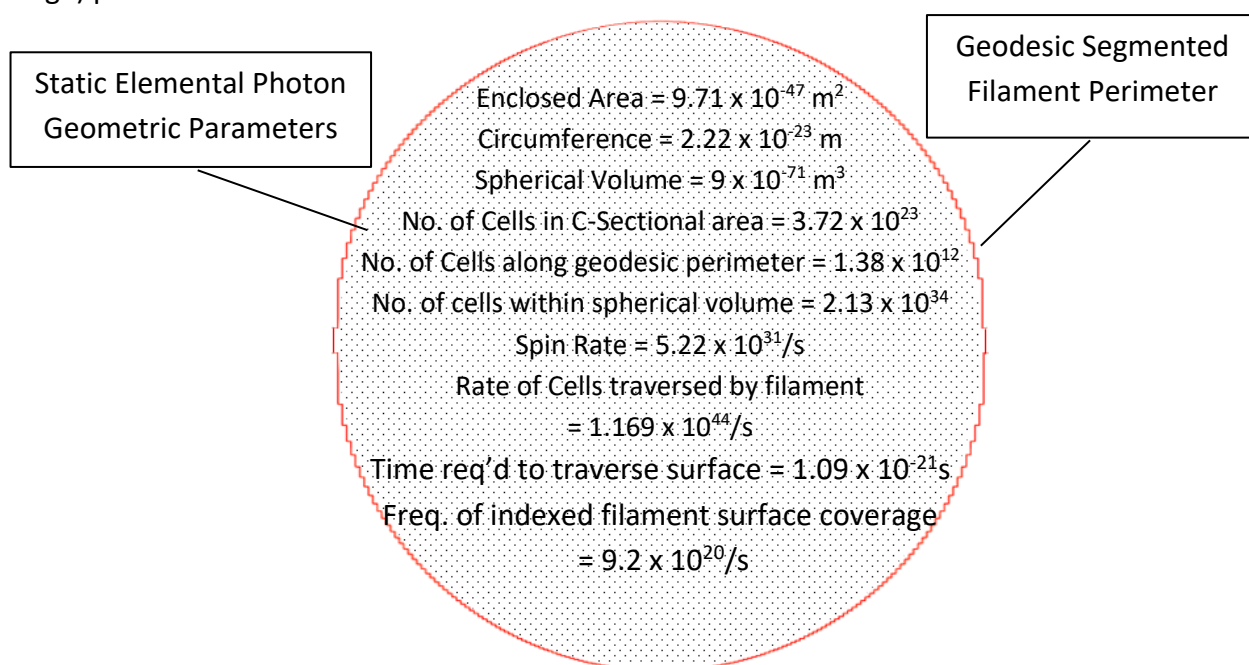
**Figure 34: Photony Static Elemental Photon**

The elemental photon has orthogonal segmented boundary filament tracings based on the Manhattan or “taxicab” metric. The geodesic polar filament tracings are indexed longitudinally to form a 3-D sphere corresponding to the depiction above.

# Chapter 4

The angular rate of indexing is of interest relative to the angular velocity of the filament. Filament tangential speeds, much greater than  $c$ , are certain since an elemental dynamic photon's forward motion as a quantum of energy traverses at velocity  $c$  in helical motion format.

The static elemental photon with an essentially spherical (but virtual shell) is shown below, and a geodesic cross-sectioned view with a 2-d view of Planck Space is shown within the cross-section. The number of enclosed Planck Cells (Cells) is illustrated to appear more in scale with the overall size as depicted in previous illustrations. The actual Planck cell density is much higher. The cross-sectional illustration below depicts the enclosed Planck cellular density. The parameters listed inside the segmented perimeter of the static elemental photon are spacetime geometric in nature. This static elemental photon geometric parameter listing is followed by a combined Primary Photons Properties Table of parameters listing the associated mass, energy, and wavelength-oriented parameters. The Primary Photons Properties Table compares the static elemental photon parameters to the EM (IR) photon parameters. Radius considerations- the radius is calculated relative to the proton radius and scaled by  $G$ . The radius =  $50 G \times r_p$  where  $r_p$  = proton radius and  $G = 6.674 \times 10^{-11}$ . The static elemental (charge) photon radius:  **$r = 2.78 \times 10^{-24} \text{ m}$**



**Figure 35: Static Elemental Photon Parameters**

The frequency of filament coverage above is notable since the **spherical shell, or virtual photon boundary, is enforced at a very high rate,  $9.2 \times 10^{20}$  filament-traced full surface coverages per second** at Planck length separation or pitch of the longitudinally advancing or indexing geodesic filament at the equator.

# Photony Foundations

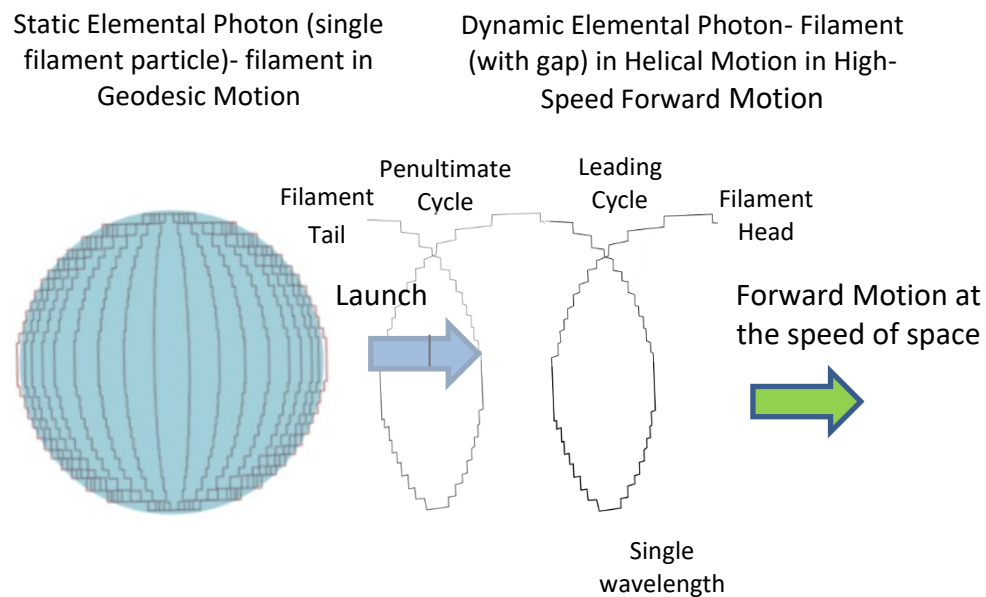
## D. Static to Dynamic Elemental Photon

### High-Speed Helical Filament Motion:

A collision that falls within the ability of the static elemental photon to restore itself to a virtual spherical (shell) shape results in its minimum energy configuration and stability returns. Otherwise, the head-to-tail overlap of the energy filament separates by at least one Planck Space unit, and a high-speed launch occurs. Once launched, the Photony energy filament engages the Planck Space cellular network in a helical motion, corresponding to the “speed of space” according to the permittivity and permeability. At this speed, the filament and the Planck Space cellular network interact in a minimum-energy mode, i.e., with minimum compression of quasi-static Planck Space proximal to the dynamic elemental photon passage.

The following filament motion illustrates helical motion—the bottom left illustration of a static elemental photon. The bottom right illustration below depicts the filament motion following the transition of indexing to the launch of a dynamic elemental photon with a segmented helical pathway (orthogonal Manhattan pathway segments), providing forward motion with traction in Planck Space.

Orthogonal Manhattan Metric Pathways in quasi-static Planck Space for both the static elemental photon traces and the dynamic elemental photon.



**Figure 36: From Static to Dynamic Elemental Photon**

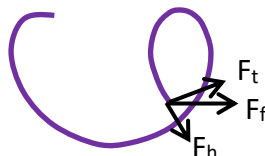
The helical motion does not appear to include extrinsic end-over-end spin since that would result in a wobbling motion and chaotic behavior and would not be sustainable in Planck Space due to not being the least energy scenario.



# Chapter 4

## E. Dynamic Elemental Photon- Filament Motion Kinematics

The head of the helical wormlike filament provides the force  $F_h$  for traction in Planck Space. Increased compression of Planck Space is due to the forward force  $F_f$  corresponding to increased curvature and therefore increased mass, commonly called inertial mass. The Photony Motion/Spin Rule forbids **transverse filament motion** directly in the direction of  $F_t$  since side-ways motion might cause either a rupture in the filament or excessive distortion of Planck Space.



EM Dynamic Elemental Photon (segmented and smooth shown): Helical traction spearheaded by the “head” of the helical “wormlike” energy filament.

**Figure 37: Helical Motion**

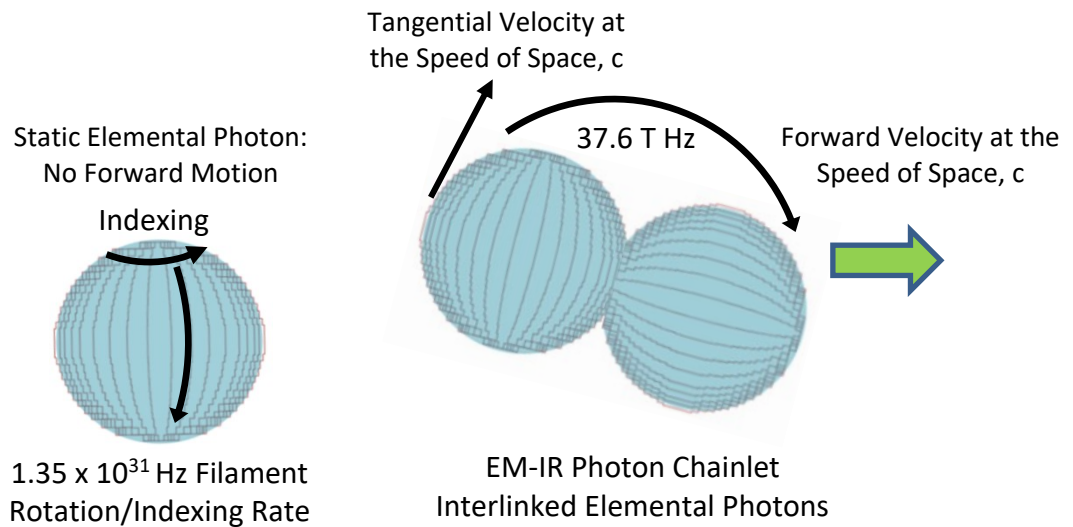
The energetic filament in dynamic elemental photons worms through Planck Space and provides traction through the Planck Network of orthogonally interconnected cells while deflecting or compressing the network proximally as the filament traverses between cells. Since the dynamic elemental photon with helical spin (spin level 1) as a whole quantum typically travels at least  $c$  (the speed of space), the filament must be traveling tangentially at much greater than light speed! The energy for a dynamic elemental photon as it maneuvers through Planck Space is conserved as follows:

- A filament of energy has a fixed quantum of energy bestowed at its creation.
- A filament's energy is a quantum and cannot be destroyed, which means that each filament is eternal and can be increased or decreased based on spin level 1 frequency.
- When directed into a helical motion (with or without end-over-end rotation) as a launch event, the filament's energy added during the launch event is maintained until a next pressure/collision event occurs resulting in deflection or interlinking with another filament.
- A filament worms its way through the quasi-static Planck Space Network of orthogonal cells attempting to maintain a helical arc motion constrained by cells as they enforce orthogonality (Photon Motion/Spin Rule).
- At each turn through the network of cells, a slight deflection occurs to the quasi-static (flexible) Planck Space Network as the filament experiences an orthogonal turn of the filament according to its helical radius following the Manhattan Metric through the network.
- The energy expended by the helical filament as its head maneuvers through and deflects the quasi-static (flexible) Planck Space is returned to the filament as the filament's tail passes through the same network cells. This produces a traveling mini-wave of space distortion (mass) in the proximity of the energetic filament that is impossible to detect as an individual dynamic quantum due to its speed and small size. Hence, the appearance of zero mass.



# Photony Foundations

## F. Static Elemental vs. EM-IR Photon Properties



**Figure 38: Elemental Photon Indexing Freq.**

**Figure 39: EM-IR Photon Frequency**

The static elemental photon's virtual spherical boundary or shell behaves like a force field that maintains the static photon's spherical integrity. The shell may deflect or deform temporarily inward and become fully restored to its original spherical shape when the interfering particle or quants passes or when the interfered photon becomes nudged away from the point of contact. If an intrusion is severe enough, the static photon transforms into a dynamic photon, with the filament achieving a helical trajectory through Planck Space. The number of Planck Cells contained within the spherical (virtual) shell of the static photon of Planck Cells  $N_p = 3.7 \times 10^{32}$ .

The static elemental photon and the EM-IR photon (infrared EM chainlet) are illustrated above, with a few of their constituent Photony energy filament tracings. The energy filament tracings are Manhattan metric segmented as dictated by orthogonal Planck space.

The calculations below show that the common transformation between the static elemental photon properties and EM-IR photon properties is  $c^2$  (both directly and inversely depending on the property) in free space. This exposes a clarification of the  $E = mc^2$  relationship. Comprehending **the composite  $c^2$  transform as tangential  $c_T$  and forward (linear,  $c_F$ )**, the Einstein relationship can be written as:

$E = m \cdot c_T \cdot c_F$  is equivalent to the original formula since  $c = c_T$  and  $c_F$ ! This is logical since a mass-to-energy transform implies a mass (static elemental photon) converting to energy (EM-IR photon) at the most fundamental and macro levels.

# Chapter 4

## G. Static Elemental Photon/Classical Photon Properties Calculations [36]

The Primary Photons Properties Table summarizes the elemental static charge photon's properties and the EM-IR photon (the lowest frequency infrared (classical) photon counterpart, an EM chainlet, moving at  $c$ . The sequenced governing equations are revealed as follows. The lead governing equation for the EM-IR (classical) photon comprised of two interlinked elemental photons is  $M = M_p/\mu^3$ , where  $\mu$  is the NIST Proton-Electron Mass Ratio = 1836.153. [37]

Elemental (Primary) Photons Properties Table:

| Property                           | Static elemental photon-             | Static Elemental Equation Seq.                 | EM-IR Photon at IR Frequency 37.6 THz | EM-IR Photon Equation Seq.           | Notes<br>Proton radius $r_p$ , $\mu = 1836$ ,<br>$M_p$ = proton mass                                              |
|------------------------------------|--------------------------------------|------------------------------------------------|---------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Radius (m)                         | $2.82 \times 10^{-24}$               | 1:<br>$r_0 = 16\pi G r_p$<br>$= r_p/c$         | Not a sphere- no radius               | N/A                                  | $G = 6.674 \times 10^{-11} \text{ m}^3/\text{Kgs}^2$<br>$r_p$ = proton radius = $0.842 \times 10^{-15} \text{ m}$ |
| Mass (kg)                          | $1.19 \times 10^{-53}$               | 4:<br>$M_0 = E_0/c^2$                          | $2.70 \times 10^{-37}$                | 1:<br>$M = M_p/\mu^3$                | EM-IR photon is $c^2/4$ more massive than the static photon mass                                                  |
| Energy (J)                         | $1.1 \times 10^{-36}$                | 3:<br>$E_0 = \lambda_0/(4c^{1.5})$             | $2.49 \times 10^{-20}$                | 2:<br>$E = Mc^2$                     | EM-IR photon is $c^2/4$ more energetic than the static photon energy                                              |
| Wavelength (m)                     | $2.25 \times 10^{-23}$               | 2:<br>$\lambda_0 = \Pi^2 * r_0$<br>$= 8 * r_0$ | $7.98 \times 10^{-6}$                 | 3:<br>$\lambda = h/(Mc)$             | EM-IR photon is $4c^2$ times longer than the static photon wavelength, $\Pi=4$                                    |
| Circumference (m)                  | $2.25 \times 10^{-23}$               | 5:<br>$C_0 = 8 * r_0$                          | $8.88 \times 10^{-23}$                | 4:<br>$C = M\lambda/E = \lambda/c^2$ | EM-IR Photon 4x larger $\lambda$ than the Static photon                                                           |
| Distance per photon revolution (m) | 0<br>(stationary)                    | N/A                                            | $7.98 \times 10^{-6}$                 | 5:<br>use $\lambda$ from 3:          |                                                                                                                   |
| Period per revolution T (s)        | $7.42 \times 10^{-32}$<br>(filament) | 6:<br>$T_0 = \lambda_0/c$                      | $2.66 \times 10^{-14}$                | 7:<br>$T = \lambda/c$                | $C/T = 1/c$                                                                                                       |
| Frequency $\nu$ (1/s)              | $1.35 \times 10^{31}$                | 7:<br>$f_0 = 1/T_0$                            | $3.76 \times 10^{13}$                 | 8:<br>$f = 1/T$                      |                                                                                                                   |

Review and compare here regarding the Photon Properties Table above:

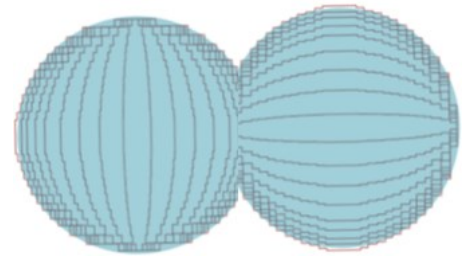
- Regarding mass, the EM-IR photon is  $c^2/4$  more massive than the static photon mass.
- Regarding energy, the EM-IR photon is  $c^2/4$  more energetic than the static photon energy.
- Regarding wavelength, the EM-IR photon is  $4c^2$  times longer than the static photon wavelength.
- Regarding period, the EM-IR photon is  $4c^2$  longer than the static photon period.  $C/T = 1/c$ , so the forward velocity of the EM-IR photon is the same as the tangential velocity!
- Regarding frequency, the EM-IR photon is  $4c^2$  lower than the static photon period.
- The EM-IR dynamic photon moves  $\lambda = 8 \times 10^{-6} \text{ m}$  during one full rotation in flight.
- If we divide the circumference ( $8.88 \times 10^{-23} \text{ m}$ ) by the time ( $2.66 \times 10^{-14} \text{ s}$ ), we get  $1/c = 3.34 \times 10^{-9}$ . This is equivalent to a rate of spin that equals the reciprocal of the forward velocity  $c$ .

# Photony Foundations

## H. Classical Photon Emergence

What a classical EM photon image might look like **if it could be at rest and viewable**- an interlinked pair of static elemental photons.

Classical Photon Filament Tracing Model:  
Two interlinked elemental photons



**Figure 40: Hypothetical Photon at Rest**

### 1. Classical Photons Background

No one person, including Einstein, can take credit for the photon revolution. The ancient Greeks studied some of the many properties of light (such as reflection and refraction). However, not much was learned about light until the seventeenth century; Galileo hypothesized that light had a measurable, finite speed. The first close estimate of the speed of light came in 1676 by Ole Romer (220,000 km/s), and it was based on the eclipse times of Jupiter's moon, IO, at various positions of Jupiter relative to earth. About 200 years later, A.A. Michelson (299,910  $\pm$ 50 km/s) accurately measured light's speed in 1879.

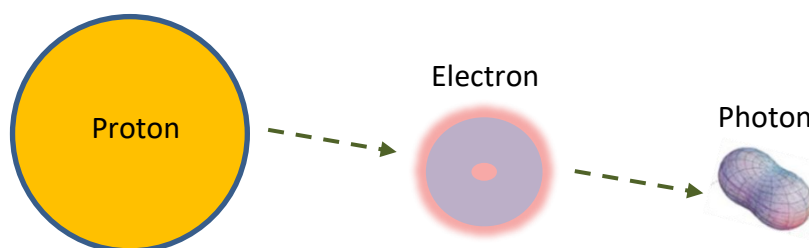
From the time that light was first created until 1879, this essential parameter of light (its velocity) was left for humans to determine through science and technological endeavors.

### 2. A brief record of classical photon theories:

- First, Descartes, in 1637, asserts that light is composed of small discrete particles called "corpuscles" (little particles), which travel with impetus along a straight line with a finite velocity.
- In 1704 the genius, Sir Isaac Newton, building on the corpuscle idea, articulated that reflection and refraction of light required particles, also referred to as perfectly elastic corpuscles.
- In 1900 Max Planck introduced Planck's postulate that electromagnetic energy could be emitted only in quantized form as a multiple of fundamental energy quanta units,  $h \cdot v$ , where  $h$  is Planck's constant, and  $v$  is the frequency of the emission.
- In 1905 Einstein published the quantum theory of light as tiny packets or particles, which he called "light quanta," unified with Planck's "quanta of heat" as  $h \cdot v$  resulting in a revolutionary new way to view the universe; as a quantum reality. Einstein's Photoelectric Effect further sealed the deal.
- It was another 21 years before Gilbert N. Lewis gave the light quantum of Einstein the name "photon" in 1926. This naming of the photon came five years after Einstein won the 1921 Nobel Prize in physics for the Photoelectric Effect, which he finally received in 1922.

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## 3. Calculating Classical Photon's Parameters from Electron's Parameters



**Figure 41: Reverse Engineering Classical Photons**

A reversed emergence analysis is appropriate here since we know much (size, shape, mass, energy, etc.) about protons and electrons, and we seek to know the equivalent information about a classical photon. A scaling constant is used to scale from larger particles to smaller particles.

Looking ahead in the electron emergence section, we find that the electron is the shape of a toroid and is comprised of a photon (gamma ray) in rotational motion acquired during an electron/positron pair production event. Since we are qualitatively sequencing up herein from smaller to larger particles, we must acquire certain parameters from the electron to build back to the classical photon. This provides the parameters needed from particles we know much about, such as the proton and electron, to determine the classical basis photon parameters. The mass ratio from a proton to an electron is 1836.15267343 from NIST. So, we anticipate using  $\approx 1836$  to calculate the basis photon mass from the electron mass. By “mass,” we mean the amount of compression of quasi-static Planck space due to the energy filaments active within these particles.

The electron is at rest as a particle even though the contained gamma ray is rotating at the speed of light with its two interlinked energy filaments tracing a toroidal virtual body or shell. All of the energy contained in the electron within this energy field comprises two interlinked elemental photons, further comprised of the energy filaments, one for each elemental photon. From the electron look-ahead<sup>13</sup>, the following **parameters of the electron**- mass  $m_e = 9.11 \times 10^{-31}$  kg, Radius  $r_e = 1.11 \times 10^{-17}$  m,  $r_e = 1/c^2$  m, Energy  $E_e = 8.188 \times 10^{-14}$  J, with  $a_1$ ,  $a_2$ ,  $b$ , and  $c$  spins. Mass divided by the radius for the electron or  $E = mc^2 = m/r_e$ , is further investigated in the QPC Compendium book.

Basis Photon Parameters:

The basis (least energy) photon mass is calculated based on electron-to-proton scaling. It is assumed to be a scaled-down electron mass, noting that the electron is a gamma ray photon with added  $c$ -spin taking the photon from forward to rotational motion, thus allowing a rest mass.

---

<sup>13</sup> In general, for the electron:  $E = mc^2$  and radius  $r_e$  equal to  $1/c^2$  so,  $c$  equals  $1/\sqrt{r_e}$  and  $E$  equals  $m \cdot (1/\sqrt{r_e})^2$  equals  $m/r_e$  where  $m$  can be the mass of any particle and  $E$  is that particle's energy!

# Photony Foundations

## 4. Elemental Photons to Classical Photons

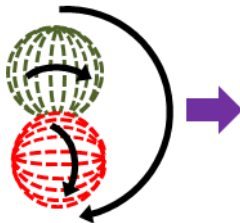


Figure 42: EM Chainlet- Classical Photon

**Electromagnetic (EM) fields and waves** have had a nascent need for aether or some other field explanation as to the medium for Maxwell's Equations' electromagnetic waves to propagate. Heretofore, classical photons have erroneously been designated as the basis (medium) of the EM field and misidentified elemental photons as virtual photons, extensions to the role of real photons. As described by many particle physicists,

*"The **photons which make up a static electric or magnetic field** are "virtual" -- their energy and momentum don't satisfy the relationship for "**real**" photons."*

Here (above), the "**real**" **photons** reference means **classical photons**, and **photons that make up static electric or magnetic fields of chains are Photony Elemental Photons**. In many cases, and ironically, elemental photons are the "virtual photons" of mainstream physics, even though they are not "really" virtual beyond the virtual shell concept.

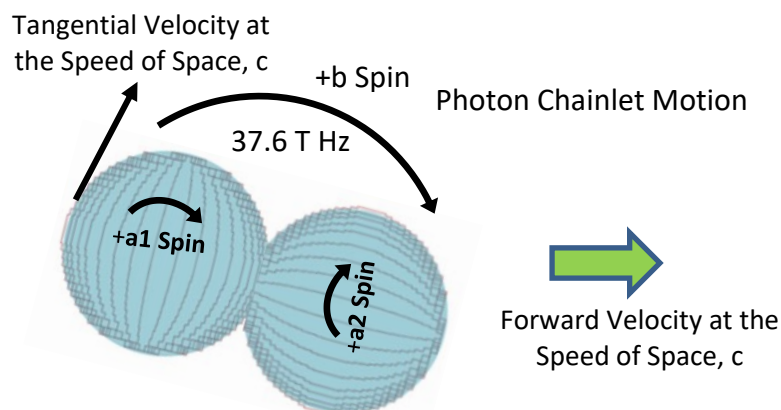
Classical photons (Photony chainlets of two interlinked elemental photons with end-over-end spin frequency/energy in the gamma ray section of the spectrum) can interact (collide) and emerge as electron/positron pairs. Free electrons transduce Photony magnetic chains as loops emerging from within a conductor. Free electrons transduce Photony electric single-ended chains at the surface of a conductor (or non-conductor). Variable-length chain loops are formed in an alternating voltage/current field based on the AC source's frequency.

EM radio waves/photons transition occurs in the gray region at the microwave/far-infrared interface. Starting with conditions at DC, the chains can be quite long. With increasing frequency and shorter wavelengths, the EM wave frequency exceeds a specific frequency limit in the terahertz region; the resulting chains become shorter with fewer interlinked elemental photons. At a limit, the chains become limited to two interlinked elemental photons (chainlets), which Photony identifies as classical photons. As the spin frequency increases, the chainlet's frequency determines its energy, maximally extending from far infrared to gamma rays.

Photons can be created/assembled in a variety of ways. Photons are emitted from atoms as electrons change energy levels (shells), high frequency (optical) transduction, thermal, particle collisions in annihilation, chain collisions in voltage breakdown, and bio/chemical reactions.

# Chapter 4

## 5. Classical Photon Propagation- IR Photon Example



**Figure 43: EM-IR Photon Chainlet Motion**

The Photony model of the classical photon has two interlinked elemental photons (one energy filament each). The frequency of the end-over-end extrinsic spin of classical photons is based on the Photony Motion/Spin Rule. The higher the spin rate, the more proximal compression occurs in quasi-static Planck Space. Planck Space compression is the measure of mass resulting from energy causing the compression based on Einstein's Energy-Mass Equivalence principle. Elemental and Classical photons are neither particles nor waves in the classical sense. Elemental and classical photons each individually possess and communicate a "particle-wave" quanta based on the motion of energy filaments.

The speed of photon propagation ( $vp$ ) is determined solely<sup>14</sup> by the permittivity of the dielectric wherein the photon is propagating. In the case of quasi-static Planck space, the velocity of propagation is the speed of light in space (the speed of space).

$$vp = \frac{1}{\sqrt{LC}} = c/\sqrt{\epsilon_r} \text{ where } L = \text{the inductance and}$$

$$C = \text{velocity of light (in local space), and } vf = 1/\sqrt{\epsilon_r}$$

where  $vf$  = velocity factor = 1 for free space

$vf$  varies based on the variability of proximal real (Planck) space within our Universe. The velocity factor of local or real space changes as the gravitational field changes due to the compression of quasi-static Planck space due to energy density (typically conflated with "mass"). This is a veiled restatement of general relativity.

<sup>14</sup> A dielectric has no magnetic properties, so the relative  $\mu$  is 1 or  $\mu=\mu_0$  (the same as free space).

# Photony Foundations

In Einstein's speech (translated) on 5 May 1920 at the University of Leiden:

*"Within matter it takes part in the motion of matter and in empty space it has everywhere a velocity; so that the ether has a definitely assigned velocity throughout the whole of space. **There is no fundamental difference between Hertz's ether and ponderable matter (which in part subsists in the ether).**"*

As did Einstein, Photony may take the reader beyond their comfort zone by defining the speed of light as the speed of the space/medium where the light (photons) travel noting that **photons are self-propelled**. The speed of light,  $c$ , varies according to the permittivity of the medium wherein it is traversing. So, the speed of light is a constant within a given medium only.

In Einstein's speech (translated) on 5 May 1920 at the University of Leiden (cont'd):

*"Such was the state of things when H A Lorentz entered upon the scene. He brought theory into harmony with experience by means of a wonderful simplification of theoretical principles. He achieved this, the most important advance in the theory of electricity since Maxwell, by taking from ether its mechanical, and from matter its electromagnetic qualities. As in empty space, so too in the interior of material bodies, the ether, and not matter viewed atomistically, was exclusively the seat of electromagnetic fields. **According to Lorentz the elementary particles of matter alone are capable of carrying out movements; their electromagnetic activity is entirely confined to the carrying of electric charges.** Thus, Lorentz succeeded in reducing all electromagnetic happenings to Maxwell's equations for free space."*

So, from Lorentz above, we see some corroboration with Photony regarding self-propelled photons (and Photony dynamic elemental photons). We also can see Einstein the use of "carrying" charge instead of Photony's concept of charge being ubiquitous elemental (charge) photons which are pulled into toroidal cavities of electrons as they transduce (interlink) the charge photons into either magnetic chain loops or single ended electric chains. If the electrons are in a unidirectional field of charge photons the electrons face align like a windsock and maintain alignment with the (DC) charge photon "wind." If the electrons are in a bidirectional field of charge photons the electrons face align and with an x-spin, like a rotating windsock and maintain alignment with the alternating (AC) charge photon "wind."

Furthermore, the ubiquitous elemental (charge) photons are attracted to (swept into) toroidal electrons as they transduce the charge photons into magnetic or electric chains. Also, unbeknownst to Einstein was the reverse process by electrons by pulling chains back into the electrons to disassemble (unlink) the chains into individual elemental (charge) photons, a Photony process called reduction.

# Chapter 4

## I. Photony Spin Levels Preview

Photony Spin levels are described in detail in the **Photony Compendium**. Photony **Spin Levels** are **previewed and summarized** as follows:

1. Spin Level 0 (a0-spin) Static elemental photon with indexing filaments

**Elemental Static Photons**; single filament, zero/low-speed forward velocity, non-zero rest mass, capable of linking with other aligned elemental photons function as chains of electrostatic or magnetostatic “lines of flux” in a static field.

2. Spin Level 1 (a1-spin, a2-spin) Chainlets

3. Spin Level 2 (b-spin) Forward motion with end-over-end chainlets at c

**Classical Photon, Electron Neutrino, Muon Neutrino, Tau Neutrino, Tau Antineutrino, Muon Antineutrino, Electron Antineutrino, Classical antiphoton**; dual interlinked (entangled) filaments or “chainlets.” Note: LEP<sup>15</sup> showed three species of light, active neutrinos leaving room for the existence of “sterile” neutrinos, interacting only very weakly with (capable of passing through) Standard Model particles. [8 total]

With two interconnected energy filaments and three spin freedoms, eight (8) possible particle types emerge, of which photons and antiphotons are only two types leaving six (6) more particle types for assignment. These are assigned herein and studied in further detail, including filament flow factor (FFF) and its relationship to mass (curvature) and stability/lifetime in the Photony Compendium. Fundamental Two-filament Particle Table:

| b-spin | a2-spin | a1-spin | Dynamic Particle      |
|--------|---------|---------|-----------------------|
| 1      | 1       | 1       | Photon                |
| 1      | 1       | -1      | Electron Neutrino     |
| 1      | -1      | 1       | Muon Neutrino         |
| 1      | -1      | -1      | Tau Neutrino          |
| -1     | 1       | 1       | Tau Antineutrino      |
| -1     | 1       | -1      | Muon Antineutrino     |
| -1     | -1      | 1       | Electron Antineutrino |
| -1     | -1      | -1      | Antiphoton            |

**Table 1: a1, a2, b Spin Table**

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<sup>15</sup> Large Electron Positron Collider in Cern



# Photony Foundations

## J. The Electron Emergence

*"You know, it would be **sufficient to really understand the electron.**"*

- Albert Einstein [38]

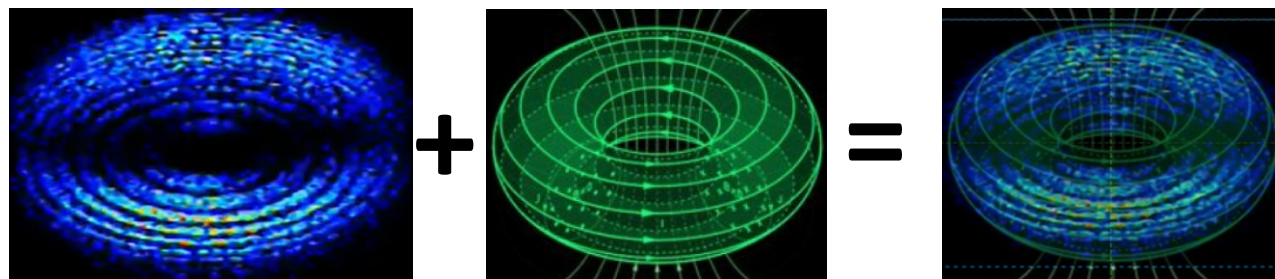
A. *Einstein and the quantum theory* – A. Pais: [39] In 1897, the electron was the first particle ever discovered. An unforeseen experimental development ended the ongoing debate at that time: “Are cathode rays molecular torrents or aethereal disturbances?” Eventually, cathode rays were determined to be neither torrents nor disturbances but a new form of matter, a particle. When first presented by JJ Thomson, there were some initial reactions of disbelief. JJ Thompson performed experiments using cathode rays (electrons) and demonstrated convincingly that atoms contain negatively charged ‘corpuscles,’ which we now call electrons. Bohr’s atom model was completely wrong [43], conceptually and in terms of agreement with experiment.

As the electrons' shape and function are studied, a great appreciation of this particle becomes possible. Photony asserts that any particle's shape is simply the superposition of energy filament tracings of a gamma ray photon's energy filaments, rotating and spinning, forming a virtual shell. This energy field outlines its outermost presence over time, establishing the “virtual” size and shape boundary. For the electron, the shape of the superimposed Photony energy filament tracings is that of a toroid. Surprisingly, the idea of toroidally shaped electrons is not new! Alternate toroidal models follow as background material for comparison with Photony theory.

When an object is said to have mass and no size (as in an overt point source), it implies that it has infinite density. The problem with this kind of theory is that it considers the combination of size, mass, AND the ability to act as a point source. The Photony electron does all three.

### 1. Electron in Motion Capture at Lund University

When the 3-D Lund Electron video frame and the Toroid Field Diagram (shown below) are superimposed, the result is stunning:



**Figure 44: Superimposed Electron Images**

LUND, Sweden, Feb. 28, 2008 -- An electron's rapid motion was filmed for the first time. Attosecond laser pulses show how an electron rides a light wave after being separated from an atom. The Lund method involved using a stroboscope and a laser that generated the extremely

## Chapter 4

short (strobed) light pulses needed to capture the movement of an electron after its bond was broken from an atom. [40] **Lund Breakthrough- video of Electron in Motion** [41]: The length of the video corresponds to a single oscillation of the light, but the speed was slowed considerably for the event to be seen by the human eye. The video shows the energy distribution of the electron (See the movie here<sup>16</sup>).

With the aid of a second laser, the Lund researchers guided the electron motion enabling the video capture of a collision between an electron and an atom. Johan Mauritsson, an assistant professor in atomic physics at the Faculty of Engineering, Lund University, is one of seven researchers in this study, which he directed, and Professor Anne L'Huillier. Concerning time frames, Johan Mauritsson said;

*"It takes about 150 attoseconds for an electron to circle the nucleus of an atom. An attosecond is  $10^{-18}$  seconds long, or expressed in another way: an attosecond is related to a second as a second is related to the age of the universe. We have long been promising the research community that we will be able to use attosecond pulses to film electron motion..."*

Lund Electron Video Frame Superimposed on Toroid Visualization (see Figures 44 and 45). LUND, Sweden, 2008 - an electron's rapid motion (video frame) was filmed for the first time with attosecond laser pulses- Lund University, Johan Mauritsson, et al.

Physicists have taken the first-ever image of a Wigner crystal, "Electron Ice" — a strange honeycomb-pattern material inside another material, made entirely out of electrons. The larger electron (circled) image above shows where the toroidal cavity is expected. Hungarian physicist Eugene Wigner first theorized this crystal in 1934.

Superimposed Lund Image and  
Toroid Visualization

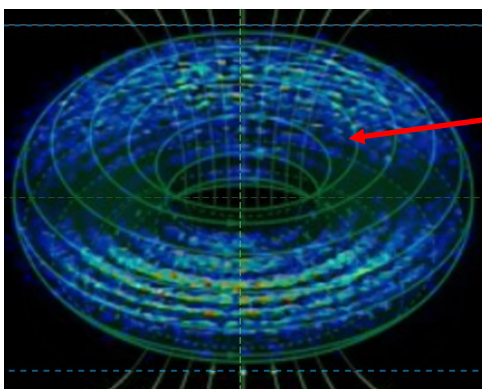


Figure 45: Superimposed Toroidal Images

Electron Ice (Image by Hongyuan Li  
and Shaowei Li, courtesy of Nature)

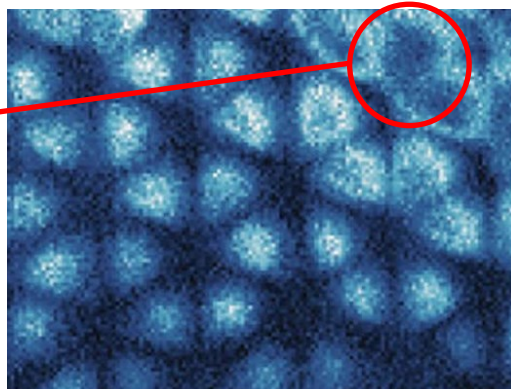


Figure 46: Electron Ice Image

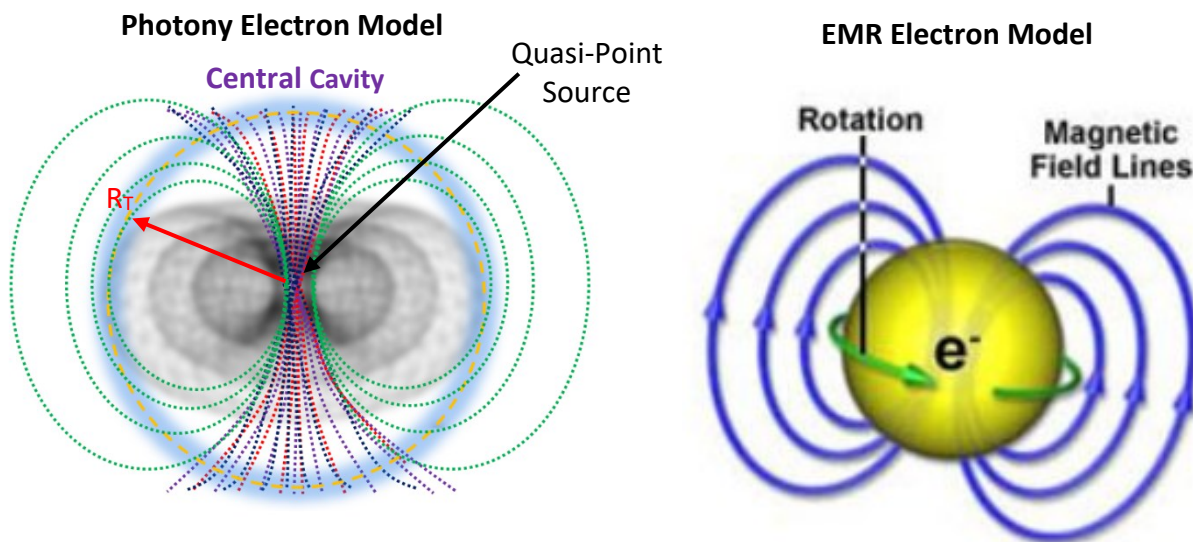
<sup>16</sup> <https://www.youtube.com/watch?v=vMVzDiBYpUY>

# Photony Foundations

## 2. Electron Toroidal and Spherical Concepts

Here, a powerful message affirms what many scientists have postured: the shape of electrons is a toroid. The shape is virtual only because the filament tracings superimposed over a short time period transcribe a toroidal shell, a kind of energy field. Figure 47 below left shows the toroidal shell with radius  $R_T$ , some magnetic field lines (Photony magnetic chains of interlinked elemental photons), and the charge field of dynamic elemental photons. Some elemental photons pass through, while others are chained (interlinked) within the central cavity. These “charges” are pulled through the cavity with spins compatible with the electron's spin. The electron's spin is merely the rotation of the confined gamma ray (photon). The spherical model for electrons with orbital charge is invalid.

The figure below-right is from the National High Magnetic Field Laboratory regarding Electron Magnetic Resonance (EMR) [42], showing an electron (albeit incorrectly drawn as spherical) with magnetic field lines and a spin direction. The Photony electron model is toroidal, acting as a point source, i.e., a quasi-point source. The spherical shape is forgiven since the presence of high-density elemental photons in a toroidal cavity would somewhat mimic this spherical model.



**Figure 47: Photony Electron Model Comparison to EMR Model**

EMR stands for electron magnetic resonance comprising electron paramagnetic resonance (EPR), electron spin resonance (ESR), and electron cyclotron resonance (ECR).

The electron is formed by a collision of two gamma ray photons in linear motion and transformed into oppositely directed rotational motion in a well-known, proven pair production phenomenon. The charge field is simply the extended central cavity zone, and the charges are attracted there in passing and not being “carried” as is popularly described.

# Chapter 4

## K. Quark Emergence

“If you’re interested in the properties of the microseconds-old universe, the best way to study it is not by building a telescope; it’s by building an accelerator,” says Krishna Rajagopal, an MIT theoretical physicist who studies quark-gluon plasma (QGP). Quarks were theoretically discovered in 1964 by Murray Gell-Mann and George Zweig and confirmed in deep inelastic scattering experiments at the Stanford Linear Accelerator in 1968. Big Bang Theory (BBT) has quark emergence by 1 microsecond before leptons and photons, which begs the question: Why?

Quark/gluon plasma (a near frictionless liquid- superfluid, not a gas) was the primordial soup of creation according to BBT. According to the standard model, quarks are elementary particles and a fundamental constituent of matter. **Matter** is (primarily) comprised of up quarks, down quarks, and electrons. Quarks combine to form composite particles called hadrons, the most stable of which are protons. Protons and neutrons are the components of atomic nuclei (nucleons). Neutrons are stable within nuclei. Quarks are not found in isolation; therefore, they are unstable in isolation. Quarks are stable only within hadrons as triplets, which include baryons (such as protons and neutrons). Mesons having two quarks, including pions, are unstable, as are quark-gluon plasmas.

At least one Big Bang Theory version has QGP first, then bosons. Quarks and anti-quarks, gluons, electrons and positrons, neutrinos/anti-neutrinos, then protons and neutrons, then, from particle annihilation, BBT has photons emerging. Interesting, but all wrong according to Photony theory. Photony holds that the Universe was created starting with the smallest elemental particles (elemental photons/anti-photons) and building all the larger quanta (classical photons/anti-photons) and particles (fermions, e.g., electrons/positrons), then quarks/anti-quarks, then hadrons/baryons and mesons paving the way for atoms and molecules.

A thought-provoking paper by Dr. Christoph Caesar has proposed quark theory as follows: [43]

*“As a working hypothesis for the structure of a quark a single spherical quantum wave is assumed. It should have a mass of around 300 MeV, a spin, show exactly three different configurations, the “colors”, and should exhibit the well-known  $\pm 1/3$  or  $\mp 2/3$  charges. ... The nucleon consists of three quarks. 3 times 6 = 18 of these 53 MeV leptons amount to 954 MeV which is very close to the proton mass; **six 53 MeV leptons therefore might generate or inhabit one quark of 318 MeV**, e.g., orbiting each other in the spherical wave. In an atomic nucleus, 8 MeV are necessary for removing one nucleon. Applying this binding energy also to a quark, this would give a total binding energy of  $2 * 8$  MeV to remove two of three quarks. 954 MeV minus binding energy now should give the nucleon mass:  $6 * 3 * 53$  MeV -  $2 * 8$  MeV = 938 MeV”*

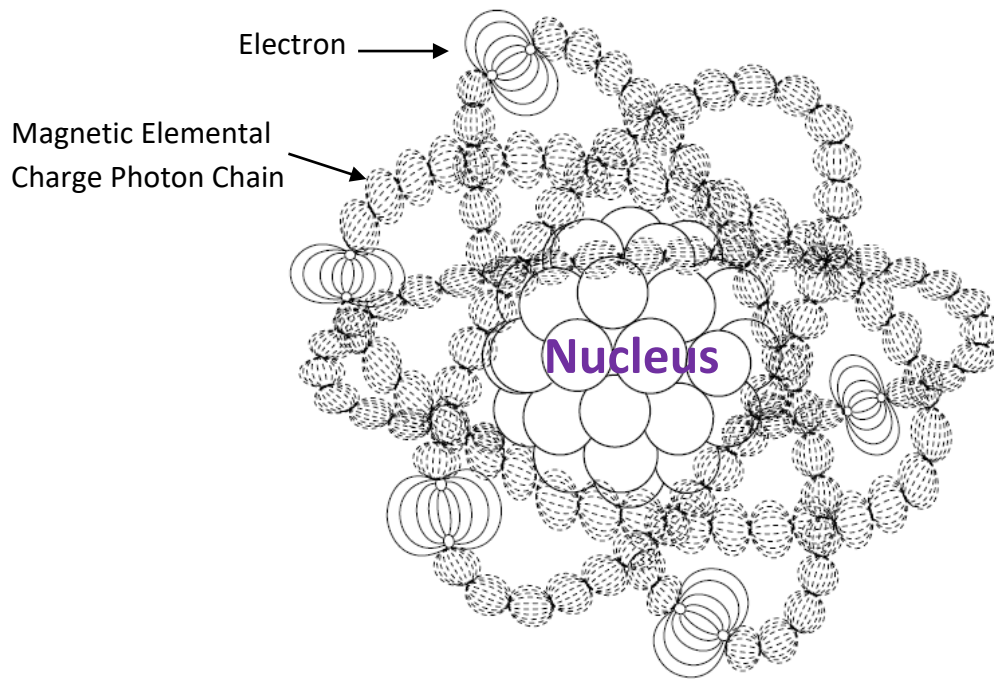
Photony provides insight into the structure of quarks (comprised of fermions) and why they only exist within nucleons. Photony examines the possibility of quarks comprising electrons and positrons with electric and magnetic chains for binding. The complete development and presentation of Photony quark theory are deferred to the Photony Compendium book, QPC.

# Photony Foundations

## L. Building Atoms with Photons

"The chains that bind us the most closely are the ones we have broken." - Antonio Porchia

Gamma ray photons with an extra c-spin comprise electrons that transduce elemental charge photons into chains ranging from classical photon chainlets to EM radio waves or, in some cases, atomic shells, as shown below.



**Figure 48: Photony Atom Overview**

The Photony Atom model illustrated above conceptually highlights the role of toroidal-shaped electrons riding on elemental charge photon chains, loosely called shells. Protons and electrons are bound by interactions with photons. [44] Elemental charge photon chains extending from protons within the nucleus (not shown) act as gluons within the nucleus and extend to the shells with some influence on the electrons/shell chains' degrees of freedom.

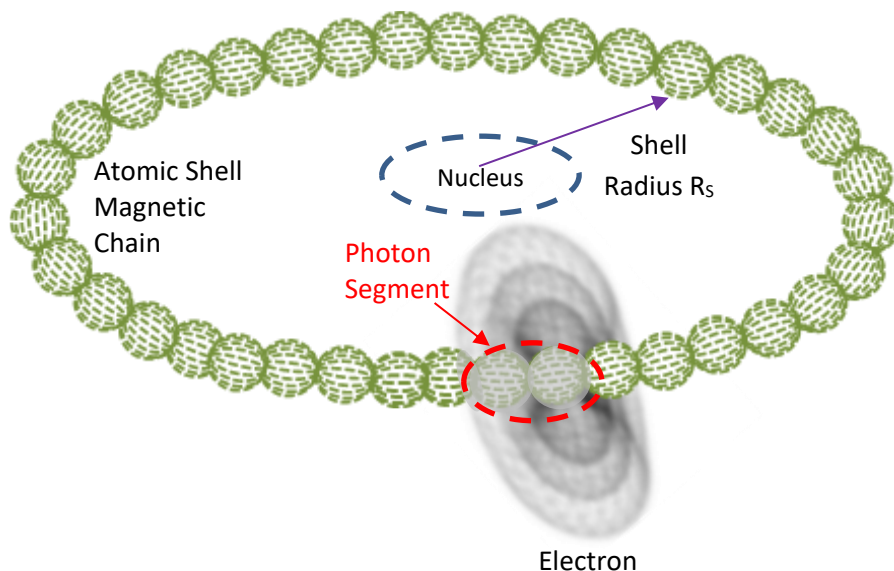
The role of Photony elemental charge photon chains, aka "lines," in fundamental physics cannot be overstated. As Photony has discovered, electrons are gamma rays with an extra spin, making them appear to be physical particles based on localization and loss of forward extended light speed. Elemental photon chains are transduced and reduced by electrons. Electric and magnetic chains provide the forces identified by Newton, Gauss, Faraday, and Maxwell. Electric fields are comprised of single-ended and terminated electric chain segments, and magnetic fields are comprised of magnetic chain loops. Magnetic chains provide more force than electric chains due to Photony spin mechanics, elemental photon linking force, and continuous-linked looping configuration vs. single-ended and double-ended terminations.



# Chapter 4

Starting with the Bohr atom, the Rutherford-Bohr model is a relatively primitive model of the hydrogen atom [45], compared to the modern valence shell atom model. The Bohr can be derived as a first-order approximation of the hydrogen atom using the broader and much more accurate quantum mechanics and is generally considered an obsolete (incorrect) scientific theory. However, because of its simplicity and its correct results for certain simple atomic systems, the Rutherford-Bohr model, unfortunately, is still commonly taught as fact to introduce students to quantum mechanics or energy level diagrams rather than more accurate but more complex, multi-shell and valence shell atomic models leading to chemistry, electromagnetics, and solid-state physics. John William Nicholson and Rutherford deserve more credit for early atomic discoveries and modeling than Bohr. Nicholson was the first to create an atomic model that quantized angular momentum as  $h/2\pi$ . [46] Nicholson was also the first to create a nuclear and quantum theory that explains spectral line radiation as electrons descend toward the nucleus, identifying hitherto unknown solar and nebular spectral lines. [47] [48]

Starting, as shown below, is a nucleus (a single proton in the case of the hydrogen atom), a Photony magnetic elemental photon chain loop (shell), and an electron riding on the chain. The photon segment is potentially two interlinked elemental photons as the electron passes over their linkage. The photon segment is emitted as a classical photon outside of the visible spectrum as debris from the electron transition. As the electron experiences a collision, it pulls against the magnetic chain (shell) until it fractures and pulls the photon segment out of the chain, allowing it to exit at light speed as a classical high-energy photon. Electric chains (not shown) extend outward from the protons in the nucleus and tend the magnetic chains of the electron shells.



**Figure 49: Electron Riding a Shell Chain**

# Photony Foundations

Electron shells are labeled K, L, M, N, . . . , corresponding to  $n = 1, 2, 3, \dots$ . For each  $n$  value, there are a number of  $\ell$  values that determine the spatial distribution of the electron. For a given  $n$ , each  $\ell$  value constitutes a subshell. For example, we often talk about 3s, 3p, 3d subshells within the M shell.

Periodic Table Considerations- Orbit/Shell ( $n$ ): The maximum number of electrons a given orbit can support,  $M(n) = 2n^2$

|                  |                           |
|------------------|---------------------------|
| K shell, $n = 1$ | $M(1) = 2 \cdot 1^2 = 2$  |
| K shell, $n = 1$ | $M(2) = 2 \cdot 2^2 = 8$  |
| K shell, $n = 1$ | $M(3) = 2 \cdot 3^2 = 18$ |
| K shell, $n = 1$ | $M(4) = 2 \cdot 4^2 = 32$ |

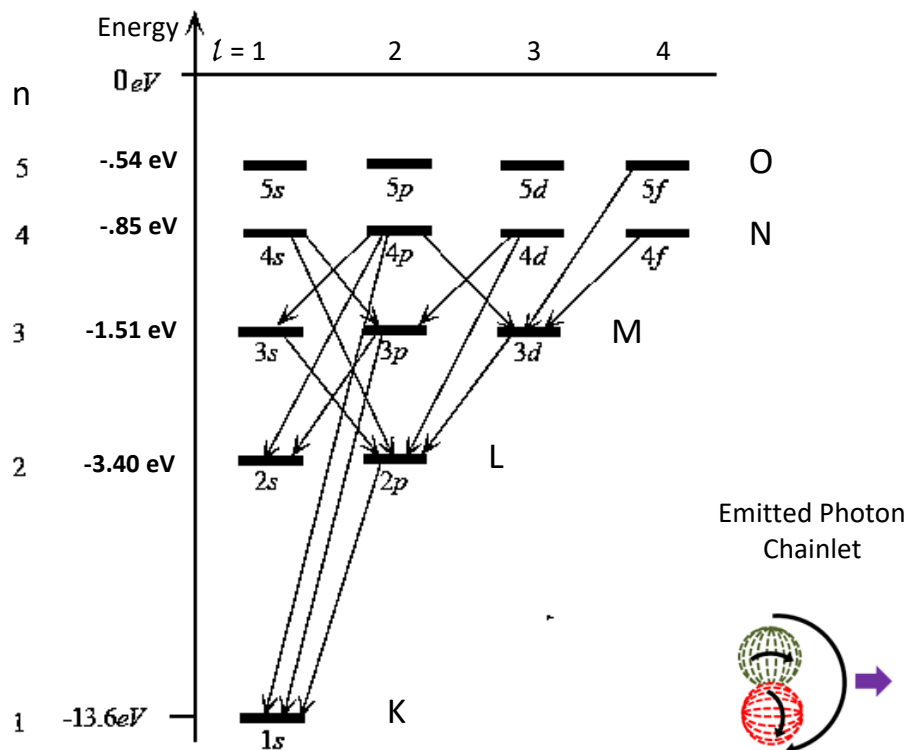


Figure 50: Electron Shell Transitions Resulting in Emitted Photons

The above electron transitions can be from spontaneous emission or stimulated emission. Spontaneous emission can be from thermal effects such as a hot metal object glowing red when sufficiently heated with the peak frequency/energy determined by Wien's Law.

Electrons can be ejected from their chains (shells) by either high-energy photons or other electrons resulting from collisions. Photony theory explains how electron shells of atoms are structured to hold electrons at certain energy levels and where and how the photons emerge when an electron loses energy by transitioning from a higher energy shell to a lower one. If the electron is ejected to a lower energy shell, the photon-segmented chainlet within the electron is emitted.

## Chapter 4

For stimulated emissions electron transitions to a lower energy level and emits at least one photon. If a high-energy photon strikes the electron, it may emit two photons, as indicated in the diagram above, at levels 4S and 4P and transition to a lower energy level.

Another possibility is when an electron absorbs a photon, causing it to transition to a higher energy level. Photon energy is related to its end-over-end spin frequency as follows:

$$\Delta E = hf \quad \text{Equation 30}$$

The energy difference,  $\Delta E$  of higher-to-lower electron-level transitions, determines the frequency of the emitted photon. The constant  $h$  is Planck's constant,  $6.62607015 \times 10^{-34}$  J·s.

Having Photony elemental (charge) photons, their ubiquitous nature, and the chain-bound nature of larger and larger particles resulting in significant energy density demonstrates what Einstein anticipated in 1920 about the ether (elemental charge photons) and ponderable mass (energy density).

This chapter has introduced the kinematics of how the smallest of quanta, Photony elemental (charge) photons, are formed, their ubiquitous nature, how they can be static, how they can be dynamic, and how they can be chained, and how particles can be built and bound by chains.

The following chapters show how elemental (charge) photons can be transduced into chains, how they can be reduced back into individual charge photons, and how classical photons are chainlets (of two interlinked elemental charge photons).

In the Photony Compendium book, **QPC**, Photony provides a set of elemental photon chain mechanics that explains the binding of electrons in shells and the binding and stability of fermion-based quarks in nucleons.





# Chapter 5

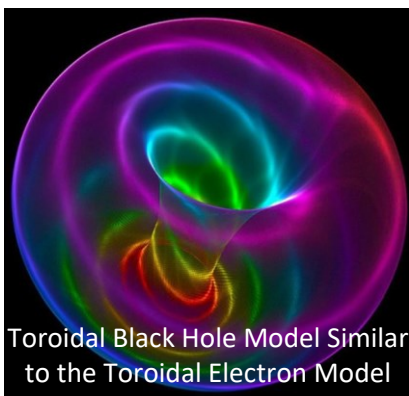
## Electron Exceptionalism

In Part I of James Maxwell's paper "On Physical Lines of Force," [49] regarding electrons, [50]

*"Maxwell treated the aspects relating to the curl of the velocity field in **a sea of tiny aethereal vortices**. [Photony dynamic elemental charge photons] ... Part III of the same paper, but in order to amalgamate the concepts, it is now proposed that **rotating electron-positron dipoles constitute dipolar aether vortices, with the electrons being sinks and the positrons being sources.**"*

### A. Electron Embodiment

The toroidal shape of electrons is functional and pleasing. Magnetic and electric fields, classically described as flux "lines," **are realized as Photony elemental photon chains**, kinematic realizations of the flexible tubes hypothesized by Faraday and Maxwell.



"An electron in the ground state will orbit at the speed of light. This is the electromagnetic equivalent of a black hole." - Richard Feynman [44]

Photony: The "orbit" here refers to the orbiting gamma ray comprising the electron

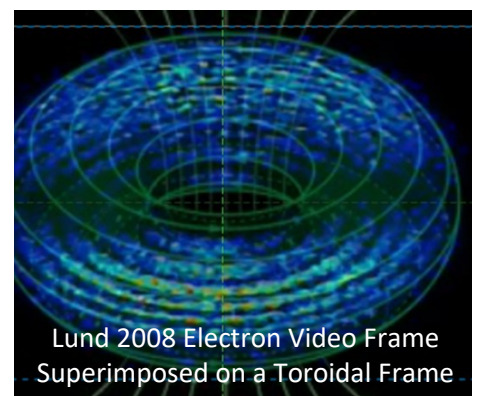


Figure 51: Black Hole vs. Toroidal Electron

Nassim Hamein describes black holes;

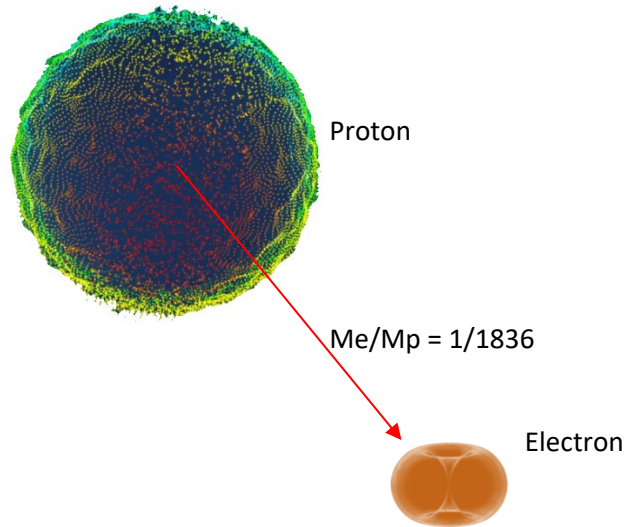
*"as having the shape and dynamics of a Torus, which allows information to both implode and expand from the central singularity."*

By some degree of coincidence, they (black holes) show the basic shape of the toroidal electron, as Photony hypothesizes. It illustrates how a gamma ray could appear after transitioning from a linear forward motion to a rotational motion based on an additional stacked c-spin. The electron also acts as a transducer taking in Photony elemental charge photons at one side of the toroidal cavity and through intense magnetic and gravitational fields at the center to interlink the elemental photons as chains.

Photony explains electric fields, magnetic fields, and electromagnetic waves and how they are generated regarding electricity and magnetism. Until charge is understood regarding; what it is and how it flows, there is no basis for voltage or current and, therefore, no basis other than empirical for static and dynamic electric or magnetic fields and no basis for electromagnetic fields.

# Chapter 5

## B. Electron Detailed Model



**Figure 52: Proton to Electron**

Photony has otherwise laid bare the reasonings and corroborations for the toroidal shape of the electron. Hopefully, it has been convincing, but with many subject matters in science, the devil is in the details that follow. Herein Photony discloses the details of the toroidal structure and how it supports the point source model of QED, plus adding the electric and magnetic chain source models of Photony. There is a particular interest in determining the size and shape of the electron.

### 1. Electron Mass and Energy

The electron's rest mass is generally accepted as proportional ( $\mu = 1836.152$ )<sup>17</sup> to the proton rest mass.

$$Mp = (1.6726 \cdot 10^{-27}) \text{ kg} \quad \text{Proton mass (kilograms)}$$

$$Me = (9.11 \cdot 10^{-31}) \text{ kg} \quad \text{Electron mass (kilograms)}$$

$$\frac{Mp}{Me} = 1.836 \cdot 10^3 \quad \text{Proton/electron mass ratio, } \mu \quad \text{Equation 31}$$

The energy of the electron is calculated using the Einstein relation,  $E = mc^2$ :

$$Ee := Me \cdot c^2 = (8.1877 \cdot 10^{-14}) \text{ J} \quad \text{Electron energy (Joules)} \quad \text{Equation 32}$$

<sup>17</sup> NIST

# Photony Foundations

Static/at rest- relatively low speed and no x-spin:

Electron at Rest (but very busy internally):

Photony Toroidal  
Electron Model

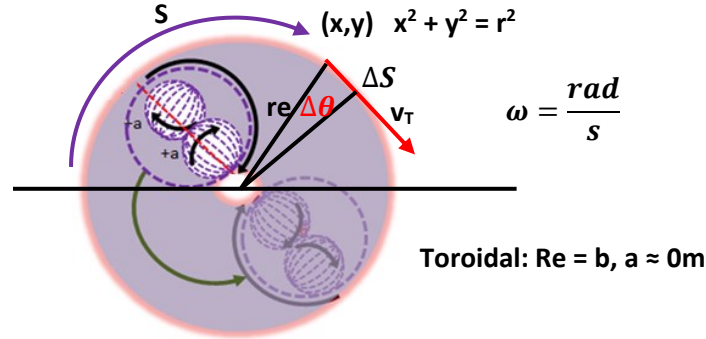


Figure 53: Photony Electron Model Parameters

$$\Delta s = c \Delta t \quad \text{Equation 33}$$

$$\text{Tangential velocity: } v_T = \frac{\Delta s}{\Delta t} = c \frac{\Delta t}{\Delta t} = c \quad \text{Equation 34}$$

$$V = \omega / re \text{ so, } re = \omega / v = 1/c / c = 1/c^2 \quad [51] \quad \text{where } 1/v \text{ units} = m/s^2$$

If  $re \approx 1/c^2$  is true,

$1/c^2 = 1.1127 \times 10^{-17}$  approximates the toroidal radius (re) of the electron at rest!  
and,

$$c \approx V_T \approx 1/(re \cdot c) \text{ must also be true.} \quad \text{Equation 35}$$

Electron Radius:

$$re = 1/(3 \times 10^8)^2 \text{ m} = 1.11 \times 10^{-17} \text{ m} \quad \text{Equation 36}$$

Electron Rest Mass:

$$9.109 \, 383 \, 7015 \times 10^{-31} \text{ kg} \quad (\text{NIST})$$

The electron rest mass is the NIST proton rest mass divided by 1836.152 (check):

$$Me := \frac{Mp}{1836.152} = (9.1093 \cdot 10^{-31}) \text{ kg} \quad \text{Equation 37}$$

Applying the fine structure constant- The 2018 CODATA value for the fine structure constant  $\alpha$   
 $= e^2 / (4\pi \cdot \epsilon_0 \hbar c) = 0.007297352 \approx 1/137.036$ , so  $1/\alpha = 137.036$ . Alpha is related to the probability that  
 an electron will emit or absorb a photon. [52]

# Chapter 5

The rest mass of the electron is defined by the fine structure constant ( $\alpha$ ), Planck's constant ( $h$ ), and Rydberg constant ( $R_\infty$ ) as follows. The Rydberg constant  $R_\infty$  links the natural energy scale of the atomic world and the SI unit system. It connects the mass of the electron, fine structure constant  $\alpha$ , Planck's constant  $h$ , and the speed of light  $c$  (in quasi-static Planck space). [53]

$M_{e0} = 2R_\infty h / c\alpha^2 = 9.1 \times 10^{-31} \text{ kg}$ , where  $m_{e0}$  is the rest mass of an electron, and,

$h = 6.62607 \times 10^{-34} \text{ J}\cdot\text{Hz}^{-1}$ ,  $R_\infty = 1.097 \times 10^7 \text{ m}^{-1}$ ,  $\alpha = 1/137 = 0.0073$  (fine structure constant)<sup>18</sup>

$$M_{ec} := \frac{\mu_0}{4 \cdot \pi} \left( \frac{e_c^2}{rec} \right) = (9.1095 \cdot 10^{-31}) \text{ kg}$$

Equation 38

Where  $rec$  = classical electron radius =  $2.8179 \times 10^{-15} \text{ m}$ , note the  $4\pi$  = the Photony compression factor

Electron at "Rest," no x-spin equivalent energy:<sup>19</sup>

Electron Energy =  $8.1871057769 \times 10^{-14} \text{ J} = 0.511 \text{ MeV}$  NIST

$$Ee := Me \cdot c^2 = (8.1877 \cdot 10^{-14}) \text{ J} \text{ check}$$

$$Ee1 := \frac{Me}{re} = (8.2072 \cdot 10^{-14}) \frac{\text{kg}}{\text{m}}$$

Equation 39

Note: the electron rest mass/radius  $\approx$  energy

Electron Size- classical electron Radius =  $2.81794 \times 10^{-15} \text{ m}$  (NIST). This is a calculated radius based on the assumption that an electron's mass-energy potential (gravitational gradient) is fully contained within a certain radius. This radius is many times too large; for example, the radius of the proton is  $0.87 \times 10^{-15} \text{ m}$  which would make the electron larger than the proton. So, the classical electron radius is either incorrect or stands for some aspect of size that needs explanation. The classical electron radius could be wrong for at least two reasons; first, the electron has the shape of a toroid, not a sphere. Second, the electron's toroidal cavity has an elemental (charge) field passing through as chains are transduced. This hourglass-shaped cavity has a different energy/mass density than the electron's toroidal body. The toroidal cavity has non-zero energy as a zero-point source and an intense gravitational and magnetic field.

$$rest := \frac{e_c^2}{(4 \cdot \pi \cdot \epsilon_0 \cdot Me \cdot c^2 \text{ k})} = (1.1099 \cdot 10^{-17}) \text{ m}$$

Equation 40

Static radius. rest, where  $k = 1836/7.232 = 253.86$ , the electron dynamic to static scaler, and the static electron mass,  $Me = 9.11 \times 10^{-31} \text{ kg}$

<sup>18</sup> In 1926, Schrödinger believed that electron waves were always spread out across all of space and that the square of the wave function gave the charge density of the electron wave in any particular location. The fine structure constant only applies to normal matter that is at zero or non-relativistic speeds.

<sup>19</sup> The electron mass/energy and the positron mass/energy are the same. Their spins are symmetrically reversed, resulting in a matter/antimatter structure.

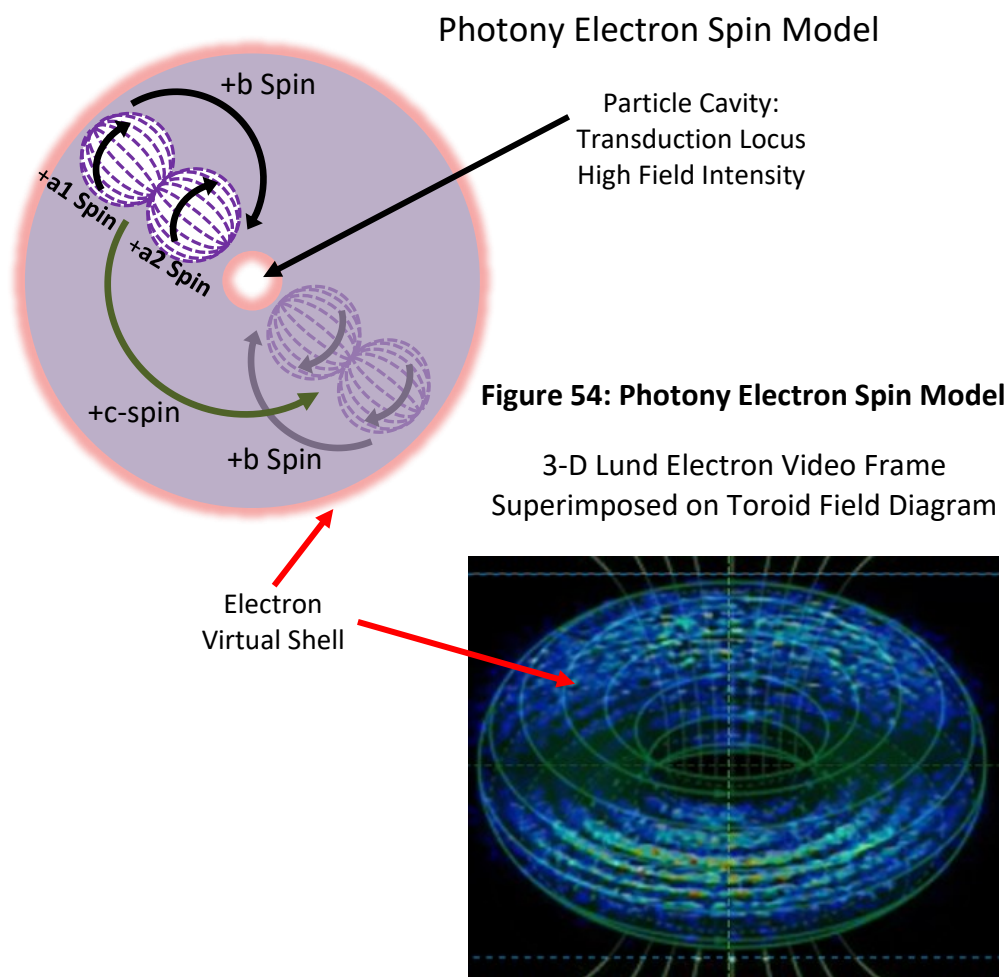
# Photony Foundations

## C. Photony Filament-Based Electron Spin Model

Spin Level 2; c-spin rotates the gamma ray tracing resulting in a virtual toroidal electron shell.

Photony Electron Spin Model:

The 2-D top view of the two interlinked energy filaments maneuvering in Planck Space based on the Photony Motion/Spin Rule is shown (below left). The internal gamma ray spin model is based on the incontestable physics of pair production and mass annihilation. The 3-D Lund Electron video frame is superimposed on a toroid field diagram (below right), illustrating the electron energy shell/containment.



**Figure 55: Electron Spin Model vs. Superimposed Toroid**

The particle cavity (doughnut hole) can be open relative to the toroidal body. This openness and ability to transduce and reduce elemental photon chains is further studied in the Photony Compendium.

# Chapter 5

## D. Electron- Elemental Charge Photon Processor

**Photony FINALLY explains what charge is- elemental charge photons;** static elemental (charge) photons, the basis for voltage as an electron/surface effect resulting in electric fields; and dynamic elemental charge photons, the basis for current as an electron/internal effect resulting in magnetic fields. **The key to understanding these phenomena is to understand the nature of electrons.** Individual electrons are free to move in conductors (free electrons) but **at a snail's pace**. Electrons are called "charge carriers," but more accurately, electrons are toroidal cyclonic charge transducers, accurately modeled as point sources. "Carried charge" is a ubiquitous collective of Photony dynamic photons continuously funneled into the toroidal opening of cyclonic electrons having virtual toroidal shells (traced out by gamma rays in rotational motion). Electrons are aligned by charge flow and process the charge flow passing through the electron's toroidal opening. In the case of electromagnets, the alignment only exists when direct current is flowing. In the case of permanent magnets, the alignment is only needed once since the electrons in ferromagnetic materials cannot move freely or linearly. Still, they can pivot to an aligned orientation with enough charge (elemental dynamic photon) flow. The electrons can hold that orientation until disturbed thermally (heating beyond the Curie point) or mechanically (as in hammering).

The charge field of Photony dynamic elemental particles produces and maintains toroidal face-alignment of free electrons in a conductor and pushes them along slowly due to interactions, including collisions. It has been easy for scientists and engineers to confuse "carrying" charge with attracting or processing charge (transducing and reducing) inside a zone within and extending around the electron's toroidal cavity. The localized charge is processed in the toroidal cavity resulting in interlinked chains called static field "lines" for direct current (unidirectional flow). If the electrons are axially aligned by and with the DC current flow internal to a conductor, this results in a magnetostatic field after an electromagnetic transient.

Free electrons cannot absorb photons. [54] Photons (gamma rays) in rotational motion ARE electrons. What a free electron in a conductor **can do** (with emphasis) at high speed is extrinsic rotation about the polar axis (extrinsic x-spin<sup>20</sup>). This is caused by alternating current flow resulting in interlinked chains of elemental photons (charge) **spewing out (transducing) chain segments that vary in length proportionally to the electron's axial rotation speed** (extrinsic x-spin). The electromagnetic field transitions to photonic (two-filament) particle spin **once the x-spin rate reaches the electron's limit to interlink no less than two Photony elemental photons** (charges).

---

<sup>20</sup> Axial end-over-end spin similar to the "x-spin;" originated by Miles Mathis' stacked spin hypothesis.



# Photony Foundations

## E. Electron Charge Photon Processing

Regarding electrons [50],

*“Maxwell treated the aspects relating to the curl of the velocity field in **a sea of tiny aethereal vortices.**”* These tiny aethereal vortices of Maxwell are Photony dynamic elemental charge photons!

As illustrated below, the electron attracts and propels charge due to the cyclonic nature of the motion of energy filaments (short-wavelength gamma ray chainlets) within the toroidal body of the electron. It has almost universally been mischaracterized as a “charge carrier.” It is better described as a charge processor (transducer and reducer), e.g., transducing some elemental dynamic photons into chains in the toroidal cavity vortex. An associated magnetic moment is produced based on intrinsic gamma ray-based axial electron spin.

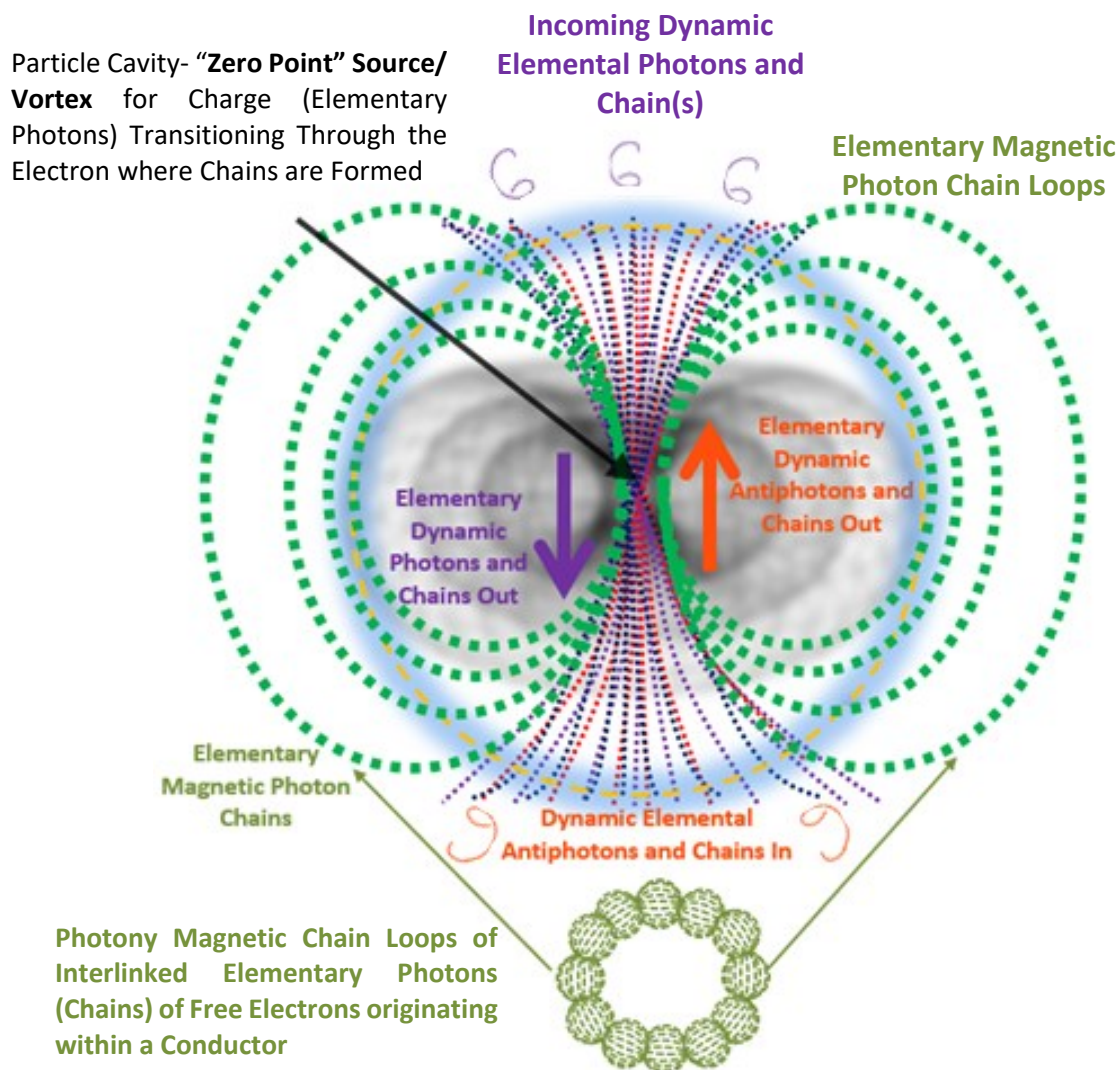
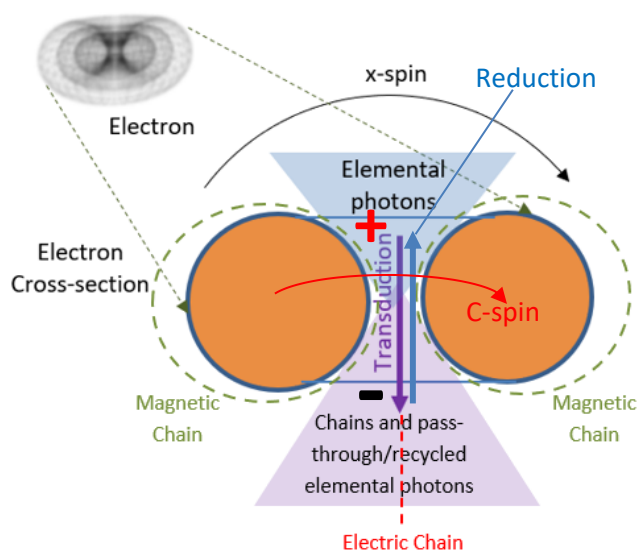


Figure 56: Elemental Charge Photon Processing by Electron

# Chapter 5

## F. Transduction and Reduction of Chains

The figure below illustrates an electron cross-section indicating alternative transduction and reduction processes. Note that electrons have a positive + and a negative - face based on the spin direction of the constituent gamma ray's C-spin as shown.



**Figure 57: Transduction vs. Reduction**

The process of transduction or reduction in an environment wherein the free electrons are transducing electric single-ended chains using static elemental charge photons or magnetic chain loops using elemental photons. Reduction is the converse of transduction and results from a decrease in accumulated static elemental charge photons as the voltage is reduced or from a decrease in dynamic elemental photons as current is reduced. Like classical photons, elemental charge photons are self-propelled and function as fuel for transduction. When the fuel runs out, the process reverses, and the E and B fields (of chains) reduce as the chains are unlinked into dynamic elemental photons by reduction within the same electrons that transduced them. The transduction/reduction processes are reversed with the presence of elemental antiphotons. The transduction of elemental antiphotons into chains from the minus to the plus face is possible such as with the negative plate of a capacitor. In other words, electric chains can be built in both directions when elemental antiphotons are present.

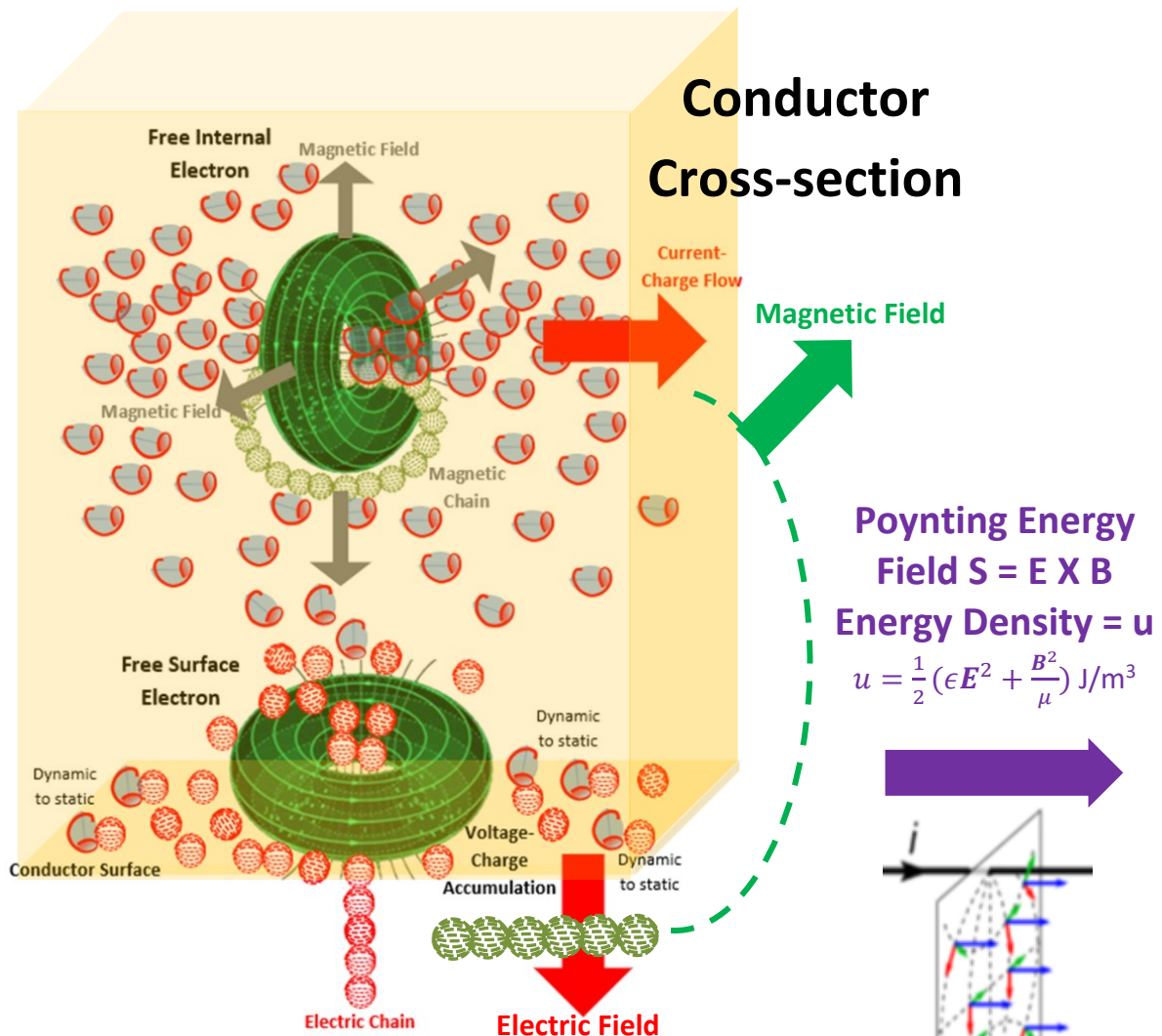
DC – the internal conductor's free electrons are aligned like windsocks by dynamic elemental photon flows in conductors depending on the polarity and current level, whereas electrons are aligned once in permanent magnets.

AC – electrons rotate (as x-spin) due to alternating flows of dynamic elemental photons (alternating polarity) while maintaining electron face (windsock) alignment as abbreviated magnetic



# Photony Foundations

and electric chains are simultaneously transduced or reduced based on increasing or decreasing current and voltage magnitudes. A graphic illustration of how elemental charge photons behave in a conductor and become transduced by free electrons into fields of chains is shown below<sup>21</sup>:



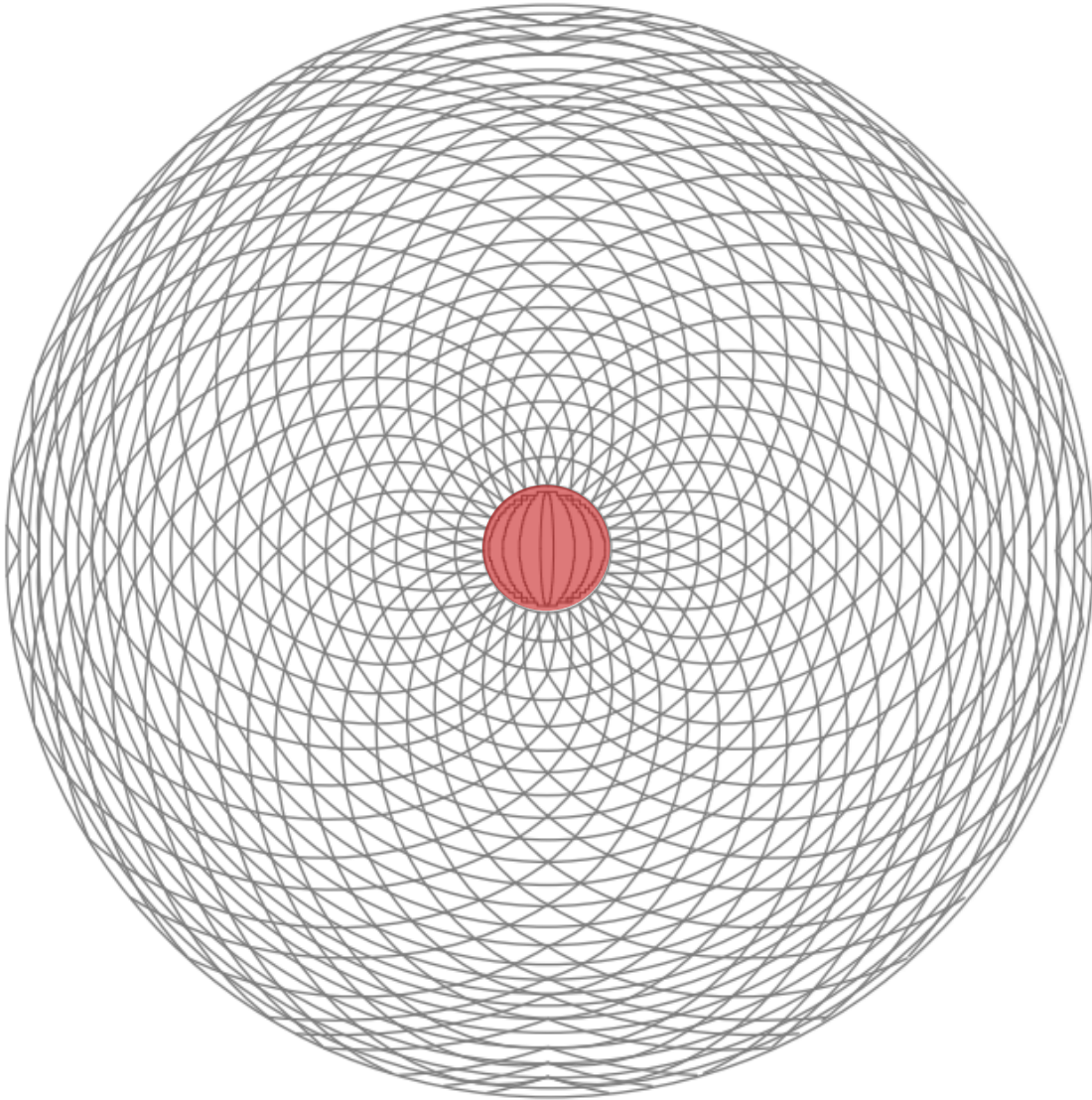
**Figure 58: Magnetic and Electric Chain Transduction**

The migration of self-propelled dynamic elemental charge photons through a conductor is the basis of current flow. Voltage is the amount of these dynamic charge photons encountering the surface and accumulating there as converted static elemental charge photons. The migration is realized by self-propelled elemental photons seeking the least energy pathways. The dynamic elemental charge photons' helical traction is insufficient to carry them beyond the conductor surface, having a very large permittivity into a medium with a relatively small permittivity. Each energy filament gap closes at the surface, resulting in static elemental charge photons with an incremental increase in voltage.

<sup>21</sup> Note that the Poynting Energy Field vector is in the same direction as the current flow.

# Chapter 5

## G. Electrons- Charge Processors (Transduction/Reduction)



**Figure 59: Electron Front View with Elementary (Charge) Photon**

The orbiting gamma ray traces are illustrated in the context of quasi-static Planck space. The figure above illustrates a front view of a Photony elemental charge photon at the central vortex of an electron cavity. Additional elemental charge photons (not shown) are typically present and queued up for transduction by the electron forces in the central vortex region of the toroidal cavity.

Electrons are charge processors (transducers), not “charge carriers.” More precisely, electrons transduce elemental charge photons into chains, called transduction, as a field is built. Alternatively,

# Photony Foundations

electrons reduce chains into elemental charge photons when a field reverses or collapses. The figure above illustrates an overview of an electron with an elemental charge photon at the center as if it is passing through and in position for transduction with other elemental charge photons to produce a chain exiting the opposing side (not shown) of the electron. Note that the electron is a gamma ray photon with an extra c-spin.

Historically, a coulomb was defined as the charge delivered by an electric current of one ampere in one second. The ampere was defined as the current needed to generate a magnetic force of 0.2 micro-newtons per meter between 2 parallel wires 1 meter apart. Under the 2019 redefinition of the SI base units, **as of May 20, 2019, the elementary charge  $e$  was assigned the exact value of  $1.602176634 \times 10^{-19}$  C**. A capacitor of one farad can hold one coulomb of charge in an electric field at a voltage of one volt. One ampere-hour equals 3600 C, hence 1 mA·h = 3.6 Coulombs of charge.

The International System of Quantities (SI) has defined 1eV as the exact value of  $1.602176634 \times 10^{-19}$  J, which is the standardized charge of 1 coulomb. The coulomb (symbol: C) is the SI-derived unit of electric charge. [55]

Free electrons in a conductor process charge through transduction to build (interlink) a chain or reduction to reverse and disassemble a chain. These electrons process charges at a high rate.

$e = 1.602 \times 10^{-19}$  C, the charge within the charge field (vortex) of an electron

The Specific charge (SC) formula is defined as a charge by the mass ratio of an electron.

$$SC = e/m_e = 1.602 \times 10^{-19} \text{ C} / 9.11 \times 10^{-31} \text{ kg} =$$

$$SC = e/m_e = 1.76 \times 10^{11} \text{ C/kg}$$

Equation 41: Specific Charge of an electron

Elementary charges (elemental charge photons processed per electron per second) is the reciprocal of the electron charge,  $e$ :

$$1/e = 1/(1.602 \times 10^{-19} \text{ C}) =$$

$$1/e = 6.241509 \times 10^{18}$$

Equation 42: Charges<sup>22</sup> per electron

Ampere is also defined as  $2 \times 10^{-7}$  N/m. A Coulomb is an Ampere-second. A Newton is defined as  $1 \text{ N} = 1 \text{ kg m/s}^2$ . Therefore, a Coulomb is: [56]

$$1\text{C} = (1\text{A}) \cdot (\text{s}) = (2 \times 10^{-7} \text{ N/m}) \times (\text{kg m/s}^2 / \text{N}) =$$

$$1\text{C} = 2 \times 10^{-7} \text{ kg/s}$$

Equation 43: Coulomb mass rate

and the rate in kg/s at which charge can be transduced by an electron is:

$$e_{mr} = (1.602 \times 10^{-19} \text{ C}) \times (2 \times 10^{-7} \text{ kg/s} / \text{C}) =$$

$$e_{mr} = 3.204 \times 10^{-26} \text{ kg/s}$$

Equation 44: Electron charge mass rate

---

<sup>22</sup> Elemental charges are Photony elemental charge photons (static and dynamic)

# Chapter 5

1. Will the real “charge” please stand up?

**Expressing the electron charge e in kg/s (charge mass rate), brings surprising facts to light. [57]**

The electron has a mass of  $9.11 \times 10^{-31}$  kg. According to the kg/s equation above, the toroidal Photony electron is receiving (through its entrance cavity) and transducing elemental charge photons every second with a collective throughput mass that is 35,170 times the electron mass.<sup>23</sup> as it transduces a chain emitting from the opposing side:

$$m_{ec} = e_{mr} / m_e = 3.204 \times 10^{-26} \text{ kg/s} / 9.11 \times 10^{-31} \text{ kg} = 35,170 \times m_e / \text{s} \quad \text{Equation 45}$$

$m_{ec}$  = Charge Mass to Electron mass Transduction Rate

$$m_{ec} = 35,170 \times m_e / \text{s}$$

$m_{pc}$  = Charge Mass to Proton mass Transduction Rate

$$m_{pc} = 35,170 \times m_e / \text{s} / 1836 = 19.16 \text{ } m_p / \text{s} \quad \text{Equation 46}$$

Elemental Charge Photon Momentum:

$$m_{sp} = 1.19 \times 10^{-53} \text{ kg Static elemental photon mass} \quad \text{Equation 47}$$

$$M_{dp} = c \cdot 1.19 \times 10^{-53} \text{ kg} =$$

$$M_{dp} = 3.57 \times 10^{-45} \text{ kg m/s} \quad \text{Dynamic elemental photon momentum} \quad \text{Equation 48}$$

Calculating interlinked elemental charge photons per meter ( $N_c$ ) based on mass:

From Equations 44 and 48:

$$N_{cm} = 3.204 \times 10^{-26} \text{ kg/s} / 3.57 \times 10^{-45} \text{ kg m/s} =$$

$$N_{cm} = 8.97 \times 10^{18} / \text{m} \quad \text{Elemental charge photon links per meter} \quad \text{Equation 49}$$

Therefore, **progressively extensive longer chains can be transduced by an electron with an intra-chain density of  $8.97 \times 10^{18}$  interlinked elemental charge photons per meter!** The question about how the chains are routed as they exit the electrons are addressed in other sections.

Calculating the distance between centers of interlinked elemental charge photon chains:

$$D_c = 1/N_{cm} = 1.115 \times 10^{-19} \text{ m} \quad \text{Equation 50: chain link distance between centers}$$

Calculating the number of elemental charge photons' transduction rate:

$$N_{cs} = N_{cm} / c = 8.97 \times 10^{18} / \cancel{\text{m}} / 3 \times 10^8 \cancel{\text{m}} / \text{s} =$$

$$N_{cs} = 2.99 \times 10^{10} / \text{s} \quad \text{Equation 51: No. charge photons transduced per second}$$

---

<sup>23</sup> Special credit goes to Miles Mathis for this determination and other insights in this section

# Photony Foundations

## 2. Transition EM Radio/Microwaves to Classical Photons with AC Potential:

Photony provides a novel inventory of elemental charge photons, classical photons, and elemental charge photon chains based on interlinking energy filament models. However, the calculations of frequency of the transition frequency between microwaves and classical photons is provided by Miles Mathis. [58]

The particle wavelength of this IR photon is the distance traveled during one revolution of the photon which is set by the revolution time of the electron.

The mass of the infrared photon  $m_{ir}$  is the proton mass  $m_{pr} / \mu^3$  (where  $\mu$  is the NIST Proton-Electron Mass Ratio = 1836.153 and  $m_{pr} = 1.67 \times 10^{-27}$  kg), or  $2.7 \times 10^{-37}$  kg, so we use Planck's relation  $E = h \cdot c / \lambda$  and the Einstein relation  $E = mc^2$  to solve for the wavelength of interest:

$$\lambda = hc/E = h\epsilon/m_{ir} c^2 = h/m_{ir} \cdot c \approx 8 \times 10^{-6}m \quad \text{Equation 52}$$

Where,  $h = 6.626 \times 10^{-34}Js$

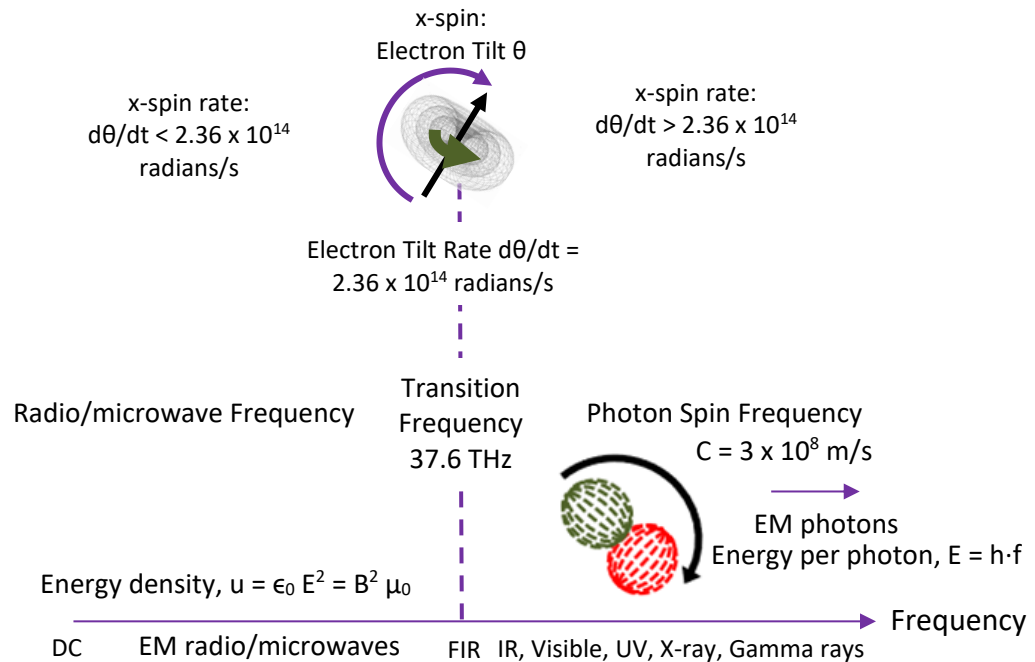


Figure 60: Photon Transition from Microwaves

The transition from radio/microwaves to photons phenomenon results from a limited transduction (production) rate of Photony elemental charge photons as the AC potential causes the transducing electrons to rotate with the x-spin (axial end-over-end). From the transitional microwave to classical photon frequency, 37.6 THz, classical photon chainlets are transduced as the electron rotates axially with the x-spin at a rate that limits the electron's ability to interlink more than two elemental charge photons. The resulting classical photons remain electromagnetic in nature since they are comprised of one elemental charge photon spinning electrically while the other one is spinning magnetically. The interaction of the two superimposed spins results in their

# Chapter 5

energy filament “keyring gaps” to remain sufficiently open to propel them with traction through Planck space with an extrinsic b-spin as they maintain forward motion.

The red and green circles in the figure below represent EM elemental charge photons. The toroidal electrons in gray rotate such that only two interlinked charge photons can make it through the electrons per x-spin 1-8 revolution positions. The electron spins the interlinked chainlets released at position 9 as it accepts two new charge photons. The newly transduced classical photon (chainlet) proceeds away from the electron at none other than the speed of light of the medium it is traveling.

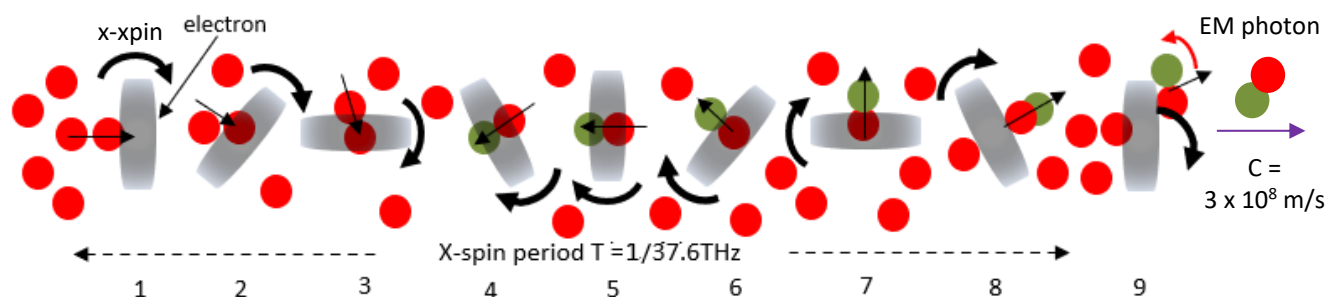


Figure 61: Classical Photon Transduction

The electron x-spin rotation period  $T_e$ :

$$T_e = 1/37.6 \text{ THz} = 2.66 \times 10^{-14} \text{ s}$$

Equation 53

The electron x-spin rotation period is also the spin period of the emitted classical photon, thanks to the position 9 rotation kick at the launch position of the photon. So, the spin rate of the emitted photon is 37.6 THz matching the electron's spin rate. The difference is the forward velocity of the electron is minuscule and negligible compared to the light speed,  $c$ , of the classical photon transduced during and emitted at the end of each revolution of electron's x-spin. However, the gamma ray that embodies the electron, travels at orbital light speed as it transcribes the energy shell and shape of the toroidal electron. Free electrons in space as opposed to a conductor do not remain spinning past the time it takes for the x-spin angular momentum to be reduced to an energy level corresponding to the ambient level aligned with the dynamic elemental charge photons of the space it is within. The windsock alignments from the dynamic elemental charge photons are the least energy circumstance the free electrons in the shared space.

Classical Photons Production Rate with AC Potential:

The rate of classical IR photons produced at this energy quanta is  $X_p$  which is simply the reciprocal of photon period  $T_p$  which is the same as the electron rotation period  $T_e$ :

$$X_p = 1/T_e = 1/2.66 \times 10^{-14} \text{ s} = 3.76 \times 10^{13} \text{ photons per second}$$

Equation 54

# Photony Foundations

## 3. Electron Exceptionalism is a Thing

As Photony has shown, electron exceptionalism is a thing. Electrons are constructed by high energy classical photons (gamma rays) with an extra c-spin.

Two well known phenomena called “pair production” and “annihilation” are seen in Feynman diagrams by Richard Feynman. The Feynman diagrams are frequently inconsistent and misunderstood. The diagram below attempts to show how two photons (gamma rays or possibly a gamma ray and an X-ray can collide to produce an electron/positron pair). Photony has an interest in these phenomena based on photon chainlets colliding at light speed with sufficient energy to add c-spin to each photon causing it to become relatively fixated in space by way of the photon converting from forward motion to rotational motion, hence the c-spin.

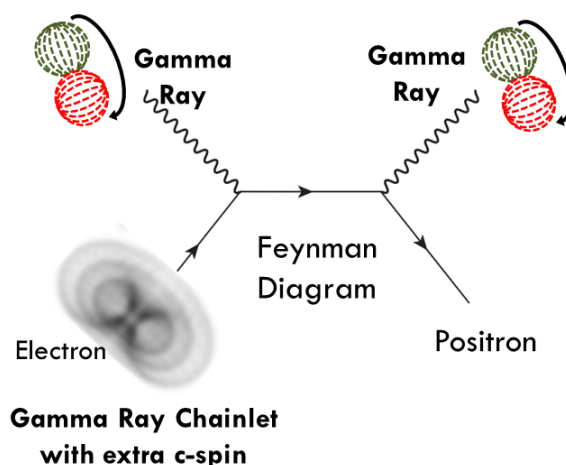


Figure 62: Gamma ray Annihilation and Pair Production

The following summarizes some of the ways electrons have played a role in physics and material science.

- In metals, conduction band electrons provide potential for electric and magnetic fields as Photony chains (lines) from voltage (static elemental charge photons) and current (dynamic elemental charge photons).
- In semiconductors (valence electrons equal four), electrons surround each atom with covalent bonding. Covalent bonding involves two atoms "sharing" a pair of electrons. Each atom forms 4 covalent bonds with the 4 surrounding atoms. Photony explains this as two toroidal electrons ride a shared elemental photon chain the extends around both atoms at the same shell/level. The common valence chain of each electron of the pair had broken and combined into a single chain since that was a least enrgy outcome with a strong covalent bond.

# Chapter 5

- In insulators there is a large band gap between the valence and conduction bands. In insulators, electrons do not have sufficient energy to crossover from the valence band to the conduction band and there is a low probability of finding an electron in the conduction band. An insulator material won't conduct any electricity at any temperature. Displacement current in insulators is based on a time varying electric field meaning it is made entirely of Photony electric chains extending and retracting through the insulator. This means that the surface electrons at the boudary of an insulator are transducing and reducing their electric chains withput being able to hop a ride on those same chains. So, the transduction and reduction of chains in an EM (AC) or slowly varying static (DC) environment are what was defined by Maxwell as “displacement current.”





# Chapter 6      Fields of Chains and Forces

## A. Fields Preeminent Research

Michael Faraday (1791-1867)- In 1831, Faraday started with the work of Oersted and Ampère on the magnetic properties of electrical currents and discovered the phenomenon of electromagnetic induction by showing that, based on mutual inductance, he could produce a current from a changing magnetic field. He demonstrated an induced current flow in the second coil or loop by moving a first wire loop (coil) carrying a substantial current or a permanent magnet in the second coil of wire.

**Faraday's introduction of the "field" concept** into physics is a significant contribution, described by Einstein as "the great change in physics" because it provided a common physical framework for electricity, magnetism, and optics. Faraday's lines of force and, more importantly, tubes were not recognized until several years later when **James Clerk Maxwell confirmed remarkably similar positions regarding field lines and tubes.**

In S.G. Starling's 1941 explanation, magnetic lines are considered "tubes of force." He explained that "pressure" exists between these *"tubes of force. ... an anti-clockwise rotation brings the direction of the current into coincidence with that of the motion, that is the cause into the direction of the effect."* [59]

James Clerk Maxwell (1831-1879)- Maxwell's Equations are his most notable accomplishment was bringing together electricity, magnetism, and light as different manifestations of the same phenomenon for the first time. Maxwell demonstrated with his "Maxwell's Equations" that electric and magnetic fields travel through space as waves at the speed of light  $c$ . One of Maxwell's innovations was the displacement current (Photony electric chains) to provide continuity of current through a dielectric, such as filling a capacitor or contributing to return current in a transmission line. [60]

Maxwell developed the famous equations by taking a set of known experimental laws (Coulomb's-Gauss' Laws, Faraday's Law, Ampere's Law) and unified them into a symmetric coherent set of Equations known as Maxwell's Equations (1864). Oliver Heaviside merged Maxwell's original 20 equations in the mid-1880s, eliminating the scalar and vector potentials into four simplified differential equations, which we know today as Maxwell's Equations. [60]

The biggest misconception about Maxwell's equations is that they prove that an electric field causes a magnetic field and vice versa. Not true! Photony and others, including Jefimenko, have proven this. Static B and E fields can be built independently at a proximal location. Without an absolute "potential," there can be no voltage (between two locations) and, therefore, no E field; without potential, there can be no current and no B field; another way of saying that the voltage and current potentials are independent.

## Chapter 6

According to J.J. Thomson, Faraday referred to lines of force as **chains of polarized particles** in a dielectric. [61] Faraday discussed them as having an existence all their own, as in stretching across a vacuum. The Photony elemental photons' interlinking processes in which electrons build (transduce) various length chains and chain types (electric and magnetic), including chainlets, are provided in greater detail in the Photony Compendium. Once an electric chain is completed between two opposed conductive surface electrons as a displacement current, with the same spin direction as chain-originating electrons, electrons can transit the chain between conductors as a real current. When spurred by a classical photon, this electron flow is called the photoelectric effect.

Due to the historical and current theoretical importance of tubes or, more specifically, Photony chains, we quote Mason's recordation of Faraday regarding "tubes of ether" and Maxwell's follow-up: [62]

*"Maxwell developed the qualitative aspects of Faraday's conception of lines of force, bringing in the ether of the wave theory of light. **Lines of force, Maxwell supposed, were tubes of ether rotating on their axes.** The centrifugal force of such rotations caused the tubes to expand sideways and contract lengthways, as Faraday had suggested in order to explain attraction and repulsion. But two neighboring tubes, rotating in the same sense, would move in opposite directions at the points where they touched, a conception which was not feasible mechanically. **Maxwell supposed, therefore, that there were layers of particles between the tubes of ether, the particles rotating in a direction opposite of that of the tubes, like ball bearings or idling wheels.** If all of the tubes of ether rotated at the same speed, the particles would not change their positions, but if not, a given particle would move linearly with a speed which would be the mean of the circumferential velocities of the tubes on either side. Thus, if by some means the rotatory speed of one tube was altered, a disturbance would be propagated throughout the system, and the particles would be set into a linear motion, rolling from tube to tube. Maxwell considered that the particles were electrical in character, and he thought, therefore, that such a motion of the particles would constitute an electric current.*

*Conversely, if a particle were displaced from its normal position, a tangential stress would be exerted on the adjacent tubes and, since such tubes were elastic, they would tend to restore the displaced particle to its normal place. Maxwell suggested that such a state of strain existed in the electrostatic field between two condenser plates, the charges on the plates displacing the electric particles, which in turn placed a stress on the ether tubes in the intervening space."*

As with Faraday's ether tube theory as lines of force, Maxwell's theory and mathematization of flexible spinning tubes could be the conceptual equivalent of elemental charge photon chains (ether tubes) with Photony kinematics. Free electrons within a conductor build magnetic chain loops. In contrast, surface electrons build single-ended electric chains, each electron operating as a toroidal transducer that performs the linking (Interlinking) within the electron's toroidal cavity. Surface electrons can slide across (ride) the electric chain it transduced within the dielectric.

# Photony Foundations

## B. Faraday and Maxwell Tubes

Faraday lines of Force - the lines are modeled as elastic spinning tubes by Faraday, followed by Maxwell a few years later. Faraday; *“observed that lines of force tended to contract longitudinally and expand laterally and modeled them as tubes with fluid inside, rotating axially and creating centrifugal bulging, a fluidic collective of dynamic vortices (or field).”* [63]

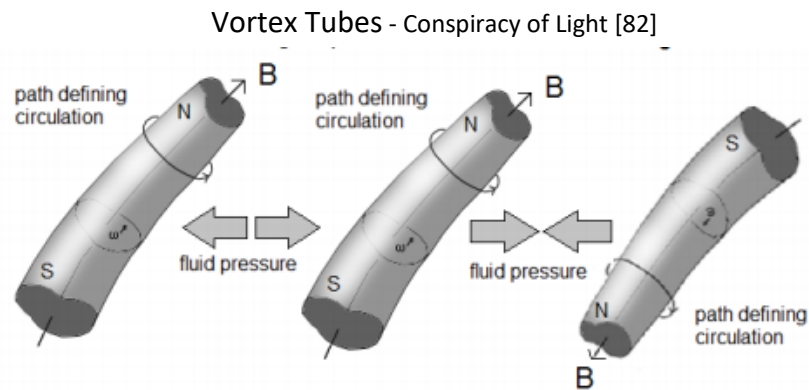


Figure 63: Faraday/Maxwell Tubes

Faraday's introduction of the “field” concept into physics is a significant contribution, described by Einstein as “the great change in physics” because it **provided a common physical framework for electricity, magnetism, and optics**. Faraday's lines of force and, more importantly, tubes were not recognized until several years later when James Clerk Maxwell confirmed similar positions regarding field lines and tubes.

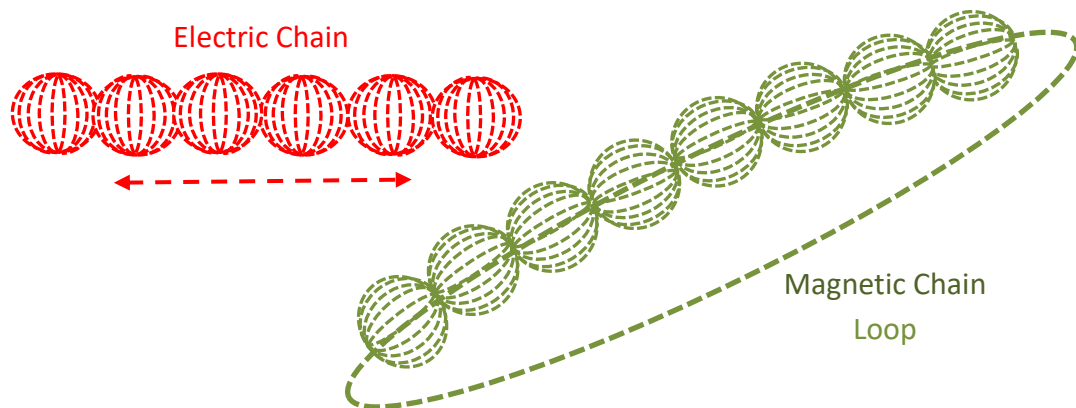


Figure 64: Photony Chains as Tubes of Force

# Chapter 6

## C. Potentials

### 1. Potential in Circuit Theory

In circuit theory, one form of potential is the voltage difference between any two points of any form of a circuit. Voltage is present within a complete circuit with transient or continuous current or partial circuits with secondary currents from reactive components with local fields. Reactive components include inductors with external magnetic fields of chains and capacitors with internal electric fields of chains. Electric/magnetic field energy can be exchanged between reactive components. Circuit elements and interconnects have various potentials from intended or unintended fields.

Voltage is typically sourced from a power supply or generator. The filter capacitor's plates in the output section (rectification and regulation) of a direct current (DC) power supply or a battery have a large capacity of electric field chains. The energy in the electric field's chains is potential energy, energy from interlinked elemental photons. In addition to the electric chains, the static charge photons that accumulate on the capacitor plates act as pressure due to their efforts to stay apart and distribute throughout the plate area and beyond. So, when a load is connected to the DC source, the static charge photons convert to dynamic charge photons and self-propel into the connected circuit as a current. As they depart the plate area, the electric pressure reduces, and electric chains retract back through the surface electrons that originally transduced them. In this reverse transduction process, the parental electrons unlink the chains one link at a time, thus replenishing the charge on the plate as it is lost to the load circuit.

Alternately the potential source can be alternating from a transformer, generator, or amplifier output as AC voltage and current. Alternating simply means periodically using voltage and current reverse directions as sinusoidal (steady state), transient, or digital waveforms. A digital waveform is a superposition of fundamental and harmonic waveforms, so an AC source is always a sinusoidal source. AC power sources are typically inductive. AC coupling of signals into a circuit is usually accomplished with a series capacitor having an inductive Norton equivalent as an inductor in parallel with the load. Alternatively, AC coupling of signals and power is accomplished with a transformer which is also highly inductive.

AC-powered circuits behave very differently from DC circuits; as voltage reduces, reverses, and increases repetitively, several AC-related phenomena result. DC voltages throughout typical circuits provide bias (voltage levels) so that AC signals can be processed without clipping and nonlinearities/distortion. However, many circuits have an AC voltage riding on a DC voltage.

Kirchhoff's Voltage Law (KVL) states that the algebraic sum of voltages in a closed path in a circuit mesh (loop) is equal to zero; the sum of source voltages is equal to the sum of voltage drops

# Photony Foundations

in that mesh of the circuit. There may be multiple meshes within a complete circuit, each having some combination of electrical/electronic components. If the current flows from a higher potential to a lower in an element, we consider it a voltage drop. If the current flows from lower potential to higher potential, we consider it a voltage rise.

Capacitors are reactive. The amount of electric field energy (Photony interlinked elemental electric photon single-ended chains) stored in a capacitor's electric field is proportional to the voltage level and time required to build the electric field chains (electric flux lines). The electric field energy can be built and **discharged as current back into the circuit** as a voltage change in DC or periodically in AC scenarios. As the electric field collapses (capacitor discharges), the electric chains unlink into elemental dynamic photons as the chains reverse flow through surface electrons (reverse transduction or reduction).

Reversing the electron from transducing to reducing results from a reduction of elemental (charge) photons available at the entrance to the toroidal electron cavity. In AC applications, reversing the electron's orientation by 180° by either current (dynamic charge flow) or voltage (static charge flow) pinches off the existing chains being transduced by electrons. It reverses the direction of transduction as the free electrons rotate.

## 2. Electric and Magnetic Potential(s)

Potentials as sources of energy exist as electric and magnetic potentials. Potentials also help to unify circuit and field theory with the Poynting Vector/Field theory. **Poynting's theory shows that energy is primarily delivered in circuits as fields outside the circuit interconnects/conductors.** There are two designations for fundamental potential; the letter A is designated for magnetic (vector) potential, and  $\phi$  is designated for electric potential as follows:

### a) Electric Potential

$\mathbf{E} = -\nabla\phi - \partial\mathbf{A}/\partial t$  E is a function of the gradient of the electric potential and the rate of change of the magnetic where **E** is the electric field (volts/meter), comprised of Photony electric chains. We see from above that **E** is ill-defined since electric flux is ill-defined. Electric flux has been ill-defined since electric "lines" of flux have been ill-defined, including electric flux lines. Photony defines electric lines as **electric chains**, thus providing meaning to electric lines of flux and the E field.

### b) Magnetic Potential

$\mathbf{B} = \nabla \times \mathbf{A}$  **B** is the curl of **A** where **B** is the magnetic field (magnetic flux density)

We see from above that **B** is ill-defined since magnetic flux is ill-defined. Magnetic flux has been ill-defined since magnetic "lines" of flux have been ill-defined, including magnetic flux lines. Photony defines magnetic lines as **magnetic chain loops**, thus providing meaning to magnetic lines of flux and the B field.

# Chapter 6

## D. Fields and Forces

Localized forces between particles, identified by the Standard Model as weak and strong forces, require Photony kinematics to identify the mechanisms, particularly with nucleons. The standard quark-gluon model and description of the Standard Model are not sustainable. The electron shell theory of the Standard Model provides reasonable explanations of electron orbital behavior but provides kinematics to govern such behavior. Again, it is missing the Photony chains of force that provide the binding forces and the elemental photons within the electrons' orbital chains to produce the classical photon releases as the electron moves down to the next orbital level. Energy is conserved as the liberated photon is released as the electron transitions to a lower energy level.

Early arguments between Bohr and Einstein regarding quantum mechanics were qualitative and philosophical in nature and unresolved. This includes the “spooky action at a distance,” as predicted by quantum mechanics and promoted by Bohr. Eventually, the terminology evolved to quantum entanglement (Verschränkung), a term coined by Erwin Schrödinger, in a letter to Albert Einstein.

Forces at a distance require extended Photony elemental photon chains as the flexible tubes of force first identified by Faraday and later advanced by Maxwell. The spin states of electrons, positrons, and possible protons can share a common elemental photon chain. By sharing, the chain can be transduced by one spinning particle, e.g., an electron. It can enter the toroidal cavity of a proximal electron or positron, followed by a separating motion of the two particles. The chain can be of almost any length while threaded through both particles. This means the particles are entangled. When one particle flips, the chain rotation reverses, and the remote particle will naturally flip, hence action at a distance. The issue of allowing influences to travel faster than the speed of light, disturbing to Einstein, whose theory of relativity prohibited any such superluminal propagation, is hereby resolved by the Photony entanglement theory. The delay of entanglement chain reversal is a topic of study in the Photony QPC Compendium book.

It should be anticipated that there should also be considerable differences in the explanations of static and dynamic fields of electric and magnetic chains of fields from electrons acting as elemental charge photon transducers/reducers in the process. The fields associated with EM radiation always oscillate as they propagate at the speed of light in space. In contrast, static fields and the fields associated with radiated particles are not always from oscillating potential.

There are only two basic forces, chain forces, and collision/momentum related forces. The Photony Compendium addresses and studies how chains maintain atomic shells and sub-shells, how photons are emitted when electrons change shells, and how photons can cause electrons to change shells. The Photony Compendium addresses and studies how chains maintain atomic nucleon/quarks binding.

# Photony Foundations

## 1. Standard Model Forces and Fields

In Universal Field Theory by John Macken, the gravitational and electromagnetic forces appear distinct. Still, these forces have been shown to be related in the tests of the electron model.

*“The wave-based model of the universe has only one fundamental force, and it is always repulsive. ... All other forces of nature are just different manifestations of this truly fundamental force” [64]*

It is common among physicists that all forces were united during the high energy conditions of the Big Bang and have since separated as the Universe expanded and the energy density decreased. In the aftermath, the forces of nature are fundamentally different and allegedly transferred by “messenger particles” such as virtual photons, virtual gluons, and gravitons.

Photony identities both repulsive forces, resulting from energy filament collisions, and compressive (depicted by Macken as “attractive”) from energy filament motions as per the Photony Motion/Spin Rule with filaments attempting smooth geodesic motion ( $\pi = 3.1415\dots$ ) in an orthogonal space governed by the Manhattan Metric with  $PI = 4$ .

From the Standard Model, there are four (4) generally accepted forces:

- Gravitational Force (Newton/static and Einstein/dynamic- Equivalence Principle)
- Electromagnetic Force (Electric- electrostatic/electrodynamic, Magnetic- magnetostatic/magneto dynamic) and electromagnetic (combinations of any number of the electric and magnetic components).
- Weak and Strong

According to the Standard QCD Model, quarks are the only particles that experience all four forces listed below. The items in red are based on Photony theory as an alternative to the Standard Model. It is possible that force carriers as proposed (black) are in error or not identified correctly. For example, gravitons have not yet been observed.

However, the first direct experimental evidence of gluons was found in 1979 when three-jet events were observed at the electron-positron collider called PETRA at DESY in Hamburg. Quantitative studies of deep inelastic scattering at the Stanford Linear Accelerator Center had established their existence a decade before. [65]

Since **no free quarks nor free gluons have ever been observed**, we can logically conclude that they must be confined as nucleons. The exception is high temperature/high pressures gluons and quarks as a plasma in which the quarks and gluons remain stable and flow like a fluid.<sup>24</sup>

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<sup>24</sup> Quark-gluon plasma has been found recently at the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratories (BNL).- CHEMEUROPE.COM

# Chapter 6

| Forces                                                | Strength                    | Range            | Force Carrier<br>(Photony in red)                                 | Quarks | Charged<br>Leptons<br>(e, $\mu$ , $\tau$ ) | Neutrinos<br>( $\nu$ ) |
|-------------------------------------------------------|-----------------------------|------------------|-------------------------------------------------------------------|--------|--------------------------------------------|------------------------|
| Weak                                                  | < Strong<br>> Gravity       | very<br>short    | 3 W and Z Bosons<br>Electric Chains                               | Y      | Y                                          | Y                      |
| Strong                                                | >> Gravity<br>>> weak >> EM | short            | Gluons?<br>Magnetic Chains                                        | Y      | N                                          | N                      |
| Electromagnetic<br>(EM)                               | > Weak                      | $\approx \infty$ | Photon?<br>Electric and Magnetic<br>Interlinked Chainlets         | Y      | Y                                          | N                      |
| Gravitational,<br>Strong: Black Hole<br>and Star Core | Weakest in<br>Nucleus       | $\approx \infty$ | Gravitons?<br>Geometric: general relativity<br>+ energy filaments | Y      | Y                                          | Y                      |

**Table 2: Fundamental Forces**

The Weak Force- while the other forces allegedly hold things together, the weak force plays a greater role in keeping nucleonic particles assembled and stable.

The Strong Force- gluons allegedly participate in strong interactions. **The gluon-gluon strong force interactions constrain color fields to string-like objects called “flux tubes”<sup>25</sup>,** which exert constant force when stretched. Due to this force, quarks are confined within composite particles called hadrons (protons and neutrons). Unlike the virtual and classical photons of QED or the three W and Z bosons of the weak force/interaction, there are eight (8) independent types of gluons posited in QCD. In QCD, quarks and antiquarks are theorized to carry three types of color charge. Gluons may be thought of as carrying both color and anticolor. This gives nine possible combinations of color and anticolor in gluons.

Gravitational Force (Gravity)- gravity is NOT overtly a force; it is a gradient of Planck space compression caused by proximal energy density. Gravitational intensity is another way of describing quasi-static Planck space compression directly related to permittivity! Gravitational force,  $F_G = m \times E_g$ , where the gravitational intensity  $E_g = [-(G \cdot M \cdot m / r^2) / m] \vec{r}$ .

The force associated with gravity is the product of the mass (curvature associated with an energy density) in the gravitational field times the gravitational field intensity,  $E_g$ . In the case of Earth, the gravitational intensity at the surface is  $g = 9.8 \text{ m/s}^2$ . This gravitational constant for Earth is quickly confirmed by reducing the force equation to  $g = GM/r^2$  and plugging in the Earth’s mass ( $5.9736 \times 10^{24} \text{ kg}$ ) into M and the Earth’s radius (6378137 m) into r with the gravitational constant G ( $6.674 \times 10^{-11}$ ).

<sup>25</sup> eaeaa1232 (<https://physics.stackexchange.com/users/291994/eaeaa1232>), How does the strong force increase in attraction as particles move farther away?, URL (version: 2021-03-14): <https://physics.stackexchange.com/q/620966>



# Photony Foundations

$$g := -G \cdot M \cdot \frac{1}{r^2} = -9.8002 \frac{m}{s^2}$$

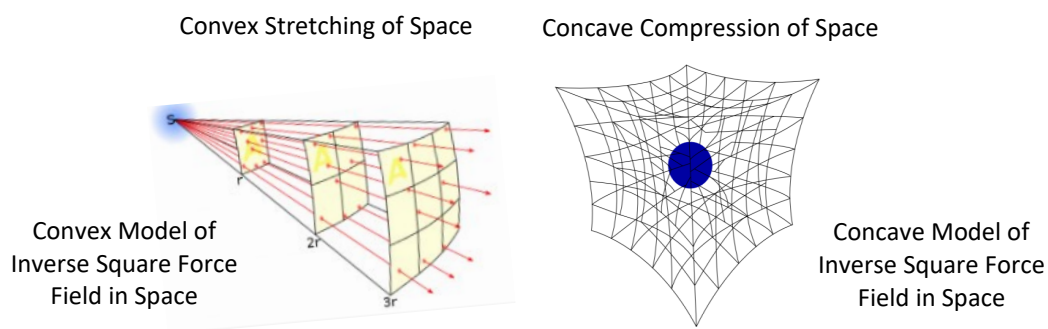
Equation 55: Earth's Gravitational Acceleration, g (above)

A popular general relativity physical interpretation is **geometric without being a true force**. [16] Photony provides an accommodating theory; the energy filaments of energy densities (masses) in space provide compressive forces that align toward (e.g., attracted to) other compressive energy densities in space (e.g., gravitational lens). **Energy filaments, passing through space, maneuver the spacetime gradients with a tendency to go toward a higher gradient of compression (gravitational intensity)**. This tendency also applies to space between subatomic particles comprised of energy filaments.

However, when the energy filaments, causing the gradients, collide or interfere, the result is repulsive and/or destructive. So, when two massive particles or objects are attracted, they may eventually move inward to interfere or collide at some momentum. Their proximal surface energy filaments in contact provide repulsive force or reflect like pool balls as energy is conserved. Damage or deformation may result from either type of repulsive contact. In addition, if the two objects are of magnetic material, the magnetic force (Photony magnetic chain-flex interactions) may add to or subtract from the gravitational forces depending on polarity and orientation.

## 2. Inverse Square Law

Interestingly, Newton's gravitational law, Coulomb's charge force law, and Coulomb's law of magnetism are similar but not overtly unified in standard theory. The standard gravitational model of the inverse square law in space is misrepresented (much like the trampoline model) as illustrated below left, relying upon the concept of pushing of Planck Space rather than the pull concept resulting from compression and the Photony (energy filament) Motion/Spin Rule. The model of Photony energy filament compression of quasi-static Planck Space is illustrated below right.



**Figure 65: Convex vs. Concave Inverse Square Law Models**

# Chapter 6

## 3. Two-body Forces<sup>26</sup>:

Two (or more) proximal bodies interact through attractive, “pulling,” gravitational forces, charge (elemental electric photon chains), and magnetic flux (elemental magnetic photon chain loops). The force magnitudes are shown below.

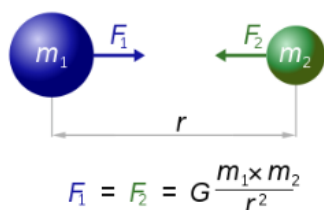
|                                                            |                                                              |                                                                     |
|------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|
| Gravitational Bodies: $m_1, m_2$<br>Force: Attractive Only | Charged Bodies: $q_1, q_2$<br>Force: Attractive or Repulsive | Magnetic Bodies: $q_{mA}, q_{mB}$<br>Force: Attractive or Repulsive |
| $F = G \frac{m_1 m_2}{r^2}$                                | $F = k_e \frac{q_1 q_2}{r^2}$                                | $F = k \frac{q_{mA} q_{mB}}{r^2}$                                   |
| Equation 56                                                | Equation 57                                                  | Equation 58                                                         |

----- Two-Body Forces -----

Charge and magnetic forces can be repulsive or attractive depending on the bodies' polarity (composite spins), like charge signs (++ and --) and poles north-north and south-south repel.

Energy density is equivalent to compression pressure. Thus, for example, photon forward pressure is always repulsive.

## 4. Gravitational Two-body Force:



Energy density is the root cause of the quasi-static Planck Space compression; the mass is merely the measure or metric of the Planck space compression caused by the energy filaments that trace out the spacetime extent of the energy density.

**Figure 66: Newton's Universal Gravitation Law**

Gravitation, which produces an attractive force between any two masses in the Universe in Newton's picture, applies to the energy causing the masses (compressions in quasi-static Planck Space). So, Newton's Law of Universal Gravitation could have the masses,  $m_1$  and  $m_2$ , replaced by  $E_1/c^2$  and  $E_2/c^2$ .

This equation was initially derived experimentally by Newton. The constant  $G$  was adjusted to calibrate the equation. Therefore, Newton's Law of Universal Gravitation was an empirical equation, not a theoretically derived equation. However, an empirical equation can be converted to a theory-supporting equation by identifying and deriving the constant(s). [66]  $G$ , in this case,

<sup>26</sup> The multi-body force relationships are deferred to the Photony EMFF Compendium book.

# Photony Foundations

represents some heuristic combination of fundamental constants or parameters, including pi (= 3.14...) and PI (= 4).

Photony has resolved the question of fundamental constants and parameters hidden within G, reduced Newton's law from mass to energy, and determined the effects of the inter-particle medium on the gravitational force. First, we study G to unlock its secrets as follows:

Starting with G, where the  $4\pi$  compression factor is built into G:

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^2 \approx 1/(16\pi \cdot c) = 1/(4\pi \cdot 4c) = 6.636 \times 10^{-11}$$

$$G = 1/(4\pi \cdot 4c) \text{ Equation 59: } G, \text{ a function of compression factor}$$

Newton's universal law of gravitation:  $F = Gm_1m_2/r^2$  could be written as:

$$F = G'(4\pi) \cdot m_1m_2/r^2 = \frac{1}{4\pi \cdot 2c} \cdot \frac{E_1E_2}{c^4r^2} = \frac{E_1E_2}{4\pi \cdot 2c^5r^2}$$

$$F = \frac{E_1E_2}{4\pi \cdot 2c^5r^2} \text{ Equation 60: Gravitational force } F, \text{ a function of compression factor}$$

Relative permeability and permittivity apply here as follows:

$$\mu = \mu_0 \mu_r, \text{ where } \mu_r \text{ is the relative permeability, and}$$

$$\epsilon = \epsilon_0 \epsilon_r \text{ where } \epsilon_r \text{ is the relative permittivity}$$

$$\text{Permeability: } \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\text{Permittivity: } \epsilon_0 = 8.854 \times 10^{-12} \approx 4\pi / \sqrt{2} \times 10^{-12} \text{ F/m}$$

The speed of light in space (the "speed of free space") is based on the properties of space- the permittivity and the permeability, as indicated. In a variable medium, the permeability,  $\mu_0$  becomes  $\mu$ , and the permittivity,  $\epsilon_0$  becomes  $\epsilon$ . Therefore, Newton's law of gravitation can vary based on the medium between the masses (energy centers).

So, the Photony  $4\pi$  force equation:

$$F = \frac{E_1E_2}{4\pi \cdot 2c^5r^2} \text{ Equation 61: Force equation with Photony } 4\pi \text{ Compression Factor}$$

This equation can have c replaced in the denominator based on:

$$c = \frac{1}{\sqrt{\mu\epsilon}} \text{ Equation 62: Velocity of Light-Based on the Medium}$$

This substitution produces an equation that is based only on energy, separation, and the characteristics of the medium! This force equation aligns perfectly with Photony theory since energy and Planck space are the primary components of the Universe. Also, it provides a computational vehicle for force (and pressure) and kinetic energy applicable to classical photons. The Standard Model fails to account for the equivalent mass of a photon (claiming zero mass), whereas Photony does provide a kinetic photon mass. The classical photon never rests, so the rest

# Chapter 6

mass is undefined and unneeded. Note that mass is only a measure of the compression of Planck space due to an energy density.

As Photony anticipates above,  $G$  varies based on permittivity and permeability, including near-fields and far-fields and various media other than space. Further indications of the variability of  $G$  based on the variability of the medium can be found in scientific data; [67]

*“In 1998, the US National Institute of Standards and Technology published values of  $G$  taken on different days, revealing a remarkable range. On one day the value was 6.73, a few months later, it was 6.64, 1.3% lower.”*

Einstein’s general relativity experiment involving gravitational force from the sun on starlight (photons) proved that gravity compresses space and sets the stage for understanding gravitational lensing based on the same phenomenon. The mechanism for this form of refraction is associated with changes in permittivity and permeability. This phenomenon affects the gravitational gradient and the “speed of space.” The speed of light is typically conflated with the speed of space based on cause/effect. Quasi-static Planck space or any other transmitting media governs the speed of quanta (e.g., classical photons), throttling their self-propelled speed at the speed of space (i.e., gravitational wave speed) as possible for least-energy (least compression) passage to maintain least energy.

## 5. The Many Faces of the Fine Structure Constant $\alpha$ (the Magic of 137)

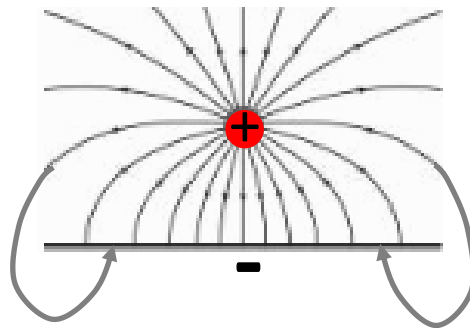
Given two hypothetical charged particles, each having Planck mass,  $m_p$ , and elementary charge,  $e$ , separated by any reasonable distance,  $\alpha$  is the ratio of their electrostatic repulsive force to their attractive gravitational force. From the energy perspective,  $\alpha$  is the ratio of the energy needed to overcome the electrostatic repulsion between two electrons a distance of  $d$  apart and the energy of a single photon of frequency,  $f$ .

The 2018 CODATA value for the fine structure constant  $\alpha = e^2/(4\pi \cdot \epsilon_0 \hbar c) = 0.007297352 \approx 1/137.036$ , so  $1/\alpha = 137.036$ . This value for  $\alpha$  gives  $\mu_0 = 4\pi \times 1.00000000054(15) \times 10^{-7} \text{ H/m}$ .



# Chapter 7 Electrostatics/Magnetostatics- DC Fields

Electric chains are transduced by free surface electrons, typically at or near the surface of conductors, but can be at the surface of non-conductors like balloons or human skin for DC. The electric chains are transduced initially single-ended as outgoing tentacles and can terminate with other incoming spin-compatible electric chains, or they can terminate at an adjacent surface with free electrons available, such as an opposing capacitor plate. Suppose there are no nearby objects with free electrons able to be mobilized to the surface. In that case, the distal ends of the electric chains diverge to minimize energy as they search for a mating electron or positron. The diagram of an elemental charge photon collection acting as a familiar “point source” in proximity to an opposing plate is shown below to illustrate convergent electric chains vs. divergent electric chains. The greater the separation, the larger number of electric chains will diverge. Some of the diverging lines (chains) may make it around to the back side of the plate, as shown.



**Figure 67: Electric Lines Depicting Terminated Electric Chains**

As Photony shows further into this chapter, if the electric field chains get too dense, a chain reaction occurs, which disintegrates the chains into chainlets, many of which are two elemental charge photon chainlets that provide light from the “static discharge.” The above figure shows how the electric chain density can be increased by simply bringing the positive charge modeled as the red circle closer to the opposing plate. At some distance, a discharge (voltage breakdown) due to electric chain collisions may occur.

In the atmosphere, free electrons and ions are produced through various phenomena, one of which is friction between air molecules. Once the electrons are freed, the ever-present ubiquitous elemental charge photons start being transduced into electric chains whose rotation corresponds to the spin direction of the electrons in a particular cloud. Once the electric chains reach ground level as leaders, the electrons in the ground migrate to align and mate with the electric chain ends. This can be a precursor of a lightning strike (voltage breakdown and chain disintegrations). The charged cloud above only needs to increase the electric chains in number to reach the density threshold for a strike.

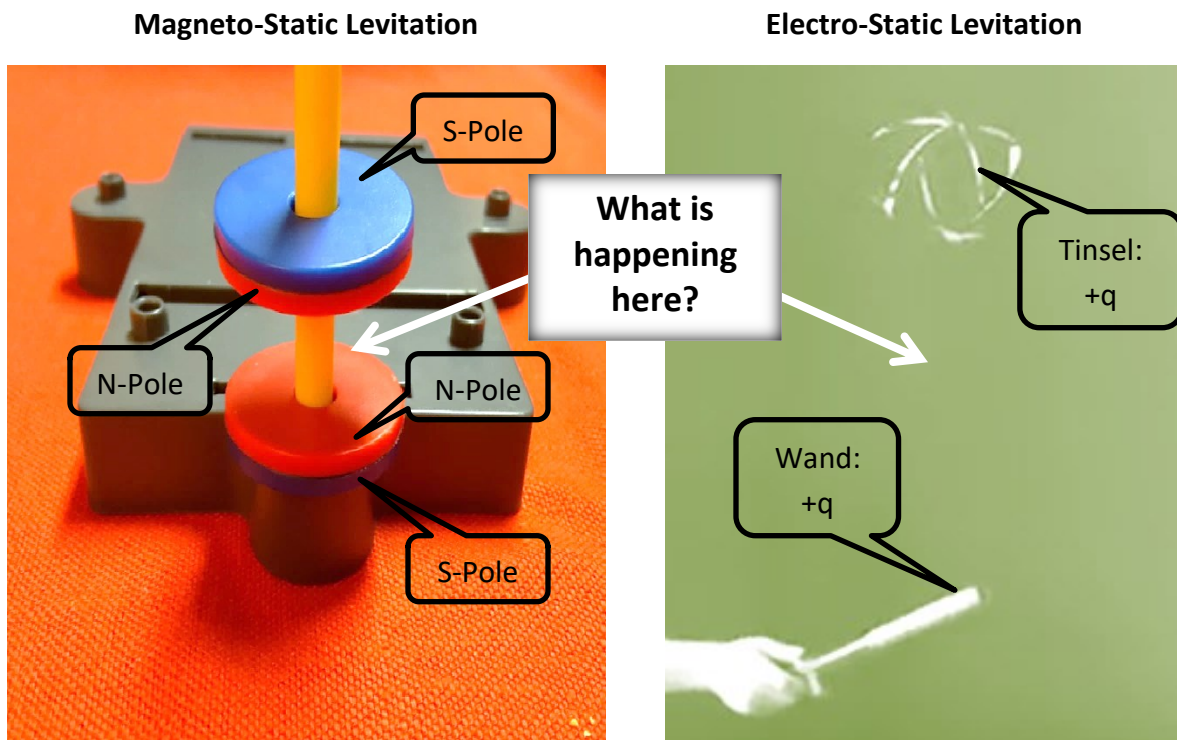
# Chapter 7

## A. Magnetostatic and Electrostatic Levitation Forces

Magnetostatic and electrostatic forces, specifically repulsion and levitation, are good topics to start considering. Levitation (on Earth) is simply the repulsion of an object in the Earth's radial direction that opposes gravity sufficiently to make the object float with no observable mechanism to rely upon.

A static electric field implies stabilized (DC) voltage at the surface of an object emitting the field. A dynamic field can build or collapse in response to varying DC voltage without reversing polarity. This can also be represented by an alternating (AC) voltage riding on a DC voltage whereby the DC level exceeds the peak AC voltage.

In order to illustrate the **wonder of magnetostatics and electrostatics**, working scientific toys are used for static field levitation examples and presented as follows:



**Figure 68: Magnetic vs. Electric Forces**

The picture on the left (above) shows two ring magnets on a post with their north poles facing each other. The resulting forces are repulsive, therefore, resulting in magnetostatic levitation. The classical answer is that like-poles in permanent magnets repel. Yes, this is true, but the underlying questions have traditionally remained unanswered.

# Photony Foundations

The picture on the right (above) shows an electrostatic (high voltage) generator wand and an expanded spherical aluminum tinsel object being forced upward as the electrostatic wand approaches from below. Note that the tinsel starts as just a shambolic pile before it gets “charged.” The resulting forces are obviously repulsive once the charge is shared between the wand and the tinsel, resulting in electrostatic levitation. The classical answer is: Like-charges in charge centers repel. Yes, this is true, but the underlying questions remain unanswered in mainstream science; **what is charge**, and how does it cause the levitating force? The fundamental answer is not found in an equation, specifically the Coulomb’s Law force equation. Equations attempt to describe or predict the outcomes of natural phenomena.

We are informed that “lines” of magnetic force or “magnetic field lines” exist, and that magnetic flux is the magnetic field through a given area. We are informed that magnetic fields are produced by currents (charge flow in a conductor), but what are the magnetic fields and lines comprised of, and how can they produce force? Why are the magnetostatic lines always in loops, and how did the loops become formed? Why are the magnetostatic field “lines” more forceful than the electrostatic field lines? Photony provides the answers to these and other questions extending to electromagnetic field lines in subsequent chapters and in the more rigorous Photony Compendium books.

When magnetic levitation is observed, it is natural to wonder what is producing the forces that overcome gravity and what is causing the “pulling” gravitational force. In the example of the levitating turntable shown below, a third force is involved that rotates the record and platter at a constant speed. So, it is evident that forces are detected, but the collective and structural composition of the collective behaviors and interlinked structural phenomena producing those forces have not been previously identified.

We detect forces, but we cannot identify the elemental particles that are everywhere and how they can be chained to produce forces. The dynamic elemental photons as particles cannot produce levitating force since they are so small and penetrative. Still, the spring-like force can be substantial if they are interlinked in magnetic chain loops (called magnetic flux lines in classical physics) as part of a magnetostatic field.

Levitating Turntable

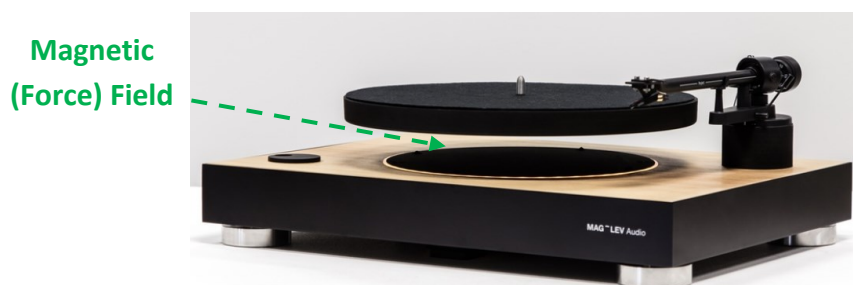


Figure 69: Levitation Forces

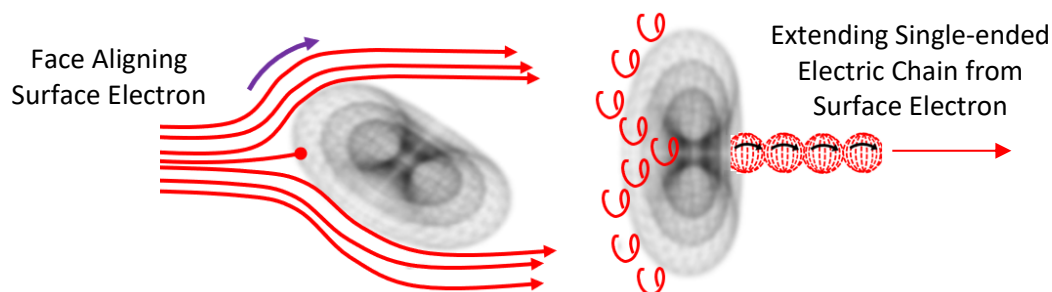


# Chapter 7

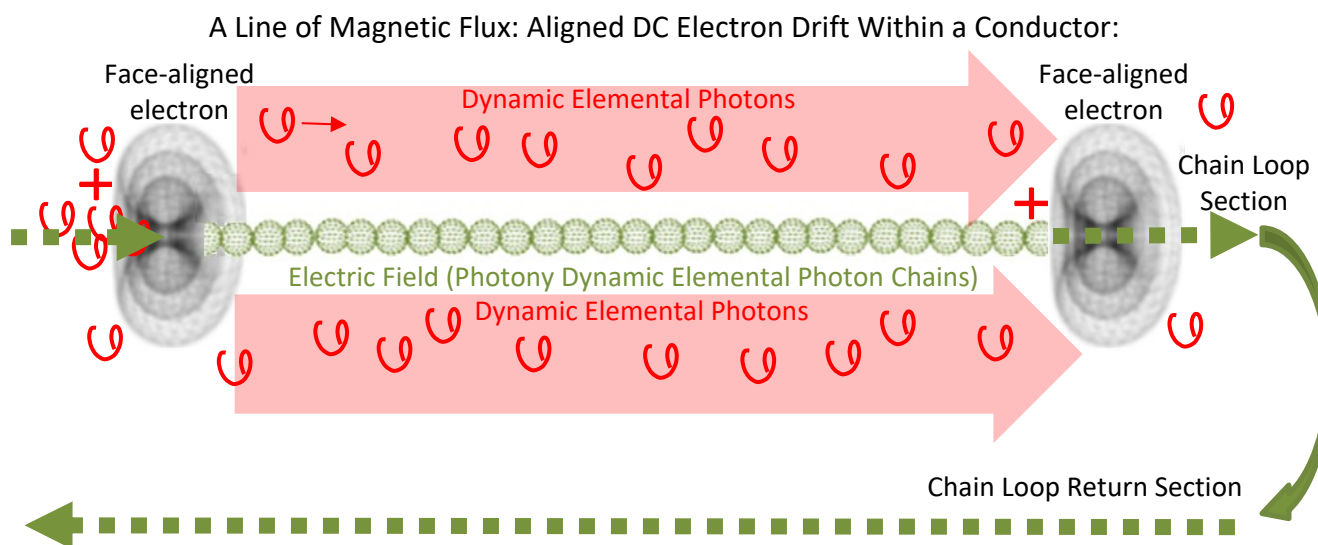
## B. Free Electron DC Field Alignment

Direct voltage and current (DC) vs. Alternating voltage and current (AC) concerning the behavior of toroid-shaped free electrons are illustrated below. The electron's intrinsic gamma ray spins align with the elemental dynamic photons (charge spin) and act as black hole-like tempests as the elemental dynamic photons pass through the toroidal cavity.

**Electrons are nudged along by dynamic elemental charge photons (charge, not classical photons)** “as a wind.” Free electrons are subject to face alignment with “the wind” of self-propelled charge (Photony dynamic elemental photons). A line of Electric Flux: Aligned with a conductor surface during transduction.



**Figure 70: Free Surface Electron Face Alignment and Electric Chain Transduction**



**Figure 71: Free Internal Electron Face Alignments and Magnetic Chain Transduction**

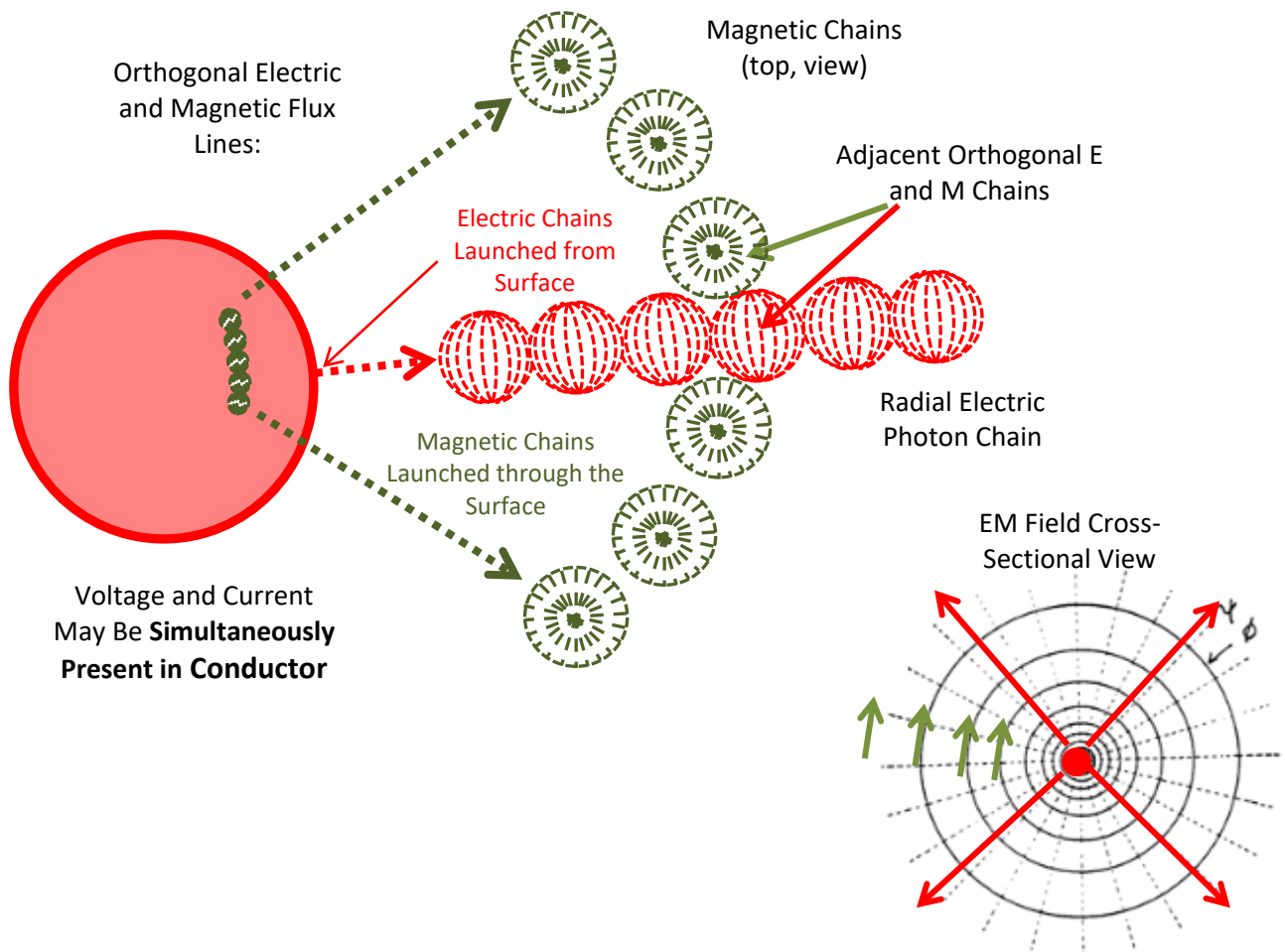
The electrons shown above within the conductor drift toward the right at a very low velocity once they have aligned with the flow of dynamic elemental charge photons (the current).



# Photony Foundations

## C. Electric and Magnetic DC Fields

Elemental magnetic photon chains are formed from the current passing through the wire. The current is composed of dynamic elemental photons that transform internally by electrons into the magnetic elemental photon chain loops as the magnetic field is exponentially built. In this scenario, electrons within the wire act as current-to-magnetic field (magnetic elemental photon chains) transducers. An energy potential (e.g., a battery) must be the driving force for current (based on Ohm's Law). Surface, face-aligned electrons present in the wire produce electric elemental photon chains as radial outreaching tentacles that remain orthogonal to adjacent magnetic chains.



**Figure 72: Orthogonal EM Chains and Fields**

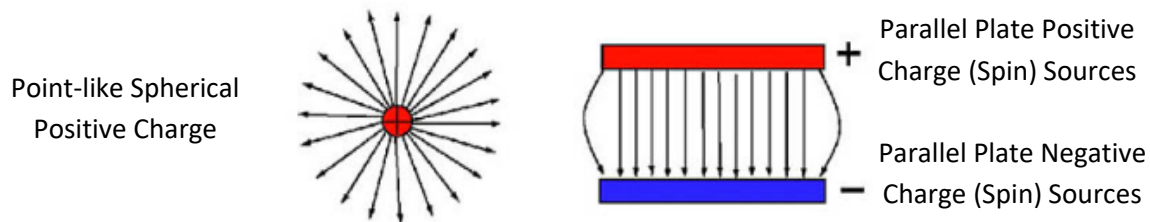
From Eric Dollard;

"Flux lines – never cross each other; they can be bunched, crowded, or thinned out over a large area. When they are bunched or crowded, each flux line has a repelling effect upon its neighboring flux line keeping them separated from one another. **A flux line has tension. It can be stretched or constricted along the direction of the lines of force.** This tension is somewhat like a stretched rubber band, prone to become as short as possible." – Gestalt Reality, Garrett M.

# Chapter 7

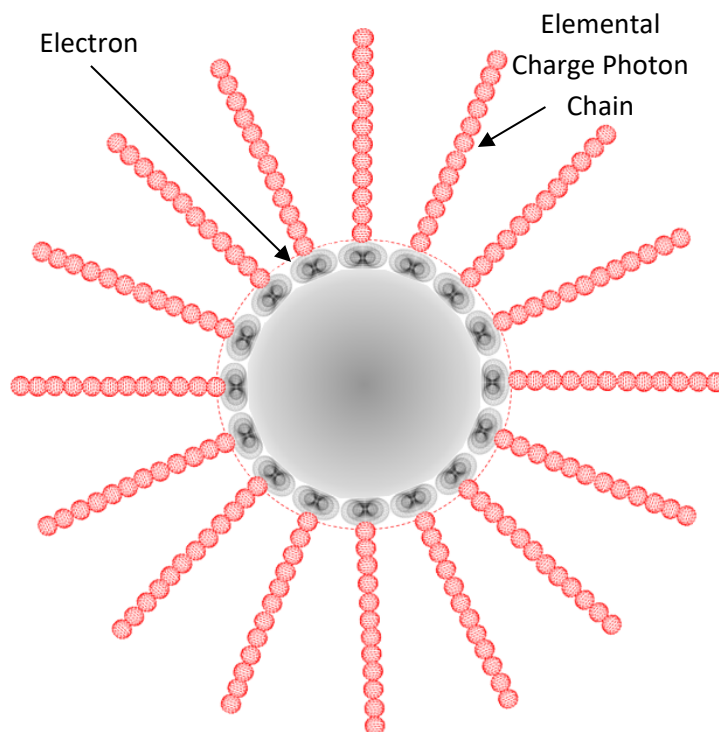
## D. DC Electric Field

The radial launch pathways from these point-like distributions become redirected and influenced by neighboring charges depending on spins. Similarly, planar launch pathways can be redirected and influenced by neighboring charges depending on spins.



**Figure 73: Classical Lines of Electric Flux**

The cross-sectional illustrations below are intended to provide insight regarding how the static charges distribute at the conductor surfaces and how free surface electrons align to transduce electric field chains outward and, in some cases, terminate at an opposing surface depending on separation and polarity (surface electron spin compatibility).

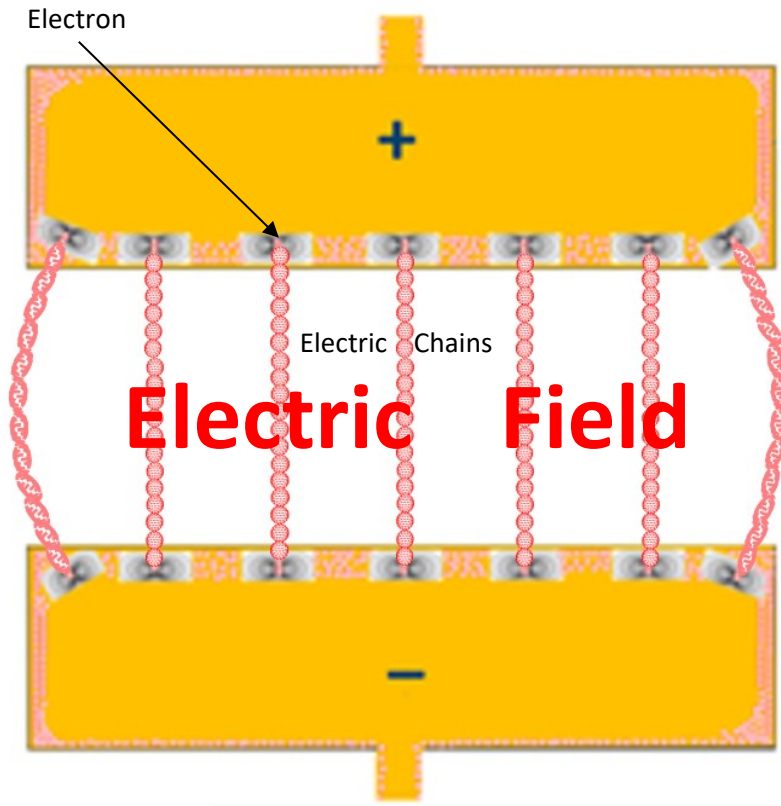


Electrons distribute evenly on the surface of the spherical body. If the spherical body is a conductor, the electric field extends radially outward in the form of Photony elemental charge photon chains transduced by the surface electrons. A positive polarity results from positive potential or voltage on the sphere. The polarity corresponds to the electrons' c-spin (pertaining to the gamma ray spin comprising the electrons). The electrons collect on the surface of the conductor due to elemental charge photons migrating to the surface where they transition from dynamic to static elemental photons.

**Figure 74: Positively Charged Sphere**

# Photony Foundations

Yes. Electrons can ride on the chain they create by transduction! In the figure below the electric field points downward from the positive to the negative capacitor plate. In this configuration electrons can travel across the gap by riding on the elemental charge photon electric chains.



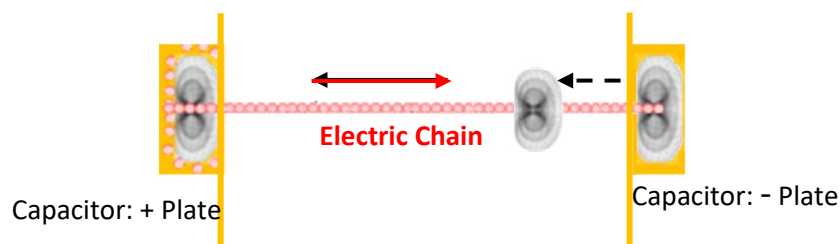
Capacitor plates- the polarity corresponds to the electrons' c-spin (pertaining to the gamma ray spin comprising the electrons). The electrons collect on the surface of the conductive plates due to elemental charge photons migrating to the surface where they transition from dynamic to static elemental photons.

The wider the plate separation, the fewer single-ended electric chains will find their way to terminate at the opposing plate. The single-ended fringe electric chains extending (lost) into space are not shown.

Too much potential or voltage or too small the plate separation can result voltage breakdown, electric elemental photon chain reactions (collisions).

**Figure 75: Electric Chains as Electric Field Lines**

For an electron to travel across the gap the electron must exceed an energy level threshold similar to that of a photoelectron in a doped semiconductor. The surface electrons shown below must be spinning in the same direction for the electric chain to terminate and provide a tether for an electron to transit across the gap between plates.



**Figure 76: Electric Field Line- Electric Chain**

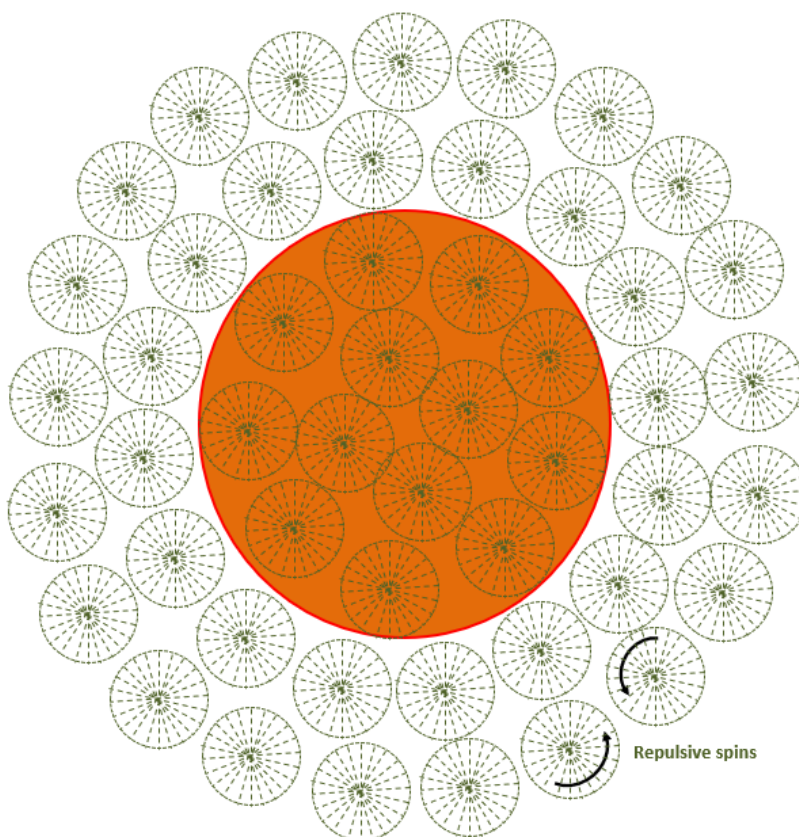
**The building of the electric chains is called displacement current** whereas the passage of the electrons on the chains is classical (real) current (charge flow).

# Chapter 7

## E. DC Magnetic Field

### 1. Low permeability magnetic field

The cross-sectional illustration below is intended to provide insight regarding how a conductor, a round wire, in this case, originates magnetic chains that build and extend longitudinally within the conductor as the free electrons align along internal pathways. The magnetic chain loops around the toroidal electron bodies join longitudinally, much as they do within a solenoid coil. As increased numbers of elongated magnetic chain loops are built, they extend in concentric layers outside the conductor with a nonlinear density gradient based on the radius squared.



**Figure 77: Cross-section of Magnetic Chains- In and Around a Conductor**

DC Current: Faraday postulated that a current-carrying wire is immersed by a near-infinite series of circular concentric lines of force, which he termed the current's magnetic field. Faraday believed that the lines were tubes of force with real physical significance. Faraday proposed that each field line of force corresponds to a unit of electric charge in an electric field. Photony provides the structure and kinematics for electric chains functioning as lines or **tubes of force**. **Lines of force, Maxwell supposed, were tubes of ether rotating on their axes.**

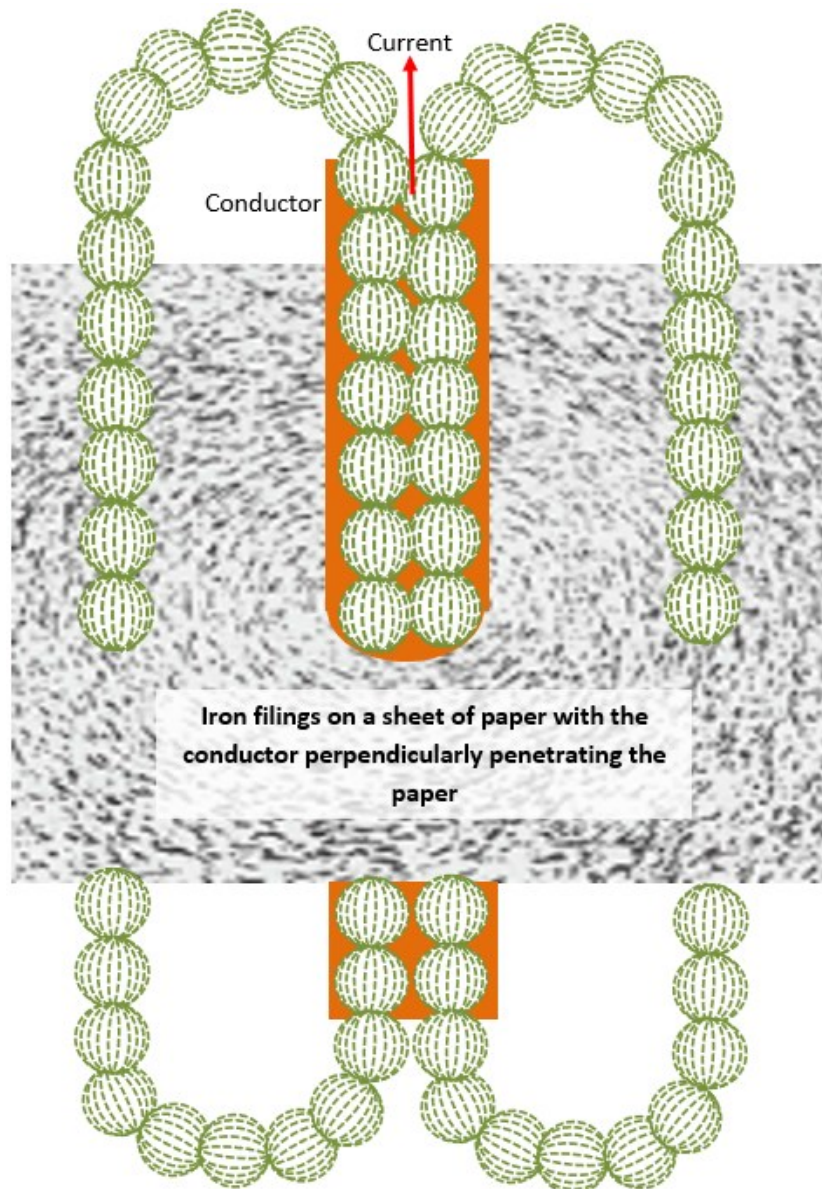


# Photony Foundations

## 2. High permeability magnetic field

Magnetic chains (two shown of many below) are always formed in a loop, beginning with the elemental photons aligned axially (sharing a common spin) according to the right-hand rule relative to the conductor's current direction.

Magnetic field lines (chains) align iron filings in circular patterns as the magnetic chains deflect the iron filings for least energy passage through the paper sheet.



**Figure 78: Iron Filings Aligned by Magnetic Chains**

# Chapter 7

## F. Voltage Breakdown



**Figure 79: Voltage Breakdown by a Spark Plug**

Voltage breakdown is the result of Photony elemental electric photon chain (“electric chain”), collisions, e.g., lightning phenomena producing a mix of elemental photons and chainlet debris (causing “static” on radio stations) and classical photons (causing heat, visible light, UV, X-rays, and gamma-rays in the upper spectrum).

Photony electric elemental photon chains can become so dense that the result can be catastrophic. Some additional examples of nature and manufactured materials come to mind.

- A static discharge (voltage breakdown) between objects, e.g., from a finger to a doorknob in very dry (low relative humidity) is exemplary of voltage breakdown.
- Voltage breakdown of an electronic component (e.g., a capacitor)- results in flash, heat, disintegration at high speed, and acoustic reports such as a loud “pop.” Capacitor breakdown can occur due to overvoltage (high electric chain density) exceeding the voltage rating of a capacitor or reversal of a polarized or an electrolytic capacitor's voltage polarity.
- Lightning- cloud-to-cloud or cloud-to-ground with results of flash, heat, ionization of air (plasma), large amounts of energy transmitted through the plasma, and acoustic reports such as thunder.
- Treeing- voltage breakdown in solid materials such as electrical insulators and crystals have been studied. Lichtenberg Figures-named after the German physicist Georg Christoph Lichtenberg. Shown above middle and right) are from positive and negative voltage surges in a dielectric material.

Voltage breakdown is the overcrowding collision-based breakup of elemental electric photon chains, as seen below- it is literally a chain reaction. The resulting chain debris/fragments (chainlets) can be comprised of any number of interlinked elemental photons. Chainlets can be classical photons with two entangled elemental photons, or three or more entangled elemental photons. The breakup of electric chains may also result in electromagnetic chains, photon chainlets, and gamma rays with an extra spin (intrinsic +/- c-spin), resulting in electrons and positrons<sup>27</sup>. [68]

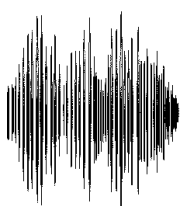
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<sup>27</sup> In research published in 2011 by the American Astronomical Society positrons were discovered originating above thunderstorm clouds.

# Photony Foundations

Most internal combustion engines require a spark plug for each cylinder. The spark is timed to ignite the fuel-air mixture based on critical timing to keep the engine running. In this case, the spark is generated by an ignition (inductive) coil, which produces high voltage pulses that pass to the spark plug terminal and through the spark plug to a pin and a spark gap to a grounded terminal. The high voltage pulse generates a spark due to voltage breakdown.

The Quasi-static Spectrum (spectral segment) begins at 0 Hz (DC) and transiently extends into the EM spectrum based on the voltage breakdown phenomena. The charge rate is generally slow and reactive in nature (as a capacitive or an inductive field). In contrast, the discharge rate is usually due to a voltage breakdown producing ionization, free electrons, and high currents, acting more like an inductor (a conductive channel of high current density plasma).



Electrical “Noise” from voltage breakdown: can be from a variety of sources.

**Figure 80: Electrical Noise Transient**

The term “static charge” implies a steady voltage potential or DC voltage resulting in a stabilized field (no charging or discharging). The term “static” may apply to an electric field, capacitive with a fixed voltage, or a stabilized magnetic field around an inductor with a constant current flowing within the conductor.

Dynamic fields include periodic EM AC fields and transient fields comprised of non-periodic exponential and harmonic potential driven fields. Some transient fields originate from natural phenomena and human-made static fields that experience voltage breakdown (e.g., engine spark plugs). Fourier analysis shows that these transients are comprised of sinusoids with superpositioned harmonics and phases. More specifically, as static fields are built, they typically build slowly with limited power. They may eventually experience high voltage breakdown violently (e.g., lightning) with high frequencies and power levels (currents) based on ionizations. This breakdown (collisions between elemental electrical photon chains) results in spectral intrusion by chain segments and chainlets (electric chains debris) into other bands throughout the spectrum.

Voltage Breakdown Manifestations: The voltage between charge centers produces Photony electric elemental photon chains (electrostatic field lines) (e.g., surface electrons on capacitor plates). Voltage breakdown is the breaking down or disintegration of these electric chains due to high-density crowding (pressure) and perturbation-induced collisions progressively resulting in a “chain reaction” as described herein and studied in detail in the **Photony Compendium**.

# Chapter 7

## G. Photony Electric Chain Collisions

Voltage breakdown (e.g., lightning, corona, sparks and arc flash, dielectric breakdown, or treeing) is literally a chain reaction from collisions of high-density electric elemental photon chains resulting in chain fragments. The longest single flash on record (shown below) was photographed by NOAA in 2020 and lasted for 17.1 seconds.



### Voltage Breakdown: Chain Collisions- Chain Reactions

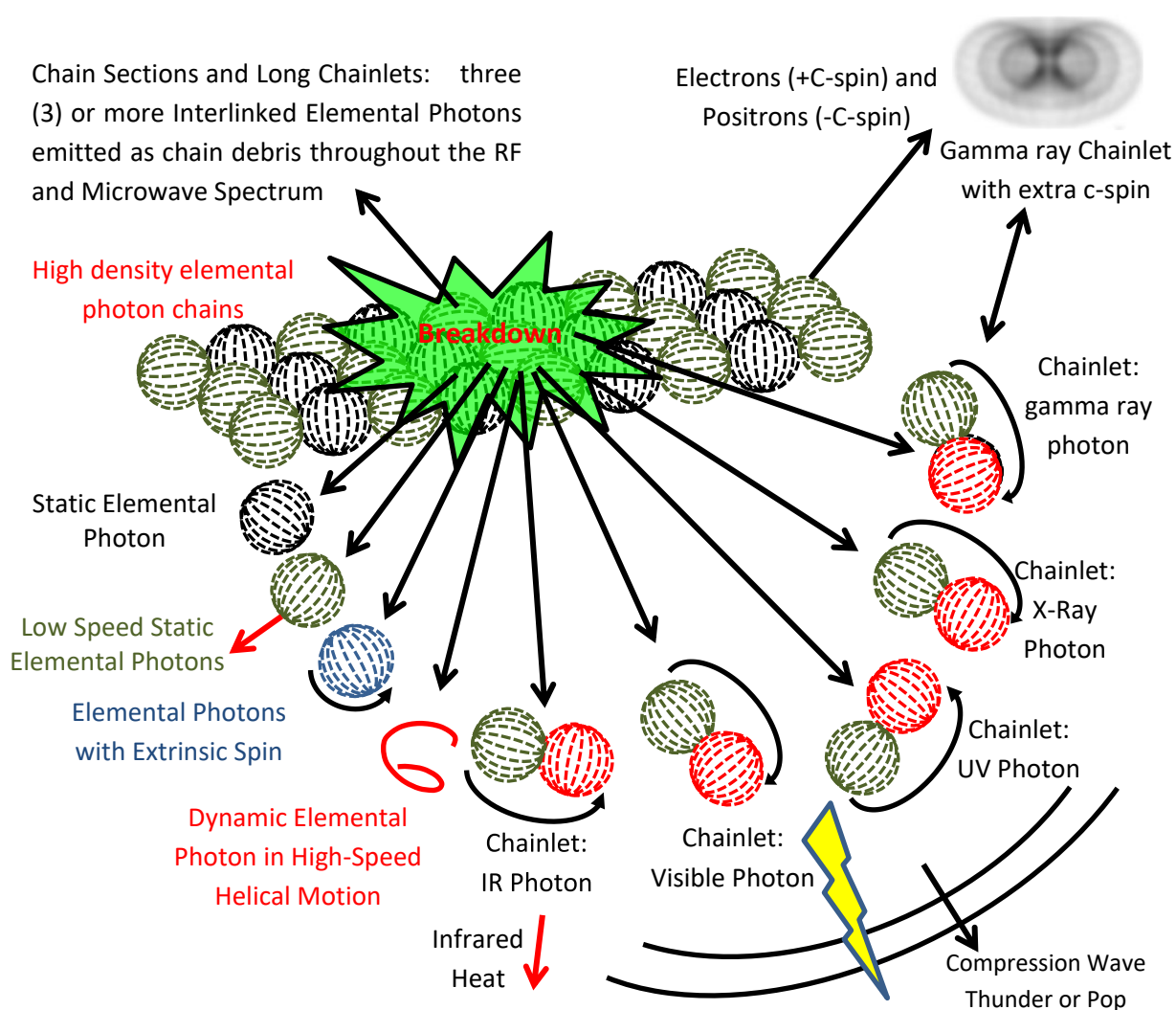


Figure 81: Voltage Breakdown- High Density Electric Chain Collisions

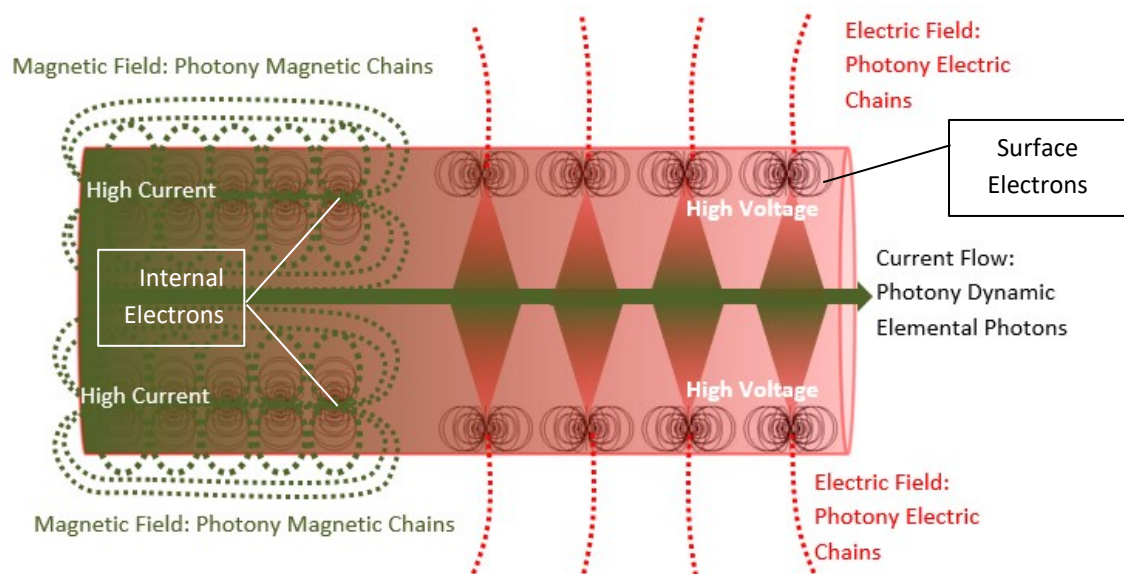


# Chapter 8 Electromagnetics- AC Fields

Electromagnetic AC Fields are synchronized, orthogonally linked electric and magnetic fields of chains, each link transduced by free electrons (outside the bondage of atomic shells), thus forming chains of interlinked elemental photons.

## A. Chain Basics in and Around a Conductor

There are two basic types of chains, electric and magnetic. Electric chains are transduced by free surface electrons, typically at or near the surface of conductors and semiconductors. The electric chains are transduced initially single-ended as outgoing tentacles, but in AC conditions, they are alternately orthogonally linked as chain loops with magnetic chain loops as they exit or launch from a sectional conductor/antenna surface below.



**Figure 82: Magnetic and Electric Chain Distributions of an Antenna Section**

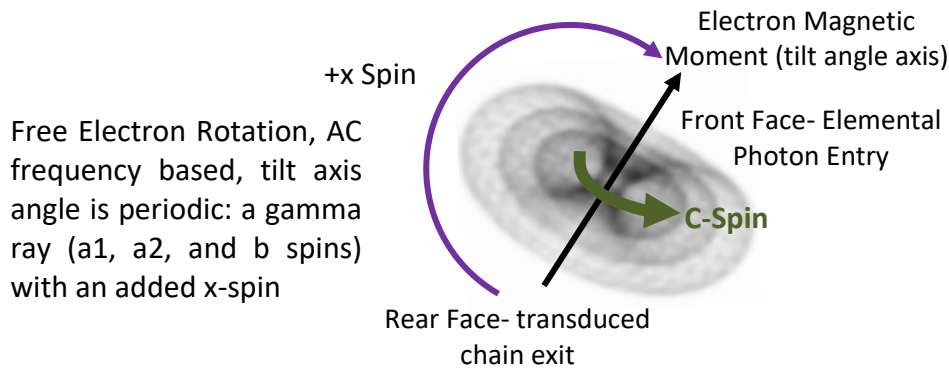
Magnetic chains (above in the high current section)- the incremental magnetic chain loop sections adjoin adjacent chain loop sections to extend the chains longitudinally within the conductive element/wire until specific forces cause the combined chain loops to exit the element/wire. The chain loops extend to wrap around the element sections where the highest electrons' highest concentrations are located (high current sections of the antenna element). This model appears similar to the magnetic field of coils of wire. Electric chains (above in the high voltage section) are transduced by surface electrons as single-ended chains seeking termination at opposing spin electrons or their electric chain leaders.

# Chapter 8

## B. Electron AC Phenomena

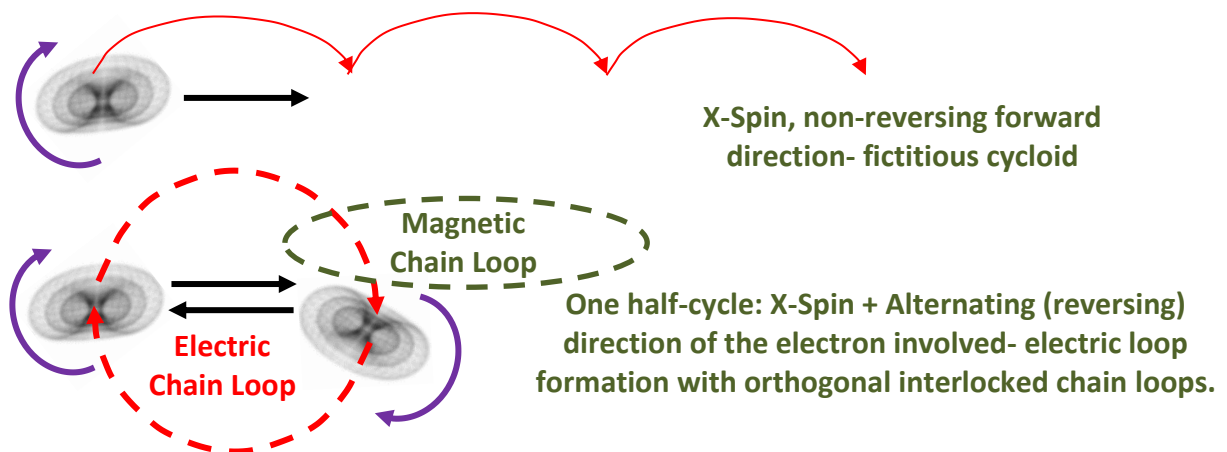
### 1. AC Induced x-spin

Electron Toroidal Face Alignment in an alternating field of dynamic elemental photons causes electron rotation or x-spin as it transduces elemental photon chains. In alternating voltage/current scenarios, the free electrons in a conductor respond by adding the x-spin (axial end-over-end) with a spin rate that matches the AC frequency of the electric field produced by AC voltage potential.



**Figure 83: Electron Tilt Axis (X-spin)**

Electrons continue to transduce chains as they rotate in an AC field of dynamic elemental charge photons, which are self-propelled and alternating in the flow direction. As the free electrons internal to a conductor rotate, they also experience longitudinal motion. As the AC reverses flow direction, so do the electrons as they continue to rotate in the same direction as the axial X-spin. This electric transduction produces chain loops as the surface electric chains are snipped and stitched into orthogonal magnetic chain loops normally produced by internal electrons as they reach the surface. Perhaps the best way to envision this is to start with a cycloid. If the surface electron were to have no reverse displacement from the flow of elemental charge photons, the transduced electric chains would appear as an ongoing cycloid, as shown below.



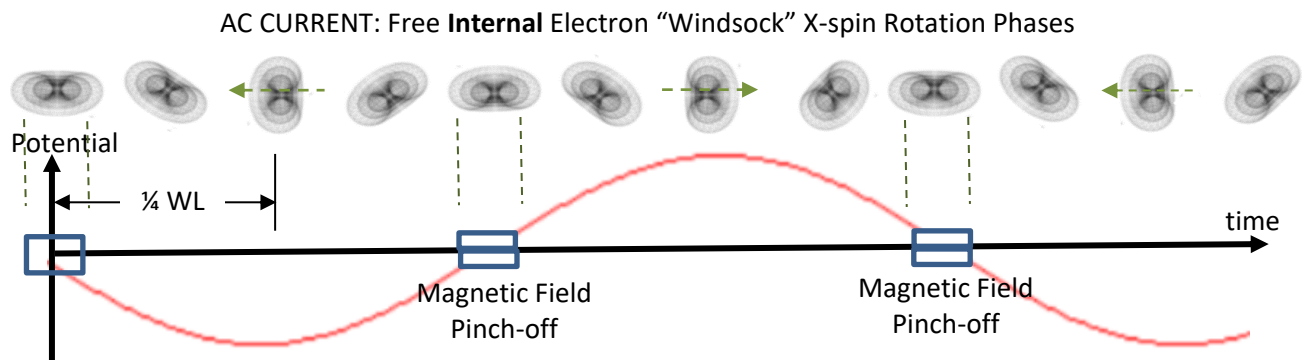
# Photony Foundations

**Figure 84: The Making of EM Chains**

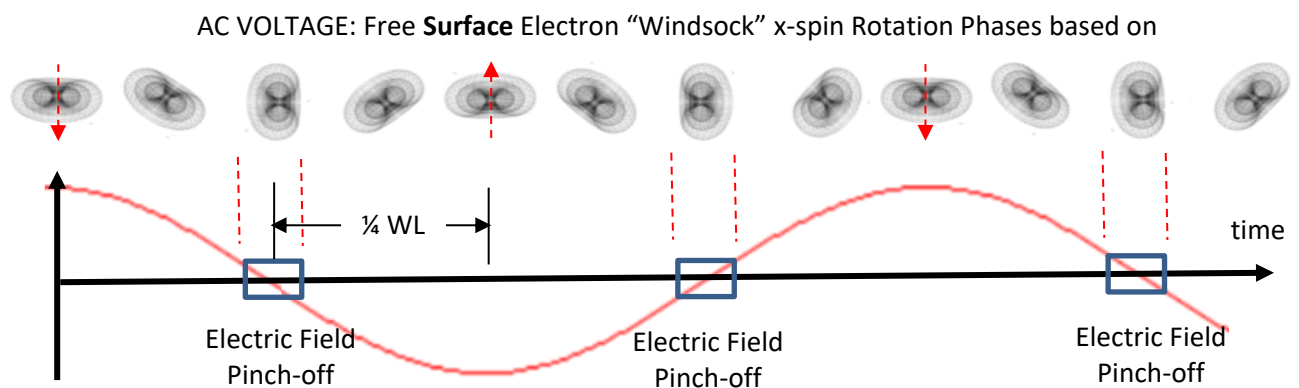
## 2. Pinch-off and EM Chain Loops

With DC and voltage, the chains build proportionally and exponentially stabilize. Free electrons are free to move along with the elemental charge photon flow. In this condition, the dynamic elemental photons spread throughout the cross-section of the conductor. When the DC signal or power is switched on or when it changes in magnitude, a transient condition occurs whereby many of the dynamic elemental charge photons migrate to the conductor's surface and are converted to static elemental charge photons. As they accumulate, the voltage builds (only) at the surface as electric chains are exponentially transduced by the surface electrons with static elemental charge photons as the fuel. These electric chains are the electric field around the conductor. A transient magnetic field is produced until the dynamic elemental charge photons stabilize.

Regarding AC, the current (internal charge photon flow) and voltage (surface charge photon flow) reverse, causing the affected free electrons' tilt axis to vary resulting in electron rotation on its spin x-axis as the transduction of chainlets continues.



**Figure 85: Magnetic Chain Pinch-off**

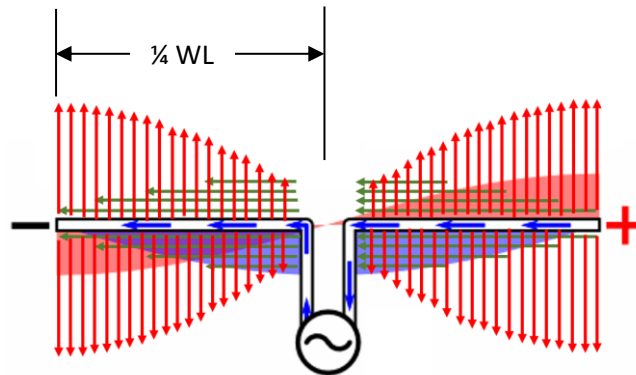


**Figure 86: Electric Chain Pinch-off**

# Chapter 8

## 3. EM Standing Waves

EM fields of standing waves of dipole antennas are informative regarding the previous section describing EM chains. With the standing wave in motion, the current peak oscillates from negative (see snapshot below) to positive (above) at the center of the antenna. The dipole shown below shows a particular time snapshot of the standing wave. At the time of this frame, the current was maximally negative and peaking at the dipole center. In contrast, the voltage is maximally positive at the right tip of the antenna and maximally negative at the left tip. The magnetic field associated with the current (dynamic elemental photon flow) density (green arrows). In contrast, the electric field associated with the voltage (surface static elemental photons) is shown below in red. The Poynting Vector is defined as  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ , where  $\mathbf{E}$  is the radial electric field vector, and  $\mathbf{H}$  is the magnetic field's auxiliary field vector ( $\mathbf{H} = \mathbf{B}/\mu$ ) which is parallel to the conductor (dipole antenna rods).



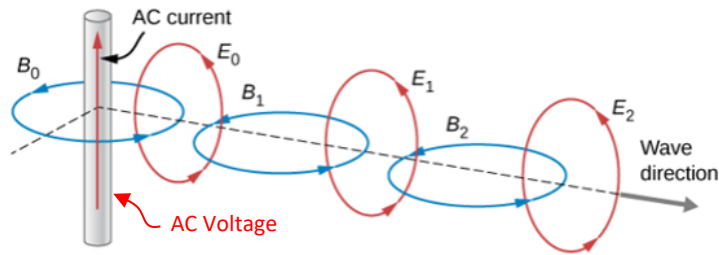
**Figure 87: Snapshot of Dipole Standing Wave**

From the dipole with a  $\frac{1}{2}$  wavelength standing wave, the sinusoidal amplitude of both voltage and current changes at the same frequency as the antenna feed signal. This effect is seen below as the voltage and current are 90 degrees out of phase in time and space. The voltage lags the current by 90 deg, as indicated by the orthogonal alternating electric and magnetic chain loops. This lag can be explained because the dipole acts as an elongated capacitor, which is normal for the voltage to lag the current by 90 degrees.

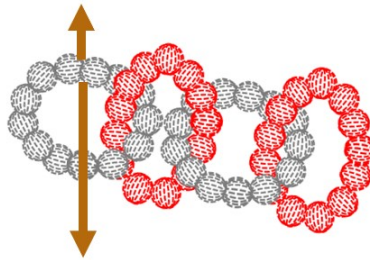
Pairs of proximal x-spinning electrons, one internal and one on the conductor surface, synch up to build orthogonal chain sections as EM radiation extends through the air, space, or any medium that is transmissive at the frequency of the radiation. Each link is produced by one-fourth of a wavelength corresponding to sequential pinch-offs whereby the electron's transduction of the chainlets transitions to interlinking of the ends as they become pinched during each transduction null signaling the next quarter cycle chain build.

# Photony Foundations

Transmissions of orthogonal electric and magnetic chains pinched off into intersecting loops are shown below.

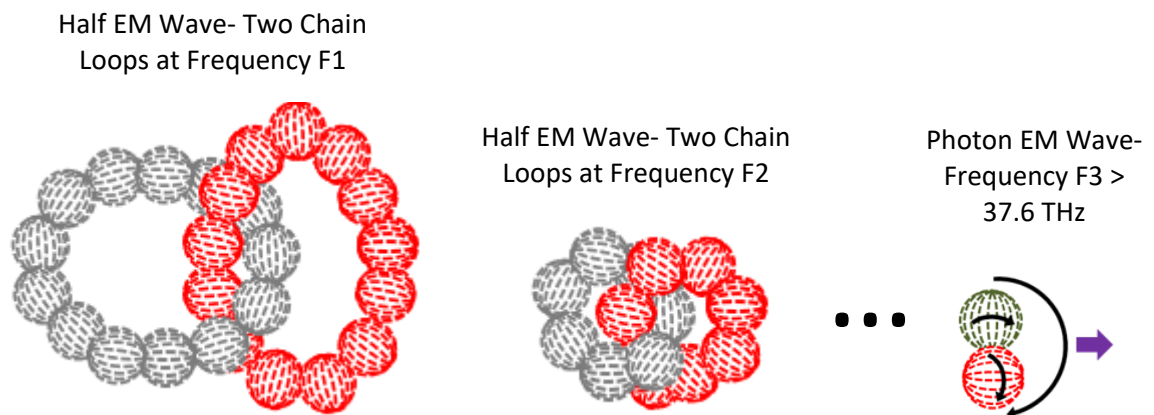


**Figure 88: Transmissions of Orthogonal Interlinked Lines/Loops**



**Figure 89: EM Transmissions of Orthogonal Pinched-off Chain Loops**

At 37.6 THz the EM orthogonal chain loops (EM Chainlets) transition to two interlinked elemental photons, thus producing classical photons as electrons can no longer act on the frequency with a corresponding x-spin rate. Photons become emitted based on various phenomena other than the EM field from AC X-spinning electrons.

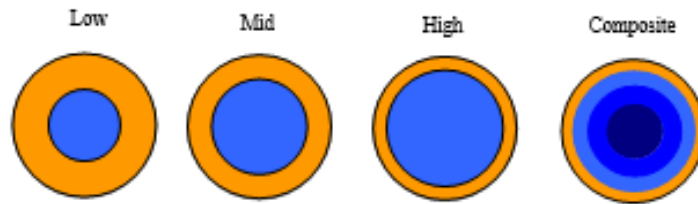


**Figure 90: EM Wave Transition to Classical Photons**

# Chapter 8

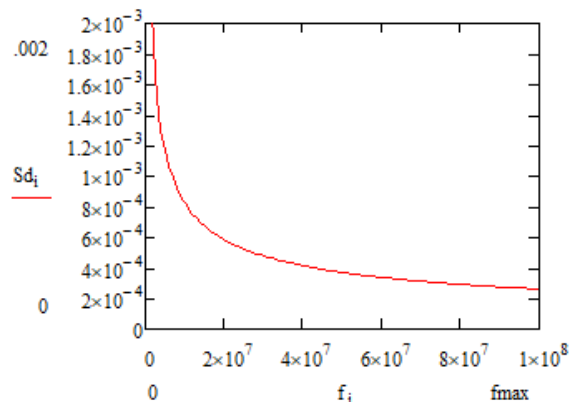
## C. Skin Effect

Skin effect for EM waves in conductors is the effect of frequency on the current distribution whereby the current density becomes increasingly concentrated near the conductor's surface based on frequency. Round conductor skin effect as a function of frequency<sup>28</sup>:



**Figure 91: Round Wire Skin Effect at Various Frequencies**

In an AC voltage/current scenario, the skin effect emerges substantially at a low frequency (kHz). As the frequency of an AC signal increases, the conductor's magnetic effects (as eddy currents) cause the current (Photony dynamic elemental photon field) to bend toward the surface as it moves longitudinally through the conductor. The current inside the conductor (dynamic elemental photons) creates a magnetic flux. This internal magnetic flux induces a field within the conductor, thus producing inductance. It can be said that an AC (alternating current) flows in the path of least inductance. The skin depth ( $\delta$ ) is inversely proportional to the square root of the frequency. 63% of the current will flow within one skin depth  $\delta$  of the surface, and 98% will flow within  $4\cdot\delta$  (four skin depths) of the surface.



**Figure 92: Skin Depth Plot as Function of Frequency**

The plot above shows skin depth in inches as a function of frequency. The maximum frequency shown is 100 MHz as skin depth reduces to 0.2614 mils or 6.64 microns.

<sup>28</sup> Composite means there is a combination of frequencies depicted.

# Photony Foundations

## 1. Skin Effect Description

Electromagnetic induction is the basis of the skin effect. An AC time-varying magnetic field caused by a current is accompanied by a time-varying induced electric field caused by potential, creating a skin effect. “What is skin effect?” For AC signals, the current density is higher near the surface than in a conductor's interior once a skin effect “start frequency” is exceeded. The skin effect is the “effective” thickness of a conductor. The current density is highest near the surface and exponentially decreases toward the center of the conductor.

The added x-spin to the electrons internal to the conductor produces eddy currents that migrate toward the secondary currents (induced currents) and the associated secondary magnetic field caused by secondary currents. We know from Lenz's law that the induced currents produce magnetic flux, which is the opposite of the external flux (which generated the induced currents), reducing the total flux. The larger the conductivity, the larger the induced currents are, and the larger the permeability, the more the flux is reduced. Consequently, both the total time-varying magnetic field and induced currents inside conductors are reduced compared with the DC case.

Skin effect begins at the cross-sectional center of a conductor at an axis whereby the largest outer dimension is referenced. At a frequency limit called the skin effect start frequency, the current (dynamic elemental photon flow) is exponentially directed away from the conductor's cross-sectional center (centroid) toward the conductor surface(s). Increased frequency increases the current density as the skin depth decreases near the surface. This process results in skin effect, which **increasingly limits the current flow to thinner skin depths at the surface based on the square root of the frequency**. Due to frequency, the current is flowing in the “path of least inductance.” The **skin effect reduces the loop area between the high current density near the surface and the return path outside of the conductor**. The smaller cross-sectional area where, the higher current density exists causes increased temperature/heat losses from the conductor and increased longitudinal voltage drop (attenuation). A conductor's central volume may be open (devoid of metal), given that the current is predominately flowing in the conductor's skin.

Photony explains how internal toroidal electrons rotate end-over-end (x-spin) as they transduce magnetic elemental chain loops. As the frequency increases, the eddy currents associated with the internal electrons help push them near the surface as the current flow of Photony dynamic elemental photons travels longitudinally in alternating directions at the speed of light (space).

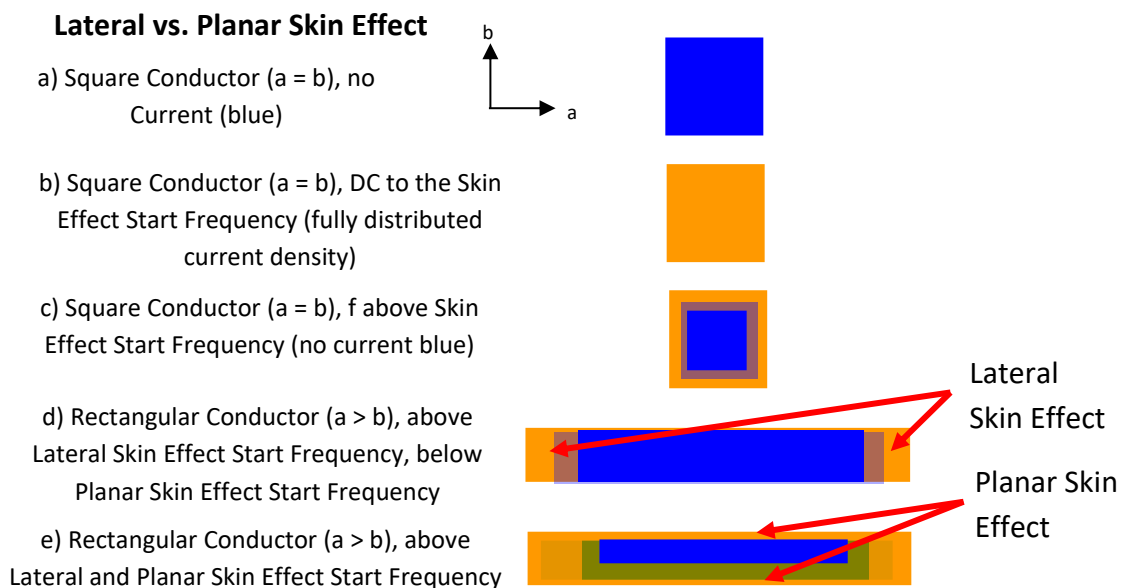
A conductor's alternating potential (voltage) causes the toroidal near-surface electrons to rotate end-over-end (x-spin) as they transduce electric elemental Photon chain tentacles that are pinched off every half cycle, forming loops that are orthogonally interlinked with the adjacent magnetic chain loops.

# Chapter 8

## 2. Rectangular Conductors- Lateral vs. Planar Skin Effect

The cross-sectional view of a square conductor is shown (below, a) with no current flowing AC or DC. The dark blue color represents no current and no associated current density. Shown (below, b) indicates an evenly distributed D.C. current is flowing throughout the square conductor's cross-section, and no A.C. current is flowing. As shown (below, c), the light orange area represents the highest current density of an A.C. current flow associated with a frequency greater than the same skin effect start frequency in both a, b dimensions. The start frequencies for rectangular conductors are calculated from the skin depth equation by setting  $\delta_{CU} = \frac{1}{2} a$  for the lateral skin effect start frequency and  $\frac{1}{2} b$  for the planar skin effect start frequency (higher than the lateral skin effect start frequency).

As the conductor changes from a square to rectangular, the start frequency changes relatively for the two (a and b) dimensions. In this case, the larger of the two, the "a" dimension, has the smallest start frequency since the start frequency has the associated skin depth in the denominator. Shown (below, d) indicates a lateral skin effect current distribution wherein the frequency is above the lateral skin effect start frequency but below the planar skin effect start frequency. Shown (below, e) indicates a current distribution wherein the frequency, f, is above the lateral skin effect start frequency and above the planar skin effect start frequency.

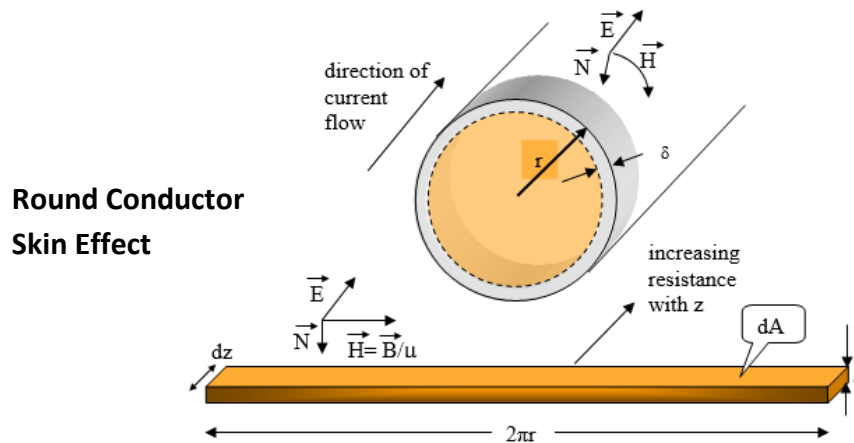


**Figure 93: Lateral vs. Planar Skin Effect**



# Photony Foundations

## 3. Skin Depth Derivation Notes<sup>29</sup>



**Figure 94: Round Conductor Skin Depth Model**

The definition of skin depth is the distance below the surface where the current density has fallen to  $1/e$  (note: the natural constant  $e$  is 2.718) or 37% of its value at the surface. The simplified Skin Depth Equation for copper is 2.6 divided by the square root of frequency (units in inches).

As shown above, the skin effect can be modeled with an unrolled strip  $2\pi r$  in length,  $dz$  in width, and  $\delta$  in thickness. The general formula for skin effect in SI units is:  $\delta$  equals the square root of  $(2/(\omega\mu\sigma))$  in meters, where  $\mu$  is permeability and  $\sigma$  conductivity. Skin depth for copper is straightforward; for example, the skin depth  $\delta_{CU}$  equals  $2.6/\sqrt{f}$  (in), where  $f$  = frequency.

## 4. Skin Effect Start Frequency

The start frequency is the frequency at which a current density void at the cross-sectional centroid of the largest cross-sectional conductor dimension emerges as a discontinuity in the homogeneous current density associated with DC. Multiple start frequencies occur in irregular or non-circular cross-sections of conductors. For example, a thin/wide rectangular conductor has two start frequencies. A rectangular conductor has two start frequencies, one from the midpoint of the sides and one from the diagonal corners.

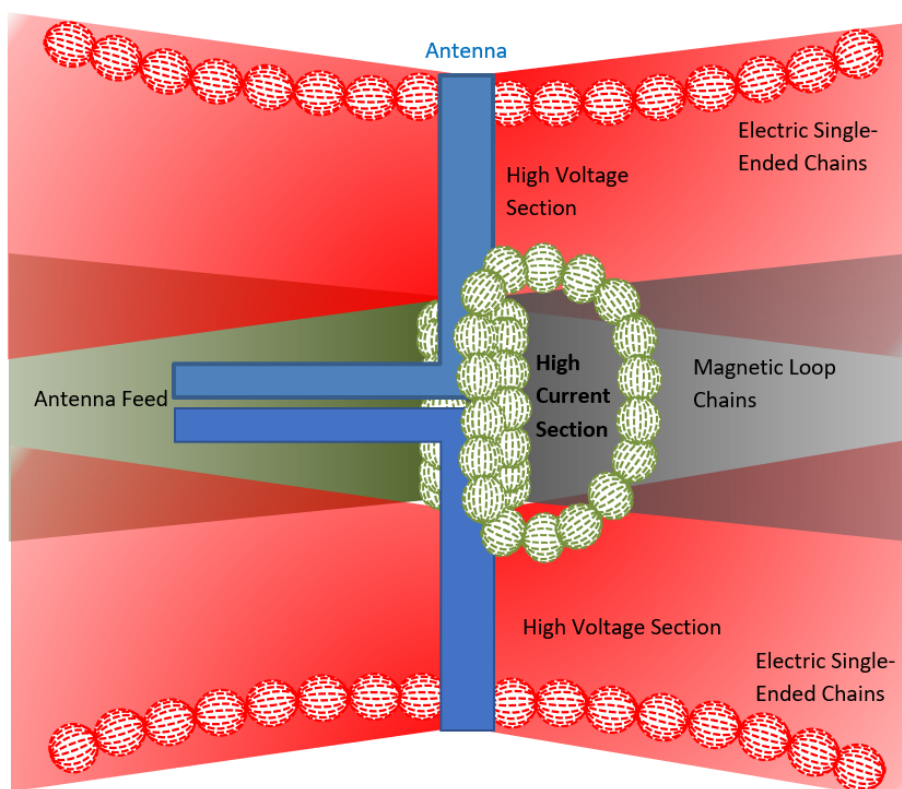
The start frequency can be calculated by setting  $\frac{1}{2}$  the largest cross-sectional conductor dimension relative to the centroid (the radius for a round wire) equal to the skin depth and solving for frequency. In a 0.2 in diameter copper wire, the skin effect start frequency =  $(2.614/0.1)^2 = 676$  Hz. The skin depth of copper at 1MHz is 2.6 mils.

<sup>29</sup> See the Photony Compendium EMFF book for a detailed derivation of skin depth and start frequency.

# Chapter 8

## D. Dipole Antenna with Snapshot of EM Field Chains

Antennas are designed to transmit electromagnetic waves from conductive constructs into air, water, or space. A standing wave is involved in maximizing the antenna efficiency. As illustrated below, a standing wave results in sections of the antenna with relatively high current with alternating transmissions in adjacent sections of relatively high voltage.



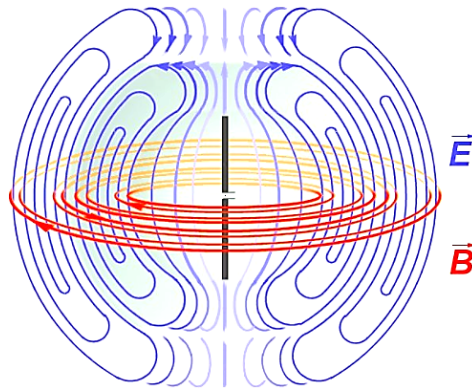
**Figure 95: Exemplary Electric and Magnetic Chains of a Dipole Antenna**

Every half-cycle transition of the alternating current and voltage conducted into and onto (respectively) the dipole antenna causes the antenna's free electrons to rotate and reverse their toroidal face resulting in an end-over-end x-spin at the same frequency as the conducted signal. The conducted signal is comprised of Photony dynamic elemental photons flowing within the antenna in high current sections through windsock-aligned electrons that transduce the elemental photons into magnetic chain loops extending from the zero-point focus within the toroidal cavity around the electron's toroidal body. On the antenna conductor surface at high voltage antenna sections, the windsock face-aligned surface electrons transduce elemental photons into single-ended electric chains extending outward like tentacles from the zero-point focus within the toroidal cavity of the surface electrons.

# Photony Foundations

## E. Transmitted Electromagnetic Field View in Half-cycles

Starting with a 3-D electromagnetic field illustration:

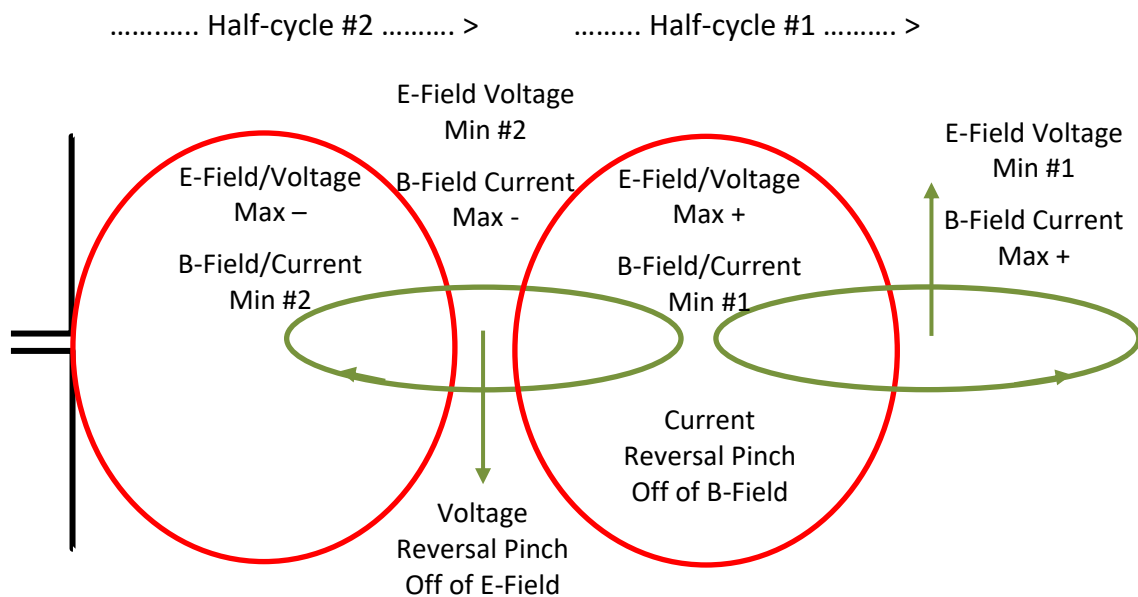


Electromagnetic Field in 3-D  
Around a Vertical Half-wave  
Dipole Antenna

Transmit Mode: Expanding  
Electric and Magnetic Field  
Distribution for Two Half-cycles in  
3-D View- Wikipedia *Near and Far  
Field*

**Figure 96: 3-D Electromagnetic Field Around a Dipole Antenna**

The view from the above 3-D electromagnetic field is limited to two half-cycles launched, showing the maximal electric and magnetic field lines (Photony chain loops). Additional lines (not shown) would concentrically exist inside of the maximal chain loops shown below. The time gaps between the E-field chain loops are due to the voltage polarity reversals that pinch off the E-field and require some time to build back the E-field in the opposite direction. The time gaps between the B-field chain loops are due to the current polarity reversals that pinch off the B-field and require some time to build back the B-field in the opposite direction. The orthogonal E-M-E-M... chain loops extend outward in lobes of electromagnetic field energy.



**Figure 97: Orthogonal Pinched-off EM Loops in Half-cycles**

## Chapter 8



# Chapter 9      Circuits and Fields of Energy

Circuits and Fields are intimately related but are often thought of as separate subject matter. Other misconceptions exist, such as E-field and M-field causality and the transport of energy within the conductors of a circuit. Photony explains how circuits and fields are unified, how E-fields and B-fields can be coincident but noncausal, and how energy flows in a circuit, predominately outside the conductors.

## A. Preeminent Research

The premise of this chapter is the relationship between circuit theory and field theory. The principals here are James Clerk Maxwell (Maxwell) get much of the credit for formulating their relationship (Maxwell's Equations<sup>30</sup>), but Oleg D. Jefimenko (Jefimenko), Nikolay Umov (Umov), John Henry Poynting (Poynting), and Oliver Heaviside (Heaviside) showed us alternative and ancillary views from the sometimes-inaccurate beliefs by others claiming to comprehend Maxwell's work.

Jefimenko said, [69]

*"...neither Maxwell's equations nor their solutions indicate an existence of causal links between electric and magnetic fields. Therefore, we must conclude that **an electromagnetic field is a dual entity always having an electric and a magnetic component simultaneously created by their common sources: time-variable electric charges and currents.**"*

Maxwell (introduced in a previous chapter) published an early form of his "Maxwell's Equations" in 1861 and 1862 that described how electric and magnetic fields are generated by charges, currents, and interactions of the electric and magnetic fields.

Umov was the first to introduce in 1873—1874 the basic concepts of speed and direction of energy motion, currently associated with Umov-Poynting vector, and density of energy at a specific location in space. The [Umov-] Poynting vector denotes the directional transfer of energy per  $\text{m}^2$  per second (energy flux) or power flow integrated over time of an electromagnetic field.

Poynting, who first derived the Poynting vector in 1884, provided a mathematical basis for the concepts introduced by Umov 10 years earlier. The Poynting vector units are  $\text{watts/m}^2$ . The Poynting vector has been used with Poynting's theorem to calculate the power flow in electromagnetic fields. Poynting's theorem is mathematically similar to the continuity equation, relating the energy stored in the EM field to work done on charge passing through energy flux. Heaviside also discovered "vector rate of energy transfer," [70] or  $\mathbf{W}$ , which is very similar to Poynting's vector/energy flow. Heaviside also rewrote Maxwell's 20 equations using the curl and divergence operators of the vector calculus to reformulate 12 of these 20 equations into four equations in four variables in a currently familiar form.

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<sup>30</sup> Maxwell's equations derivation in the Photony Compendium EMFF Book

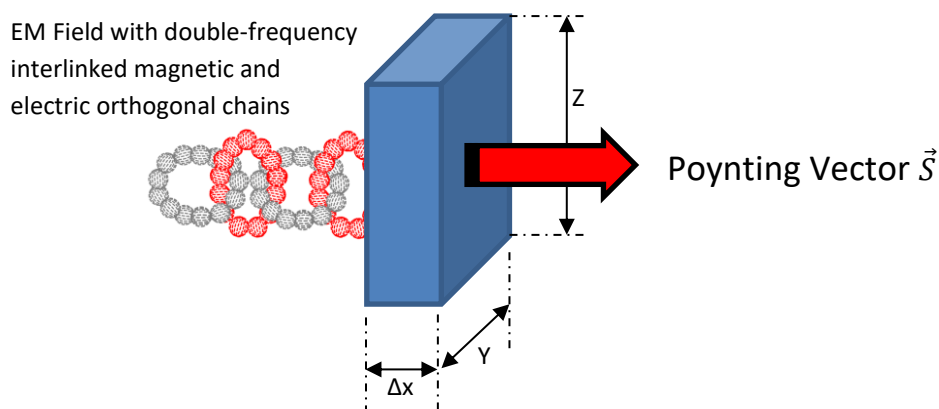
# Chapter 9

## B. Poynting Energy Fields

Poynting Field Theory provides the necessary tools to explain electromagnetic energy flow, including the transient phase of DC energy. The term “real power” is used for power that flows in one direction past a given cross-section, whereas reactive power is the power that flows back and forth in alternating directions with a net flow of zero in any one cycle. A resistor radiates, conducts, and convects real energy, whereas an antenna radiates only real energy, traveling outward in lobes.

Make no mistake; a circuit is an antenna that, in most EMC cases, is an unintended antenna. The “Poynting Field” incorporates the product of the E-field by the B-field (vector cross-product in the proximity of an antenna and beyond). Surprising to many, the **electromagnetic field energy primarily flows outside of conductors in circuits and antenna elements.**

The **energy density ( $U$ )** in the electric field and the magnetic field of an EM wave are of the same magnitude within a differential volumetric reference (DVR) space in the general proximity of the source. The electric energy density ( $U_E$ ) is a function of the permittivity and the electric field squared. The conductor surface voltage is squared and divided by the distance squared from the surface. The magnetic energy density ( $U_B$ ) is a function of the magnetic field generated by the current of the source, squared, times the permeability. The superposition of the two energy densities resulting in  $U$ , and when multiplied by the DVR, results in differential energy which can be divided by the differential time required to pass through the DVR to result in the Poynting Power in joules per second. The Poynting vector points in the direction of power/energy flow illustrated in the x-direction.



**Figure 98: EM Field Through a Differential Volume**

The Poynting vector  $S$  always points in the direction of propagation while oscillating in magnitude. The time-averaged electric field magnitude portion of the Poynting vector description follows: A phasor,  $E_m$ , represents a sinusoidally varying field whose instantaneous amplitude  $E(t)$  follows the real part of  $E_m e^{j\omega t}$ , where  $\omega$  is the angular (radian) frequency of the sinusoidal E-field wave. [71]

# Photony Foundations

## C. The Poynting Energy Fields of a Simple Battery/Resistor Circuit

The Poynting energy field is visualized with a simple battery/resistor circuit by superimposing the magnetic and electric field frames. The bottom-right diagram shows the Poynting energy flowing in and around this simple circuit. Poynting Energy Flow Diagram provides a graphic insight into the Unification of Circuits and Fields<sup>31</sup>. Progressive Poynting Energy Flow Diagram:

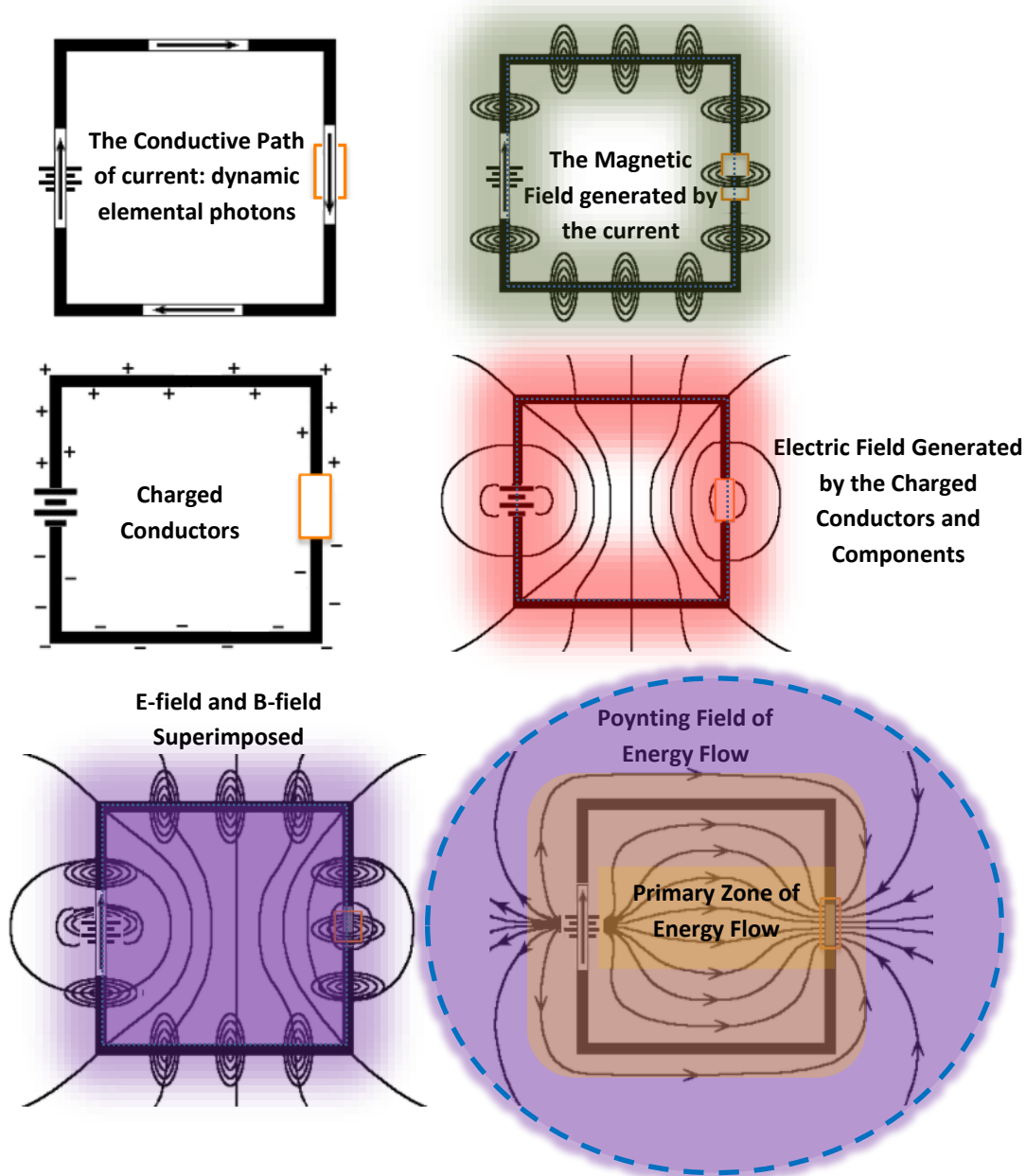


Figure 99: Poynting Field of Energy Flow (identifying in the primary zone)

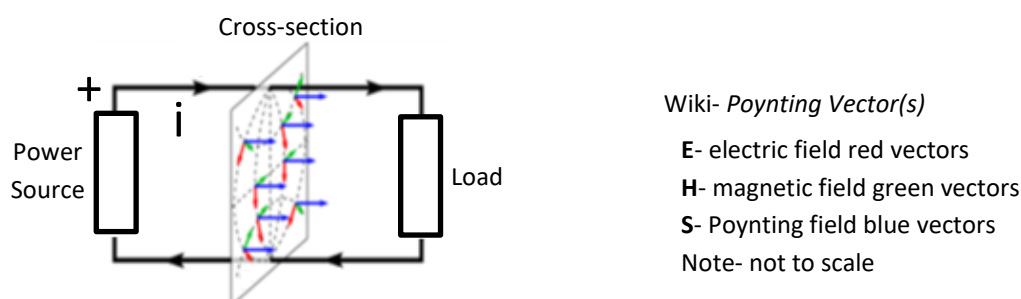
<sup>31</sup> Note: there is a dissipative portion to the Poynting Field based on  $I^2R$  heating.

# Chapter 9

Electric and magnetic fields store energy. The (volumetric) energy density is given by:

$u = \frac{1}{2} (\epsilon \mathbf{E}^2 + \frac{\mathbf{B}^2}{\mu})$  J/m<sup>3</sup>, where E is the electric field, B is the magnetic field, and  $\epsilon$  and  $\mu$  are the permittivity and permeability of the medium, respectively.

The 3-D detailed DC circuit below comprises a battery supplying a voltage, V, and current, I, and a resistor, R. The E, H, and S field vectors encompass the battery, resistor, and a cross-section of the circuit. The cross-sectional plane of the circuit encompasses the supply and return currents. The E, H, and S field vectors located at the cross-sectional plane indicate where the energy flows, identified as the power or Poynting Field. The Poynting vectors at the battery are pointed radially outward, indicating that the energy is leaving the battery into the fields around the battery. The Poynting vectors at the resistor indicate that the field energy flows into the resistor, which is converted to heat and lost through conduction, convection, and radiation. The energy flows through the cross-sectional Poynting plane between the battery and the resistor. It flows primarily in the fields outside the conductors from the battery towards the resistor. In other words, the Poynting or power flux is in the direction of the resistor.



**Figure 100: 3-D Poynting Field Around a Battery Circuit**

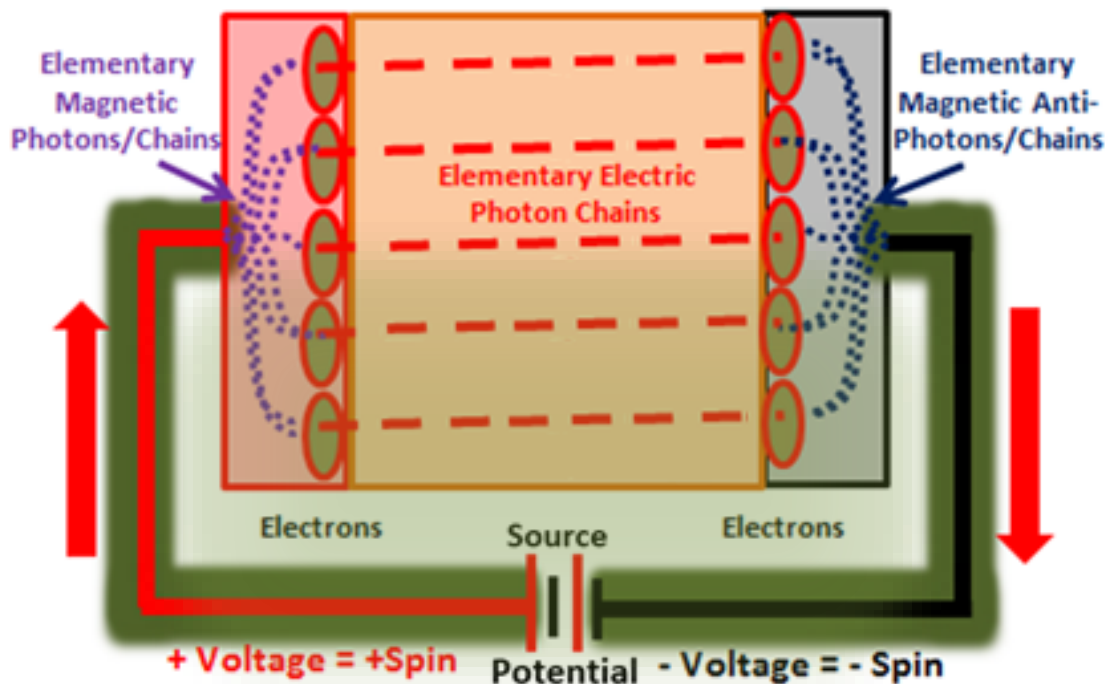
The Poynting Vector is defined as  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ , where E is the electric field vector, and H is the magnetic field's auxiliary field vector ( $H = B/\mu$ ). As seen above, the Poynting vector is parallel to the conductor and perpendicular to the current flow in the resistor, as shown. If a conductor has a significant resistance, then the Poynting vector would be tilted toward and impinge upon the conductor near the conductor's surface. If the conductor is somewhat resistive, the Poynting vector impinges, meaning energy would flow within the wire proportionally to its resistance. This phenomenon provides insight into the flow of energy through a circuit. Although magnetic fields originate within the conductors, they readily escape as the magnetic loops expand through the surface. The electric fields egress from free surface electrons as they transduce electric chains, so **only a small percentage of elemental photon chains' electric and magnetic (electromagnetic) fields are within the conductors**. Therefore, **the preponderance of the AC energy associated with a circuit flows external to the conductors!** What better unification of circuits and fields could exist than Poynting external energy flow?



# Photony Foundations

## D. Poynting Energy Fields of a Simple Capacitor Circuit

As shown below, we start by charging a capacitor with a DC source. Current leads voltage by 90 degrees in a capacitor. It takes time for the electric field chains to be transduced by elemental charge photons flowing through free surface electrons (windsock aligned by the same elemental photons). The free electrons transduce the elemental charge photons into single-ended electric chains.



**Figure 101: Chains Associated with a Capacitor Circuit**

When the source is switched on (switch not shown), large numbers of static elemental photons from within the source are mobilized by their nature to nudge and spread out on conductor surfaces. The closed series switch (not shown) acts as an open gate for the high density of static elemental photons to escape the source and minimize their (energy) density since this action corresponds to minimum energy. There is a surge of current at the source just after opening the gate. A wavefront consisting of dynamic elemental photons develops as the energy filaments' keyring overlaps oof the static elemental charge photons are forced open, giving them traction in quasi-static Planck space, and proceeds down the conductor and into the capacitor plate. The magnetic field shown in green is present when the current (dynamic elemental charge photons) is flowing. The magnetic field dissipates as the electric field stabilizes at any fixed voltage (accumulation of static elemental charge photons converted from dynamic elemental photons at the plate surface).

## Chapter 9

In other sections of this book, it is shown how toroidal electrons can ride across the gap between the plates as low speed electron flow. The electric field (comprised of electric chains) building role of electrons is called displacement current flow across insulative gaps.

Electrons can flow across capacitor plates as the capacitor builds an electric field of elemental electric photon chains. An electron can move along an encompassed electric chain extending across the capacitor plates as it rotates on its axis in the same spin direction if not in synchronization with the spin of the electric chain. So, more electrons can move across capacitor plate separations (below) as the electric field of electric chains increases.



**Figure 102: Electrons Can Ride on an Electric Chain**



# Chapter 10 Photony Energy Science

“No philosopher has been able with his own strength to lift this veil stretched by nature over all the first principles of things. Men argue, nature acts.” - Voltaire (1764)



Solar and Wind Energy

By watching carefully how nature acts, humans and later AI-based entities will be capable of lifting the veil and learning the true nature of Nature. **Photony maps the natural progression** from the Planck-based fabric (as they say) **of quasi-static Planck space and the origins of energy to a comprehensive energy science, Photony Energy Science.** The natural progression is accomplished in this chapter and in the Energy Science Compendium book. As we as scientists continue to argue, nature is acting. By ignoring science, the energy production companies, energy-hog corporations, and their bought politicians continue to make our offspring and theirs' and theirs' and so on inherit a toxic world. The weather of nature is throwing natural tantrums and reminding those who will listen to hurry and make energy science a solution for reversing the process. So, Energy Science is more than important; it is existential.

A multi-disciplined effort is required, from the origins of energy to harnessing energy. Solutions are realizable from material science, physics, optics, mechanical engineering, chemistry, and mathematics to legal and political disciplines. We talk of renewable energy as opposed to fossil fuel-based energy. Renewable energy is collected from renewable resources that are naturally replenished on a debatable timescale. Renewables include sunlight, wind, hydropower, and geothermal heat sources. Nuclear energy is renewable for fission, assuming a long-term supply of fissionable materials (e.g., uranium). Nuclear energy is renewable for fusion assuming it comes online as a follow up to the fusion breakthrough at the Lawrence Livermore National Laboratory in California in December 2022.

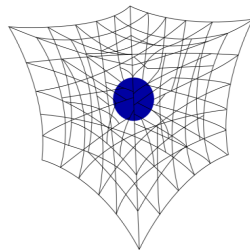
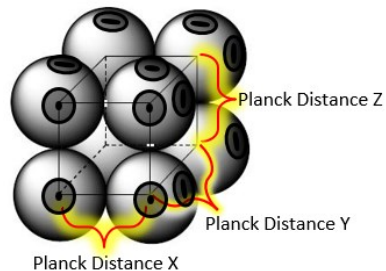
To cover all the renewable energy sources would require thousands of pages and, therefore, would be futile. So, the focus herein regarding energy science is to show the progression of energy from the origins of energy to how energy can be harvested, converted, transferred, stored, and used. This process

# Chapter 10

starts in quasi-static Planck space (reviewed below) and ends when a switch is turned on for light or heat, a process we call “Photony Energy Science.”

Flexible Planck network cells below illustrate the Photony form of interconnectivity via cell junctions of Planck space. Each spherical Planck Cell model connects six neighbors (four sides plus top and bottom) for an outer cubical structure with orthogonal inter-cellular portals. The interconnected centroids of the eight adjacent Planck Cells form a Planck Cube. The Planck Cells are flexible, somewhat like a structured membrane. The relaxed, minimum energy, non-stressed intercellular distance, or Planck length  $L_P$  [9], is defined by combining other constants of nature such that their units combine to give a length.

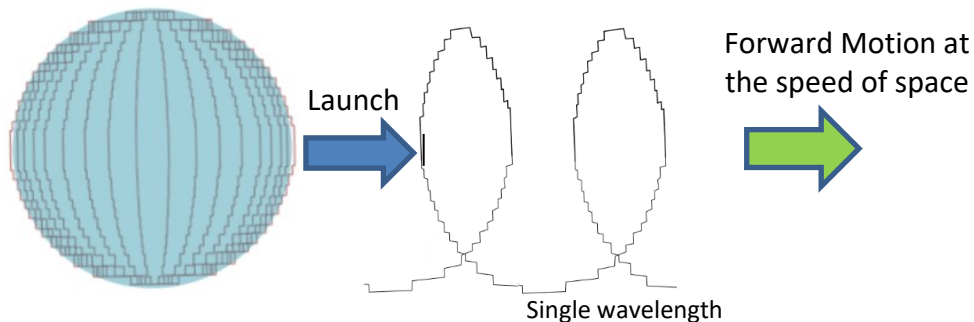
Planck Space Cubic  $L_P$  Section



Geodesics are "straight lines in curved space." Energy filaments in constant motion compress proximal quasi-static Planck space.

Static Elemental Photon (single filament particle)- filament in Geodesic Motion

Dynamic Elemental Photon- Filament (with gap) in Helical Motion in High-Speed Forward Motion



# Photony Foundations

## Photony Energy Science: The Electromagnetic Spectrum

Spectroscopy deals with the interactions between matter (solid, liquid, or gas) and energy (EM waves or photons). The electromagnetic spectrum has two primary sections, an EM wave (radio/microwave) section and a photon section with a transition range in the far-infrared, wherein the electromagnetic chain's transition from a multiplicity of interlinked elemental charge photons to chainlets of two and only two interlinked elemental charge photons. However, various frequency/wavelength partitions or bands in the EM wave section have differing spectroscopic characteristics regarding transmission, penetration, reflection, refraction, diffraction, and scattering. At the molecular and atomic levels, various frequencies involve various effects. Lower frequency EM radio waves can affect nuclear spin transitions in molecules. Microwaves can affect the rotational motion of molecules.

Infrared radiation is emitted or absorbed by molecules when changing rotational-vibrational modes. Infrared photons cause molecules and atoms to vibrate in a solid. The reverse is true, whereby the temperature of the solid matter is related to the frequency and energy of the photons well beyond the infrared range as per Wien's law. Ionizations occur throughout the upper end of the spectrum, starting with high-energy ultraviolet photons. The remainder of the spectrum (visible, UV, x-rays, and gamma rays) are studied, used, and written about in great depth and breadth.

### A. Preeminent Research

In airwave transmissions, for example, EM waves in the HF band (see Augmented Spectrum below) will bounce off and refract within the ionosphere (as Arthur Kennelly and Oliver Heaviside independently proposed in 1902), allowing long-distance HAM radio communications "over the horizon." Higher frequencies will pass through the atmosphere and continue propagating in space.

The energy of EM waves is measured based on their voltage and current amplitude and phase in RMS as transmitted circuit signals or power delivery systems directed through conductors. The energy outside of conductors, including antennas, is measured and calculated as the cross product of the E (electric field) and the B (magnetic field) as taught by Poynting.

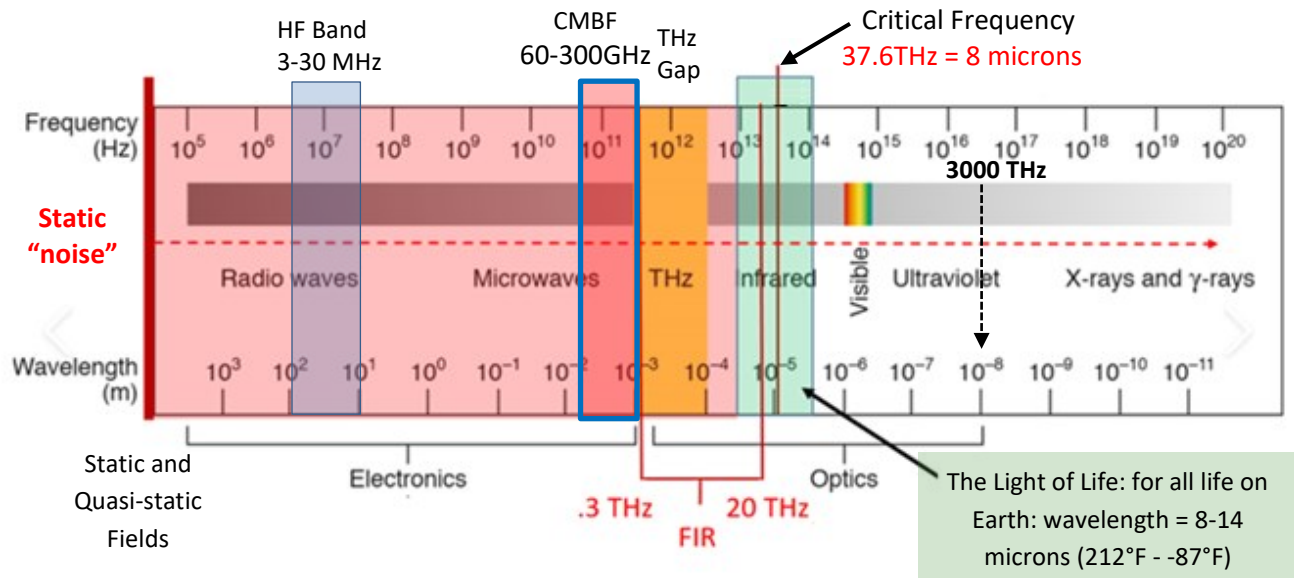
Astronomer Sir William Herschel discovered infrared radiation in 1800. [72] Through Herschel's studies, it was eventually found that slightly more than half of the Sun's energy arrived on Earth in the form of infrared. **The balance between absorbed and emitted infrared radiation has an important effect on Earth's climate.**

Johann Wilhelm Ritter discovered ultraviolet (UV) light in 1801. The UV band is between visible light and x-rays. X-rays were discovered in 1875 by Wilhelm Rontgen, Philip Lennard, and Johann Hittorf. Gamma Rays were discovered by Antoine Henri Becquerel in 1900.

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## B. The Augmented Spectrum

The Photony rendition of the Augmented Spectrum in the figure below is comprised of **DC/Quasi-static Fields, Electromagnetic Fields, and Classical Photon frequencies and wavelengths**:



**Figure 103: Augmented Spectrum**

Spectrum Sections Highlights:

Quasi-static fields- As direct current slowly varies, so do chain lengths and numbers of chains of energy filaments comprised of elemental photons resulting in static fields.

Static Fields- Stabilized electrostatic and magnetostatic fields correspond to fixed/steady DC voltage or current producing constant numbers of fixed-length chains.

Electronics including radio waves and microwaves- EM waves are fields of orthogonal sinusoidally transduced electric and magnetic chains.

The cosmic microwave background frequency range is largely 60-300 GHz. The terahertz region (0.1 THz -20 THz)- includes the far-infrared, which is largely unknown and unexplored.

Transition Frequency- from EM waves to classical photons, the infrared photon spin frequency is 37.6 THz.

Human body temperature is 98.6 °F which translates to 9.3 microns and 18.2 THz using Wein's Law.

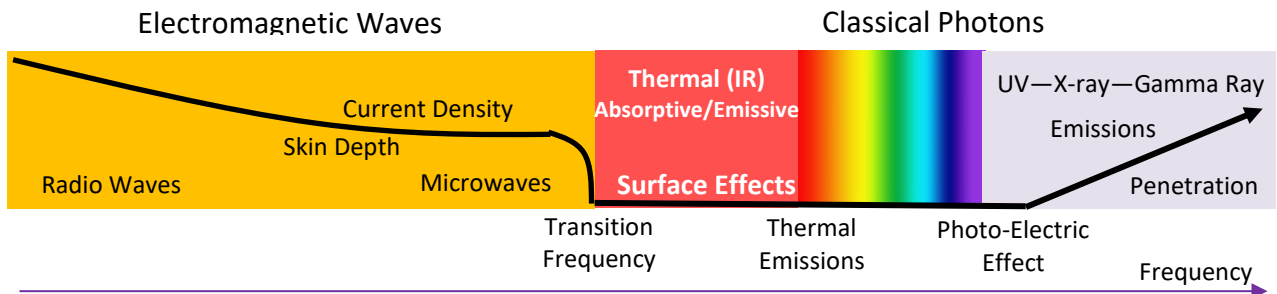
Optics frequency range (0.3THz – 3000THz).

See the EMFF Compendium Book for additional spectrum material.

# Photony Foundations

## C. Microwave/Infrared Transition

There is a critical transition frequency regarding metals in the EM Spectrum from microwave to infrared, whereby the nature of the EM wave transitions to EM classical photons.



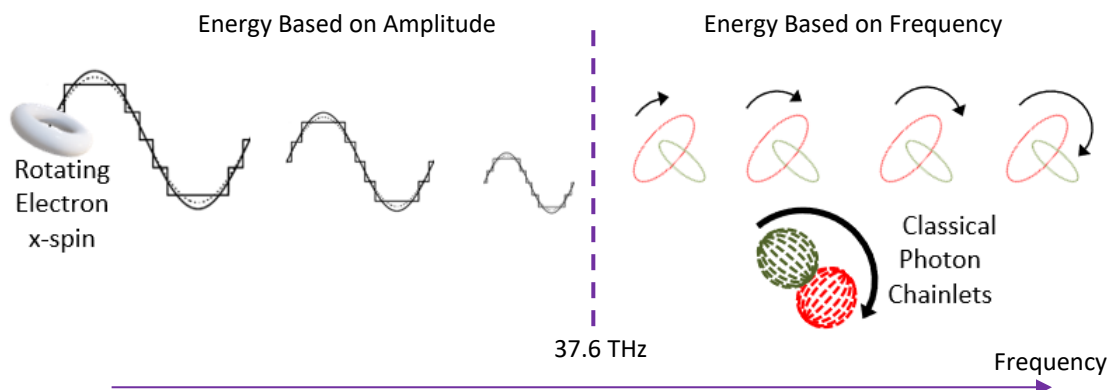
**Figure 104: Skin Depth-Surface Effects-Penetration for Metals**

### 1. EM Microwaves to IR Photons Transition

The microwave/IR transition frequency (37.6 THz) in the EM Spectrum has several associated characteristics and coincidental realities, including:

- The frequency whereby EM radio/microwaves transition to photonic EM structure
- The **optical opaque threshold frequency whereby skin effect discontinues since the conductor becomes opaque to microwaves. The conductor radiates due to temperature from this frequency and beyond.**
- A frequency associated with the Wein temperature of the boiling point of water
- The temperature at which vibrating molecules start emitting infrared energy (photons).

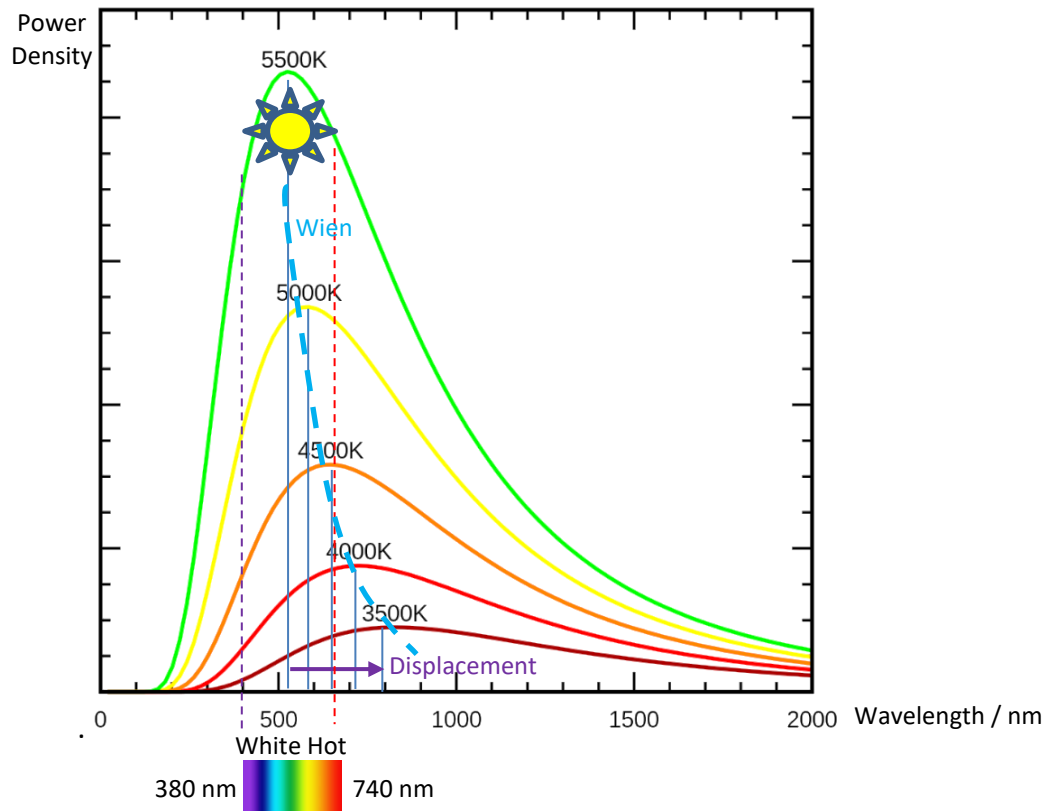
EM Waves of chains vs. classical photons chainlets transition at an infrared frequency (37.6 THz):



**Figure 105: EM Wavelets Transition to IR Photon Chainlets**

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## D. Solar Spectrum and Wein's Displacement Law



**Figure 106: Wien's Law Peak Displacements**

Wien's law states how the wavelength of the maximum is displaced when the temperature is changed. Wien used the Deutsch word "verschieben" which was translated into English as "displace," but in German, it has synonyms of move, postpone, and shift, depending on the connotation. Perhaps "shift" would be a better word in the context of curve shifting, but historically it was translated as "displacement." We can see the shift of the peak in the above diagram as temperature increases which is described as the peak being directly proportional to the temperature.

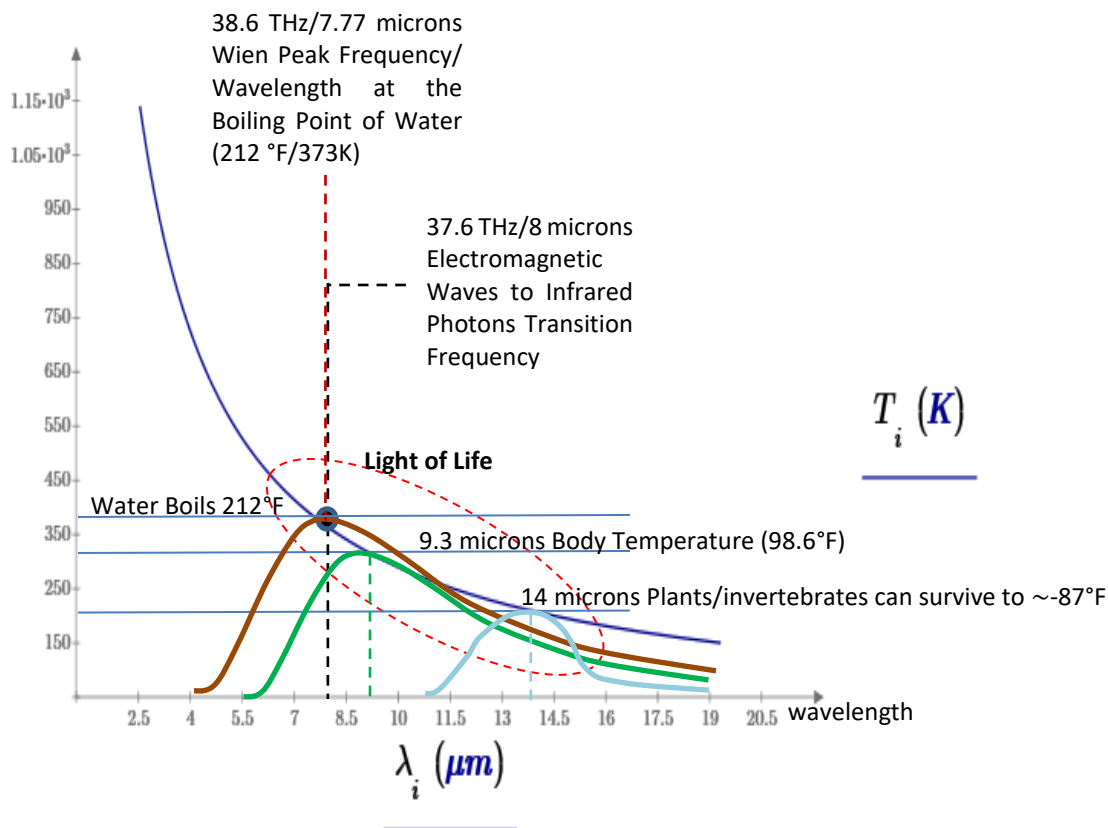
At 1200 to 1800 degrees Fahrenheit, metal glows orange and can be forged and hardened. For example, heated metal can become "red hot" at 800 to 1200 degrees Fahrenheit. From 1800 to 2600 degrees Fahrenheit, metal glows white-hot, as indicated above.

The Sun provides the life-supporting light and heat needed for life on Earth. The temperature of the Sun's surface is 5800K. The sun emits most of its radiation at visible wavelengths, particularly yellow (580 nm). The Sun's radius is  $7.0 \times 10^8$  m, and at a distance from Earth of  $1.5 \times 10^{11}$  m provides 1000 - 1400 W/m<sup>2</sup> of power to the Earth's surface depending on atmospheric conditions. Of this energy, 527 watts are infrared radiation, 445 watts are visible light, and 32 are ultraviolet radiation. [73] Nearly all the infrared radiation in sunlight is near-infrared, shorter than 4 micrometers.



# Photony Foundations

## E. The Light of Life



**Figure 107: Wien's Law Informed Light of Life**

The Light of Life upper limit for biology to survive and planets to retain water is (212°F or 373K), peaking at 38.6 THz. This frequency is very close to the Photony calculated spin frequency of elemental dynamic photons (8 microns/37.6 THz), curiously at the transition from EM waves to classical photons.

Based on Wien's Law and reference to the Spectrum, the "Light of Life" (LOL) for Earth is between 38.6 THz/7.77 microns and 21.4 THz/14 microns (shown in green), corresponding to 212°F through -87°F (373K – 207K). This range is approximately centered within the mid-infrared range extending from 3 – 50 microns.

Light is required for life within limits. Lynn Rothschild, Ph.D.;

*"Temperature is a critical parameter because it determines whether liquid water is present. If temperature is too low, enzymatic activity slows, membrane fluidity decreases. Below freezing ice crystals form that slice through cell membranes. High temperatures can irreversibly alter the structure of biomolecules such as proteins and increase membrane fluidity. The solubility of gasses in water is correlated with temperature, creating problems at high temperature for aquatic organisms requiring oxygen or carbon dioxide." [74]*

# Chapter 10

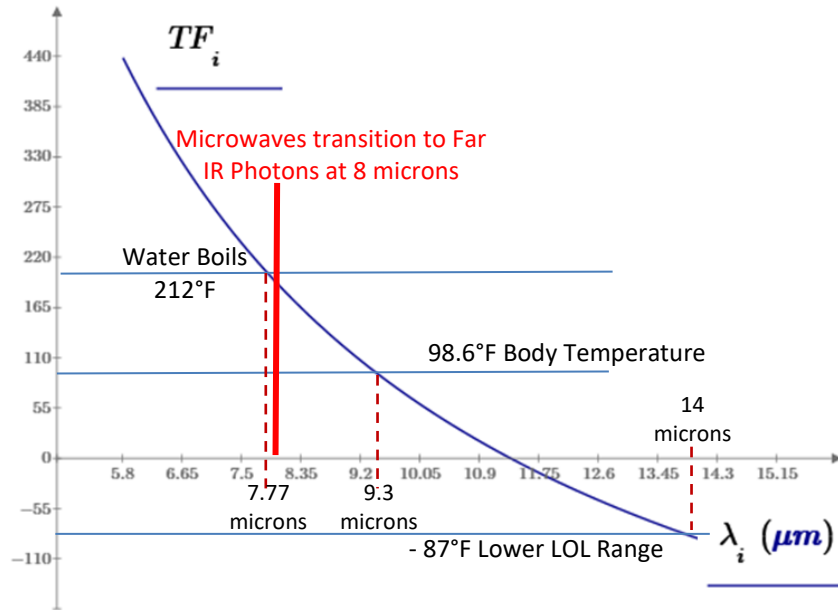


Figure 108: Light of Life Detailed Plot

Please consider the fantastic coincidence of the independently computed Electromagnetic Waves crossover to Infrared Photons' Transition Frequency (37.6 THz)/wavelength (8 microns) with the Wien's law calculated 7.77 microns peak corresponding closely to the boiling point of water.

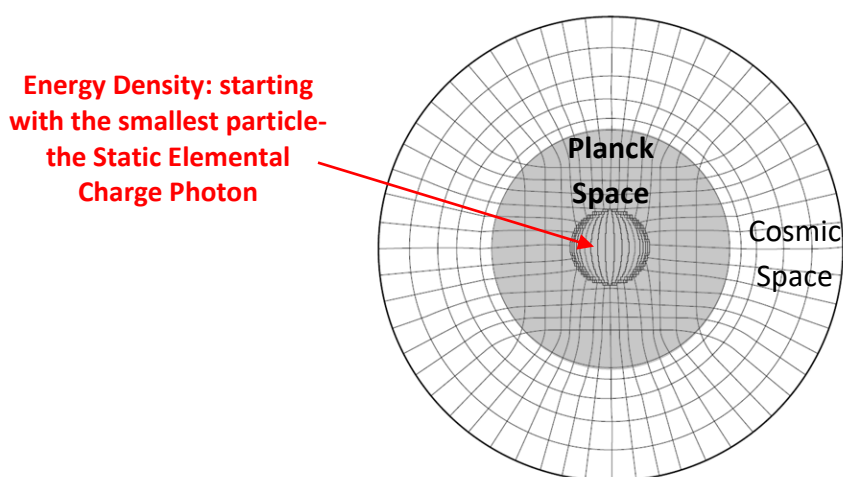


# Photony Foundations

## Photony Energy Science: The Origins of ENERGY

Photony Energy Science begins at the smallest of spaces, Planck space. The origin of energy is called the Photony Dawn. The evolution of energy from elemental photons to classical photons to particles is called Photony Emergence. Big Bang Theory suggests that all the energy and mass of the Universe were introduced in a short period of time from a singularity point. The Big Bang theory timeline has classical photons coming into existence after quarks. In some ways, Big Bang Theory contradicts quantum mechanics. If space is the “Dirac Sea,” teeming with energy, is there a need for a big bang? (or) Did the Big Bang charge the Dirac Sea? Based on previous chapters’ content regarding quasi-static Planck space, energy filaments, and elemental quanta and particles, a generalized macro-science of energy is possible by building on the Photony Planck scale space-energy-mass principles and elemental particles. As the density and number of Photony interconnected or interlinked energy filaments, the energy density transitions from subatomic and fundamental particles to massive particles with high energy density (HED).

The figure below depicts how a mass (energy density) compresses Planck space governed by general relativity and Photony principles. The figure below illustrates a monocular magnification from cosmic space into quasi-static Planck Space, where we can discover Photony elemental charge photons hiding in plain sight. This view is like the standard model’s conceptualization of gluons and dark matter; they must be there but cannot be observed.



**Figure 109: A Monocular View of Cosmic Space into Planck Space**

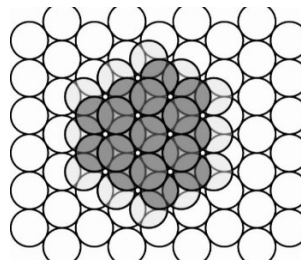
Photony has given lines and tubes structure and the means of producing the structure by free electrons as single-ended electric and magnetic chain loops. What if chains of elemental photons are the gluons? What if the spinning tubes of Faraday and Maxwell, the agents of force, were elemental photon chains?

### A. Concepts of Cellular Space

# Chapter 10

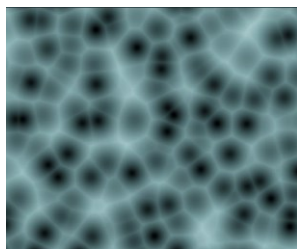
Quantum theory refers to space as the “Dirac Sea,” which is teeming with energy (vibrating with tremendous intensity [75]), popping into and out of existence with virtual particles in the mix that supposedly do not violate conservation and uncertainty principle by having short squinting “existences.” Quantum modeling (zero-point energy, vacuum energy, and vacuum fluctuations) is limited to Planck length and time by the uncertainty principle.

Alternate Cellular Space Concept Examples:



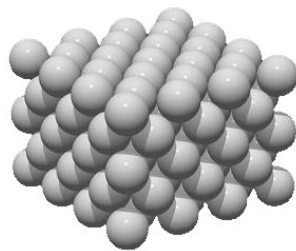
“The universe is extremely simple. Three types of particles (grey: quarks, electrons, electron neutrinos) form the lattice structure of matter and three types of particles (white: gluons, photons, gravitons) hold it together.”  
- Laszlo Petruska, Philosopher of Natural Sciences

**Figure 110: A lattice Structure**



Quantum foam- Physicist Steven Carlip suggests 'quantum foam' may explain away huge cosmic energy [88] —“it may be hidden by effects that are canceling it out at the Planck scale.”

**Figure 111: Quantum Foam**

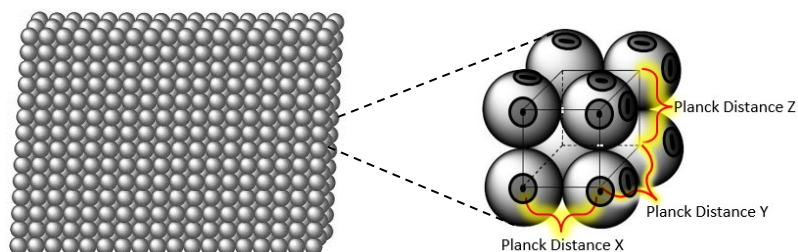


Illustrative hexagonal close-packed cellular structure

**Figure 112: A Hexagonal Cell Structure**

The space concepts above have largely six-sided cells- alternatively, Photony offers mechanical models and kinematics to support inter-cell communication of energy filaments. Photony supports what has been theorized in relativity, EM theory, and quantum theory and what has been verified experimentally. Many verified theories of the Standard Model have no plausible underlying mechanics explaining why the experimental results match the theory.

# Photony Foundations



**Figure 113: Cellular Planck Space**

Photony models the vacuum as a physical network of interconnected cells in which the orthogonal network of Planck space guides energy filaments. The interaction of the energy filament and quasi-static Planck space network results in a compression compromising the orthogonality characteristic (whereby  $\pi = 4$ ) preference of the network and a bending of the geodesical (classical  $\pi = 3.1415...$ ) preference of the energy filaments.

Since matter is 99.9% space (void) and since space is spread out over the cosmos, it would be easy to perceive that matter, as well as energy, is a direct product of space. Since time is based on the motion of mass, then spacetime is included automatically in the discussion. Gravity must be included since gravity is the primary product of space. Gravity can be understood through basic algebraic and geometric equations [75], but Photony provides the mechanism for gravitational pull. It's all about the compressive effect of an energy density vs. its mobility and tangential momentum. Energy is swimming through space (self-propelled by energy filaments) and swimming (as a current) toward higher compressed space caused by other energy, one spin stroke at a time (so to speak).

Photony is compatible with several aspects of quantum theory concerning spacetime. When real energy and mass comes forth- for example, Photony does not preclude latent energy (and associated mass) but proposes that most, if not all, of the energy and mass of the Universe, were introduced "into the void" in an extended time frame of Creation. BBT has the Planck Epoch, Grand Unification, and Inflationary Epoch all in  $< 10^{-32}$  seconds (essentially immediate). Does not pass the giggle test!

Photony holds that space is quasi-static and reactive to energy resulting in compression and the perception of mass. The dynamic elemental photons and classical photons (energy quanta) interact with quasi-static Planck space as they move at the speed of light in that spacetime structure. Static elemental photons are static/slow speed in nature. Chain structures of electric and magnetic chain loops are capable of low or high-speed motion. In DC applications, the chains build fast into space and any low permittivity materials, but the EM field building process stabilizes when the potential stabilizes. In AC applications, the fields of chains can move at near light speed as Poynting Fields.

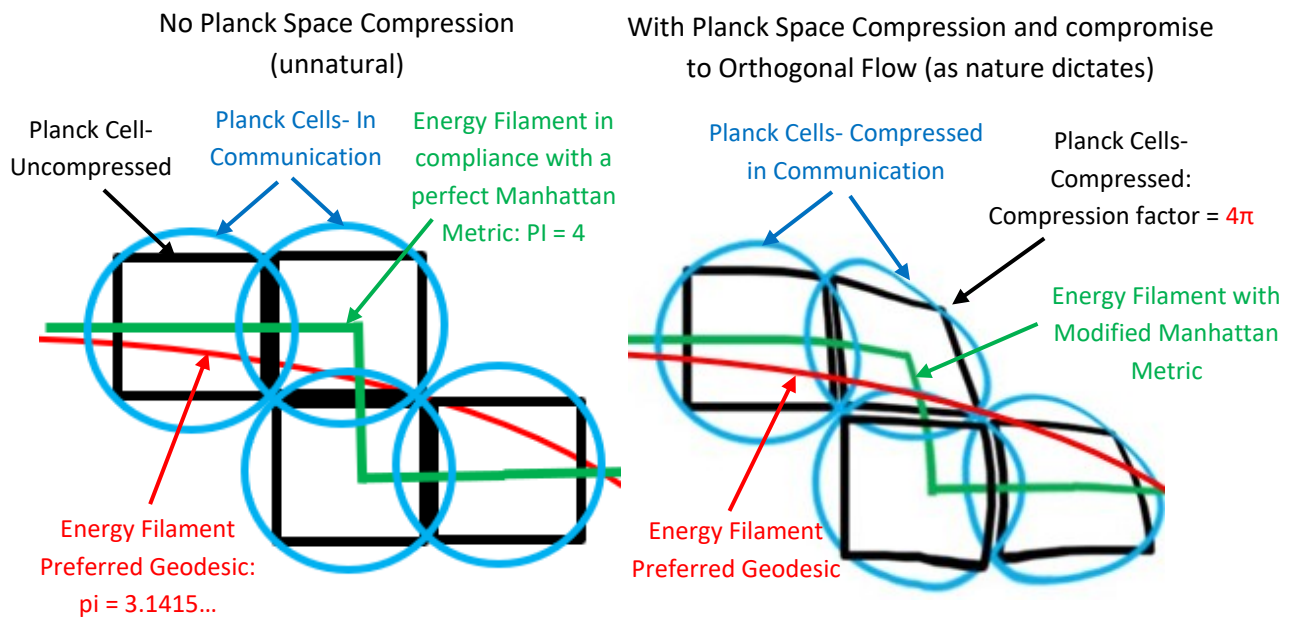
The fundamental quantum of energy is photonic, including dynamic elemental photons and classical photons. Photony elemental photons containing a single energy filament are the ultimate particle when static and the ultimate quanta when dynamic. Mass is localized energy as photons with an added c-spin (for example, electrons and positrons from gamma rays) and nucleons/atoms/molecules bound by elemental photon chains.

# Chapter 10

## B. Energy Density of Quasi-Static Planck Space

A 2-D cross-sectional graphic of how general relativity originates as compression of quasi-static Planck Space at the smallest volumetric scales, Planck Length cubed ( $\approx 10^{-105} \text{ m}^3$ ). The “syrup” of the Higgs field is comparable to energy filament motion in quasi-static Planck Space. The 2-D diagrams (before and after quasi-static Planck Space compression) below illustrate the heart of Photony, Unification, and the basis of Photony Theory of Everything (Big ToE) in ways that is studied further in the **Photony Compendium**.

Below, the black outlines illustrate orthogonal Planck Space cell boundaries. The blue outlines illustrate a spherical representation of cells within the boundaries with interconnecting channels appearing as Venn overlaps, the red line of the energy filament attempting to remain circular in 2-D (geodesic in 3-D), and the green Manhattan Metric pathway. The compression and curvature are caused by the energy filament attempting to maintain an analog radius in a digital, quasi-static (flexible) Planck Space network. It attempts to enforce the orthogonal Manhattan Metric pathways maintaining least-energy interactions with a compression factor of  $4\pi$ .



**Figure 114: Planck Cells Before Compression**      **Figure 115: Planck Cells After Compression**

Planck Space is a structured (but flexible) cubic portal interconnected network of cellular intersections. In contrast, Photony energy filaments are predisposed to geodesic spin motion; natural stress exists on each as they coexist. Therefore, some compression results within the Planck Space, and some orthogonality results in the otherwise extended arc-shaped filaments attempting geodesic motion. This  $4\pi$  interplay is loosely comparable with orthogonal Planck space as an analog-to-digital converter that becomes distorted as the signal (filaments) is converted with distortion.

# Photony Foundations

## C. Impedance of Space

The constant, **c**, is **not a unique speed of light**; it is the speed of light in space (or any transparent medium) with varying permittivity and permeability. Furthermore, the speed of space depends entirely on its permittivity and permeability, which varies throughout space (free space is only an ideal). The intrinsic impedance of space  $Z_s$  is defined as the impedance an electromagnetic wave or photon faces while traveling. The speed of light slows, and time dilation occurs due to gravitational fields. In this regard, the permittivity and permeability of free space depend on gravity, so they are variable. [29]

Regarding the impedance of space, the definition of impedance and the definition of space/spacetime must be coordinated with mutual regard. When the term “free space” is used, it is associated with “spacetime,” but noting that free space is only an ideal condition of space. The term “free space” is approximately the same as space since space is mostly empty (or is it? what does empty mean?). The Photony model of space is quasi-static Planck space, a flexible network of Planck cells arranged in a 3-D matrix. The matrix of interconnected cells is physically orthogonal in structure but is compressed by energy (Photony energetic filaments) proportional to the energy density. The compressibility of space itself, particles, and ponderable mass (objects and materials) is the basis of Einstein’s relativity principles and Lorentz’s time dilation and length contraction principles noting that physical objects and materials are “mostly space.”

The concept of impedance must be considered from three basic and distinct points of view:

1. High-speed quanta “energy.”

In this category, we include Photony elemental photons (possibly another name for axions or Higgs basis particles), classical photons, and EM waves (chains of Photony elemental photons). The masses of these quanta are immeasurable for several reasons, including their size and speed. As their energy filaments engage Planck space for traction, they achieve high-speed motion. They leave undetectable levels of compressed space in their wake.

2. Relatively slow-moving matter

In this category, relatively slow-moving matter with high energy density (HED) objects with ponderable mass range from massive particles to objects to celestial bodies. HEDs have static mass (gravitation) as well as dynamic mass (inertial mass), contributing to their total mass (space compression) in motion.

3. Compression of spacetime itself

Quasi-static space compression results from transitory dynamic energy flow and lower speed energy densities (including black holes). Gravitational waves are produced at detectable levels when energy densities collide as their speed escalates. Gravitational compression waves travel at the speed of space based on permittivity and permeability and are impeded by the compressibility of quasi-static Planck space.

# Chapter 10

The impedance of space can be extended from optical to EM transmissive materials (e.g., metals) by including the factor  $\sigma$  (conductivity).

The intrinsic impedance of space is a measure of the ratio of the electric field to the magnetic field. Intrinsic impedance is calculated the same way as any transmission line. Intrinsic impedance is given by the complex impedance:

$$Z = \sqrt{j\omega\mu/(\sigma + j\omega\epsilon)} \quad \text{Equation 63}$$

The **real part of  $Z$  is responsible for attenuation**, the energy loss based on heating. The **imaginary part is responsible for the phase shift**, so there is a noticeable similarity to the circuits and transmission line impedance.

Electromagnetic waves in space have an intrinsic ratio between their electric field (**E**) and magnetic field (**H**), which we call impedance.  $Z = E/H$  gives that impedance, and it is a fundamental constant; It's only when EM waves travel through some medium other than a vacuum of free space wherein an increased permittivity decreases the impedance, and an increased permeability increases the impedance.

The kinematics of energy filaments comprising elemental photons are interlinked by electron transduction into EM chains or classical photon chainlets depending on the frequency of end-over-end electron x-spin in response to the AC signal.

The impedance of space is real to an electromagnetic wave but not dissipative (like a transmission line) in the typical sense. The impedance units (of space) are Ohms because E is measured in Volts/meter, H is measured in Amperes/meter, and 1 Volt/Ampere is defined as an Ohm. This concept of impedance does not imply that space "resists" electromagnetic waves and dissipates them as heat (energy quanta) as a resistor in an electrical circuit would. As light comprising waves or collections of photons travel cosmic distances through space, their frequency (energy) is gradually reduced.

Ohms law also works in space as the intrinsic impedance of free space (Planck Space with no compression- no proximal energy density):

$$Z_0 = E/H \quad \text{Equation 64: Impedance of Free Space}$$

where E = electric field strength,

H =  $B/\mu_0$  magnetic field strength,

$\mu_0$  = permeability of free space =  $4\pi \times 10^{-7}$  n/m<sup>2</sup>,

$\epsilon_0$  = permittivity of free space =  $8.854 \times 10^{-12} \approx 4\pi/\sqrt{2} \times 10^{-12}$  F/m and

$4\pi$  = the Photony Planck space compression factor



# Photony Foundations

Other relationships **relating the impedance of free space to the speed of light in free space (“the speed of free space”)**:

$$Z_0 = \frac{E}{H} = \mu_0 c_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = \frac{1}{\epsilon_0 c_0}$$

Equation 65: EM Impedance and the Speed of Light in Free Space

The permittivity of free space (a vacuum):

$\epsilon_0$  is approximately  $8.85 \times 10^{-12}$  farad per meter (F/m) and

$c_0$  = the speed of light in free space (the speed of energy-free space itself)

The approximate speed of light in space is  $c = 3 \times 10^8$  m/s.

Based on the above equations,  $Z_0 = 376.73$  ohms is the impedance of free space and the impedance encountered by EM radiation (e.g., radio waves) and photons. This impedance encountered by EM waves ( $Z_0$ ) is  $4\pi$  times the impedance encountered by gravity waves:  $Z_0 = 4\pi \cdot Z_s$  with units of kg/s. The Planck impedance  $Z_p = 1/4\pi\epsilon_0 c = 29.98 \Omega$  ( $\approx 30$ ) converts directly to the strain impedance of space ( $Z_p \rightarrow Z_s$ ) as follows:

$$Z_s = Z_0/4\pi = 376.73/4\pi = 29.98 \Omega \quad \text{Equation 66: Strain Impedance of Space}$$

The strain (deformation/compression) impedance  $Z_s$  of quasi-static Planck space is approximately 30 ohms which is also the well-known Planck impedance. This impedance involves radial-longitudinal compression waves known as gravitational waves and the impedance that massive particles and objects see during movement and motion. Suppose a medium (e.g., space) has an impedance. In that case, the inference is that the medium is elastic and can realize energy density in the form of compression or expansion (strain) from the native or intrinsic state.

The phase velocity of an electromagnetic wave depends on gravitation. From the equation of the phase velocity of an electromagnetic wave, it is predicted that the  $\epsilon_0 \mu_0$  factor of the velocity of free space equation,  $c_0 = \sqrt{1/(\mu_0 \epsilon_0)}$ . However, the impedance and speed of light vary in real space according to  $c = \sqrt{1/(\mu \epsilon)}$ .

Gravity affects the value of the permittivity and permeability of free space to isotropically decrease the speed of light (that is, the increase of a refractive index). [29] It is known that the speed of light depends on the gravitational potential. In relatively high gravitational fields, the speed of light becomes slow, and time dilation occurs.

For example, as the permittivity increases, the impedance becomes smaller, and the speed of light becomes slower near the surface of the Earth than at some altitude. By this same phenomenon, **gravity critically relates to the phase velocity of electromagnetic waves including GPS signals.**

# Chapter 10

## Photony Energy Science: Energy Conversion

Energy Conversion, in one sense, could be defined as the transformation of energy from forms provided by nature to forms usable by humans. The various forms of natural energy have various challenges to harvest, convert, transport, store, and use without negatively impacting the environment. Many energy sources are not renewable and harmful to the environment. Renewable energy is energy derived from Earth's natural resources, noting that most of that energy originated from the sun. Traditional energy relies on fossil fuels, which are harmful to the environment.

“Be the Photon” is encouraged by Photony. To be the photon on this journey provides a unique view of the photon’s formation in the Sun, its travel to Earth, its pathway to a solar panel on the surface, and how it can be either converted to a potential (in a doped semiconductor called a solar cell) or be absorbed and converted to heat. Photony focuses on solar energy since this type of renewable energy involves photons from the sun. Helium fusion reactions in the Sun generate atoms and photons of various energies (a spectrum). The sun emits most of its radiation at visible wavelengths, particularly yellow (580 nm). The Sun emits approximately at 5778K as a black body.

Photons are a byproduct of primarily hydrogen to helium fusion in the Sun’s core as larger atoms are formed such as oxygen, carbon, nitrogen, and silicon. Photons take a long time to get to the Sun’s surface due to random collisions, so it is possible that the photons that arrived today on Earth were born at a time when humans first started wearing clothing. From the Sun to the Earth’s outer atmosphere, it only takes 490 to 507 seconds, depending on the distance.

The radiation reaching the upper atmosphere of the Earth is essentially constant, however the radiation reaching some point on Earth surface is random in nature. The main cause of the randomness is that various gases (water vapor, CO<sub>2</sub>, etc.) within the atmosphere absorb or scatter solar radiation at different wavelengths, and clouds and dust particles also affect it. On average, the Earth reflects (albedo, reflective power) about 29% of the incident solar irradiance back to space.

Light is absorbed as it passes through the atmosphere, and at the same time, it is subject to scattering. One of the mechanisms for light scattering in the atmosphere is known as Rayleigh scattering caused by molecules in the atmosphere. Rayleigh scattering is particularly effective for short wavelength light (that is, blue light) since it has a  $\lambda^{-4}$  dependence. In addition to Rayleigh scattering, aerosols and dust particles contribute to scattering atmospheric photons, known as Mie scattering.

Scattered light in the atmosphere is undirected, so it appears to be coming from any region of the sky. This light is called "diffuse" light. Since diffuse light is primarily "blue" light, the light that comes from regions of the sky away from the sun, it appears as blue. Without scattering in the atmosphere, the sky would appear black, and the sun would appear as a singular light source. On a clear day, about 10% of the total incident solar radiation is diffuse.

# Photony Foundations

## A. Photo-electric Effect

The photoelectric effect is essential to Photony Energy Science in two ways; the first is related to electrons emanating from conductor surfaces, and the other is electron-hole pairs in semiconductors.

### 1. Electrons Emanating from Conductor Surfaces

The electric field between  $V^+$  and  $V^-$  is a function of differential voltage ( $\Delta V = V^+ - V^-$ ), separation ( $s$ ), and position between plates, all of which contribute to the density of electric charge photon chains between the plates. Free surface electrons transduce the electric chains and provide a guided pathway for electrons to traverse the separation.

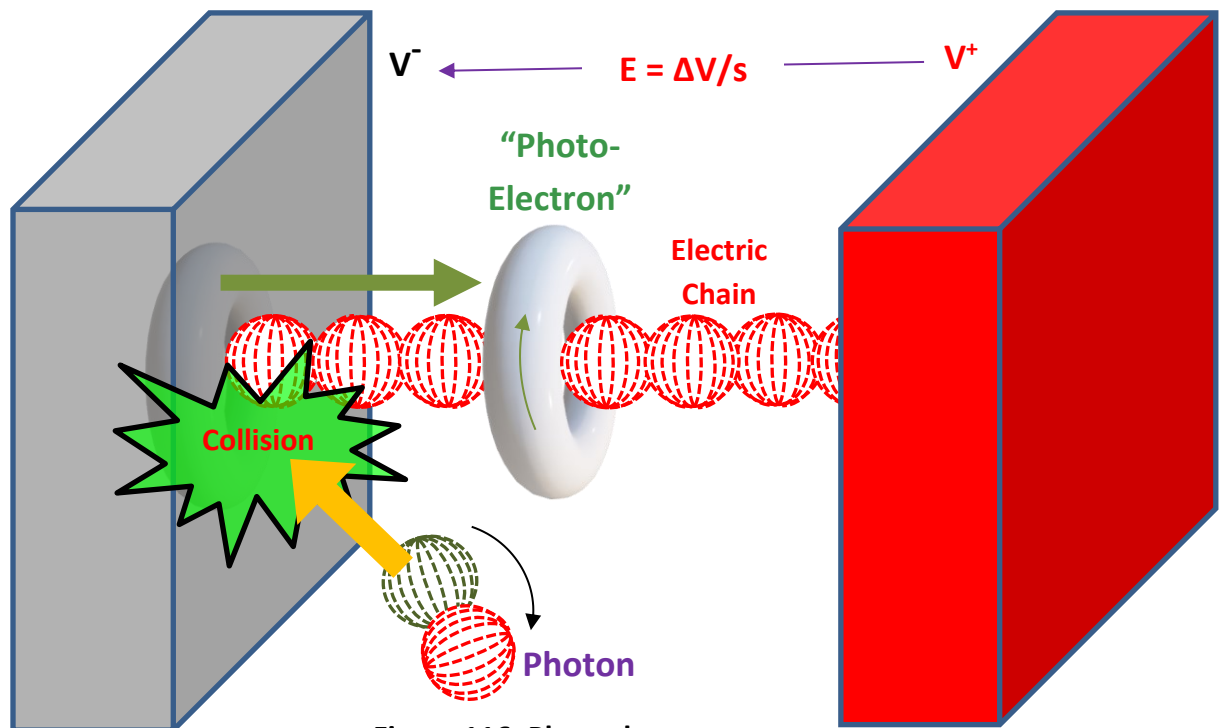


Figure 116: Photoelectrons

Photoelectric (photovoltaic) effect: An incoming light quantum (photon) collides with an electron proximal to the surface of a metal and transfers **its energy to the electron**. A certain amount of energy, called the work function, is needed simply to overcome the force to eject the electron, so there cannot be any photoelectric effect unless this threshold is reached. Above and beyond the work function, any energy leftover from the exchange appears as kinetic energy (energy of motion based on the Photony Motion/Spin Rule) of the ejected electron. Increased frequency (energy) of incident radiation beyond the work function significantly impacts photons in a collision, resulting in a more energetic photoelectron. Increasing the radiation intensity (the number of light quanta per unit area) does not affect individual electrons' energy since each electron is ejected by a single quantum (photon).

# Chapter 10

## Key Points Regarding the Photoelectric (Photovoltaic) Effect:

*“Based on the wave model of light, physicists predicted that increasing light amplitude would increase the kinetic energy of emitted photoelectrons while increasing the frequency would increase measured current. **Contrary to the predictions**, experiments showed that increasing the light frequency increased the kinetic energy of the photoelectrons, and increasing the light amplitude increased the current. Based on these findings, Einstein proposed that **light behaved like a stream of particles called photons**, (each) with an energy of  $E=h\nu$ . The work function is the minimum amount of energy [frequency] required to induce photoemission of electrons from a metal surface, and the value of depends on the metal. The energy of the incident photon must be equal to the sum of the metal's work function and the photoelectron kinetic energy.” [76]*

Significant advancement of photoelectron theory is the Photony principle that free electrons can transit while riding on an electric chain of transduced (interlinked) elemental photons produced by that same electron and followed by other electrons that become aligned with the same or similar electric chains at the source (e.g., the negative plate of a capacitor).

Electrons at a negative potential spin in the same direction as the axis perpendicular to the surface. Electric fields are built by free electrons from the surfaces of conductors. As elemental charge photons are pulled into the polar vortex of the entrance to the toroidal cavity of an electron, they are interlinked (transduced) by the electron into an electric chain that extends through the electron and out into space as a single-ended chain. As more and more chains are built by more surface electrons, they repel each other because they spin in concert with the electron's spin (comprised of a gamma ray in rotational motion). The single-ended electric chains tend to extend radially unless influenced by other electric chains or proximal sinks (receiving electrons) for chains. Electric chains either repel or attract each other based on their spin. Like spins repel, whereas opposite spins attract. When two chains with opposite spins from opposite directions make contact, they join and connect their opposite polarity “charge sources,” as indicated by electric field lines in physics literature.

An electric field between two plates of a capacitor is built as chains are transduced by free surface electrons. Each completed chain is stored energy. Chains are typically arrayed between the plates in an orderly and even distribution in the minimum energy configuration. However, at the edges of the plates, the electric chains tend to flare out as fringe fields of arc-shaped chains connecting the plates but with longer chains separated further in space in the fringe field.

# Photony Foundations

## B. Semiconductors and Photovoltaic Conversion

### 1. Doping and p-n Junctions in Semiconductors

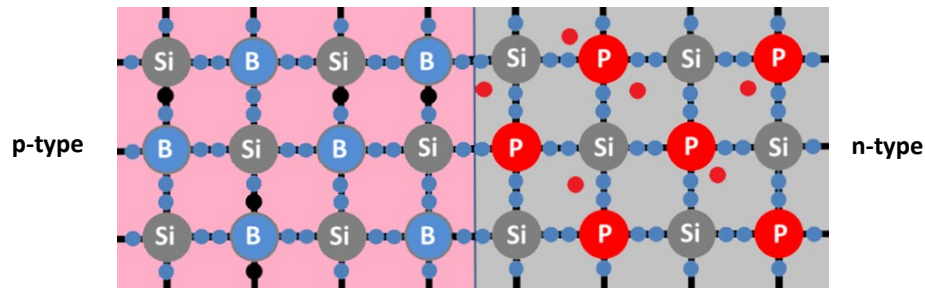


Figure 117: p-n Junction Doping

Silicon: Metalloid, Group 14, Period 3, Block p, atomic number (protons)-14, Neutrons- 14, Electrons-14, Shell model- (2,8,4), Valence Electrons- 4

Phosphorous: Pnictogens, Group 15, Period 3, Block p, atomic number (protons)-15, Neutrons- 15, Electrons-15, Shell model- (2,8,5), Valence Electrons- 5

Boron: Metalloid, Group 13, Period 2, Block p, atomic number (protons)-5, Neutrons- 6, Electrons- 5, Shell model- (2,3), Valence Electrons- 3

The n-type doping involves incorporating small quantities of elements like phosphorus (shown as P in red above) into the silicon. Electrons are shown as smaller red circles outside of the lattice structure. Phosphorous atoms migrate into the crystalline silicon structure because each has five electrons in its outer orbitals. The fifth electron allows an electric current to flow through the semiconductor because it has nothing to bond to and is free to move around and x-spin. The p-type doping uses boron (shown as B in light blue above) as the dopant. Holes are shown as smaller black circles within the lattice structure. Since boron has three electrons in its valence orbitals, it forms holes when integrated into the silicon structure. As a result, some electrons become more mobile, and the holes they create will move in the opposite direction of the electrons' movement.

The charge carriers that are present in large quantities are called the majority charge carriers. The majority charge carriers carry most of the electric charge or electric current in the semiconductor. Hence, the majority charge carriers are mainly responsible for electric current flow in the semiconductor. The pentavalent doping atoms such as Phosphorus or Arsenic are added to the intrinsic semiconductor; an n-type semiconductor is formed. In an n-type semiconductor, large quantities of free electrons are present. So, free electrons are the majority charge carriers in the n-type semiconductor. The free electrons (majority charge carriers) carry most of the electric charge or electric current in the n-type semiconductor.

The charge carriers in a small quantity are called minority charge carriers and carry small quantities of electric charge or electric current in the semiconductor. In an n-type semiconductor, small numbers of holes are present. Therefore, holes are the minority charge carriers in the n-type semiconductor.

# Chapter 10

## 2. Creation of electron-hole pairs in semiconductors

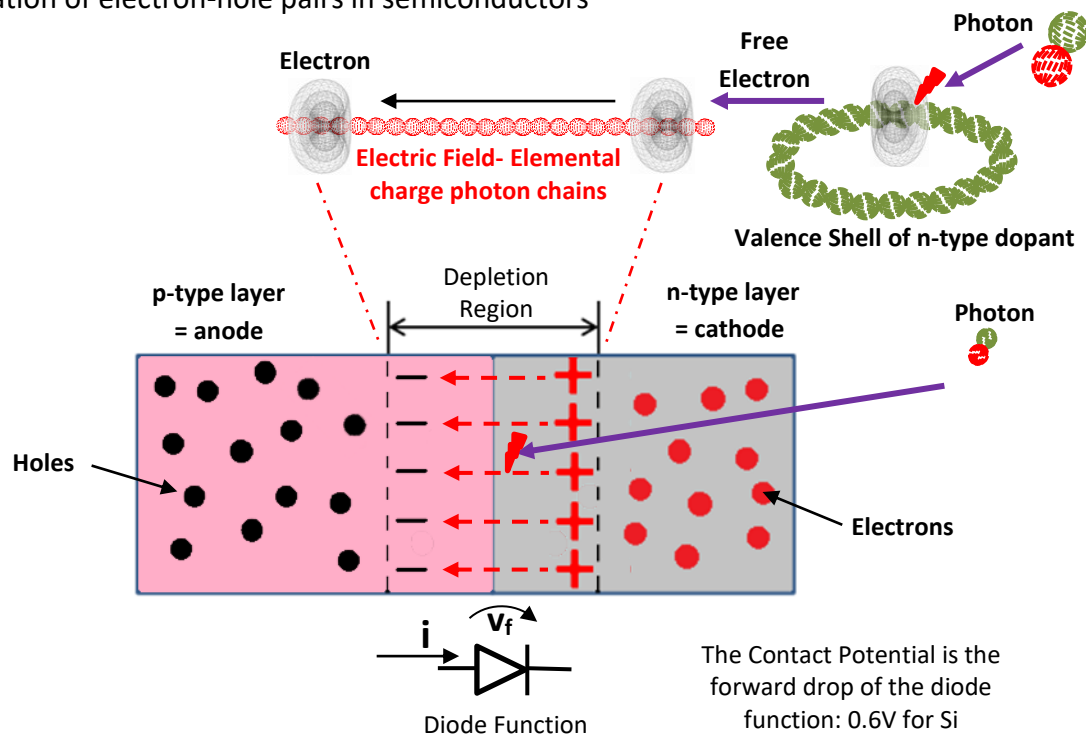


Figure 118: Photony p-n Junction

Some fraction of the photons that strike near or within the depletion region produce electron/hole pairs. This occurs as the magnetic elemental photon chain of the valence electron is broken, freeing the electron to be further influenced by the electric field chains. This type of photoelectric effect comprehends photodetectors, etc., and photovoltaic solar cells.

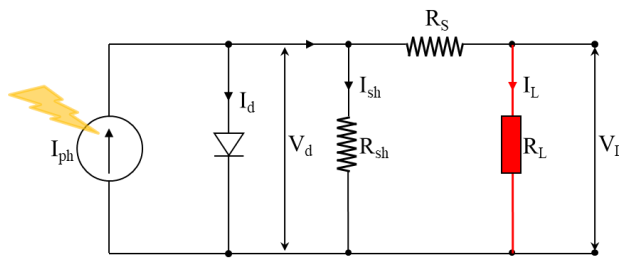
The absorption (photon-electron collisions) of incident photons of certain energies creates electron-hole pairs, provided that the incident photons have an energy greater than the band gap. However, electrons (in the p-type material) and holes (in the n-type material) are meta-stable. They can only exist for an average time in an excited state after electron-hole generation. The effect is lost if the excited electron recombines with any hole. Once the electron/hole separation has begun, the p-n junction prevents this recombination by spatially separating the electron and the hole further by the effect of the electric field (Photony electric chains) existing at the p-n junction. The less abundant charge carriers are called minority carriers; **in n-type semiconductors, they are holes, while in p-type semiconductors, they are electrons**. If a photon-generated minority carrier reaches the p-n junction, it is swept across the junction by the electric field at the junction, making it a majority carrier.

The diode current,  $i$ , is "conventional current" flow which is opposite to the electron flow direction. The **holes do not exit the semiconductor since conductors do not support hole flow**.

# Photony Foundations

## 3. Photovoltaic (PV) Solar Cells

Equivalent Circuit:



$I_{ph}$  = current source from photo electric effect  
 $I_D$  = current flowing internally through the p-n junction  
 $R_{sh}$  = shunt resistance (high)  
 $I_{sh}$  = shunt (electron-hole recombination) current  
 $R_s$  = series resistance (small)  
 $R_L$  = load resistance  
 $V_L$  = output or load voltage

### Photocell Energy Efficiency

Weaker (infrared) photons with energy less than the band gap energy are absorbed. More energetic photons (with energy more than the band gap energy) will deliver only a portion of the energy to create electron-hole pair. The rest of the energy is wasted as heat.

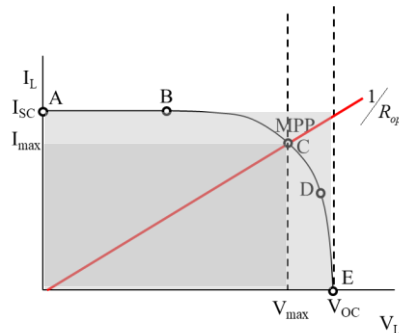
For maximum efficiency we need to have a maximum load current,  $I_{max}$ , simultaneous with a maximum load voltage ( $V_{max}$ ). For this to happen an optimal load  $R_{opt}$  must be determined to operate at a maximum power point (MPP) as follows:

$$R_{opt} = V_{max} / I_{max}$$

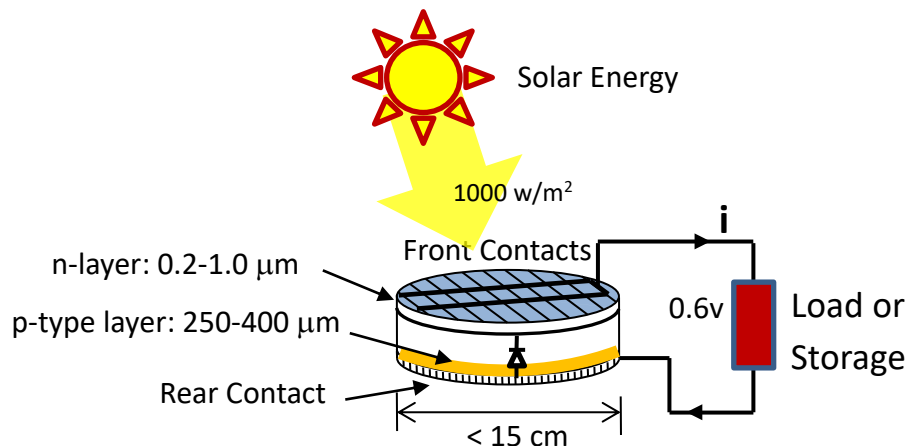
$1/R_{opt}$  is represented by the red line below:

$$\text{Fill Factor (FF)} = FF = \frac{V_{max} \cdot I_{max}}{V_{oc} \cdot I_{sc}}$$

FF > 0.7 is preferable for a PV panel or cell



Example Cell: Silicon, n-doped with phosphorous, p-doped with boron, on a sunny day:

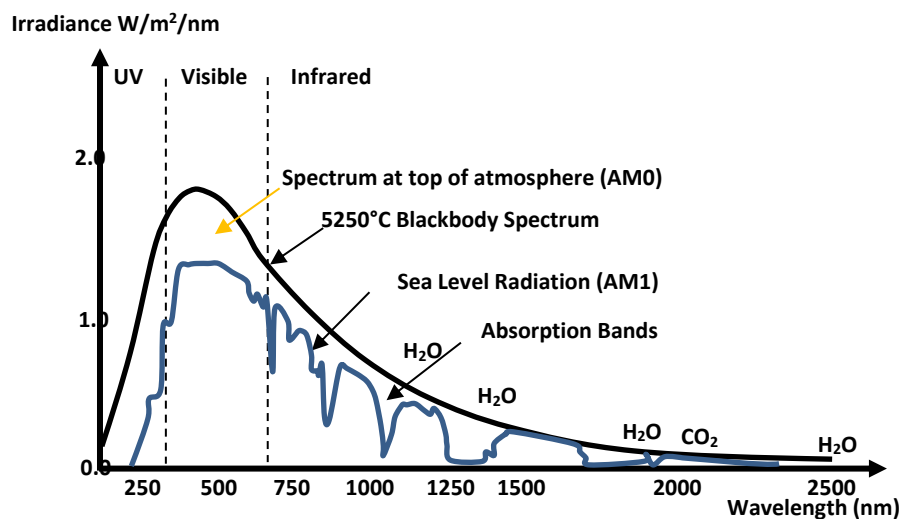


# Chapter 10

## C. Solar Energy

Regarding photovoltaics (PV), energy from the sun (solar energy) is of great interest as an energy source. The sun is a sphere of superheated gas whose internal temperatures reach over 20 million Kelvin due to nuclear fusion reactions at the sun's core. The fusion converts hydrogen to helium. Heat is transferred through the sun's photosphere surface layer by convection and radiated out into space from a surface temperature of approximately 5800 K. The sun emits most of its radiation at visible wavelengths, particularly yellow (580 nm). The resulting peak radiated energy (frequency) is determined by Wien's law:  $T = b/\lambda = b f/c$ , where  $b = 2.88 \times 10^{-3} \text{ m K}$

The Sun's radius is  $7.0 \times 10^8 \text{ m}$ . At a distance from Earth of  $1.5 \times 10^{11} \text{ m}$ , the Sun provides approximately  $1360 \text{ W/m}^2$  above the atmosphere and  $950\text{-}1000 \text{ W/m}^2$  of power to the Earth's surface, depending on atmospheric conditions. Of this energy, approximately 527 watts are infrared radiation, 445 watts are visible light, and 32 are ultraviolet radiation. Nearly all the infrared radiation in sunlight is near infrared, with wavelengths shorter than 4 microns. The radiation spectrum reaching the outer edge of Earth's atmosphere is called "AM0." The radiation spectrum reaching the Earth's surface is called "AM1."



**Figure 119: Solar Irradiance Above and Below the Atmospheric Edge**

Solar radiation can be absorbed, reflected, or absorbed and then re-emitted by carbon dioxide and water vapor in different bands. Various frequency bands (ranges of wavelength/frequency) of solar energy are reduced or redirected due to absorption, scattering, and reflection in the atmosphere affecting the spectral content of the energy reaching Earth and introducing an indirect diffuser component into the solar radiation. Variations in the atmosphere (such as clouds, dust, and pollution) limit solar energy's received power, spectrum, and directionality.



# Photony Foundations

## 1. Atmospheric Scattering Cross-sections

The collisional cross-section is an "effective area" that quantifies the likelihood of a scattering event when an incident species strikes a target species. In a hard object approximation, the cross-section is the area of the conventional geometric cross-section. The collisional cross-sections are typically denoted  $\sigma$  and measured in units of area. [77]

Rayleigh scattering results from the electric polarizability of the particles. The oscillating electric field of a light wave acts on the charges within a particle, causing them to move at the same frequency. The particle becomes a small radiating dipole whose radiation we see as scattered light. The size of a scattering entity may be characterized by  $x_s = 2\pi r/\lambda$ , where  $r$  is the particles or object's effective radius, and  $\lambda$  is the wavelength of light.  $x_s$  is a dimensionless stricture that characterizes the larger particles/object's interaction with the incident radiation such that particles with  $x_s \gg 1$  scatter photons based on their projected effective cross-sectional area.

Rayleigh scattering applies to the case when the scattering particle is very small ( $x_s \ll 1$ , with a particle size  $\approx < 1/10$  of wavelength), and the whole surface re-radiates with the same phase. The scattered light arrives at a particular random particle. Therefore, the light is incoherent, and the resulting intensity is just the sum of the squares of the amplitudes from each particle, making it **proportional to the inverse fourth power of the wavelength and the 6th power of its size**. [Wikipedia- Rayleigh Scattering]

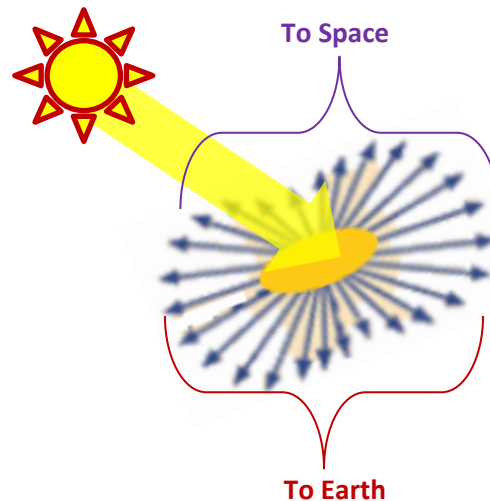
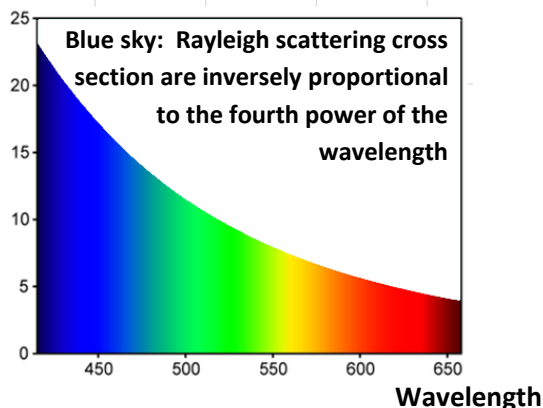


Figure 120: Scattering Particle

### % Scattering



Collision Cross-sections  $\sigma$  of atmospheric gas molecules: [89]

Air =  $0.36 \text{ nm}^2$

$\text{H}_2\text{O} = 0.27 \text{ nm}^2$

$\text{H}_2 = 0.27 \text{ nm}^2$

$\text{N}_2 = 0.43 \text{ nm}^2$

$\text{O}_2 = 0.4 \text{ nm}^2$

$\text{CO}_2 = 0.52 \text{ nm}^2$

Collisional cross-section =  $\pi(2r)^2$

## 2. Rayleigh Scattering- Photons/Water Molecules

The parameter  $r$  is an equivalent radius given that the molecule is polarized and in motion. The hydrogen bond angle is  $104.5^\circ$ .

NOTE: the wavelengths shown are not shown to scale to simplify the illustration.

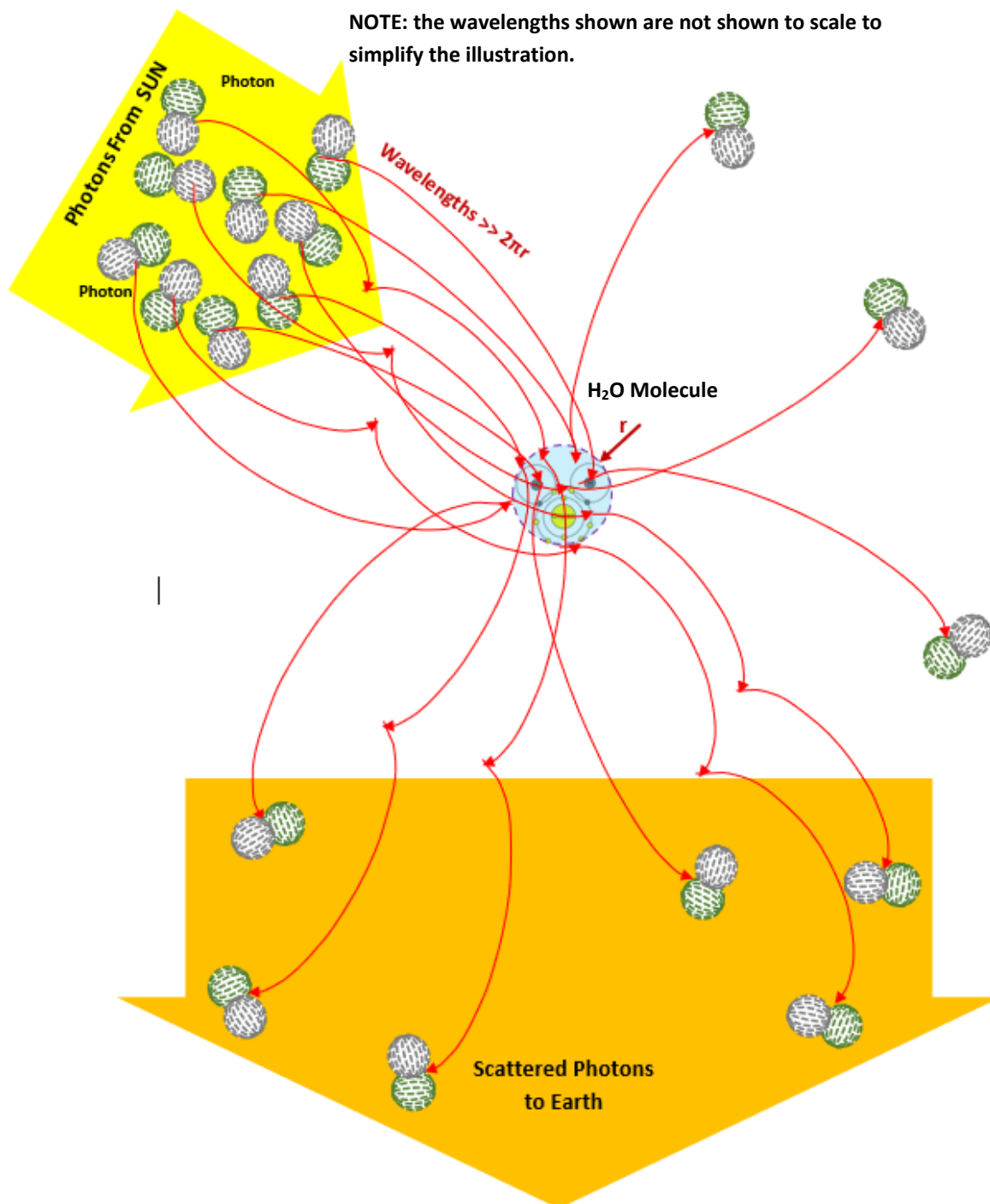


Figure 121: Atmospheric Rayleigh Scattering of Water Molecules

# Photony Foundations

## 3. Rayleigh Scattering- Photons/Carbon Dioxide Molecules

The parameter  $R$  is an effective radius given that the molecule is polarized and in motion.

NOTE: the wavelengths shown are not shown to scale to simplify the illustration

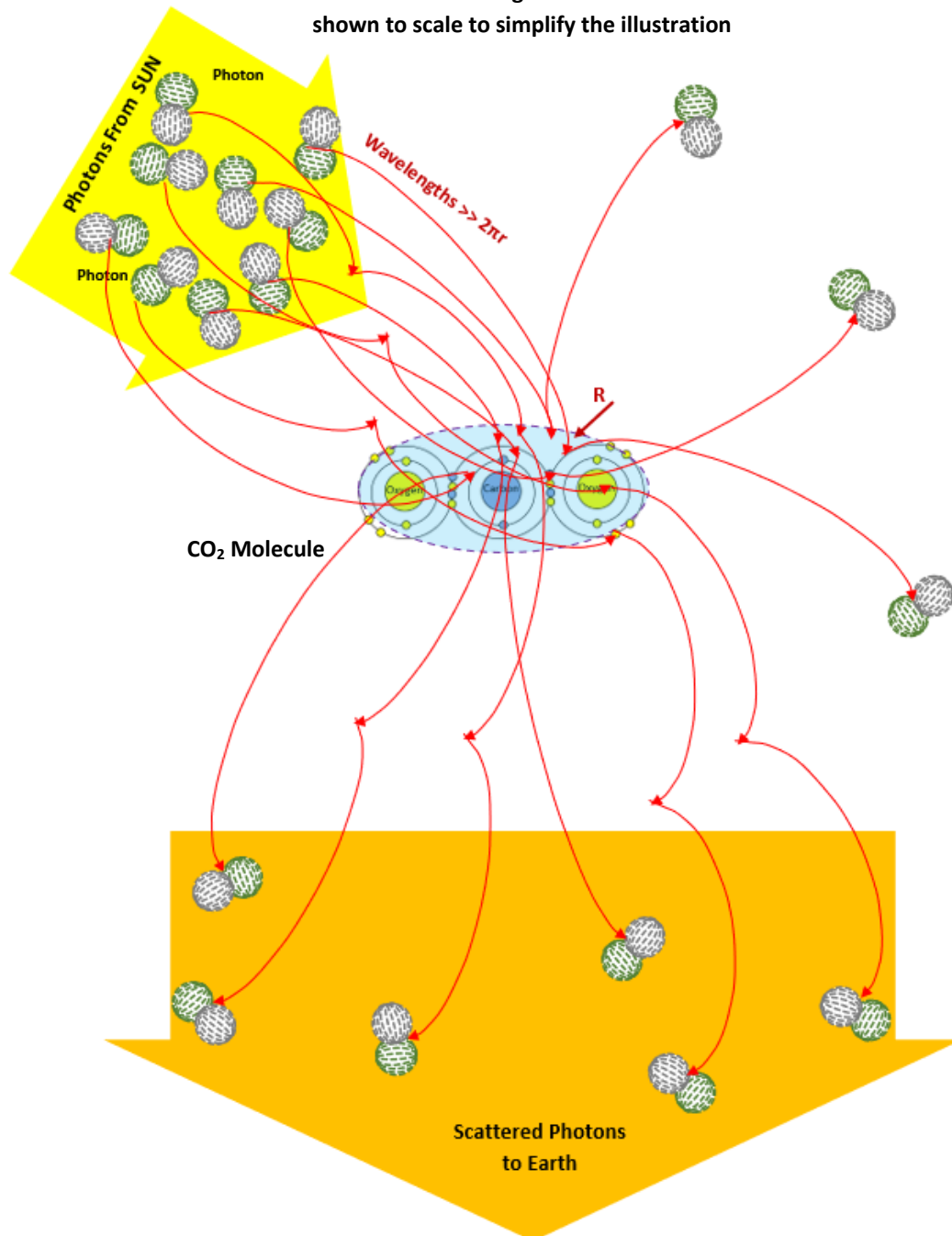


Figure 122: Atmospheric Rayleigh Scattering of Carbon Dioxide Modules

# Bibliography

## 4. Solar Energy Making it to Harvest

Solar energy is the focal point of Photony Foundations since the Earth, and all its creatures are doomed without the right amount of it. This section explains how Photony principles support existing theory with models, mechanisms, and kinematics for harvesting solar energy. Elemental photons, when interlinked by electronic transduction for EM chains, as described in other sections, provide the energy generated and released from the sun as EM waves. The EM fields are comprised of radio, microwave, infrared, visible, UV, X-ray, and gamma ray spectral travelers extending throughout our solar system. Here, we are interested in how these self-propelled travelers can be harvested, converted, and stored for our energy needs. Since most of the solar energy received is photonic rather than radio/microwaves, we focus on photons as energy quanta. In referring to photons, we mean classical photons, each comprised of two interlinked elemental (charge) photons making them non-spherical and, therefore, capable of polarization with filters.

Once a photon is released from the confines of an electron (particle annihilation), or any other mechanism, it uses the traction provided in space to travel self-propelled along a straight line except for sectors of space with increased permittivity due to proximal gravitational fields (general relativity: Remember Einstein's proof of starlight refraction during a solar eclipse in 1919?). As a photon enters Earth's gravitational field, it may experience minor gravitational and magnetic influences, but it is mostly affected by its trajectory and energy/wavelength.

An interesting way to proceed is to **"be the photon"** and see various atmospheric conditions through the rhetorical eye of an infrared photon, " $Q_{IR}$ ," a visible photon, " $Q_V$ ," a UV photon, " $Q_{UV}$ ," an X-ray photon, " $Q_X$ ," and a gamma ray photon, " $Q_G$ " as follow:

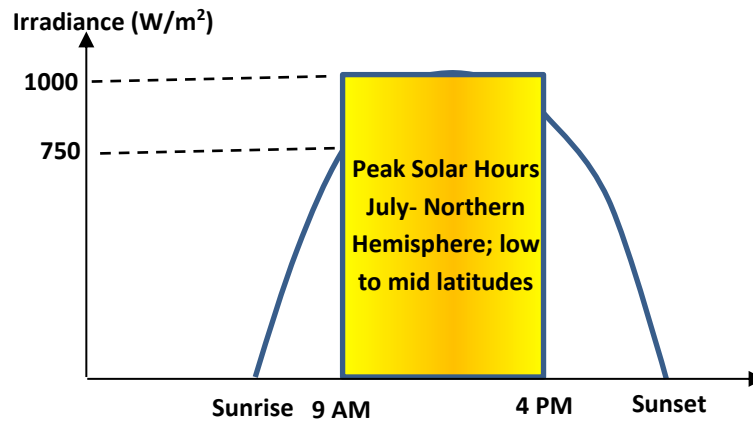
Larger particles, water droplets (clouds), and large particulate-laden airmasses absorb IR at dawn and dusk. For a  $Q_{IR}$  photon, the IR wavelength is 700 nm (430 THz) and longer, so air molecules other than  $CO_2$  do not absorb IR photons. For a  $Q_V$  photon, the visible wavelength is between 380 and 700 nm. This range is based on the adaptation of human eyes for sight, not nature's fundamental tenant. The  $Q_V$  photon passes through the atmosphere freely except for clouds and except for Rayleigh scattering in the blue part of the spectrum. The ozone  $O_3$  molecules absorb the  $Q_{UV}$  photon in the upper atmosphere. The  $Q_X$  photon emitted from the sun's corona, which is hotter than the sun's surface, is absorbed by the Earth's atmosphere. The  $Q_G$  photon was not emitted by the sun since it was produced in its core and became captive.

On average, the Earth reflects (albedo) about 29% [78] of the incident solar irradiance back to space. Collision cross sections between two essentially spherical particles A, B, with respective radii  $r_A$  and  $r_B$  are:

$$\sigma_{AB} = \pi(r_A + r_B)^2 \quad \text{Equation 67: Collision cross-section A-B}$$

# Photony Foundations

## 5. Peak Solar Energy Harvesting



**Figure 123: Solar Irradiance at Earth's Surface**

The solar energy available at Earth's surface after the atmospheric gauntlet is, on average, somewhere between 160 and 350  $\text{W/m}^2$ , so during 9 AM - 4 PM daylight (apart from dusk and dawn), it would range from 750 to 1000  $\text{W/m}^2$ . Of the solar energy reaching Earth's surface (AM1), 49.4% is infrared, 42.3% is visible, and about 8% is Ultraviolet. [79] Solar panel design must comprehend these energy percentages and associated frequencies and wavelengths to be optimal in converting to electrical and thermal energy. This process of harvesting solar energy is called hybrid conversion.

Roughly 10-15 percent of all 9-4 daylight energy available at the Earth's surface is diffuse. Raleigh Scattering, which makes the sky appear blue. The scattered light is undirected, and so it appears to be coming from any region of the sky. This light is called "diffuse" light. Diffuse light comes from regions of the sky other than where the sun is and is further diffused and reflected by terrestrial features. A large percentage of diffuse light energy is often lost (not converted) in typical PV panel fields.

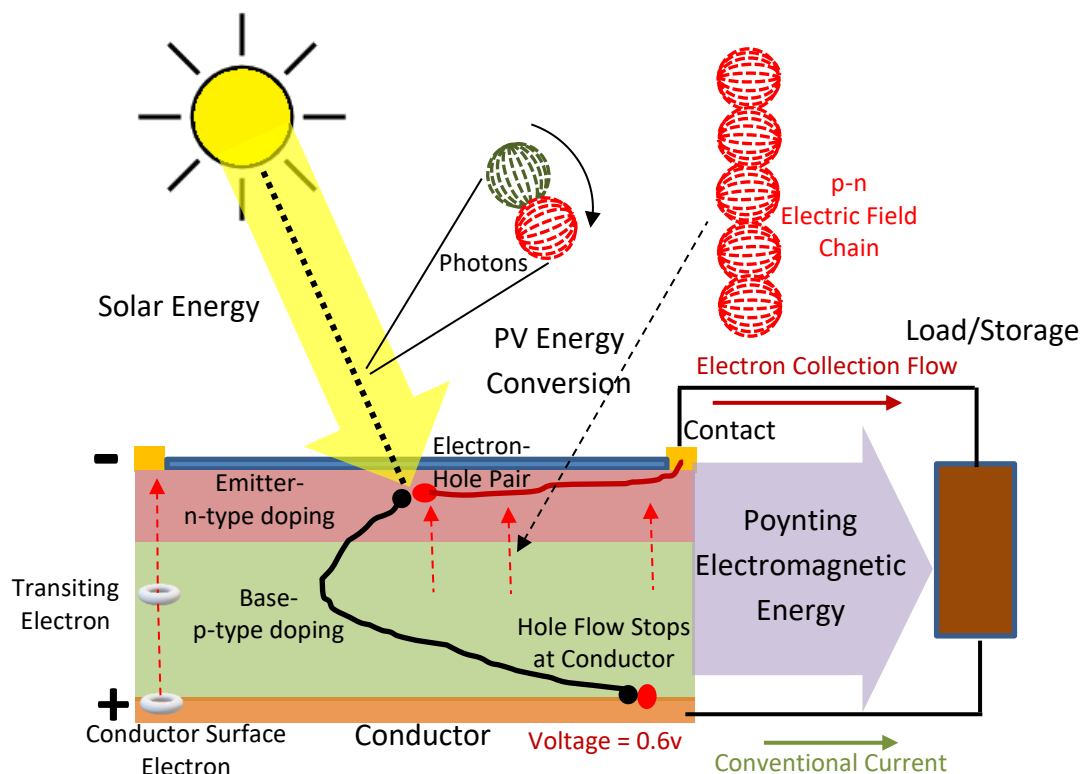
Photony provides a model for classical photon motion, including collisions. So, with Photony, we can better understand the absorption and scattering of classical photons. As the photons finally make it to solar panels, we can also use quantum mechanics, as first described by Richard Feynman [80] to determine the optimum thickness(es) of solar panel panes as a function of frequency to align with the AM1 spectrum.

# Bibliography

## 6. Solar Energy Conversion

Solar panels must be constructed uniquely to efficiently receive the solar energy through their transparent surface glass (panes), trap the energy, convert it to electrical energy using photovoltaic (PV) coating, and pass the received IR-driven solar thermal energy along with the energy lost in the photoelectric process of the PV through the PV coating into a tank wall. The tank contains a single- or two-phase coolant, transferring the converted thermal energy to a centralized processing and storage facility.

Aside from detecting photons as signals, photovoltaic conversion in solar panels is directed at harvesting solar energy.



**Figure 124: Silicon Solar Cell Cross-section During Conversion**

Each electron transports its “charge”  $e = 1.602176634 \times 10^{-19}$  coulombs of Photony elemental (charge) photons as it transduces the charge into electric field chains.

For example, a current of one amp for one hour is  $1 \text{ A} \cdot \text{h} = 36 \text{ Coulombs}$  of charge that can be delivered to a load or an energy storage device such as a battery. It would require  $2.24 \times 10^{20}$  electrons per hour from solar energy conversion by doped silicon to produce  $1 \text{ A} \cdot 0.6 \text{ v} \cdot \text{h} = 0.6 \text{ Wh}$  of energy or  $3600 \text{ J/Wh} \cdot 0.6 \text{ Wh} = 2160 \text{ joules}$ .

# Photony Foundations

## D. Hydro and Wind Energy

*"Men argue, Nature acts."*

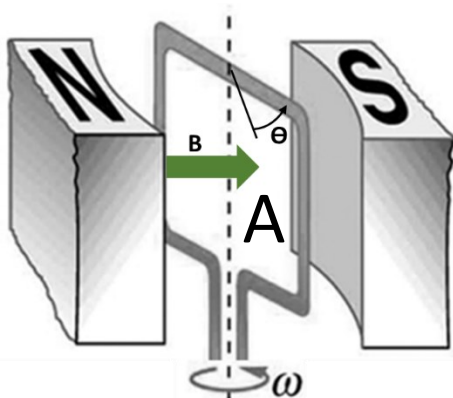
- Voltaire, 1764

Hydroelectric and wind (fluid) powered generation are combined in this section since they both use fluids to turn shafts that "in turn" produce electromagnetic energy. The shafts rotate when a moving fluid interacts with angled or curved rotor blades such as propellers or impellers with symmetrical sets of radial blades that produce torque on a shaft. The shaft torque is used to rotate magnetic rotors that induce currents in stators to produce typically AC power. The AC power is regulated and transformed to transmission voltage to supply the grid.

From a Photony perspective, the interesting parts of fluid power are the electromagnetic aspects which includes electron transduction and reduction of elemental charge photon chains comprising EM fields (of chains). These Photony EM processes provide the models and means by which fluid power is converted, transformed, and transmitted. The mechanical to electrical conversion, voltage and current transformation, and the transmission (originated as fluid energy) involves Photony EM processes.

Regarding fluid power conversion, the Photony EM processes are based on electrons as transducers of electric and magnetic chains from elemental charge photons and reducers of these same chains to backfill the flow of elemental charge photons in conductors as the AC current and voltage potentials reverse the flow magnitude and directions of the elemental charge photons in a conductor. The basis of Faraday's Law is a basic law of electromagnetism defining the interaction between a magnetic field with an electric circuit resulting in electromotive force or electromagnetic induction and highly coupled magnetic fields with mutual inductance which is the operating principle of generators and transformers respectively.

Faraday's Law of electromagnetic induction encompasses the principle that magnetic flux (magnetic elemental charge photon chain density) becomes electromotive force (emf) in a conductor loop with area  $A$  when the loop rotates at a rate  $\omega$ . We start with a rotatable conductive loop and a magnetic flux comprised of Photony magnetic elemental charge photon chains.



The magnetic field  $B$  in Tesla shown here is limited to the magnetic flux  $\Phi$  passing through the loop area  $A$ . So,  $\Phi = B \cdot A \cdot \cos(\theta)$  Weber with each Weber assigned to  $10^8$  lines. The force associated with  $\Phi$  is  $F = qvB$ . The period  $T = 2\pi/\omega$  and frequency  $f = 1/T = \omega/2\pi$

Figure 125: Generator Basics



## Bibliography

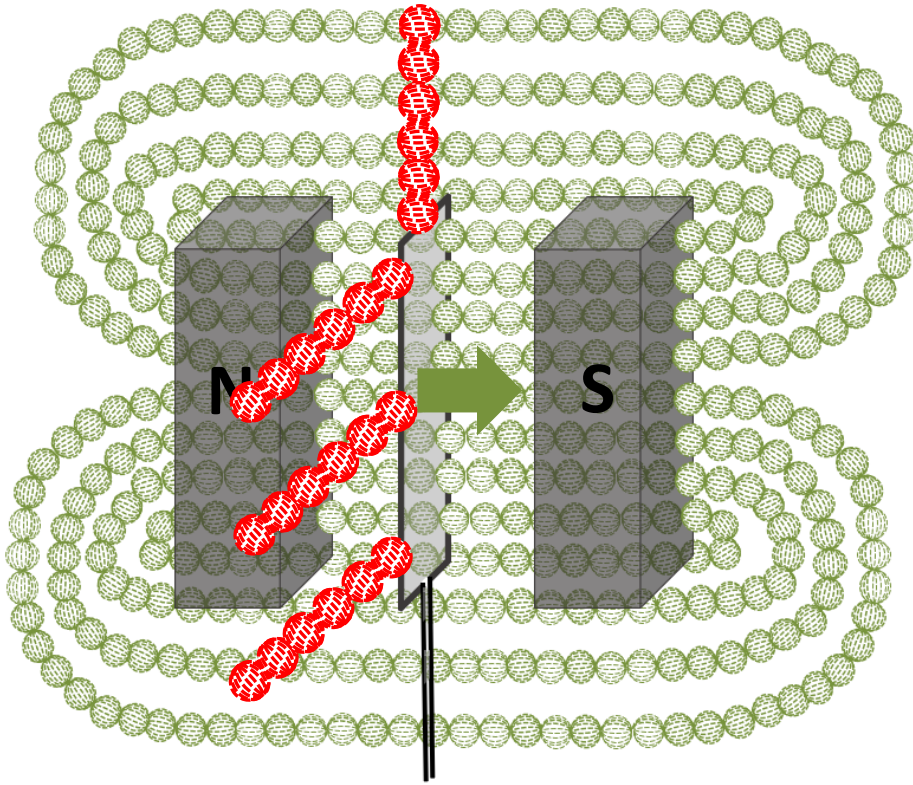


Figure 126: Photony Model of Generator Fields

The figure above shows some exemplary magnetic and induced electric elemental charge photon chains resulting from rotation of the conductor loop. As the conductor loop increases in angular velocity, the magnetic chains break and disassemble into dynamic elemental charge photons within the conductor and are converted to static elemental charge photons, distributed evenly on the surface of the wire loop conductor. If the wire loop is terminated with either a short or a load, the dynamic elemental photons will be absorbed by the conductor and load material as atomic/molecular vibrational energy and be radiated as thermal IR photons. We see here how induction works.





## Compendium Abstracts

Photony provides essential material for supplementary papers in the family of Photony books called the “Photony Compendium.” This foundational book provides a basis for in-depth Photony Theory, which is delegated to the more mathematically rigorous books of the Photony Compendium.

The Photony essential material in the Compendium books is based on three **Unification Classes** and Unification as follows:

- **Space, Gravity, Energy, Mass, and Time {SGEMT}**
- **Quanta, Particles, and Chains {QPC}**
- **Electricity and Magnetism, Fields, and Force {EMFF}**

Unification Classes Venn Diagram

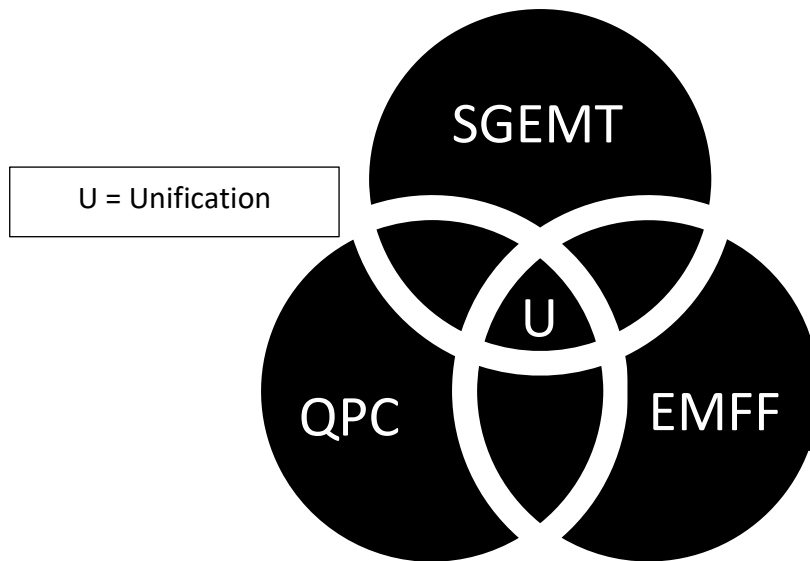


Figure 127: Photony Unification Diagram

- The 5<sup>th</sup> Photony Compendium Book, Vol. V is called **Unification**.

The Photony Compendium Books investigate the Unification Classes separately and with overlapping shared material.

# Bibliography

## **A. Space, Gravity, Energy, Mass, and Time, Compendium Vol. I**

### **SGEMT ABSTRACT:**

The SGEMT (Space, Gravity, Energy, Mass, Time) Unification Class in "Photony Foundations" proposes a novel perspective on the fundamental structure of the universe. The theory challenges the perception of smooth, curved motion, suggesting it's an averaging effect masking the movement of 'Photony energy filaments' within a quasi-static Planck Space network. This network, the 'fabric of space,' is modeled as a flexible, cellular structure that interacts with energy filaments, leading to spacetime compression and curvature, which manifests as gravity. The theory posits that Planck Space, predating energy/mass and time, provided the framework for the universe's existence and natural laws. This model offers a fresh perspective on the interplay of space, gravity, energy, mass, and time, potentially revolutionizing our understanding of the universe's fundamental building blocks.

## **B. Quanta, Particles, and Chains, Compendium Vol. II**

### **QPC ABSTRACT:**

The QPC (Quanta, Particles, and Chains) Unification Class in "Photony Foundations" explores the genesis and behavior of fundamental energy structures in the universe. It posits that Photony energy filaments, modeled as keyrings moving geodesically through Planck Space, are the basis of all quanta and particles. The theory differentiates between static and dynamic elemental charge photons, the former being the building blocks of matter and the latter representing energy quanta. The interaction of these photons with free electrons leads to the transduction of electric and magnetic chains, forming the basis of electromagnetic fields. The theory further elucidates the transition from electromagnetic waves to classical photons at a specific frequency threshold, highlighting the quantized nature of energy at higher frequencies. This model offers a novel perspective on the emergence and interactions of quanta, particles, and fields, providing a unified framework for understanding the fundamental building blocks of the universe.

## **C. Electricity and Magnetism, Fields, and Force, Compendium Vol. III**

### **EMFF ABSTRACT**

The EMFF (Electricity, Magnetism, Fields, and Force) Unification Class within the Photony framework redefines the nature of charge, current, and voltage. It proposes that static and dynamic elemental photons constitute charge, and that electric current arises from the flow of these dynamic elemental photons. The theory further posits that voltage is the accumulation of static elemental photons at surfaces. The interaction of these photons with free electrons leads to

# Photony Foundations

the transduction of electric and magnetic fields, visualized as chains of elemental photons. The model challenges conventional understanding by suggesting that electromagnetic energy flow occurs primarily outside of conductors, guided by these transduced chains according to the Poynting Vector Field. This offers a novel perspective on the relationship between circuits and fields, emphasizing the role of elemental photons in mediating electromagnetic phenomena.

## **D. Energy Science, Compendium Vol. IV**

### Energy Science ABSTRACT

Photony Theory provides a new paradigm for understanding energy science. It begins by exploring the fundamental building blocks of matter and energy, proposing that Photony energy filaments, existing at the Planck scale, are the basis of all energy and mass. The theory's mathematical framework and implications for energy density and different forms of energy are thoroughly discussed. The book delves into the nature of the electromagnetic spectrum, reinterpreting phenomena like electricity, magnetism, and the photoelectric effect through the lens of Photony Theory. It also examines the structure of matter, chemical bonding, and reactions, viewing them as rearrangements of Photony elemental charge photon chains. Much of the material focuses on reimagining energy science, exploring the Photony Theory's implications for materials science, photovoltaics, renewable energy harvesting, photosynthesis, energy conversion, storage and transmission, nuclear energy, and quantum thermodynamics.

## **E. Unification, Compendium Vol. V**

### Unification ABSTRACT

This 5th Photony Compendium book (Volume V) shows how the Unification Classes tie together to unite the Photony theory into a “big” theory of everything (Big ToE) while supporting as many classical and quantum theories as possible. Rather than refute the Standard Model, Photony provides the models, mechanics, and kinematics needed to bolster generally accepted theories. Nature has already united everything. We must comprehend and articulate nature’s unification work, including spacetime, quanta, particles, waves, forces, and fields. Scientists have identified classical physics, quantum physics, and relativity with little success in unifying these theories. Attempts to stitch together these disciplines using the Standard Model, String Theory, and Universal Field Theory have failed. No plausible fundamental models or mechanics have emerged that fit into Planck Space as a basis for Unification. Photony introduces a quasi-static Planck space and physical models (including fundamental quanta, particles, and chains) that unify its SGEMT, QPC, and EMFF classes with unification implications extending to quantum mechanics and relativity.

# Bibliography

## Appendix 1: Photony Postulates

### A. The Photony Spacetime Postulate

**Spacetime tells energy how to move; energy tells spacetime how to curve; curvature is what we perceive as mass.**

Explanation, illumination, and interpretation:

In this framework, spacetime is not a passive stage on which events unfold but an active participant, akin to a vast network of interconnected Planck cells. Energy, including mass, dictates how these Planck cells are arranged, causing them to compress and cluster around regions of high energy density. This localized compression is what we perceive as the curvature of spacetime. Much like a photon passing the Sun follows a curved path, objects in spacetime follow the "straightest possible" paths dictated by this curved geometry. Geodesics define how objects move under the influence of gravity.

The compression of spacetime due to energy density is not merely a mathematical abstraction; it is the essence of gravity based on permittivity. What we perceive as the force of gravity is, in fact, the result of objects following these curved paths within the compressed spacetime fabric. The concept of spacetime compression due to Photony filament interactions offers a deeper understanding of the underlying mechanisms behind gravity. This profound insight offered by general relativity has revolutionized our understanding of gravity. It reveals that gravity is not a force acting at a distance but rather an emergent property arising from the interaction between energy and spacetime geometry. It also highlights the interconnectedness of energy and spacetime, where the presence of one inevitably affects the other.

This intimate relationship between energy and spacetime curvature culminates in our perception of mass. In Photony Theory, mass isn't an inherent property of matter but a consequence of the spatial distortion induced by energy. The denser the concentration of Photony filaments within a region, the more spacetime compresses, and the greater the perceived mass.

To illuminate this concept, the Photony interpretation challenges the conventional notion of mass as a fundamental property. Instead, it posits mass as an emergent phenomenon resulting from the interplay between energy and the fabric of spacetime. This perspective aligns with Einstein's famous equation,  $E=mc^2$ , demonstrating the equivalence of mass and energy.

This revolutionary perspective on mass deepens our understanding of gravity and opens up new avenues for exploring the fundamental nature of matter and energy.

# Photony Foundations

## B. The Photony Motion/Spin Rule Postulate

**Energy filaments cannot transit laterally (broadside) through Planck Space without fracturing the filament or excessively compressing quasi-static Planck Space. Instead, the filament moves through the Planck Space network more like a worm with head-first motion.**

Explanation, illumination, and interpretation:

*Self-Propelled Dynamic Elemental Charge Photons:*

**Energy Filament Motion:** The dynamic elemental charge photons exhibit the helical motion of their energy filaments. This helical motion is not a simple rotation but a complex interplay of longitudinal and transverse components, allowing the filament to move forward through Planck Space while adhering to the PHOTONY Motion/Spin Rule.

**Propagation:** The helical motion of the energy filament propels the dynamic elemental charge photon through space. The energy filament does not rotate or vibrate laterally like a jump-rope (disallowed and counter to string theory), but for a static elemental (charge) photon, the energy filament scans spherically by longitudinally shifting geodesic paths (indexing) at or near a polar index zone of Planck cells. Indexing of multiple geodesic filament pathways traces and embodies a virtual shell (a spherical elemental force field), making it an elementary particle.

*Static Elemental Charge Photons:*

Unlike their dynamic counterparts, static elemental charge photons have energy filaments that exhibit primarily rotational motion. This rotation occurs within a localized region of Planck Space, effectively creating a stable, spherical structure.

The filament's indexed longitudinal rotational motion generates a spherical force field surrounding the particle. This force field interacts with other charged particles through minor low-energy collisions.

Photony relates to science by offering alternative explanations for established scientific principles. For example, it reinterprets Faraday's concept of lines of force and Maxwell's concept of tubes of force as chains of elemental charge photons. This provides a more tangible and quantifiable understanding of these abstract concepts.

Photony also introduces new concepts that could potentially expand our understanding of the Universe. For instance, it proposes that charge is not just a property of particles but is composed of elemental charge photons. This could have significant implications for our understanding of electricity, magnetism, and the fundamental forces of nature.

# Bibliography

## C. The Photony Rotational Motion Postulate

**Fundamentally, matter (mass) is energy in a state of rotational motion. This rotation manifests at various scales, from the Planck scale, where energy filaments within Planck space rotate and index to form elemental charge photons, to macroscopic scales. At the Planck Scale, the rotational motion of two interlinked elemental charge photons gives rise to classical photons. Adding c-spin to high-energy photons results in the rotational motion, thus embodying electrons and positrons (pair production) as toroidal energy shells wherein the rotating photon resides.**

Explanation, illumination, and interpretation:

This postulate provides a fundamental connection between energy, motion, and the existence of matter, offering a new perspective on the nature of particles and their interactions in the Universe. Matter, as observed in the Universe, is fundamentally energy in a state of rotational motion. This rotation manifests at various levels, from the Planck scale to macroscopic objects.

- **Planck Scale:** At the smallest levels, energy filaments within Planck space rotate geodesically and index longitudinally to form elemental charge photons, the basic building blocks of energy.
- **Classical Photons:** When two elemental charge photons interlink, they form a classical photon chainlet, exhibiting rotational motion as it moves forward.
- **Pair Production:** A collision of two high-energy photons (gamma rays) creates an electron and a positron. This process involves adding c-spin rotation to each photon, transforming their linear motion into rotational motion, and forming toroidal energy shells.
- **Atomic Scale-** Bound by magnetic chain loops, electrons rotate around a nucleus in distinct energy levels or shells.

The Photony model provides a comprehensive framework for understanding the behavior of electrons in both bound and free states. By considering the electron's interaction with elemental charge photons, we can gain deeper insights into the fundamental processes that govern atomic structure, chemical bonding, and the interaction of matter with electromagnetic fields.

Electrons exhibit rotational motion, whether bound to atoms in shells or in a free state as free electrons. Bound electrons rotate in shells around atomic nuclei, while free electrons rotate with x-spin (end-over-end) when excited. In the context of atomic structure, electrons predominantly exist in two states: bound or free. Bound electrons are associated with a specific atom, occupying discrete energy levels or shells around the nucleus. On the other hand, free electrons are not bound to any

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particular atom and can move freely within a material. The Photony model provides a unique perspective on electron behavior in both states. In this model, electrons are viewed as toroidal structures with a central vortex interacting with elemental charge photons. These elemental charge photons are transduced into chains, which form the basis of electric and magnetic fields.

In their bound state, electrons are held in specific shells around the nucleus by the electromagnetic force, which is mediated by the exchange of elemental charge photons between the electron and the nucleus. The specific configuration and motion determine the electron's energy level. When an electron absorbs energy, it can transition to a higher energy level, moving to an outer shell. This transition is accompanied by a change in its elemental photon chain configuration. Conversely, when an electron emits energy, it returns to a lower energy level, releasing a photon chainlet as it does so. This photon emission is the release of a chainlet of elemental charge photons from the electron's vortex.

The Photony model also offers a unique perspective on "free" electrons. While traditionally considered unbound, Photony suggests that even free electrons are in constant communication with a photon chainlet within their toroidal vortex. This provides a mechanism for the electron's motion and its interaction with magnetic chains, blurring the lines between bound and free states.

**Energy Density and Rotation:** The Photony Motion/Spin Rule establishes a direct link between a system's energy density and the rotational motion of its constituent photons. This rotational motion is the primary mechanism for localizing energy and creating the conditions necessary for matter to exist.

**Localized Energy and Curvature:** Effective mass curves the surrounding spacetime by collecting and concentrating energy through rotational motion. Since energy (filament) motion cannot be stopped, the only way it can become massive is through localized rotational motion, resulting in spacetime curvature. The Photony Motion/Spin Rule governs these motions in quasi-static Planck space, dictating that a system's energy density is directly proportional to the rate of rotation of its constituent elemental charge photons. For matter to exist, with its associated curvature of space and perceived mass, this energy density must be localized through rotational motion and maintained through a relatively low speed of collective motion within an inertial frame.

This curvature is what we perceive as mass, and the effect is what we perceive as gravity. The degree of curvature is directly proportional to the proximal energy density of the matter. Einstein rings are a direct consequence of the effect of energy density on spacetime, as explained by general relativity and further elaborated upon in Photony theory. In general relativity, mass (energy density) curves spacetime, and light (photons) follow these curved paths. When a massive (high energy density) object, such as a galaxy or black hole, lies between Earth and a distant light source, the light from the source is bent around the massive object due to its gravitational field. This bending of light can create a ring-like image of the source, known as an Einstein Ring.

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## D. The Photony Electron Passage Postulate

**Electrons may ride captive on transduced elemental charge photon chains extending through the toroidal cavity. The passage can be point-to-point on electric chains as with lightning and capacitors or looped on magnetic chains as in orbitals of atoms obeying the Pauli exclusion principle.**

Explanation, illumination, and interpretation:

The toroidal shape of electrons is functional and pleasing. Magnetic and electric fields, classically described as flux “lines,” are realized as Photony elemental photon chains, kinematic realizations of the flexible tubes hypothesized by Faraday and Maxwell.

Free electrons can transit while riding on a chain of electrically transduced (interlinked) elemental charge photons produced by that same electron. These photons are followed by other electrons that become aligned with the same electric chain at the source (e.g., the negative plate of a capacitor). This electron flow in space or in an insulator constitutes an electron displacement current.

DC Current: Faraday postulated that a current-carrying wire is immersed by a near-infinite series of circular concentric lines of force, which he termed the current's magnetic field. Faraday believed that the lines were tubes of force with real physical significance. Faraday proposed that each field line of force corresponds to a unit of electric charge in an electric field. Photony provides the structure and kinematics for electric chains functioning as lines or tubes of force. Lines of force, Maxwell supposed, were tubes of ether rotating on their axes.

Faraday's introduction of the “field” concept into physics is a significant contribution. Einstein described it as “the great change in physics” because it provided a common physical framework for electricity, magnetism, and optics. However, Faraday's lines of force and, more importantly, tubes were not recognized until several years later when James Clerk Maxwell confirmed remarkably similar positions regarding field lines and tubes.

Forces at a distance require extended Photony elemental photon chains as the flexible tubes of force first identified by Faraday and later advanced by Maxwell. The spin states of electrons, positrons, and possible protons can share a common elemental photon chain. By sharing, the chain can be transduced by one spinning particle, e.g., an electron. It can enter the toroidal cavity of a proximal electron or positron, followed by a separating motion of the two particles. The chain can be of almost any length while threaded through both particles. This means the particles are entangled. When one particle flips, the chain rotation reverses, and the remote particle will naturally flip, hence action at a distance. The issue of allowing influences to travel faster than the speed of



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light, disturbing to Einstein, whose theory of relativity prohibited any such superluminal propagation, is hereby resolved by the Photony entanglement theory.

Within the Photony Theory framework, electrons exhibit a remarkable ability to move along elemental charge photon chains that they themselves have transduced, or converted, from elemental photons. This movement can manifest in two distinct ways. First, electrons can travel point-to-point along electric chains, as observed in dramatic natural phenomena like lightning and in engineered systems such as capacitor plates. Second, electrons can also move in a looped fashion along magnetic chains. This behavior is fundamental to the structure of atomic orbitals and the Pauli exclusion principle, which dictates that no two electrons within an atom can occupy the same quantum state.

As described by Photony Theory, this dual behavior of electrons offers a compelling explanation for a wide range of electromagnetic phenomena. The point-to-point movement along electric chains elucidates the rapid charge transfer observed in lightning and the controlled current flow in capacitors. Meanwhile, the looped movement along magnetic chains provides a mechanistic model for electron orbitals in atoms, accounting for their quantized energy levels and angular momentum.

In essence, Photony Theory provides a unified framework for understanding the behavior of electrons in both macroscopic and microscopic systems. By positing that electrons can ride on transduced elemental charge photon chains, the theory offers a novel perspective on charge transfer, atomic structure, and the underlying principles of electromagnetism. This model aligns with established scientific principles and opens up new avenues for exploring and potentially manipulating electromagnetic phenomena at both fundamental and applied levels.

Free surface electrons attract static elemental charge photons that collect and distribute on conductive and non-conductive surfaces as the toroidal electron face cavity vortex pulls them inward and transduces them into electric elemental charge photon chains that extend outward from the opposing toroidal side. The electric chains are single-ended as they extend to an opposed polarity surface, meaning that the surface electrons face inward relative to the surface, making their spin compatible with approaching electric chains. So, the concept of polarity in capacitor plates for example, is based on the electron facial spin direction. This is also visualized by looking at the electron spin direction based on the surface view of each plate. At the positive plate, the electron facial cavities are hidden as it spins clockwise, and the chain is extruded toward the observer. Now, the observer reverses the viewing direction by  $180^\circ$  to view the negative plate as the electrons are spinning counter-clockwise and, therefore, compatible with receiving the spinning extruded chain. So, polarity is all about which direction the surface or near surface electrons are spinning. Another example is cloud-to-ground vs. ground-to-cloud lightning leaders and bolts.

Electric and magnetic elemental charge photon chains are built and are transmitted at the speed of light within a given medium by electrons. The electrons may ride on these chains at a relatively slow speed, whereas the fields comprised of these chains are transmitted at light speed.

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## E. The Photony Microwave/Photon Transition Frequency Postulate

The Photony microwave/photon transition frequency of 37.6 THz marks a pivotal point in the electromagnetic spectrum where the nature of energy transmission undergoes a fundamental shift. At this critical juncture, electromagnetic waves, characterized by oscillating electric and magnetic chains, transition into classical photon chainlets as discrete packets of energy. This transition is not merely a change in nomenclature but signifies a shift from energy being carried by continuous waves to being carried by quantized particles.

Explanation, illumination, and interpretation:

This frequency also coincides with the upper limit of the "light of life" range, essential for biological survival and water retention on planets. It marks the threshold where conductors become opaque to microwaves and begin to radiate thermally. Moreover, it aligns with when the thermal vibrational frequency of atoms and molecules emits infrared energy, highlighting its significance in the interplay between matter and energy.

The Photony microwave/photon transition frequency of 37.6 THz is a calculated value derived from fundamental constants and the properties of the electron. It represents the threshold at which the electron's axial spin rate, induced by alternating current, reaches a limit where it can no longer transduce electromagnetic chains with more than two interlinked elemental photons. This limitation results in the formation of classical photons, each composed of two entangled elemental charge photons, marking the transition from electromagnetic waves to quantized particles.

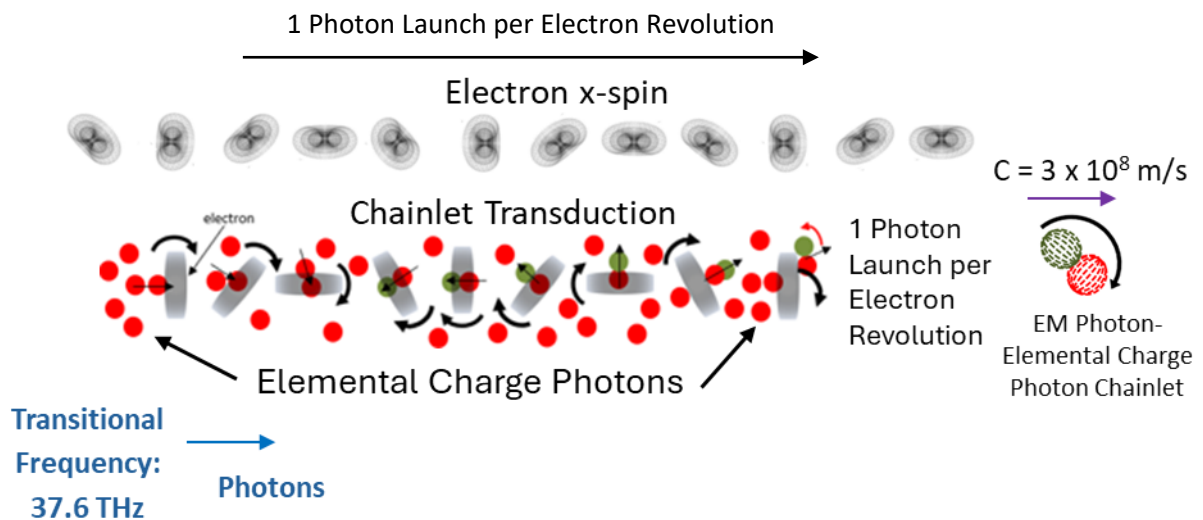
The calculation of this critical frequency involves several steps. First, the mass of the infrared photon ( $m_{ir}$ ) is determined by scaling down the proton mass ( $m_{Pr}$ ) using the proton-electron mass ratio ( $\mu$ ), which is approximately 1836.153. This yields a value of  $2.7 \times 10^{-37}$  kg for  $m_{ir}$ . Next, Planck's relation ( $E = h \cdot c / \lambda$ ) and Einstein's mass-energy equivalence ( $E = mc^2$ ) are combined to solve for the wavelength ( $\lambda$ ) of the transitional photon. Substituting the calculated  $m_{ir}$  and the values for Planck's constant ( $h$ ) and the speed of light ( $c$ ), we obtain a wavelength of approximately  $8 \times 10^{-6}$  meters. Finally, the frequency ( $f$ ) is calculated using the relationship  $f = c / \lambda$ , resulting in the Photony microwave/photon transition frequency of 37.6 THz. This calculated value holds profound implications for our understanding of energy transmission and the nature of light. It delineates the boundary between the continuous wave nature of electromagnetic radiation and the quantized particle nature of photons. Furthermore, it provides a theoretical basis for explaining the observed phenomena at the microwave/infrared interface, such as the cessation of the skin effect in conductors and the onset of thermal radiation. Therefore, the Photony microwave/photon transition frequency serves as a cornerstone in the Photony Theory, bridging the gap between classical and quantum descriptions of electromagnetism.

# Photony Foundations

## F. The Photony Single (Classical) Photon Launch Postulate

Above the 37.6 THz threshold, the rotational frequency of free electrons within a conductor exceeds their capacity to transduce extended chains of elemental charge photons. Consequently, per x-spin revolution, each electron launches (emits) a single classical photon, defined as a chainlet of two interlinked elemental charge photons. The energy of this emitted photon is then determined by its intrinsic spin frequency, marking a fundamental shift in energy transmission from amplitude-based waves to frequency-based photons.

Explanation, illumination, and interpretation:



In Photony Theory, a classical photon is modeled as a chainlet consisting of two transduction-interlinked elemental charge photons. The time constraint imposed by the electron's rotation rate at frequencies above 37.6 THz is a key factor in the Photony model.

In Photony Theory, the transduction of elemental charge photons into chains (and ultimately chainlets/photons) occurs within the toroidal cavity of the electron. This process takes a finite amount of time. As the frequency of the electromagnetic wave increases, the time available for transduction decreases.

Beyond the 37.6 THz threshold, the electron's rotation becomes too rapid to transduce longer chains of elemental photons. At these higher frequencies, the time it takes for elemental charge photons to enter the vortex of the facial cavity and be interlinked is insufficient.

Consequently, the electron can only transduce chainlets consisting of two interlinked elemental charge photons, which are then emitted as classical photons. This transition from longer chains to chainlets marks the spectrum shift from electromagnetic waves to classical photons.

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Above the 37.6 THz threshold, the rotational frequency of free electrons within a conductor, driven by alternating current, surpasses their capacity to transduce extended chains of elemental charge photons. This limitation arises due to the Photony Motion/Spin Rule, which governs the kinematics of energy filaments in quasi-static Planck space (spacetime). As the electron's rotational frequency increases, the time available for interlinking elemental photons into chains decreases. Consequently, each electron can only transduce a chainlet of two interlinked elemental charge photons per its axial spin (x-spin) revolution.

This transition from extended chains to chainlets marks a fundamental shift in the nature of energy transmission. Below 37.6 THz, energy is primarily carried by electromagnetic waves, characterized by the amplitude and phase of oscillating electric and magnetic fields. These fields are composed of chains of elemental charge photons transduced by the collective motion of electrons within the conductor. However, above the 37.6 THz threshold, the energy is carried by discrete packets called photons.

The energy of each photon is determined by its intrinsic spin frequency, as described by Planck's relation ( $E = hf$ ), where  $h$  is Planck's constant, and  $f$  is the photon's extrinsic spin frequency.

This shift from amplitude-based to frequency-based energy transmission has profound implications for our understanding of the electromagnetic spectrum. It explains the transition from radio waves and microwaves, characterized by their wavelengths and amplitudes, to infrared, visible, ultraviolet, and higher energy photons, characterized by their frequencies and energies. The 37.6 THz threshold, therefore, represents a critical boundary in the electromagnetic spectrum, where the continuous wave nature of electromagnetic radiation gives way to the quantized particle nature of photons.

**Quantum Mechanics:** This transition frequency supports the fundamental principle of quantum mechanics: quantization of energy. It demonstrates that at higher frequencies, energy is transmitted in discrete packets (photons), each with energy proportional to its frequency. This cornerstone of quantum theory has far-reaching implications for our understanding of the microscopic world.

**Electromagnetic Spectrum:** This transition provides a deeper understanding of the electromagnetic spectrum. It explains the shift from the continuous wave nature of radio waves and microwaves to the quantized nature of higher-frequency radiation like infrared, visible light, and ultraviolet. This has practical implications for technologies that utilize different parts of the spectrum, such as telecommunications, imaging, and spectroscopy.

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## G. The Photony Compression Factor ( $4\pi$ ) Postulate

The Photony compression factor,  $4\pi$ , is a fundamental constant in nature arising from the interaction between Photony energy filaments and the quasi-static Planck space network. This interaction distorts both the orthogonal structure of Planck space and the geodesic paths of the energy filaments, leading to a compression of space that manifests as gravity.

Explanation, illumination, and interpretation:

Planck Space is a structured (but flexible) portal interconnected network of orthogonal cellular intersections. In contrast, Photony energy filaments are predisposed to geodesic spin motion; natural stress exists on each as they coexist. Therefore, some compression results within Planck Space while the arc-shaped energy filaments favoring geodesic motion distort to attempt orthogonality.

The compression factor  $4\pi$  is a factor in all fundamental equations associated with spacetime. Sometimes not obvious but based on permeability, the compression factor (sometimes called the induction constant) is the basis of the origin and effects of energy in Planck space and many scientific equations. Furthermore, the Photony compression factor  $4\pi = 12.56637061436$  represents multiple significant digits (12.5664) of the permeability of free space,  $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$ .

The energy density of Photony filaments distorts space (and time), attracting other localized (or passing through) filaments. Filaments are forever swimming in Planck Space and naturally swim downhill relative to the spatial compression gradient. The effect of this compression phenomenon is what we call gravity and underlies Newton's Law of Gravitation.

The interaction of energy filaments and quasi-static Planck space is comparable to the interactive product of  $4 \cdot \pi$  whereby the "4" represents the attempt at non-distorted Manhattan Metric orthogonal pathways, and " $\pi$ " represents the energy filament's smooth, attempted/preferred and fully accommodated geodesic pathway shown in red in the above figure. The result of the interaction with Planck space is the distorted green pathway. The orthogonality of Planck space is also distorted, as shown in black. In the compression (interaction) factor  $4 \cdot \pi$ , the 4 is PI according to the Manhattan metric, and  $\pi$  is the classical  $\pi = 3.1415...$  Since both are affected by the interaction, the product of the two ( $4 \cdot \pi$ ) becomes theoretically fundamental.

The Photony compression factor,  $4\pi$ , is perhaps the most important of Photony discoveries since it comprehends the interaction between energy and quasi-static Planck space at the Planck Length scale. The interaction or product of  $4 \cdot \pi$  is prolific in scientific equations and may be fundamental to nature. As an energy density (a collective mass/spacetime compression) attempts to move through quasi-static Planck space, the energy filaments within the energy density each follow the Photony Motion/Spin Rule as they maneuver in Planck space. Each energy filament is distorted along with proximal Planck space accordingly.

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